

throughout CSF pathways.

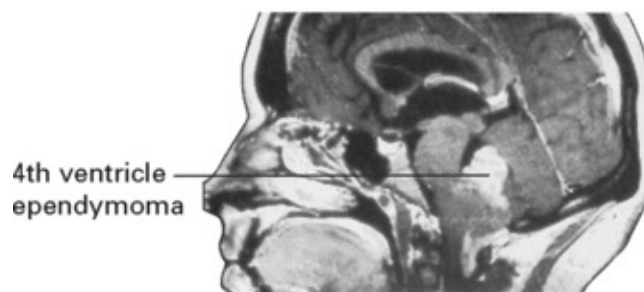
In the *4th ventricle*, ependymomas present with cerebellar signs or, more commonly, with signs and symptoms of raised intracranial pressure from CSF obstruction. *Vomiting* is often an early feature from direct brain stem involvement.

CT scanning shows an isodense mass, with or without calcification, lying within the 4th ventricle and usually enhancing with contrast. *MRI* more clearly delineates the anatomical relationships.

Management

The aim is complete operative removal, although infiltration of the floor of the 4th ventricle may prevent this. Most clinicians advise radiotherapy postoperatively, but its value is limited in the low grade tumours.

CSF metastases are treated by total neuraxis irradiation.



Prognosis

Despite relatively slow growth, results are often disappointing with 5-year survival ranging from 20–50%. Poor prognostic factors include incomplete resection and age < 2 years.

CHOROID PLEXUS PAPILLOMA

Rare, benign tumour with a granular surface and a gritty texture.

They develop from the choroid plexus – in the *4th ventricle* – *adults*, – in the *lateral ventricle* – *children*.

Malignant forms occasionally occur in children. Most patients present with hydrocephalus, either due to obstruction or to excessive CSF secretion from the tumour. CT scanning shows a hyperdense mass within the ventricular system. Operative removal gives good results.

COLLOID CYST OF THE THIRD VENTRICLE

A benign cyst, containing a mucoid fluid may arise from embryological remnants in the roof of the third ventricle. When of sufficient size (about 2 cm) it occludes CSF drainage from both lateral ventricles through the foramen of Munro.

Clinical features: Many patients exhibit no symptoms. In others, symptoms occur intermittently, possibly due to a ball-valve effect – headaches, episodes of loss of consciousness or even sudden death.

CT scan shows a small round mass of increased density, lying level with the foramen of Munro, causing lateral ventricular dilatation. The cyst wall will enhance following contrast on MRI. When *symptomatic*, operative removal through a transcallosal or transventricular approach carries relatively little risk. These cysts can be drained through a stereotactically placed needle or an endoscope, but with this treatment, recurrence almost inevitably occurs. In *asymptomatic* patients, the risk of sudden death is so small (4 deaths in 1800 patients in 5 years) operative treatment is rarely justified.

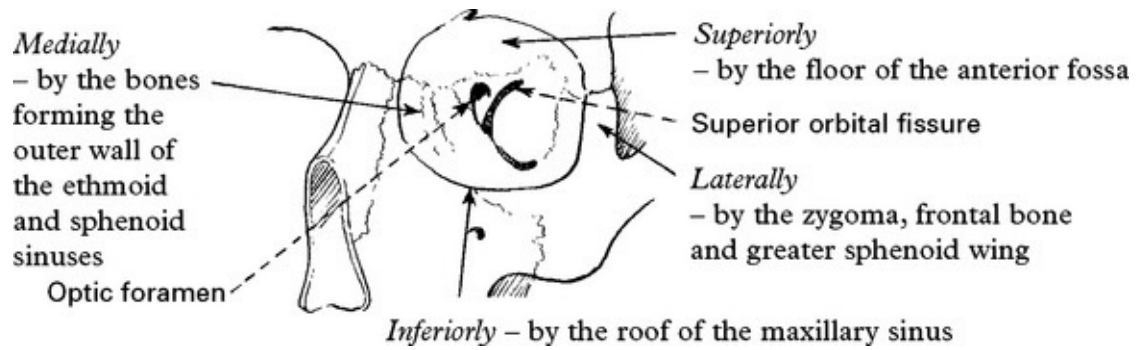


MENINGIOMA: rarely arises in the *lateral* ventricles. Often symptoms are mild and long standing. Operative removal only becomes necessary when symptoms and signs appear.

GERMINOMA } see Pineal region tumours, page 349.
TERATOMA }

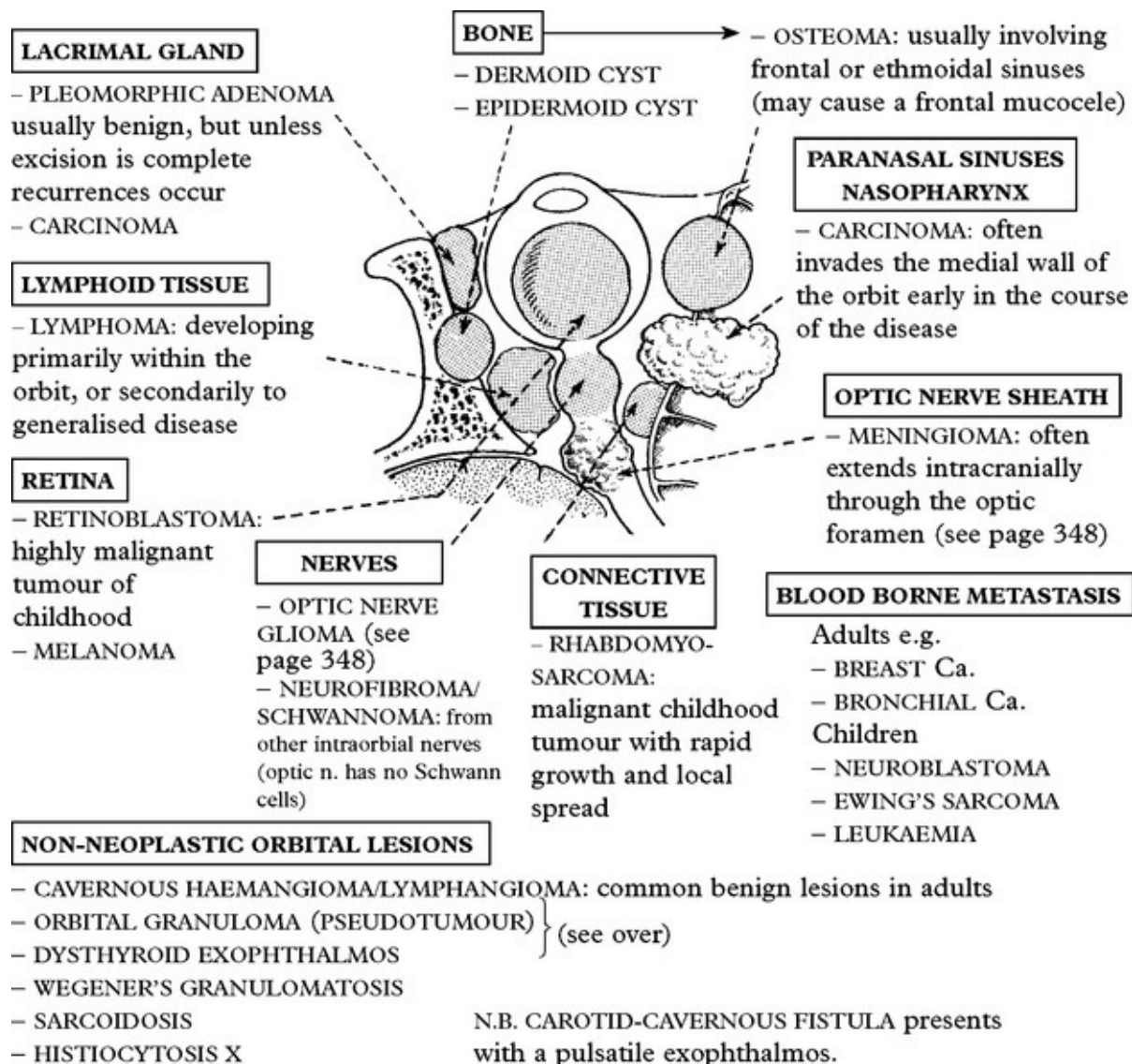
TUMOURS OF THE ORBIT

The orbital cavity is bounded –



PATHOLOGY

Tumours may arise from any of the structures lying within or around the orbit.



CLINICAL SYMPTOMS AND SIGNS

Orbital pain: prominent in rapidly growing malignant tumours, but also a characteristic feature of orbital granuloma and carotid-cavernous fistula.

Proptosis: forward displacement of the globe is a common feature, progressing gradually and painlessly over months or years (benign tumours) or rapidly (malignant lesions).

Lid swelling: may be pronounced in orbital granuloma, dysthyroid

exophthalmos or carotid-cavernous fistula.

Palpation: may reveal a mass causing globe or lid distortion – especially with lacrimal gland tumours or with a mucocele. *Pulsation* indicates a vascular lesion – carotid-cavernous fistula or arteriovenous malformation – listen for a bruit.

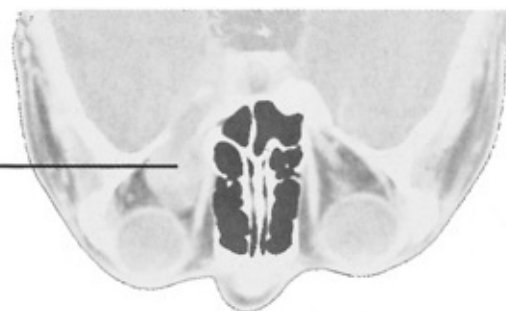
Eye movements: often limited for mechanical reasons, but if marked, may result from a dysthyroid ophthalmoplegia or from III, IV or VI nerve lesions in the orbital fissure (e.g. Tolosa Hunt syndrome) or cavernous sinus.

Visual acuity: may diminish due to direct involvement of the optic nerve or retina, or indirectly from occlusion of vascular structures.

INVESTIGATIONS

CT scan with a fast helical scanner is the investigation of choice for bone lesions. It will demonstrate the exact relationship of the lesion to surrounding structures and will show the presence of any intracranial extension.

Axial view showing an optic nerve glioma. Coronal views are of value in assessing the size of the optic nerve and extraocular muscles and the floor and roof of the orbit



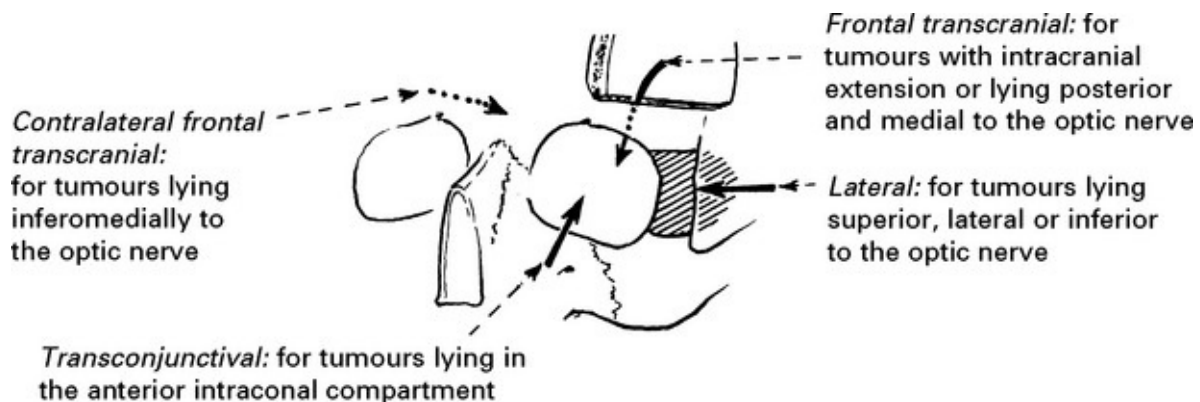
MRI shows the orbital anatomy in detail, but eye movements may cause artefacts.

MANAGEMENT

BENIGN tumours: require excision, but if visual loss would inevitably result, the clinician may adopt a conservative approach.

MALIGNANT tumours: require biopsy plus radiotherapy. Lymphomas may also benefit from chemotherapy. Occasionally localised lesions (e.g. carcinoma of the lacrimal gland) require radical resection.

Operative approach



NON-NEOPLASTIC ORBITAL LESIONS

ORBITAL GRANULOMA (pseudotumour)

Sudden onset of *orbital pain* with *lid oedema*, *proptosis* and *chemosis* due to a diffuse granulomatous infiltrate of lymphocytes and plasma cells involving multiple structures within the orbit.

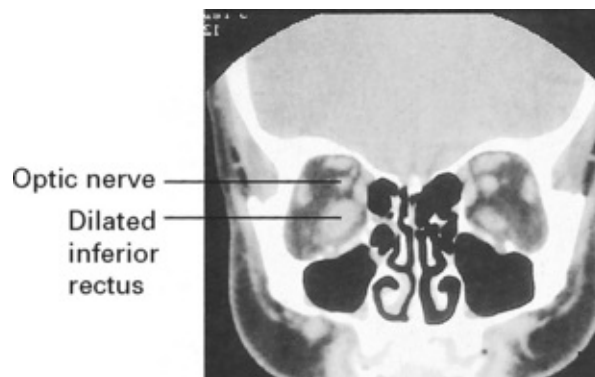
This condition usually occurs in middle age and seldom occurs bilaterally. CT scanning or MRI shows a diffuse orbital lesion, although one structure may be predominantly involved, e.g. optic nerve, extraocular muscles or the lacrimal gland. If diagnostic doubt remains, a biopsy is required. Most patients show a dramatic response to high dose *steroid therapy*. If symptoms persist, the lesion should respond well to radiotherapy.

DYSTHYROID EXOPHTHALMOS

The thyrotoxic patient with bilateral exophthalmos presents no diagnostic difficulty, but dysthyroid exophthalmos, with marked lid oedema, lid retraction and ophthalmoplegia *may occur unilaterally* without evidence of thyroid disease.

Coronal CT scanning establishes the diagnosis by demonstrating enlargement of the extraocular muscles – primarily the medial and inferior recti. MRI shows a similar appearance.

Circulating thyroid hormone levels are often normal. Thyroid releasing hormone stimulation or thyroid suppression tests may support the diagnosis.



Management

Steroids should help. A few patients require orbital decompression in an attempt to prevent corneal ulceration, papilloedema and blindness.

TUMOURS OF THE SKULL BASE

MALIGNANT

CARCINOMA

Carcinoma of the nasopharynx, paranasal sinuses or ear may extend intracranially either by direct erosion or through the skull foramina. It frequently penetrates the dura (in contrast to metastatic carcinoma of the spine) and may involve almost any cranial nerve. Symptoms of nasopharyngeal or sinus disease are often associated with facial pain and numbness. Spread to the CSF pathways leads to *carcinomatous meningitis* and may cause multiple cranial nerve palsies. *Skull X-rays, CT scan and MRI scan* will demonstrate a lesion involving the skull base. CT scanning most clearly shows the bone involvement. *Treatment* is usually restricted to retropharyngeal biopsy plus radiotherapy.

CHORDOMA

Rare tumours of notochordal cell rests arising predominantly in the sphenoido-occipital (clivus) and sacrococcygeal regions. Although growth begins in the midline, they often expand asymmetrically into the intracranial cavity. Chordomas may present at any age, but the incidence peaks in the 4th decade. They are locally invasive and rarely metastasise.

Clinical: most patients develop nasal obstruction. Cranial nerve palsies usually follow and depend on the exact tumour site.

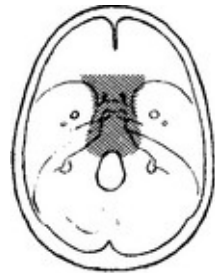
Skull X-ray shows a soft tissue mass with an osteolytic lesion of the sphenoid, basi-occiput or petrous apex.

CT scan confirms the presence of a partly calcified mass causing marked bone destruction and extending into the nasopharyngeal space.

MRI scan more clearly demonstrates the structural relationships.

Management: the tumour site usually prevents complete removal. Extensive debulking (often through the transoral route) is combined with

radiotherapy. Most patients die within 10 years of the initial presentation.



Chordoma – sites of intracranial lesion

BENIGN

GLOMUS JUGULARE TUMOUR (syn: chemodectoma, paraganglioma)

Rare tumour arising from chemoreceptor cells in the jugular bulb or from similar cells in the middle ear mucosa. This tumour extensively erodes the jugular foramen and petrous bone; many patients present with cranial nerve palsies, especially IX–XII. Chemodectomas occasionally arise at other sites and metastasis may occur.

X-ray and *CT scan* demonstrate an osteolytic lesion expanding the jugular foramen.

MRI shows the anatomical relationships.

Angiography reveals a vascular tumour, usually only filling from the external carotid artery, but occasionally from vertebral branches.

Management: tumour vascularity makes excision difficult. Selective embolisation may considerably reduce the operative risks or provide an alternative treatment. The value of radiotherapy is uncertain, but radiosurgery could be considered for tumours < 3 cm in size.

OSTEOMA

Rare tumours, usually occurring in the frontal sinus and eroding into the orbit, nasal cavity or anterior fossa. If sinus drainage becomes obstructed, a *mucocoele* develops, often with infected contents. These lesions require excision, either through an ethmoidal approach or through a frontal craniotomy.

INTRACRANIAL ABSCESS

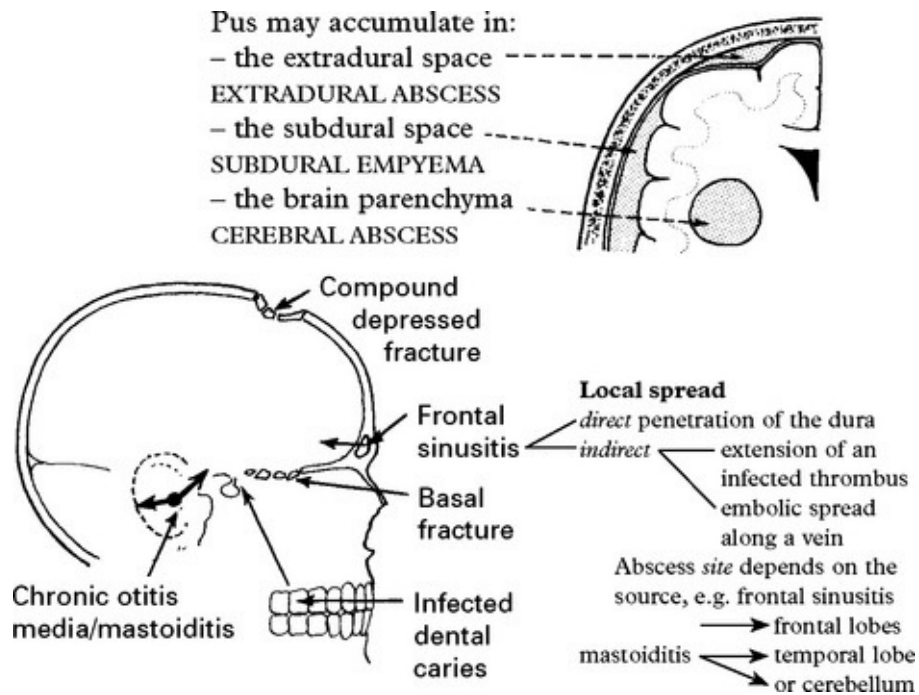
The advent of antibiotics and improved treatment of ear and sinus infection has led to a reduction in intracranial abscess formation but the incidence still lies at 2–3 patients per million per year.

CEREBRAL ABSCESS

Source of infection

Haematogenous spread

- Subacute bacterial endocarditis
- Congenital heart disease (especially right to left shunt)
- Bronchiectasis or pulmonary abscess



Organisms: Improved aerobic and anaerobic culture techniques now reveal the responsible organism in over 80% of patients. These depend on the source –

Middle ear – *Strep. milleri*, *Bacteroides fragilis*, *E. coli* } often
Proteus, *Strep. pneumoniae*. } mixed

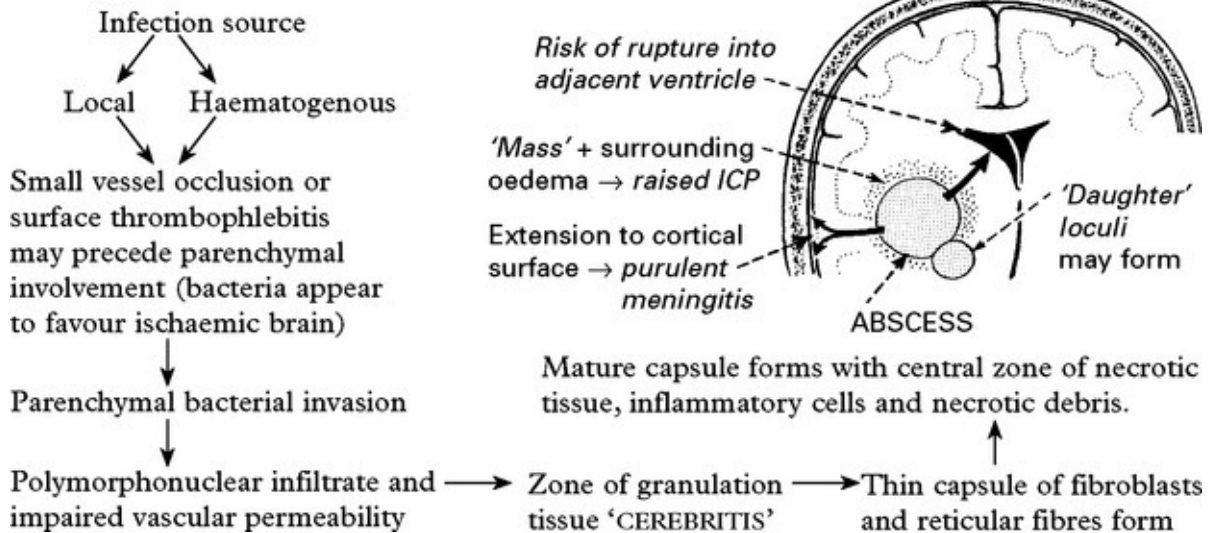
Sinus – *Strep. pneumoniae*, *Strep. milleri*.

Blood – *Strep. pneumoniae*, *Strep. milleri*, *Staph. aureus*.

Accidental or surgical trauma – *Staph. aureus*.

Immunocompromised patients – *Toxoplasma*, *Aspergillus*, *Candida*, *Nocardia* (see page 514)
– *Listeria* (microabscesses)

Pathogenesis



Clinical effects

Symptoms and signs usually develop over 2–3 weeks and progress. Occasionally the onset is more gradual, but features may develop acutely in the immunocompromised patient. Clinical features arise from:

- Toxicity – *pyrexia*, *malaise* (although systemic signs often absent).
- Raised intracranial pressure – *headache*, *vomiting* → *deterioration of conscious level*.
- Focal damage – *hemiparesis*, *dysphasia*, *ataxia*, *nystagmus* – *epilepsy* – partial or generalised, occurring in over 30%
- Infection source – *tenderness over mastoid* or *sinuses*, *discharging ear*.
bacterial endocarditis – *cardiac murmurs*, *petechiae*, *splenomegaly*.

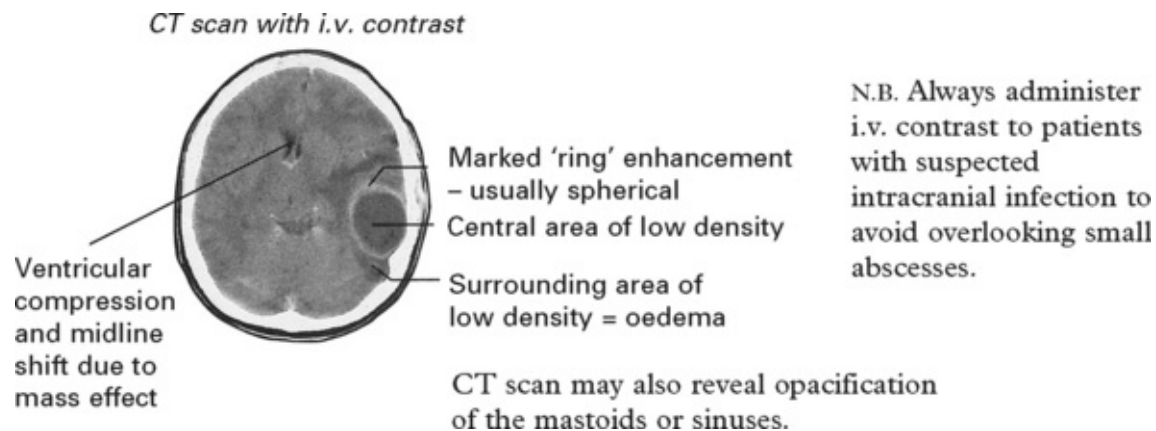
– Neck stiffness due to coexistent meningitis or tonsillar herniation occurs in 25%.

N.B. Beware attributing patient's deteriorating clinical state to the primary condition, e.g. otitis media, thus delaying essential investigations.

Investigations

X-rays of the sinuses and mastoids: opacities indicate infection.

CT scan: in the stage of 'cerebritis' the CT scan may appear normal or only show an area of low density. As the abscess progresses, a characteristic appearance emerges:



If abscesses occur at multiple sites, suspect a haematogenous source.

MRI: will more readily detect the 'cerebritic' stage, but does not distinguish infection from other pathologies.

Lumbar puncture is contraindicated in the presence of a suspected mass lesion, but if CSF is obtained inadvertently, this will show ↑ protein e.g. 1 g/l, ↑ white cell count (several hundred/ml) – polymorphs or lymphocytes. The Gram stain is occasionally positive.

Peripheral blood – may show ↑ ESR, leucocytosis. Blood culture is

positive in 10%.

Management:

1. Antibiotics

Commence i.v. antibiotics on establishing the diagnosis (prior to determining the responsible organism and its sensitivities). Antibiotics are selected on an empirical basis depending on the likely source of the infection and their ability to cross the blood–brain barrier and to achieve therapeutic concentrations in intracranial pus.

Use combined therapy: (note adult doses indicated)

– CEFTRIAXONE	i.v. 3–4 g/day
– METRONIDAZOLE	i.v. 500 mg tds
for a <i>middle ear source</i>	
+ AMOXICILLIN	i.v. 2 g 4 hourly
if <i>endocarditis or congenital heart disease</i>	
+ BENZYL PENICILLIN	i.v. 1.8–2.4 g 6 hourly

If a *penetrating trauma source*

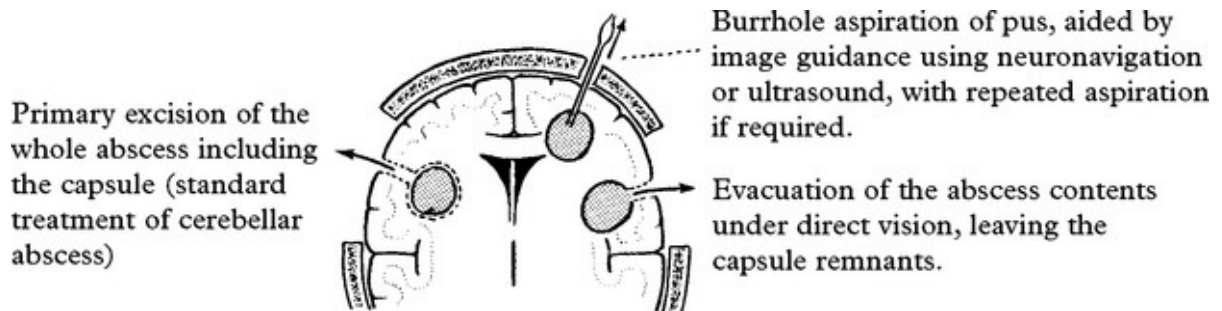
– FLUCLOXACILLIN	i.v. 2 g 4 hourly
± GENTAMICIN	i.v. 5 mg/kg/day (+ monitor levels)

In immunocompromised patients – see [page 514](#).

Later determination of the organism and its sensitivities permits alteration to more specific drugs. Intravenous antibiotics should continue for 2–3 weeks followed by oral medication for a further 3–4 weeks.

2. Abscess drainage

Various methods exist:



Burr hole aspiration is simple and relatively safe. Persistent reaccumulation of pus despite repeated aspiration requires secondary excision. Primary excision removes the abscess in a single procedure, but carries the risk of damage to surrounding brain tissue. Open evacuation of the abscess contents requires a craniotomy, but minimises damage to surrounding brain.

3. Treatment of the infection site

Mastoiditis or sinusitis requires prompt operative treatment, otherwise this acts as a persistent source of infection.

Steroids help reduce associated oedema but they may also reduce antibiotic penetration and impede formation of the abscess capsule. Their value in management remains controversial.

Conservative management: In some situations the risks of operative intervention outweigh its benefits. In those patients, treatment depends on i.v. antibiotics.

Indications:

- small deep abscesses, e.g. thalamic (although stereotactic aspiration

may help).

- multiple abscesses.
- early ‘cerebritic’ stage.

Prognosis

The use of CT scanning in the diagnosis and management of intracranial abscesses and the recognition and treatment of pathogenic anaerobic organisms have led to a reduction in the mortality rate from 40% to 10%. In survivors, focal deficits usually improve dramatically with time. Persistent seizures occur in 50%.

SUBDURAL EMPYEMA

Subdural empyema occurs far less frequently than intracerebral abscess formation. Infection usually spreads from infected sinuses or mastoids, but may arise from any of the aforementioned sources. The responsible organism is usually *Strep. pneumoniae*, *Strep. milleri* or *Staph. aureus*. Clinical features match those of intracerebral abscess but since rapid extension occurs across the subdural space, overwhelming symptoms often develop suddenly. Seizures occur in 70% at onset.

CT scan shows a low density extracerebral collection with mass effect, often with enhancement on the cortical surface; occasionally isodense lesions make identification difficult.

Management: Intravenous antibiotic treatment is combined with evacuation of pus either through multiple burr holes or a craniotomy flap. Despite active treatment, the mortality rate still runs at approximately 20%.

GRANULOMA

TUBERCULOMA

Although tuberculomas still constitute an important cause of mass lesions in underdeveloped countries (20% in India), they are now rare in Britain. The lesions may be single or multiple. They often lie in the cerebellum, especially in children.

Clinical features are those of any intracranial mass; alternatively tuberculoma may present in conjunction with tuberculous meningitis.

CT scan clearly demonstrates an enhancing lesion – but this often resembles astrocytoma or metastasis; tuberculomas have no distinguishing features. *MRI* is even more sensitive and may show additional lesions.

Other investigations: ESR, chest X-ray often fail to confirm the diagnosis. A Mantoux (PPD) test is usually positive but a negative test does not eliminate the diagnosis.

Management: When tuberculoma is suspected, a trial of antituberculous therapy is worthwhile. Follow up CT scans should show a reduction in the lesion size. Other patients require an exploratory operation and biopsy followed by long-term drug treatment.

SARCOIDOSIS

Sarcoidosis is a multisystem disease process of unknown cause whose pathogenesis involves formation of an inflammatory lesion known as a granuloma. Nervous system involvement occurs in 8% and may dominate the presentation.

When sarcoid infiltrates the central nervous system it usually

involves the meninges. In some patients mass lesions may arise from the dura, but more commonly signs and symptoms relate to an adhesive arachnoiditis involving the skull base, cranial nerves and pituitary stalk. Mass lesions may occasionally arise within the brain and spinal cord without obvious meningeal involvement.

Investigation: MRI (T1 weighted) shows either a hyperintense mass or multiple periventricular foci. The use of gadolinium and FLAIR (fluid-attenuated inversion recovery) increases the sensitivity of MRI. A definitive diagnosis is based on clinical and radiological evidence of multisystem disease confirmed by characteristic histology.

The diagnosis is often elusive and suggested by clinical presentation supported by some of the following.

- elevated serum and CSF angiotensin converting enzyme (ACE),
- elevated serum immunoglobulins,
- elevated serum calcium,
- elevated CSF cell count (monocytes), IgG, Ig index, and presence of oligoclonal bands.

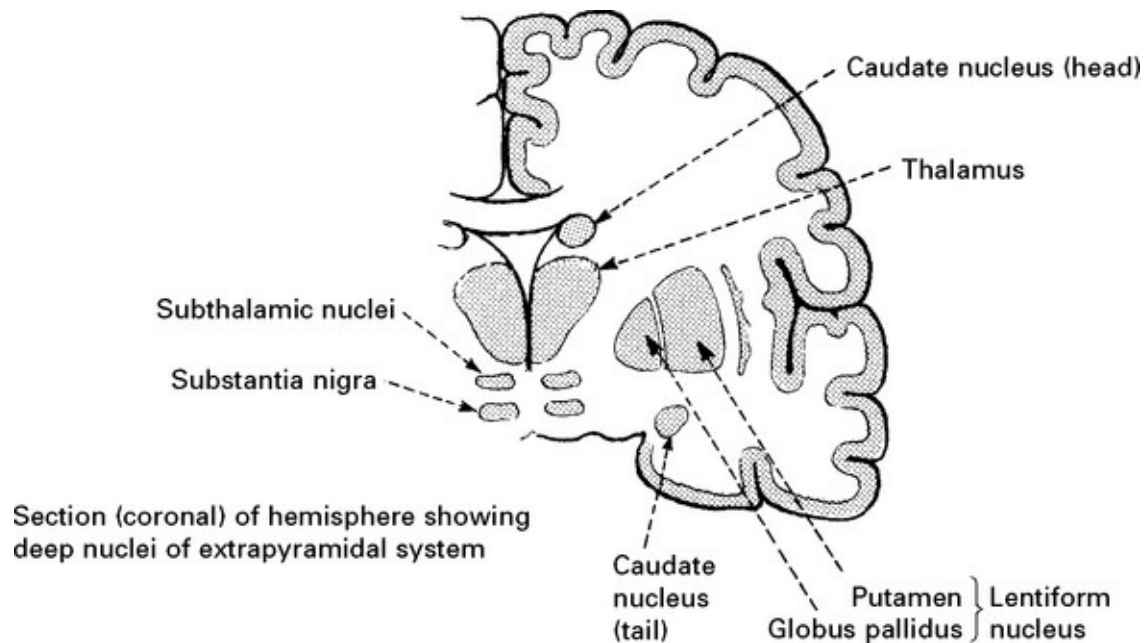
Management: Immunosuppression with corticosteroids is usually indicated and long-term therapy required. In exacerbation, intravenous pulsed methylprednisolone is used. Success in resistant cases is reported with each of the following – azathioprine, cyclophosphamide, methotrexate, cyclosporin or irradiation.

MOVEMENT DISORDERS – EXTRAPYRAMIDAL SYSTEM

The control of voluntary movement is effected by the interaction of the

pyramidal, cerebellar and extrapyramidal systems interconnecting with each other as well as projecting to the anterior horn region or cranial nerve motor nuclei.

The extrapyramidal system consists of paired subcortical masses or nuclei of grey matter **basal ganglia**.



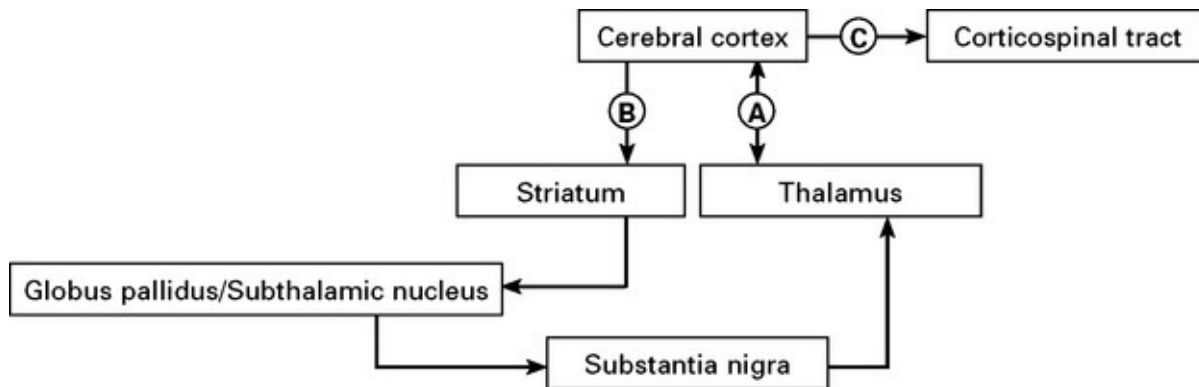
The caudate nucleus and putamen are collectively referred to as the STRIATUM.

Interconnections of the deep nuclei

The connections between components of the extrapyramidal system and other parts of the brain are complex. However, certain simple observations can be made:

- Ⓐ The thalamus plays a vital role in projecting information from the basal ganglia to the motor cortex and back
- Ⓑ The cortex projects through the striatum to other basal ganglia

© The final common pathway for basal ganglia motor function is the corticospinal or pyramidal tract



NEUROPHARMACOLOGY

The observation that drugs such as reserpine and phenothiazines regularly produce extrapyramidal syndromes has clarified the neurochemical basis of movement disorders and delineated the role of neurotransmitters.

Neurotransmitter substances

are synthesised and stored presynaptically. When released by an appropriate stimulus they cross the synaptic gap and combine with specific receptors of the postsynaptic cell,

e.g. – acetylcholine – serotonin
 – dopamine – glutamate
 – γ -aminobutyric acid

Acetylcholine

- Synthesised by small striatal cells
- Greatest concentration in striatum
- Excitatory effect.

Neuromodulator substances

diminish or enhance the effects of neurotransmitters in the basal ganglia,

e.g. – substance P
 – enkephalin
 – cholecystokinin
 – somatostatin.

Dopamine

- Synthesised by cells of substantia nigra (pars compacta) and nigral projections in striatum.
- Greatest concentration in substantia nigra.
- Inhibiting effect.

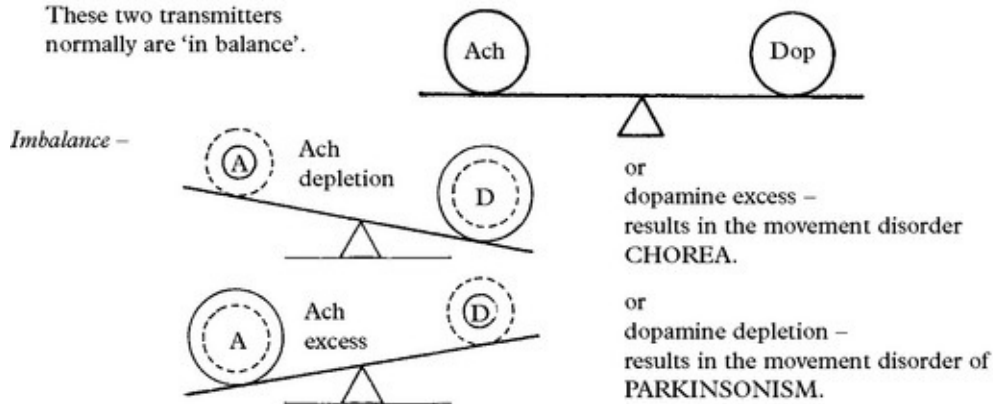
Acetylcholine

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Dopamine

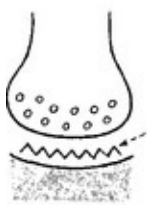
- Synthesised by cells of substantia nigra (pars compacta) and nigral projections in striatum.
- Greatest concentration in substantia nigra.
- Inhibiting effect.

These two transmitters normally are 'in balance'.



γ -Aminobutyric acid (GABA) is synthesised from glutamate in the striatum and globus pallidus. It has inhibitory actions and deficiency is associated with Huntington's disease.

Drugs may produce movement disorders by interfering with neurotransmission in the following ways:



1. – By reducing transmitter presynaptically
e.g. tetraabenazine reduces dopamine.
 2. – By blocking the receptor site postsynaptically
e.g. phenothiazines block dopamine receptors.
- Both reduce effective dopamine and create a relative excess of acetylcholine
- ↓
- Parkinsonism

CLINICAL FEATURES

The effects of disease of the extrapyramidal system on movement can be regarded as *negative* (hypokinetic) and *positive* (hyperkinetic).

Negative features

Bradykinesia: - a loss or slowness of voluntary movement.

A major feature of Parkinson's disease and produces:

- reduced facial expression (mask-like)
- reduced blinking
- reduced adjustments of posture when seated.

When agitated the patient will move swiftly – 'kinesia paradoxa'.

Postural disturbance: most commonly seen in Parkinson's disease.

Flexion of limbs and trunk is associated with a failure to make quick postural or 'righting' adjustments to correct imbalance. The patient falls whilst turning or if pushed.

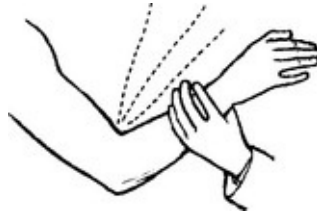
Positive features

Involuntary movements:

- tremor
- chorea (irregular, repetitive, jerking movements).
- athetosis (irregular, repetitive, writhing movements).
- dystonia (slow, sustained, abnormal movement).
- ballismus (explosive, violent movement).
- myoclonus (shock-like jerks).

Chorea and athetosis may merge into one another – choreoathetosis.

Rigidity



Stiffness felt by the examiner when passively moving a limb. This 'resistance' is present to the same degree throughout the full range of movement, affecting flexor and extensor muscle groups equally and is described as PLASTIC or LEAD PIPE rigidity. When tremor is superimposed upon rigidity it produces a COGWHEELING quality.

In Parkinson's disease both positive features, *e.g.* tremor, and negative features, *e.g.* bradykinesia, occur.

In Huntington's disease positive features, *e.g.* chorea, predominate.

PARKINSON'S DISEASE

Described by James Parkinson (1817) in 'An essay on the shaking palsy'.

Recognised as an extrapyramidal disorder by Kinnier Wilson (1912).

Annual incidence: 20 per 100 000. Prevalence: 190 per 100 000.

Sex incidence: male:female – 3:2

Age of onset: 50 years upwards. Incidence peaks in mid-70s then declines.

Familial incidence occurs in 5%.

AETIOLOGY

The cause(s) of Parkinson's disease is unknown. Gene mutations have been identified in young onset and familial cases (synuclein, parkin and LRRK2).

The observation that 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine

(MPTP), a meperidine analogue derived during illicit drug production, produces Parkinson's disease in humans and animals has resulted in increased interest in the role of toxins and an animal model for developing new treatments.

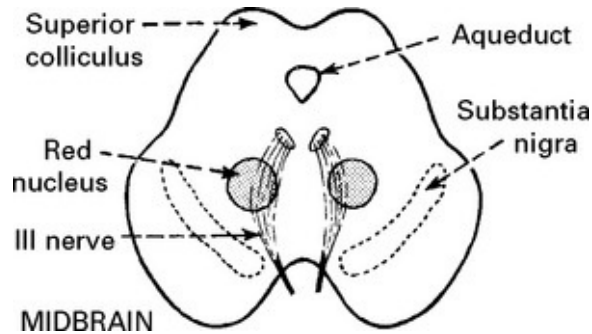
Parkinsonian features may be present in many disorders and are not always treatment (L Dopa) responsive. These disorders usually share features of slowness and rigidity (akinetic rigid syndromes).

Parkinson's disease	Mimics	
		– Multiple system atrophy (MSA) – Progressive supranuclear palsy (PSP) – Corticobasal ganglionic degeneration (CBD) – Diffuse Lewy body disease (DLBD)

Secondary Parkinsonism

- Drug induced (dopamine receptor blockers-antipsychotics/antiemetics; sodium valproate)
- Post traumatic (pugilist's encephalopathy)
- Vascular disease (small vessel multi-infarct state)
- Infectious (post encephalitic/prion disease/HIV)
- Miscellaneous: hydrocephalus/parathyroid/paraneoplastic

PATHOLOGY of idiopathic Parkinson's disease



The substantia nigra contains pigmented cells (neuromelanin) which give it a characteristic 'black' appearance (macroscopic). These cells are lost in Parkinson's disease and the substantia nigra becomes pale.

Remaining cells contain atypical eosinophilic inclusions in the cytoplasm – *Lewy bodies* – although these are not specific to Parkinson's disease. Lewy bodies may be found in the cerebral cortex especially when dementia is present (diffuse Lewy body disease).

Changes are seen in other basal nuclei – striatum and globus pallidus.

Radiolabelled ligand studies have identified two dopamine receptors on striatal cell membranes – D₁ – D₂ receptors.

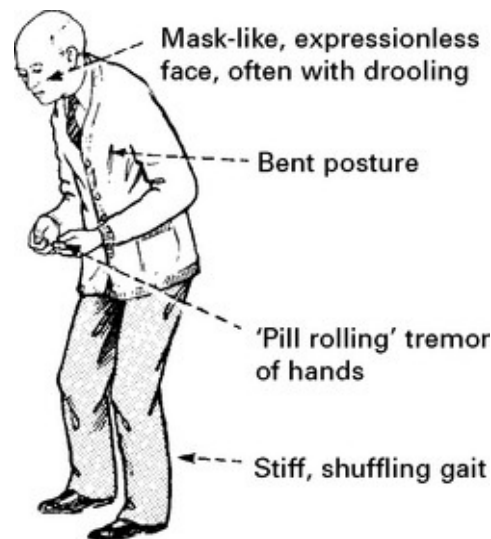
CLINICAL FEATURES

Initial symptoms are vague, the patient often complains of aches and pains.

A coarse TREMOR at a rate of 4–7 Hz usually develops early in the disease. It begins unilaterally in the upper limbs and eventually spreads to all four limbs. The tremor is often '*pill rolling*', the thumb moving rhythmically backwards and forwards on the palm of the hand. It occurs at rest, improves with movement and disappears during sleep.

RIGIDITY is detected by examination. It predominates in the flexor muscles of the neck, trunk and limbs and results in the typical '*flexed*

posture'.



BRADYKINESIA: This slowness or paucity of movement affects facial muscles of expression (mask-like appearance) as well as muscles of mastication, speech, voluntary swallowing and muscles of the trunk and limbs. Dysarthria, dysphagia and a slow deliberate gait with little associated movement (e.g. arm swinging) result.

Tremor, rigidity and bradykinesia deteriorate simultaneously, affecting every aspect of the patient's life:

Handwriting reduces in size.

The gait becomes shuffling and festinant (small rapid steps to 'keep up with' the centre of gravity) and the posture more flexed.

Rising from a chair becomes laborious with progressive difficulty in initiating lower limb movement from a stationary position.

Eye movements may be affected with loss of ocular convergence and upward gaze.

Excessive sweating and greasy skin (seborrhoea) can be troublesome.

Depression occurs in about 50%.

As the disease progresses the frequency of drug-induced confusional states and dementia increases, with 80% developing dementia after 20 years of disease (if they survive).

Autonomic features occur – postural hypotension, constipation.

REM sleep behaviour disorder – where patient acts out dreams and may hurt themselves or their sleep partner. May precede onset of motor symptoms.

Time of onset is mid-late fifties with increasing incidence with increasing age. Juvenile presentation can occur, when presentation and disease progression is often atypical; a genetic basis is more often found.

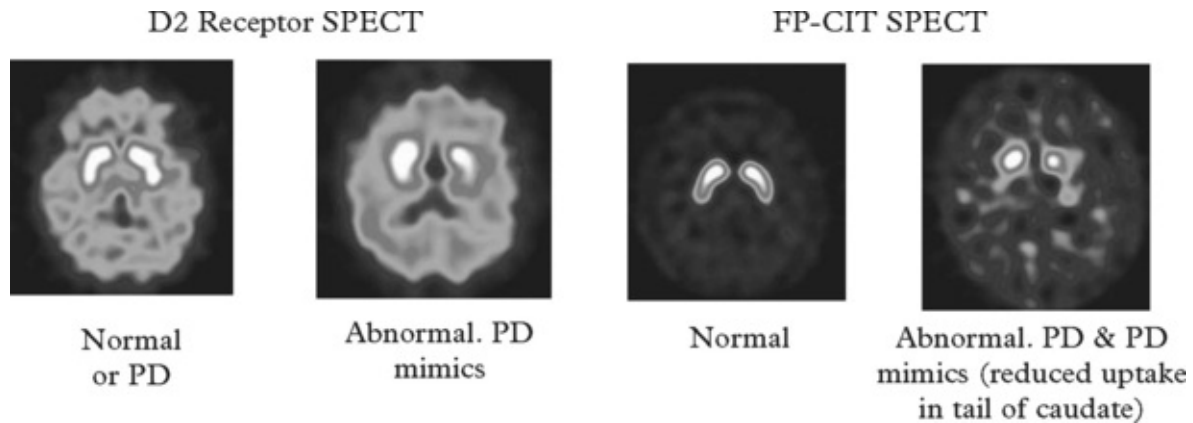
DIAGNOSIS

The diagnosis of PD in the early stages is difficult. Post-mortem data from the London Brain Bank shows this to be incorrect in 25% of those diagnosed in life.

New tremor in middle age causes particular difficulty – senile/essential & metabolic tremor is generally absent at rest and worsened by voluntary movement.

The diagnostic use of a L-dopa or dopamine agonist (apomorphine) challenge has declined due to concerns that it may increase the risk of subsequent drug induced dyskinesia.

Functional imaging (SPECT & PET) should improve diagnostic accuracy and ensure that persons with conditions unresponsive to treatments (PD mimics) are not unnecessarily exposed to them.



The ligand I-IBZM demonstrates the degree of D2 receptor binding. This is normal in PD but reduced in its mimics (MSA/PSP)

The ligand FP-CIT demonstrates the integrity of the presynaptic dopamine terminals. These are normal in essential tremor and reduced in PD and its mimics (MSA/PSP).

PD MIMICS

Multiple system atrophy occurs in two forms, a relatively symmetrical extrapyramidal syndrome associated with autonomic failure, usually postural hypotension and bladder symptoms, and a cerebellar syndrome with bilateral upper motor neuron signs. Both progress over 5–10 years.

Progressive supranuclear palsy (PSP) is characterised by gaze palsies, extrapyramidal features, axial dystonia (truncal dystonia), progressive upper motor neuron syndrome and dementia. Onset in the 5th to 6th decade. The key feature is the supranuclear gaze palsy: downward eye movement is impaired followed by all other voluntary eye movement which can be overcome by doll's head manoeuvre (a supranuclear palsy). Lid retraction is common. Levodopa gives disappointing results. Progression is relentless with death in 3–7 years.

Vascular Parkinsonism usually presents with gait disturbance with step wise deterioration. It tends to predominantly involve the lower

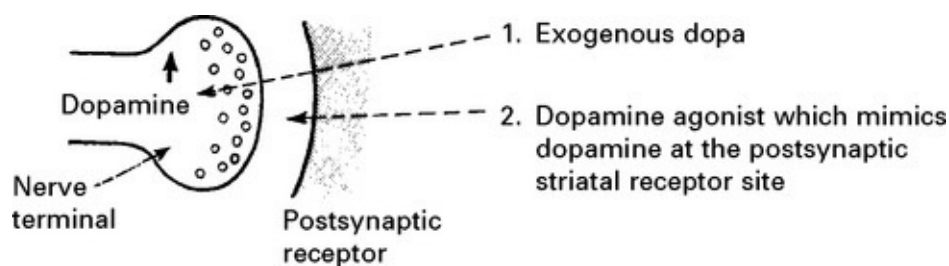
limbs and have a partial response to L-dopa. Brain imaging is helpful in diagnosis.

Wilson's disease (see [page 373](#))

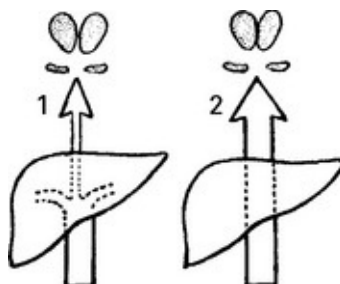
Corticobasal degeneration (CBD) is rare and presents with an asymmetric akinetic-rigid syndrome associated with marked dyspraxia, myoclonus and dementia. Patients may have an 'alien hand', where the hand moves purposely without conscious control. There are no specific treatments.

TREATMENT is symptomatic and does not halt the pathological process. No agents have yet demonstrated convincing neuroprotective effect.

Levodopa/Dopamine agonists



Exogenous dopa



Levodopa is given with a decarboxylase inhibitor, which prevents peripheral breakdown in the liver (as in 1) allowing a higher

concentration of dopa to reach the blood–brain barrier (as in 2) and reduces the peripheral side effects (nausea, vomiting, hypotension).

Central side effects: confusion, depression, dyskinetic movements and following long-term treatment – ‘On/Off’ phenomenon (see later).

Rapid onset or longer action can be achieved using dispersible or controlled-release preparations.

Exogenous dopa improves bradykinesia, rigidity and, to a lesser extent, tremor, but in 20% the response is poor. Dopa has relatively less effect on non-motor symptoms.

A new preparation of dopa is available for continuous infusion via jejunostomy in severe disease.

Dopamine agonists: Now used earlier in disease management, they act directly on the dopamine receptor independent of degenerating dopaminergic neurons. It is not clear if patients do better in the long term if dopamine agonists or dopa are used first. There are two types of dopamine agonists, ergot derived, including pergolide, cabergoline, apomorphine, and non-ergot derived, such as ropinerole, pramipexole, rotigotine (available as a transdermal patch). Ergot agonists are now avoided because of the high rate of fibrotic reactions, with up to 25% of patients developing cardiac valve fibrosis. Apomorphine is given by continuous infusion or intermittent injection and is useful late in the disease.

Side effects: postural hypotension, hallucinations & psychosis, sedation and agonist specific complications (erythromelalgia/pulmonary fibrosis).

COMT inhibitors: Entacapone reduces the metabolism of levodopa and is used as adjunctive treatment. Tolcapone is an alternative that can

cause hepatic toxicity; it requires close monitoring.

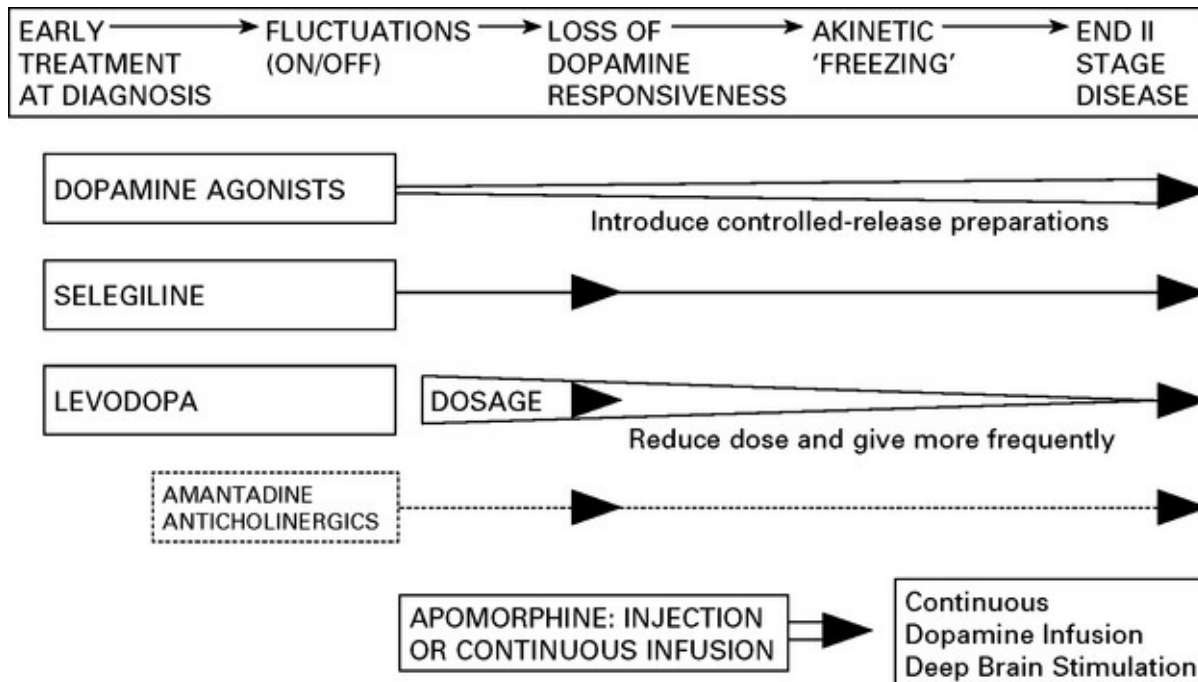
Selegiline and Resagiline are monoamine oxidase (MAO) type B inhibitors which slow breakdown of dopa. Its usage results in increased dopamine levels.

Amantidine, is useful in reducing dyskinesias late in the disease.

Deep brain stimulation: for patients with normal cognitive function who remain responsive to medication but have significant on/off phenomenon despite optimum medical therapy, the insertion of deep brain electrodes into the subthalamic nucleus can provide useful clinical benefits. Long-term studies are ongoing to determine how best to use surgery. Complications include dysarthria and visual field defects.

Human fetal and medullary transplantation: experimental evidence shows that transplantation to the striatum of tissue capable of synthesising and releasing dopamine reverses the motor symptoms of Parkinson's disease. This treatment remains experimental.

Regime of treatment (Drug therapy becomes more complex as disease progresses)



Additional measures

<i>Nausea:</i> _____	domperidone (peripheral dopamine antagonist)
<i>Hypotension:</i> _____	tilt bed head, elastic stockings + mineralocorticoid
<i>Peak dose dyskinesia:</i> _____	lower levodopa dose
<i>End dose dyskinesia:</i> _____	add dopamine agonist
<i>Nocturnal pain/immobility:</i> _____	add controlled-release levodopa at night
<i>Confusion/aggravated dementia:</i> ____	reduce dopamine agonist first, then levodopa consider anticholinesterase (rivastigmine) or quetiapine or clozapine (dopa antagonists)

CHOREA

An involuntary, irregular, jerking movement affecting limb and axial muscle groups. These movements are suppressed with difficulty and are incorporated into voluntary gestures resulting in a 'semipurposeful' appearance, *e.g.* crossing and uncrossing of legs.

Causes of chorea

Hereditary: – Huntington's disease – Benign chorea	Metabolic: – Hyperthyroidism – Hypocalcaemia
Drugs: – Antiparkinsonian drugs – oral contraceptives	Immunological: – Systemic lupus erythematosus – Polyarteritis nodosa
Toxins: – alcohol – carbon monoxide poisoning	Miscellaneous: – Chorea gravidarum – Polycythaemia rubra vera
Infections: – Sydenham's chorea – encephalitis	

HUNTINGTON'S DISEASE

Huntington disease (HD) is inherited as an autosomal dominant disease that gives rise to progressive, selective (localized) neural cell death associated with choreic movements and dementia. It is associated with increases in the length of a CAG triplet repeat present in a gene called 'huntingtin' located on chromosome 4p16.3. Huntington disease has a frequency of 4 to 7 per 100000 persons. The condition shows 'anticipation', becoming more severe in each succeeding generation.

Pathology: Neuronal loss in the striatum is associated with a reduction in projections to other basal ganglia structures. In addition, cells of the deep layers of the frontal and parietal cortex are lost

(corticostriatal projections). The neurochemical basis of this disorder involves deficiency of gamma aminobutyric acid (GABA) and acetylcholine with reduced activity of enzymes glutamic acid decarboxylase (GAD) and choline acetyltransferase (CAT).

Symptoms and signs: The classic signs of Huntington disease are progressive chorea, rigidity, and dementia. Typically, there is a prodromal phase of mild psychotic and behavioural symptoms, which precedes frank chorea by up to 10 years.

Chorea – may be the initial symptom. This progresses from mere fidgetiness to gross involuntary movements which interrupt voluntary movement and make feeding and walking impossible.

Dementia – this is of a subcortical type (see [page 126](#)).

Behavioural disturbance – personality change, affective disorders and psychosis occur.

Hypotonicity often accompanies fidgety, choreiform movements.

Primitive reflexes – grasp, pout and palmomental – are usually elicited. Eye movements are disturbed with impersistence of gaze.

Diagnosis: MRI shows an increase in the T2 signal in the caudate nucleus. Positron-emission tomography (PET scanning) demonstrates loss of uptake of glucose in the caudate nuclei. Genetic testing is diagnostic, but given the significance of the diagnosis to both patients and their family, usually requires informed consent. If the patient is too demented to consent discussion with family members is advised.

Prediction of disease: Identifying the CAG repeat provides a reliable method of detecting the disease. Presymptomatic testing is now available in many centres. These tests raise ethical issues but also the possibility of neuroprotective therapy.

Treatment: Mainly supportive. Phenothiazines (risperidone), haloperidol or tetrabenazine, may reduce abnormal movements in early disease. SSRIs help affective disturbance.

SYDENHAM'S CHOREA

Rare in an age of antibiotic therapy, this condition (also known as St Vitus' dance) followed *streptococcus pneumoniae* infection. Unlike arthritis and carditis, symptoms developed weeks or months after primary infection. Movements are diffuse and often associated with florid behavioural changes.

Pathology: Necrotising arteritis in thalamus, caudate nucleus and putamen.

Diagnosis is confirmed by elevated ESR and ASO (antistreptolysin) titre.

Treatment: symptomatic with phenothiazines. The condition may recur during pregnancy, or with intercurrent infection.

CHOREA GRAVIDARUM

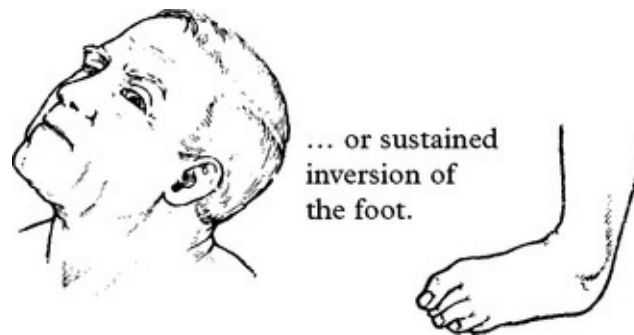
Acute onset in pregnancy, usually the first trimester or whilst on oral contraceptive. It can be restricted to face or generalised and may represent a reactivation of Sydenham's chorea. If occurring whilst on the oral contraceptive this should be stopped; risperidone can be used to control symptoms.

SENILE CHOREA

Begins in late middle age unaccompanied by family history or behavioural change. Some patients do have caudate or putaminal atrophy and occasionally test positive for Huntington's disease.

DYSTONIA

Dystonia manifests as a sustained abnormal posture produced by contraction of large trunk and limb muscles, *e.g.* sustained head retraction ...



Dystonias may be classified by distribution:

generalised, focal, when limited to one area of the body or *task-specific* such as writer's cramp.

And by aetiology:

Primary, often genetically proven, or secondary, to drugs, metabolic disorders and other neurodegenerative disorders.

PRIMARY DYSTONIA

Primary Generalised: Idiopathic Torsion Dystonia

The first gene identified for idiopathic torsion dystonia, DYT 1, is located on 9q34. The disorder is inherited as an autosomal dominant with reduced penetrance. It is responsible for early-onset generalized dystonia in Ashkenazi Jews. Initially, a flexion deformity of leg develops when walking.

Movements then become generalised but ultimately constant. Despite

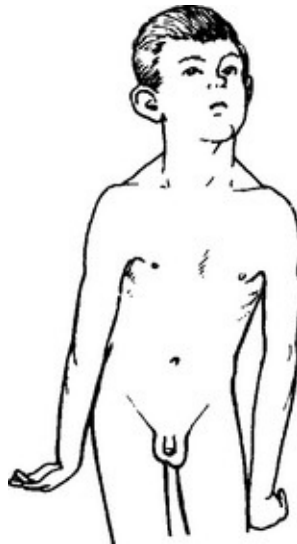
eventual gross contortion the postures disappear during sleep.

Diagnosis is made on clinical grounds and by exclusion of other disorders. – EMG studies show inappropriate co-contraction of antagonistic muscle groups.

Pathology: No known pathological substrate.

Treatment: levodopa or carbamezapine are of benefit in some patients; anticholinergics help in others. A small proportion are dramatically dopa-responsive.

Pallidal stimulation may benefit (see [page 387](#)).



DYSTONIAS – FOCAL AND SEGMENTAL IDIOPATHIC

PRIMARY FOCAL DYSTONIAS

CERVICAL DYSTONIA OR SPASMODIC TORTICOLLIS

Unilateral deviation of the head.



Aetiology is unknown. Vestibular abnormalities occur on testing, but it is uncertain whether these cause torticollis or result from the abnormal head posture. Familial spasmodic torticollis may be a restricted form of idiopathic torsion dystonia.

Dystonic contraction of the *left* sternomastoid produces head turning to the *right*.

Pressure of the index finger on the right side of the chin may turn the head back to the neutral position (*geste antagoniste*).

Turning of the head is specially noticeable when the patient is walking. Eventually hypertrophy of involved muscles occurs.

Pathology: unknown. **Diagnosis** is based on clinical findings.

Treatment: anticholinergics produce limited benefit in a few patients. Regular injection of *Botulinum* toxin into the overactive muscles gives good symptomatic control.

OROMANDIBULAR DYSTONIA

Constant involuntary prolonged tight eye closure (blepharospasm) is associated with dystonia of mouth, tongue or jaw muscles (jaw clenching and tongue protrusion). Response to treatment is poor though phenothiazines should be tried. Section of the nerves to orbicularis oculi muscles will relieve blepharospasm. *Botulinum* toxin injection is also effective.

When oromandibular dystonia occurs with cervical dystonia this segmental dystonia is called Meige's syndrome.

PRIMARY TASK SPECIFIC DYSTONIA: WRITER'S CRAMP

Muscles of the hand and forearm tighten on attempting to write and pain may occur in the forearm muscles. Previously regarded as an 'occupational neurosis' but now classified as a partial dystonia.

Treatment: may respond to botulinum toxin injection. Avoiding the

activity is most successful.

Other task specific dystonias relate to other repeated movements and include golfer's yips.

SECONDARY DYSTONIAS

DOPAMINE RESPONSIVE DYSTONIA (DRD)

This disorder presents in childhood and generally involves the legs only. Falls are frequent and the response to levodopa is maintained over many years. DRD may be the result of a developmental reduction in the number of dopaminergic nerve endings in the striatum and maps to the same region of 14q as does the gene for the enzyme GTP cyclohydrolase 1 (GCH1) implicated in a hyperphenylalaninaemia.

DRUG INDUCED DYSTONIA

Acute adoption of abnormal dystonic posture – usually head and neck or oculogyric crisis (upward deviation of eyes) – caused by phenothiazines, e.g. haloperidol, metoclopramide.

Anticholinergics, e.g. benztropine for 24–48 hours helps symptoms settle.

OTHERS

Dystonia is a feature of other conditions, most commonly Parkinson's disease on treatment, but also PSP, Wilson's disease (see [page 366](#)).

OTHER MOVEMENT DISORDERS

TICS

Abrupt jerky movements affecting head, neck and trunk. Tics can be

voluntarily suppressed and often take the form of winking, grimacing, shoulder shrugging, sniffing and throat clearing.

Gilles de la Tourette syndrome is characterised by motor and vocal tics, copropraxia (making obscene gestures), coprolalia (obscene utterances) and obsessive behaviour. Onset is in childhood, males are more often affected and the condition may be inherited. However results of a systematic genome screen were negative. A population study showed that 3% of all children and that up to 25% of children requiring special education may have mild to moderate Tourette's syndrome.

The dopaminergic systems in the basal ganglia appear involved, dopamine D2 receptor antagonists improving and dopamimetic agents worsening symptoms. Clonidine helps control tics with few adverse effects.

TARDIVE DYSKINESIA

This is a consequence of long-term treatment with neuroleptic drugs – phenothiazines, butyrophenones – and results from the development of drug-induced supersensitive dopamine receptors.

Involuntary movements in the face, mouth and tongue (orofacial dyskinesia) as well as limb movements of a *choreoathetoid nature* occur.

This movement disorder may commence even after stopping the responsible drug and can persist indefinitely.

Prevention

Incidence may be reduced by:

1. Using newer atypical antipsychotic agents.
2. Early recognition and drug withdrawal.

The practice of increasing the dose of the offending drug when movements occur should be avoided. This will improve movements initially, but they will 'break through' later.

Treatment

Discontinue neuroleptic. If not possible, continue on lowest possible dose. Drugs which increase acetylcholine (anticholinesterases), reduce catecholamine release (lithium), or deplete dopamine (reserpine) are variably effective.

ATHETOSIS

Athetosis presents in childhood and appears as a slow writhing movement disorder with a rate of movement between that of chorea and dystonia. It usually involves the digits, hands and face on each side.

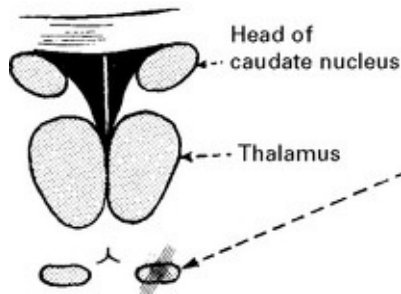
These abnormal movements may result from:

- Hypoxic neonatal brain damage,
- Kernicterus,
- Lipid storage diseases.

Response to anticholinergics is variable and occasionally dramatic.



HEMIBALLISMUS



This is a movement disorder characterised by unilateral, violent flinging of the limbs. This involuntary movement is occasionally severe enough to throw the patient off balance or even from his bed.

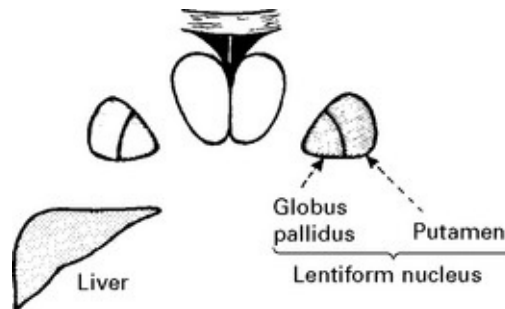
The anatomical basis is a lesion of the *subthalamic nuclei* or its connections contralateral to the abnormal movement. It usually results from vascular disease (posterior cerebral artery territory), but occasionally occurs in multiple sclerosis.

Drug treatment is ineffective. The condition often settles spontaneously.

MALIGNANT NEUROLEPTIC SYNDROME AND SEROTONIN SYNDROME – see [page 534](#).

WILSON'S DISEASE (hepatolenticular degeneration)

An autosomal recessive disorder characterised by the build-up of intracellular copper with hepatic and neurological consequences.



Pathology

Cavitation and neuronal loss occurs within the putamen and the globus pallidus.

The liver shows the appearances of coarse cirrhosis. Copper accumulates in all organs, especially in Descemet's membrane in the eye, nail beds and kidney.

Biochemistry

There is deficiency of α_2 globulin – Ceruloplasmin – which normally binds 98% of copper in the plasma and transfers copper to enzyme (cytochrome oxidase). This results in an increase in loosely bound copper/albumin, and deposition occurs in all organs. Urinary copper is increased.

Clinical features

There are two clinical forms:

1. *Acute*

Bradykinesia
Behavioural change
Involuntary movements
Liver involvement common

Untreated: death in 2 years
from hepatic and renal failure

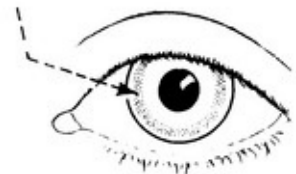


2. *Chronic*

Marked proximal 'wing beating' tremor
Dysarthria, dystonia and rigidity
Choreoathetoid movements
Psychosis, behavioural disorders and dementia
Liver involvement less severe

Untreated: death in 10 years

The deposition of copper in Descemet's membrane produces the golden brown *Kayser-Fleischer* ring, which when seen by naked eye or slit-lamp is diagnostic.



Diagnosis

Should be considered in any patient with unusual hepatic and/or neurological features.

Supported by biochemical evidence of abnormal copper metabolism:

- Low ceruloplasmin (less than 20 mg/dl)
- Elevated unbound serum copper
- High urinary copper excretion

- Liver biopsy and copper metabolism tests with radioactive ^{64}Cu .
- MRI (T2) shows thalamic and putaminal hyperintensity.

In families, biochemical tests will identify low ceruloplasmin in carriers and in presymptomatic patients. Over 20 mutations in copper transporting ATPase have been identified. Diagnostic genetic testing is not available.

Treatment

Low copper diet and a chelating agent, *e.g.* penicillamine 1–1.5g daily. Side effects such as anaphylaxis, skin rash, bone marrow suppression and glomerulonephritis are common in which case trientine is an effective alternative.

Therapy is necessary for the rest of the patient's life. Adequate treatment is compatible with normal life expectancy. Kayser-Fleischer rings will disappear with time.

HYDROCEPHALUS

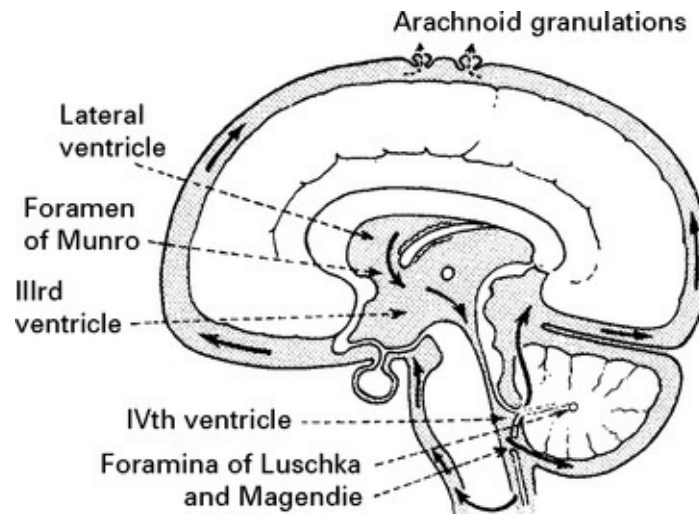
DEFINITION

Hydrocephalus is an active distension of the ventricular system of the brain arising when an imbalance exists between cerebrospinal fluid (CSF) production and absorption. This definition excludes ventricular expansion secondary to brain shrinkage from a diffuse atrophic process (hydrocephalus ex vacuo).

CSF FORMATION AND ABSORPTION

CSF forms at a rate of 500 ml/day (0.35 ml/min), secreted predominantly by the choroid plexus of the lateral, third and fourth

ventricles. CSF flows in a caudal direction through the ventricular system and exits through the foramina of Luschka and Magendie into the subarachnoid space. After passing through the tentorial hiatus and over the hemispheric convexity, absorption occurs through the arachnoid granulations into the venous system.



CLASSIFICATION

‘*Obstructive*’ *hydrocephalus* – obstruction of CSF flow *within* the ventricular system.

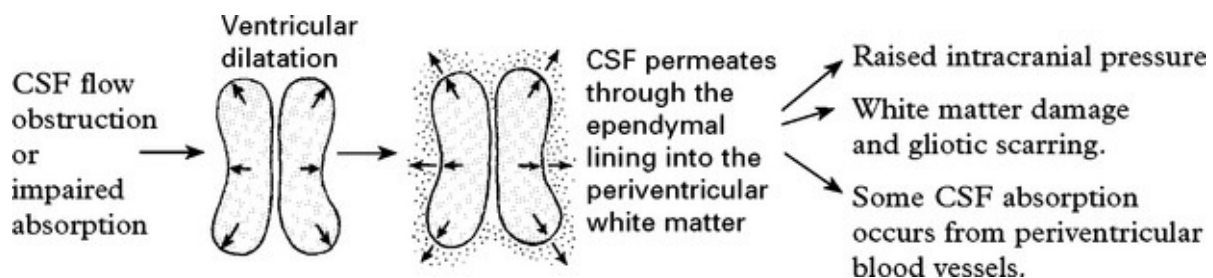
‘*Communicating*’ *hydrocephalus* – obstruction to CSF flow *outwith* the ventricular system *i.e.* ventricular CSF ‘communicates’ with the subarachnoid space.

CAUSES OF HYDROCEPHALUS

Obstructive		Communicating
Acquired	– Acquired aqueduct stenosis (adhesions following infection or haemorrhage)	Thickening of the leptomeninges and/or involvement of the arachnoid granulations

	– Supratentorial masses causing tentorial herniation – Intraventricular haematoma – Tumours – ventricular, <i>e.g.</i> colloid cyst – pineal region – posterior fossa – Abscesses/granuloma – Arachnoid cysts	– infection (pyogenic, TB, fungal) – subarachnoid haemorrhage – spontaneous – trauma – postoperative – carcinomatous meningitis Increased CSF viscosity, <i>e.g.</i> high protein content Excessive CSF production – choroid plexus papilloma (rare)
<i>Congenital</i>	– Aqueduct stenosis or forking – Dandy-Walker syndrome (atresia of foramina of Magendie and Luschka) – Chiari malformation	

PATHOLOGICAL EFFECTS

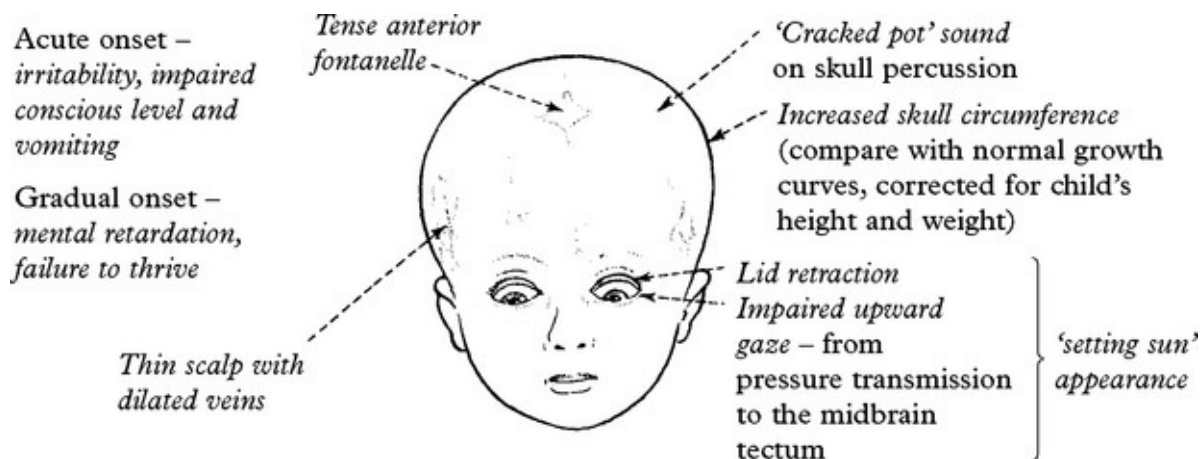


In the infant, prior to suture fusion, head expansion and massive ventricular dilatation may occur, often leaving only a thin rim of cerebral ‘mantle’. Untreated, death may result, but in many cases the hydrocephalus ‘arrests’; although the ventricles remain dilated, intracranial pressure (ICP) returns to normal and CSF absorption appears

to balance production. When hydrocephalus arrests, normal developmental patterns resume, although pre-existing mental or physical damage may leave a permanent handicap. In these patients, the rapid return of further pressure symptoms following a minor injury or infection suggests that the CSF dynamics remain in an unstable state.

CLINICAL FEATURES

Infants and young children



Juvenile/adult type hydrocephalus

Acute onset – signs and symptoms of $\uparrow$ ICP $\leftarrow$ headache, vomiting, papilloedema.
 – impaired upward gaze $\leftarrow$ deterioration of conscious level

Gradual onset – dementia
 – gait ataxia
 – incontinence

This triad of symptoms may occur despite an apparently 'normal' CSF pressure, i.e. NORMAL PRESSURE HYDROCEPHALUS (see page 130)

The condition often relates to previous trauma, meningitis or subarachnoid haemorrhage.

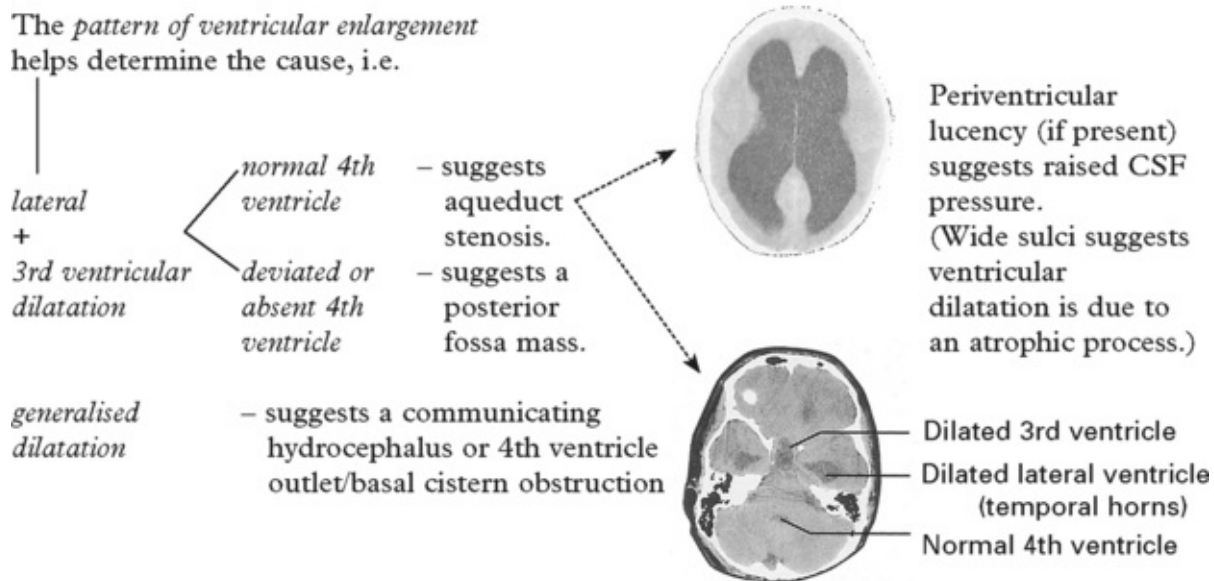
INVESTIGATIONS

Skull X-ray

Note:

- skull size and suture width.
- evidence of chronic raised pressure – posterior clinoid erosion, ‘copper beating’.
- associated defects – platybasia, basilar invagination.

CT scan



Ultrasonography through the anterior fontanelle, usefully demonstrates ventricular enlargement in infants and allows safe serial measurements.

MRI shows similar ventricular expansion, but may more clearly demonstrate periventricular lucency or a neoplastic cause of the obstruction.

ICP monitoring: used in some patients to determine whether symptoms relate to the enlarged ventricular size and to investigate patients with suspected normal pressure hydrocephalus (see [page 131](#)).

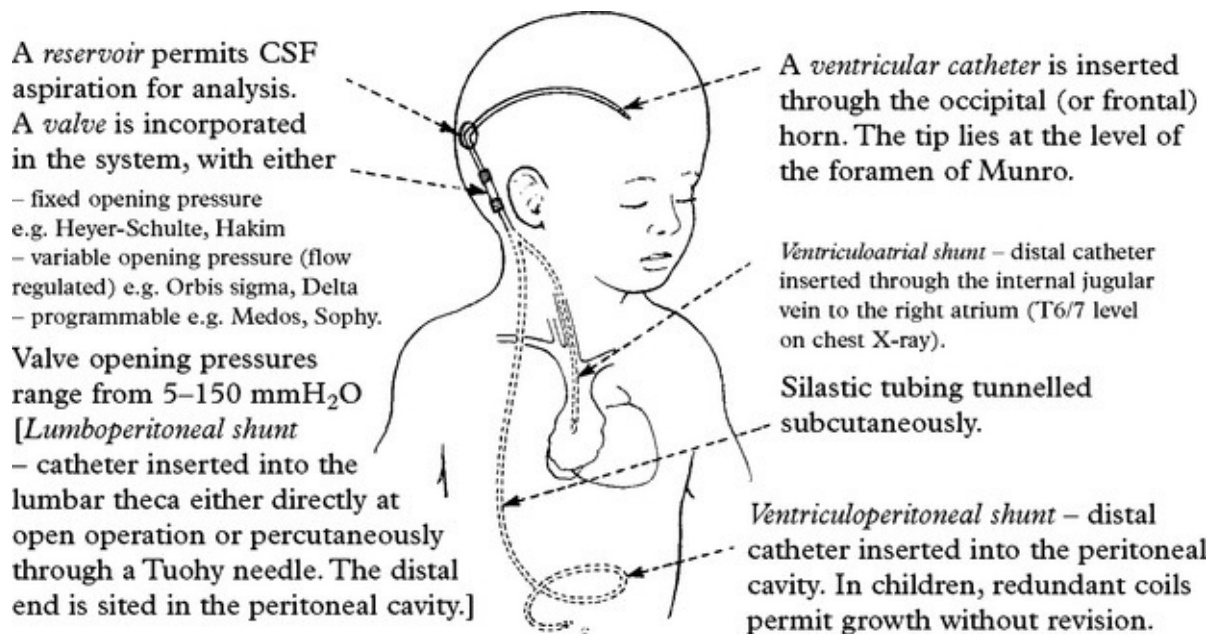
Developmental assessment and psychometric analysis detect

impaired cerebral function and provide a baseline for future comparison.

MANAGEMENT

- Acute deterioration*
 - ventricular drainage or
 - ventriculo-peritoneal (VP) shunt or
 - 3rd ventriculostomy (if tri-ventricular – obstructive hydrocephalus)
 - lumbar puncture – if communicating hydrocephalus, e.g. following subarachnoid haemorrhage.
- Gradual deterioration*
 - VP shunt (lumboperitoneal shunts are occasionally used for communicating hydrocephalus) or 3rd ventriculostomy.
 - removal of a mass lesion if present – this may obviate the need for a shunt.
- 'Arrested hydrocephalus'* – symptomless ventricular dilatation requires no treatment, but regular developmental or psychometric assessment ensures no ill effects develop from this potentially unstable state.

Shunt techniques



Complications of shunting

Infection: results in meningitis, peritonitis or inflammation extending along the subcutaneous channel. With a V-A shunt, bacteraemia may lead to shunt 'nephritis'. *Staphylococcus epidermidis* or *aureus* are usually

involved, with infants at particular risk. Minimise the risk of infection with prophylactic antibiotics and in neonates, with antibiotic impregnated shunt systems. When established, eradication usually requires shunt removal.

Subdural haematoma: ventricular collapse pulls the cortical surface from the dura and leaves a subdural CSF collection or tears bridging veins causing subdural haemorrhage. The risk may be reduced with a variable pressure or programmable valve.

Shunt obstruction: blockage of the shunt system with choroid plexus, debris, omentum or blood clot results in intermittent or persistent recurrence of symptoms. Demonstration of an increase in ventricular size compared to a previous baseline CT scan confirms shunt malfunction. Over a third require revision within 1 year and 80% within 10 years.

Low pressure state: following shunting, some patients develop headache and vomiting on sitting or standing. This low pressure state usually resolves with a high fluid intake and gradual mobilisation. If not, insertion of an antisyphon device or conversion to a high pressure valve is required.

Third ventriculostomy: Suitable for patients with tri-ventricular hydrocephalus e.g. obstructive hydrocephalus caused by aqueduct stenosis or a pineal or posterior fossa tumour occluding the posterior end of the 3rd ventricle/aqueduct. By using a flexible or rigid endoscope introduced through a frontal burrhole, a fistula is created in the floor of the 3rd ventricle. This provides an alternative method of treatment, which if successful, avoids the above problems of shunt insertion. About $\frac{2}{3}$ of patients obtain permanent benefit.

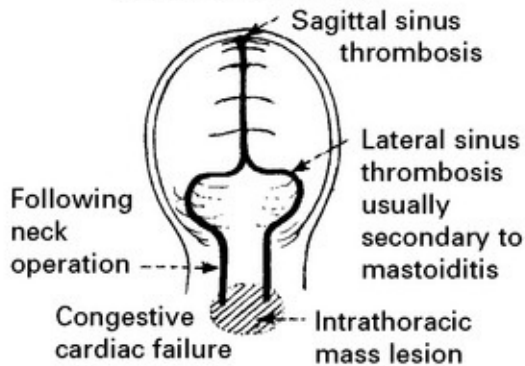
Prognosis: Provided treatment precedes irreversible brain damage,

results are good with most children attaining normal IQs. Repeated complications, however, particularly prevalent in infancy and in young children carry a significant morbidity.

IDIOPATHIC INTRACRANIAL HYPERTENSION

Idiopathic intracranial hypertension (previously benign intracranial hypertension or pseudotumour cerebri) is characterised by increased intracranial pressure *without* evidence of an intracranial space-occupying lesion, obstruction to CSF pathways, infection, or hypertensive encephalopathy.

Diagnosis is especially dependent on excluding –
VENOUS OUTFLOW OBSTRUCTION
TO CSF ABSORPTION



CLINICAL FEATURES

Age: any age, but usually in 3rd and 4th decades.

Sex: female >> male

Symptoms

Headache

Visual obscurations

Impaired visual acuity

Diplopia

Signs

Obesity

← Papilloedema

VI nerve palsy

In women the condition is often associated with – recent weight gain, fluid retention, menstrual dysfunction, the first trimester of pregnancy and the postpartum period.

Investigations

CT/MRI brain and orbit (ventricles usually small)

MRV/Venography (to exclude sinus thrombosis)

Visual field charting (enlarged blind spot & peripheral constriction)

Lumbar puncture (measure pressure)

Where this has been ruled out the cause is obscure but a variety of factors are associated –
DIET – obesity.

– hyper/hypovitaminosis A.

ENDOCRINE – pregnancy, menarche, menstrual irregularities, Addison's disease.

HAEMATOLOGICAL – iron deficiency anaemia.
– polycythaemia vera.

DRUGS – oral contraceptives.

– steroid withdrawal.

– tetracycline (minocycline)

– nalidixic acid.

Various mechanisms have been postulated.

– BRAIN SWELLING

– ↓CSF

ABSORPTION

– ↑CSF

SECRETION

} Different studies support different mechanisms. The link with obesity suggests an underlying endocrine basis, but, except in Addison's disease, endocrine assessment has failed to reveal abnormalities.

TREATMENT

Indicated to prevent visual loss, which may develop insidiously if untreated:

– Discontinue causative medication if known

– Weight loss

– Acetazolamide (a carbonic anhydrase inhibitor)

} If these measures

fail, consider – Repeated LPs

– Lumboperitoneal shunt

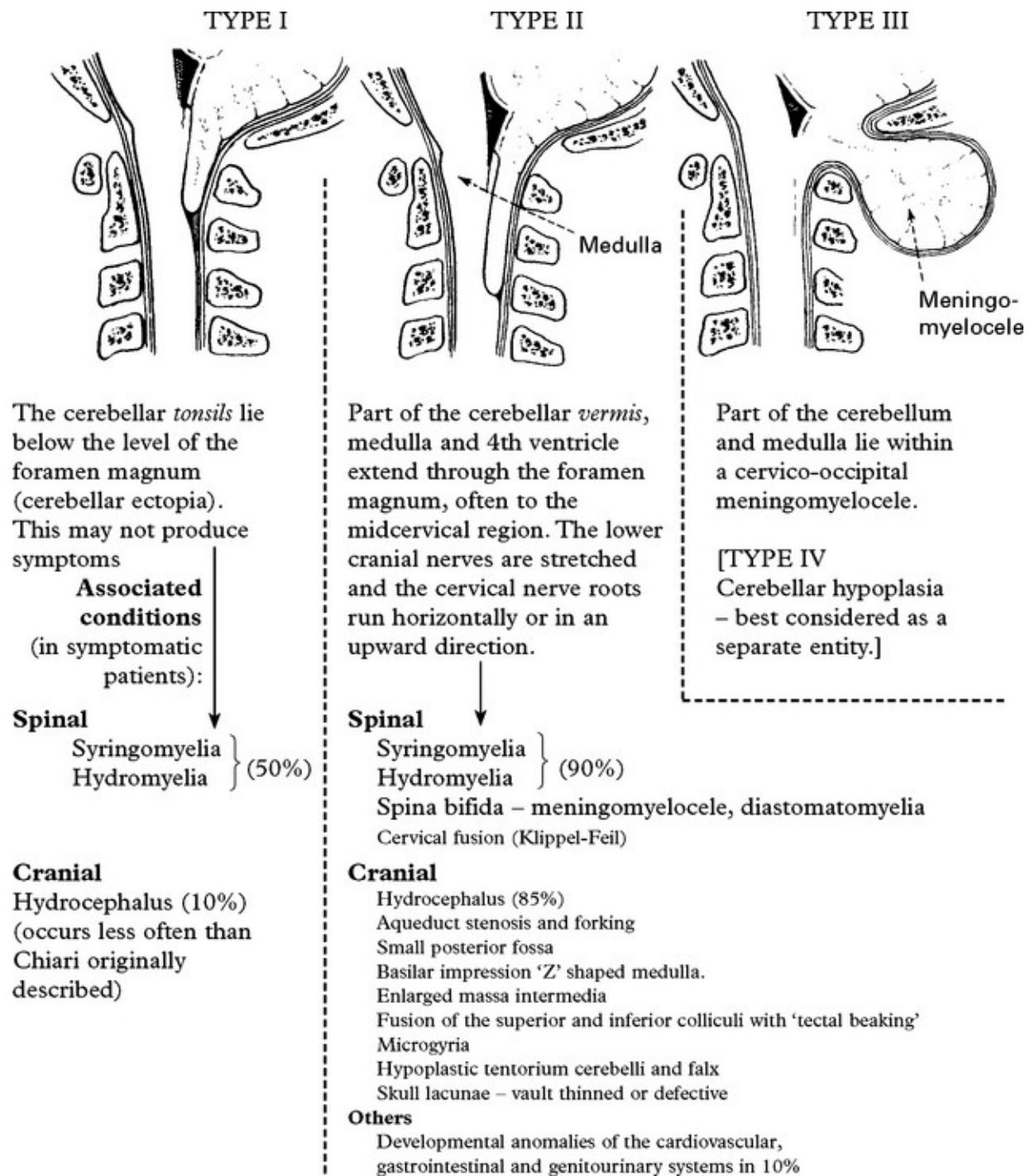
– Bariatric surgery to aid weight loss

PROGNOSIS

Generally improves with interventions described above, particularly weight loss. Minority of patients have visual loss.

CHIARI MALFORMATION

Although the names of two authors (Arnold and Chiari) were originally linked to the description of malformations at the medullary-spinal junction, Chiari must take most credit for providing a detailed description of this condition.



PATHOGENESIS

Several hypotheses have been proposed to explain the pathological findings of these malformations. Gardner suggested that downward

pressure from *hydrocephalus* played an important role in displacing the posterior fossa structures and, when associated with a patent central canal, explained the high incidence of syringomyelia (page 401). Others supposed that *traction* from a tethered spinal cord (dysraphism), or a *CSF leak* through a myelocoele into the amniotic sac in fetal life resulted in caudal displacement of the posterior fossa structures. Of these theories, none provides an entirely satisfactory explanation; a more realistic view attributes the hindbrain deformity to *maldevelopment* during early fetal life. This would explain the presence of other developmental anomalies.

CLINICAL PRESENTATION

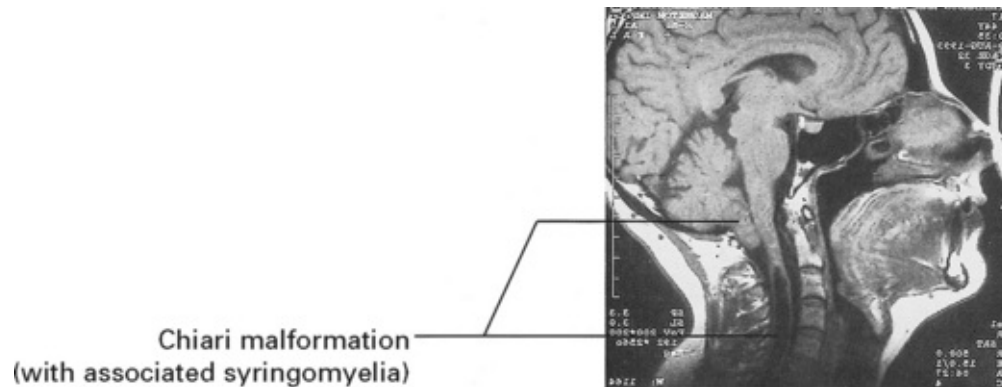
Depends on age

INFANCY	{ Severe type II (or III) deformities present with <i>respiratory difficulties</i> and <i>lower cranial nerve palsies</i> . Death may result from <i>aspiration pneumonia</i> or <i>apnoeic attacks</i> , or from complications of associated malformations, e.g. <i>spina bifida</i> . In milder forms, <i>nystagmus</i> (horizontal), <i>retrocollis</i> (neck extension) and <i>spasticity</i> predominate.																		
CHILDHOOD	{ With increasing age, <i>gait ataxia</i> may become evident. Features of an associated syringomyelia – <i>dissociated sensory loss</i> and <i>spastic quadraparesis</i> often contribute to the clinical problems.																		
ADULT	{ Only patients with a type I or a mild type II deformity present in adult life – <i>Occipital headaches</i> are induced by coughing or straining <table><tr><td><i>Nystagmus</i> – downbeat</td><td>rotatory</td><td rowspan="3">{ may result from</td></tr><tr><td>(on looking down)</td><td>(on lateral gaze)</td></tr><tr><td colspan="2"></td></tr><tr><td colspan="2"></td><td rowspan="3">{ medullary compression or from an associated syringomyelia (see page 401).</td></tr><tr><td colspan="2"><i>Ataxia</i></td></tr><tr><td colspan="2"><i>Spastic quadraparesis</i></td></tr><tr><td colspan="2">Progression may eventually lead to severe bulbar symptoms – <i>lower cranial nerve palsies, respiratory difficulties.</i></td></tr></table>			<i>Nystagmus</i> – downbeat	rotatory	{ may result from	(on looking down)	(on lateral gaze)					{ medullary compression or from an associated syringomyelia (see page 401).	<i>Ataxia</i>		<i>Spastic quadraparesis</i>		Progression may eventually lead to severe bulbar symptoms – <i>lower cranial nerve palsies, respiratory difficulties.</i>	
<i>Nystagmus</i> – downbeat	rotatory	{ may result from																	
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		{ medullary compression or from an associated syringomyelia (see page 401).																	
<i>Ataxia</i>																			
<i>Spastic quadraparesis</i>																			
Progression may eventually lead to severe bulbar symptoms – <i>lower cranial nerve palsies, respiratory difficulties.</i>																			

INVESTIGATIONS

Magnetic resonance imaging (MRI) is the investigation of choice. T1

weighted sagittal and axial scans most clearly demonstrate cerebellar ectopia and the presence or absence of an associated syringomyelia.

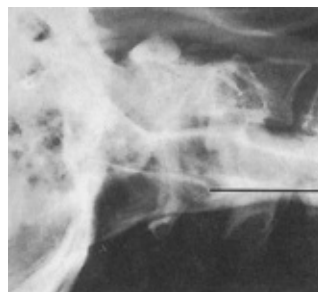


Straight X-rays

- Skull: note the presence of platybasia, basilar impression or lacunae (vault defects).
- Cervical spine: note increased canal width or fusion of vertebrae (especially C2,3) – Klippel-Feil syndrome.
- Lumbosacral spine: note any associated spina bifida.

Myelography (if MRI unavailable)

CT scan: difficult to interpret at the cervico-medullary junction, but shows soft tissue filling the spinal canal at this level.



Contrast run up to the foramen magnum with the patient in the supine position outlines a posteriorly situated filling defect.

MANAGEMENT (see also syringomyelia, page 401)

In patients with hydrocephalus	→	Ventriculoperitoneal or atrial shunt may significantly improve signs and symptoms attributed to the Chiari malformation.
--------------------------------	---	--

and signs and symptoms of raised intracranial pressure.		
In patients with other symptoms and signs	→	<i>Posterior fossa decompression</i> – by removing the posterior rim of the foramen magnum and the arch of the atlas. For more severe cases, the dura is opened and a graft is inserted. Attempts at freeing tonsillar adhesions should be resisted. An apnoea monitor in the initial postoperative period helps detect potentially fatal apnoea, especially during sleep. In some instances, patients with minimal symptoms or with no evidence of progression may warrant a conservative approach.

PROGNOSIS

Patients with mild symptoms and signs often respond well to operation, but those with long-standing neurological deficits rarely improve. Treatment should aim at preventing further progression.

Further deterioration eventually occurs in one-third, despite operative measures.

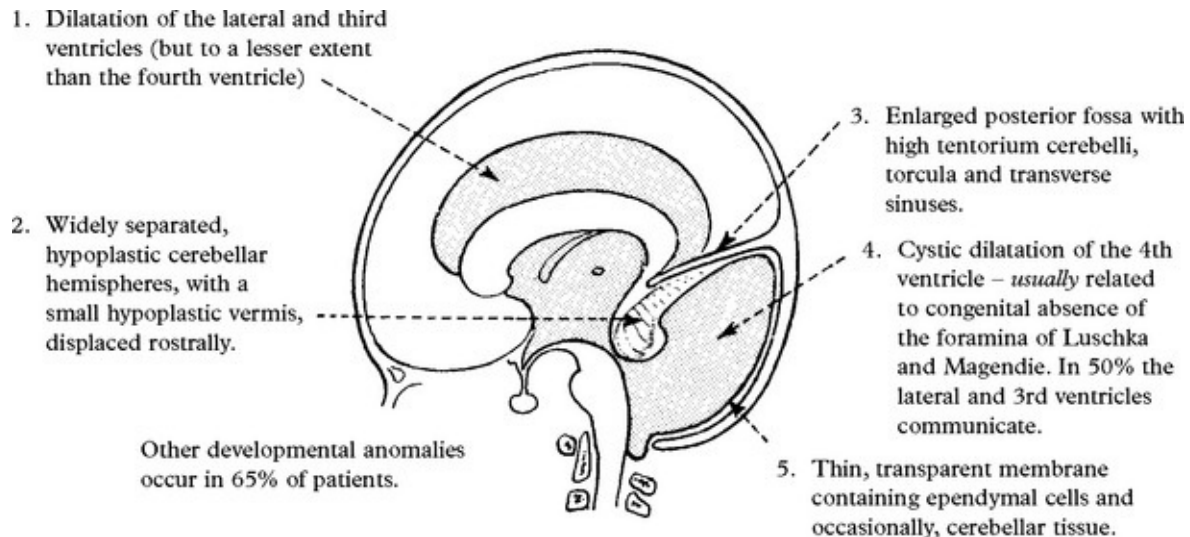
SYRINGOBULBIA

Extension of a syringomyelic cavity upwards into the medulla may produce signs and symptoms which are difficult to distinguish from those of medullary compression in the Chiari malformation:

- difficulty in swallowing, dysphonia, dysarthria, vertigo, facial pain
- nystagmus, palatal and vocal cord weakness, occasional facial and tongue weakness.

DANDY-WALKER SYNDROME

This rare developmental anomaly comprises:



CLINICAL PRESENTATION

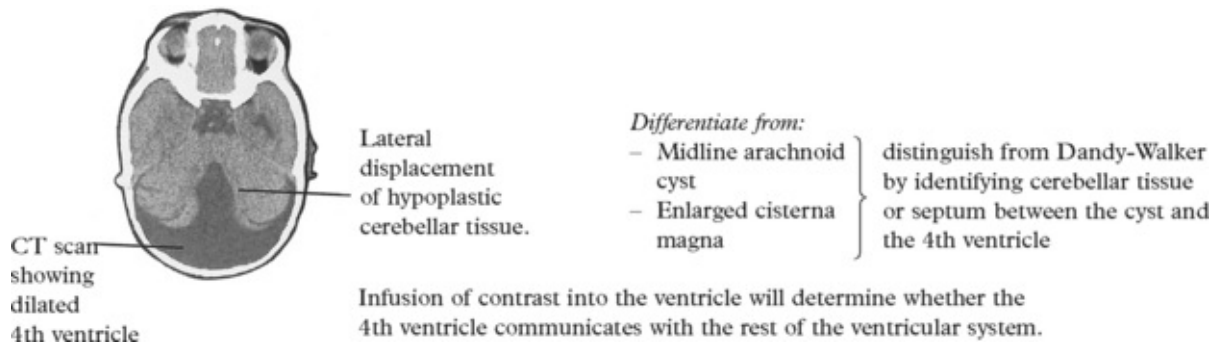
Infancy: Symptoms and signs of hydrocephalus ([page 375](#)) combined with a prominent occiput.

Childhood: Signs of cerebellar dysfunction with or without signs of hydrocephalus.

INVESTIGATIONS

Skull X-ray: Usually shows elevation of the transverse sinuses and occipital bulging, confirming the presence of an enlarged posterior fossa.

CT scan or MRI:



MANAGEMENT

When the dilated 4th ventricle communicates with the rest of the ventricular system, a cystoperitoneal shunt suffices and helps maintain a patent aqueduct. When a ‘two-compartment’ hydrocephalus exists, both the encysted 4th ventricle and the other ventricles require drainage (i.e. with a cystoperitoneal and a ventriculoperitoneal shunt).

Excision of the cyst membrane (‘marsupialising’ the 4th ventricle) is no longer thought to normalize CSF flow.

PROGNOSIS

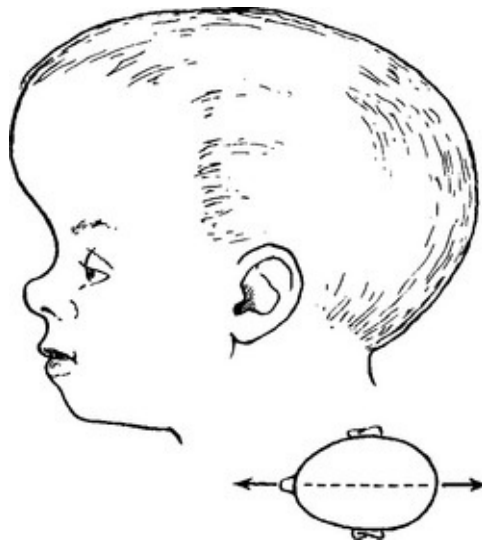
Marked neurological impairment prior to treatment carries a poor outlook. In less impaired patients, the prognosis relates more to the presence of other developmental anomalies.

CRANIOSYNOSTOSIS

In normal childhood development, the cranial sutures allow skull enlargement as the brain grows. *Premature fusion of one or more sutures* results in restricted growth of bone perpendicular to the suture and exaggerated growth parallel to the suture. The effect depends on the site and number of sutures involved. Sagittal synostosis is the most frequently occurring deformity.

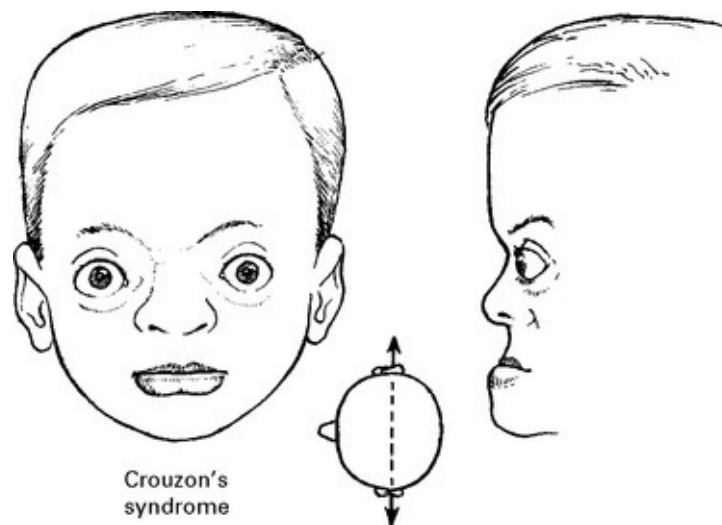
SAGITTAL SYNOSTOSIS

Lateral growth is restricted, resulting in a long narrow head with ridging sagittal suture (scaphocephaly). **Treatment:** wide excision alone does not allow for lateral expansion of the vault. Either removal of horizontal strips of bone or the use of a 'helmet' aids the remodeling process.



CORONAL SYNOSTOSIS

Bilateral or unilateral.



Expansion occurs in a superior and lateral direction (brachiocephaly). This produces a short anterior fossa, shallow orbits and hypertelorism (widening of the interocular distance). Exophthalmos, elevated ICP and visual impairment from papilloedema may result. Bilateral coronal synostosis commonly occurs as one of several congenital defects incorporated in *Crouzon's* and *Apert's syndromes*.

Involvement of several sutures (oxycephaly) results in skull expansion towards the vertex, the line of least resistance.

PANSYNOSTOSIS (all sutures affected) results in failure of skull growth with a symmetrical abnormally small head and raised intracranial pressure. ICP monitoring or a progressive reduction in normal circumferential growth distinguishes pansynostosis from microcephaly due to inadequate brain development.

Treatment of coronal, metopic and pansynostosis involves extensive craniofacial surgery correcting both cranial and orbital deformities.

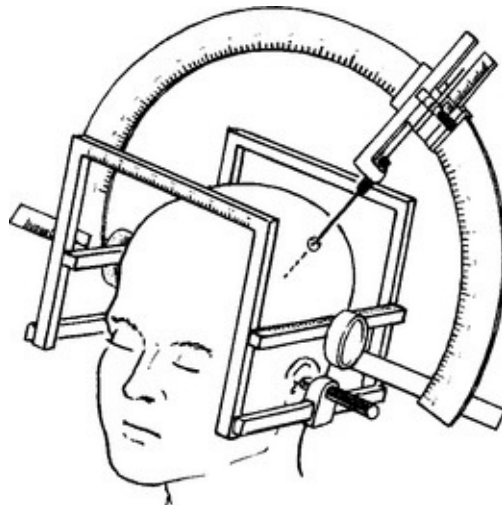
Indication for operative treatment is primarily cosmetic when only one suture is involved, but with involvement of two or more sutures operation is also aimed at prevention of visual and cerebral damage from raised ICP.

Posterior plagiocephaly (flattening of the back of the head) An increasing number of infants present with this condition perhaps resulting from the 'back to sleep' campaign. Now thought to be due to benign positional moulding rather than a true lambdoid synostosis. Very few of those who develop a progressive skull deformity require surgical treatment.

STEREOTACTIC SURGERY

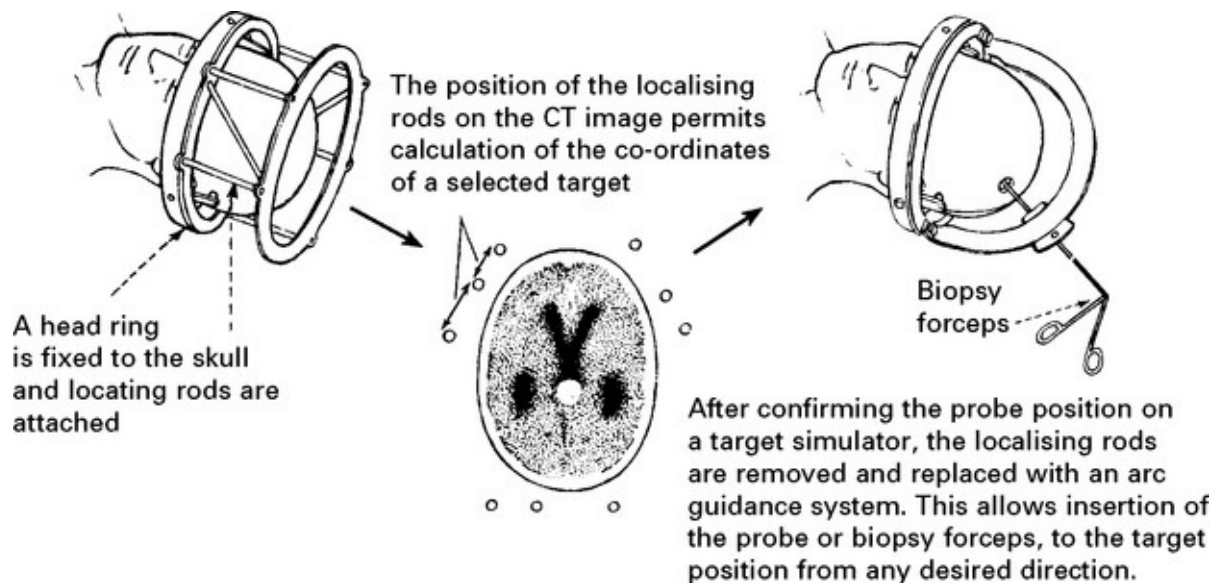
Stereotactic techniques developed initially for lesion making, enable precise placement of the tip of a cannula or electrode to a predetermined target site within the brain with the least risk.

Many different stereotactic frames have been developed, *e.g.* Leksell, Todd-Wells, Guiot. These, combined with radiological landmarks (the third ventricle) and a brain atlas, provide anatomical localisation to within $\pm 1\text{mm}$. Since some functional variability occurs at each anatomical site, electrode localisation is also based on the recorded neuronal activity and on the effects of electrical stimulation.



CT/MRI STEREOTACTIC SYSTEM

CT and MRI compatible stereotactic systems allow cannula insertion to any point selected on the image. They are all based on the concept of identifiable external reference (fiducial) markers, *e.g.* Codman-Robert-Wells (CRW) system:



CT/MRI stereotactic surgery provides the optimal method for the biopsy or aspiration of *small, deeply situated tumours* or *abscesses*. Many now use stereotactic biopsy, for larger tumours. It carries lower risk than handheld biopsy and allows selection of specific areas within the tumour. Although improved resolution now available with CT/MRI scanning has led to sufficient anatomical detail for accurate lesion making, functional stereotaxy, e.g. thalamotomy, deep brain stimulation, still requires electrical stimulation for the final target localisation.

METHODS OF LESION MAKING

<i>Heat</i>	– radiofrequency current delivered through a fine electrode	} lesion size determined by temperature change and duration.
<i>Cooling</i>	– with a cryogenic probe	
<i>Radiation</i>	– implantation of radioactive seed, e.g. yttrium ⁹⁰ – focused beam from cobalt ⁶⁰ rods (sited on a specially adapted Leksell frame) or from a linear accelerator.	

USES OF STEREOTACTIC SURGERY

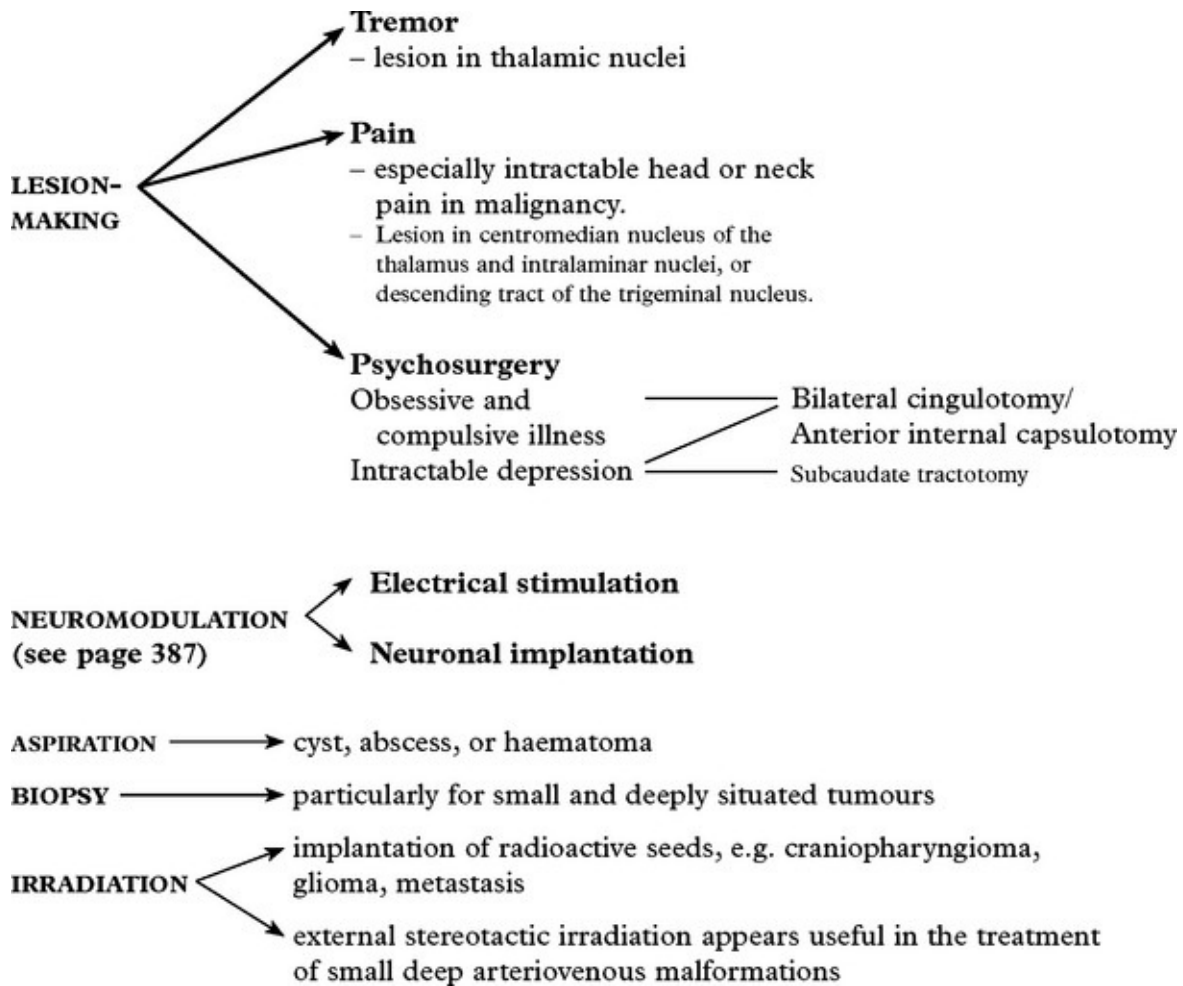
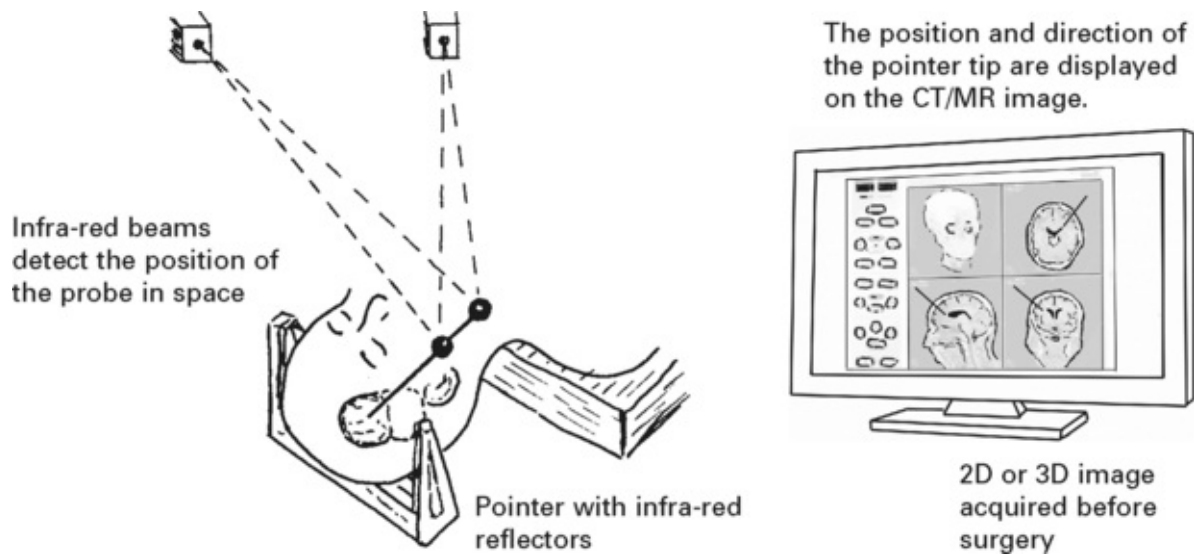


IMAGE GUIDED ‘FRAMELESS’ STEREOTAXY

Neuronavigation uses a combination of modern imaging and elaborate computer software to permit the surgeon to determine how the direction and tip of a pointer lying outwith or within the skull, relates to a two or three dimensional CT or MR image.

The accuracy of the technique depends on the quality of the digitised image and on the methods used to *register* the patient’s head to the image. The registration of recognisable skin points (e.g. nasion, orbital margins, inner canthus) on the patient to the CT/MR image provides an accuracy of 2–3 mm and this is sufficient for most purposes.



Uses of 'frameless' stereotaxy

Aids accurate positioning of burrholes and bone flap, and planning the safest approach to the lesion.

TUMOURS	ARTERIOVENOUS MALFORMATION
– biopsy	– localisation of lesion and the feeding vessels
– resection: locates, then identifies the tumour margins and the position of important adjacent structures	
	ABSCCESS
– brachytherapy (see page 314)	– aspiration
EPILEPSY	ORBIT
– defining the extent of resected tissue	– location of intraorbital lesion
– placement of depth electrodes	
	SPINE

Deficiencies of ‘frameless’ stereotaxy

The accuracy averages 2–3 mm and although adequate for the above, it is insufficient for most functional procedures.

On opening the skull, brain shift can occur, adding to any registration inaccuracy. Only real time imaging (ultrasound or CT/MRI) can overcome this difficulty (see [page 313](#)).

NEUROMODULATION

Definition: Neuromodulation is the alteration of the central, peripheral or autonomic nervous system for therapeutic benefit by electrical or pharmacological stimulation. In the central nervous system, the definition should also include the experimental implantation of foetal or stem cells.

For many years electrical stimulation of the dorsal columns has been used in the treatment of chronic pain, but with technological improvements in implantable devices – stimulators and pumps, the field of neuromodulation has rapidly developed, particularly in relation to deep brain stimulation. In contrast to ablative procedures previously forming the basis of ‘functional’ neurosurgery, neuromodulation techniques are *reversible* and by using external computers the amount of stimulation or the dose of drug can be tailored to the individual patient’s needs.

Potential Uses of Neuromodulation

* = treatment established, remainder still under assessment

ELECTRICAL STIMULATION

- Brain
- Motor cortex
 - Periventricular grey matter
 - Sensory relay nucleus thalamus
 - Postero-medial hypothalamus
 - Posterior hypothalamus
 - Subthalamic nucleus (STN)*
 - Globus pallidum internus (GPi)*
 - Ventralis intermedius nucleus thalamus (ViM)*
 - Zona incerta*
 - Anterior thalamic nuclei
 - Anterior limb internal capsule
 - Nucleus accumbens
 - Subgenual cingulate gyrus
- Nerve
- Vagus*
 - Occipital
- Spinal cord
- dorsal columns*
 - T5–T6 level
- Sacral nerve roots*

- Chronic pain
- “
- “
- “
- Cluster headache
- Tremor Parkinson’s disease
- Dyskinesia “ “ /Dystonia
- Tremor (Non-Parkinsonian)
- “ “ “
- Epilepsy
- Obsessive compulsive disorder/
- Depression
- “ “
- Depression
- Epilepsy/Depression
- Occipital neuralgia/migraine
- Chronic pain
- Angina
- Bladder/erectile function in paraplegics
- Interstitial cystitis/urge incontinence

DRUG INFUSION

- Spinal cord – intrathecal Baclofen*

Spasticity

NEURONAL IMPLANTATION

- Brain
- Embryonic stem cells
 - Foetal/adult brain neural stem cells

Parkinson’s disease/Huntington’s disease

“ “ “ “

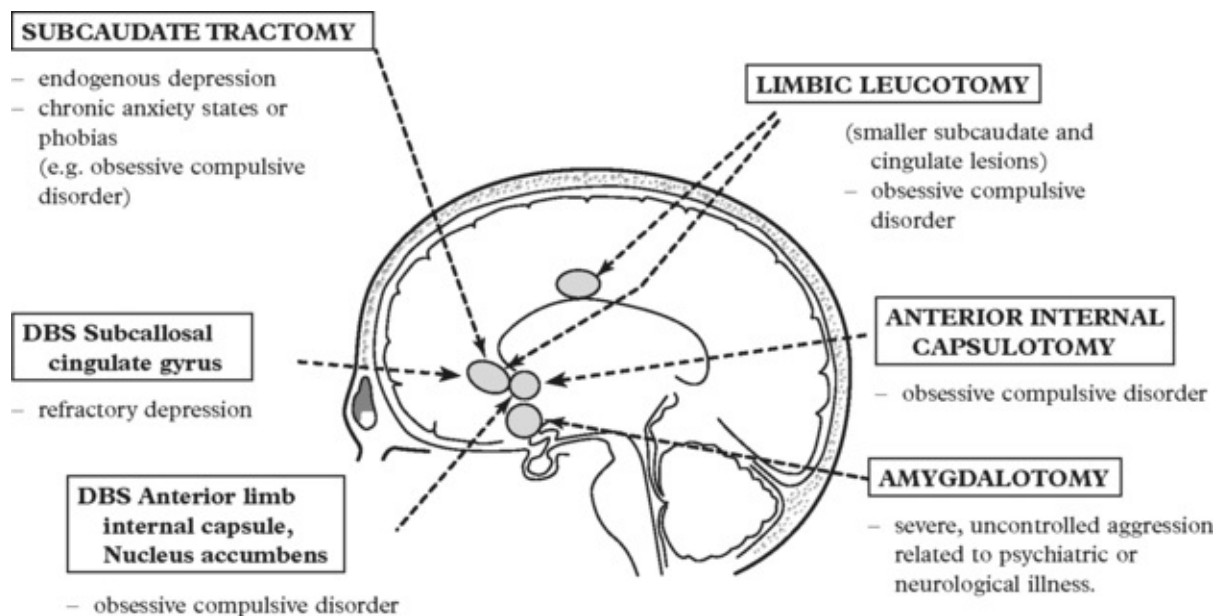
PSYCHOSURGERY

In 1935, observation of behavioural changes in chimpanzees following bilateral ablation of the frontal association area, led to the introduction of lesion-making for psychiatric disease (Moniz). In Britain, between 1940 and 1955, neurosurgeons performed over 10 000 *prefrontal leucotomy* operations. It became evident that patients with affective problems – depression, anxiety and obsessive compulsive disorder – showed better results than those with schizophrenia. Due to the introduction of *chlorpromazine* in the 1950s, and the operative

complications prefrontal leucotomy fell into disrepute, but despite pharmacological improvements, some patients developed chronically disabling conditions and the need for a surgical procedure persisted in those where drugs had little effect.

Stereotactic surgery provides a low risk method of lesion-making and is now generally accepted as a suitable treatment in *selected patients where drug treatment has failed*. Since issues of ‘informed consent’ for such procedures in the mentally ill are often ethically difficult, careful assessment by a multidisciplinary team of psychiatrists and neurosurgeons is essential. In recent years, interest has focussed on deep brain stimulation (DBS) as an alternate to an irreversible ablative lesion and early results are encouraging (see [page 387](#)).

INDICATIONS FOR STEREOTACTIC SURGERY AND LESION SITE



Results

Depression/anxiety states	– up to two-thirds benefit from subcaudate tractotomy.
Obsessional neurosis	– 80% improve following limbic leucotomy and deep brain stimulation.

SECTION IV

LOCALISED NEUROLOGICAL DISEASE AND ITS MANAGEMENT B. SPINAL CORD AND ROOTS

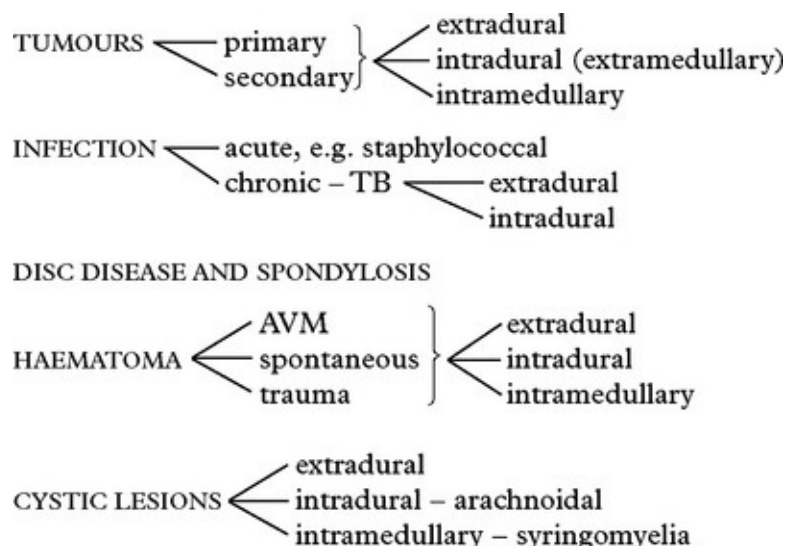
SPINAL CORD AND ROOTS

Disorders localised to the spinal cord or nerve roots are detailed below, but note that many diffuse neurological disease processes also affect the cord (see [Section V](#), e.g. multiple sclerosis, Friedreich's ataxia).

SPINAL CORD AND ROOT COMPRESSION

As the spinal canal is a rigidly enclosed cavity, an expanding disease process will eventually cause cord and/or root compression.

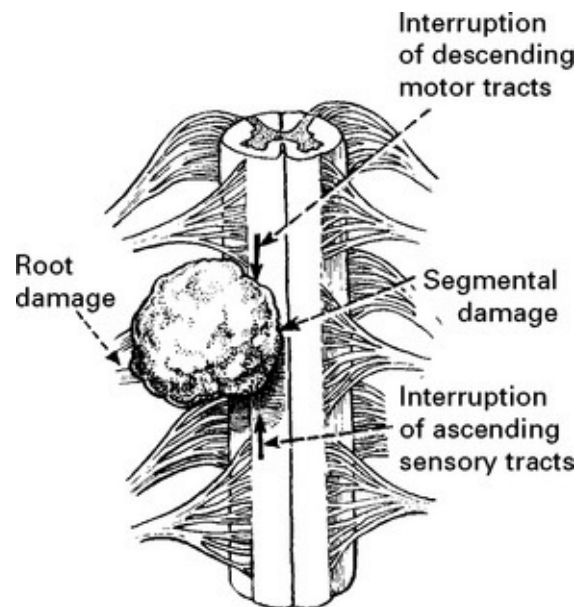
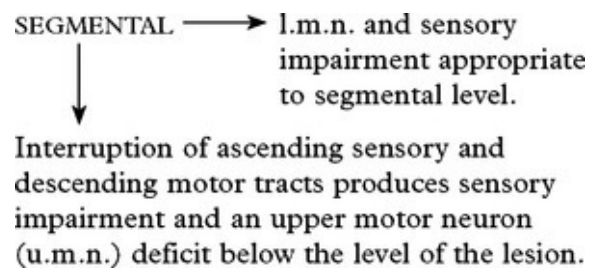
Causes



Manifestations of cord or root compression depend upon the following:

Site of lesion within the spinal canal: an expanding lesion outside the cord produces signs and symptoms from root and segmental damage.

ROOT – lower motor neuron (l.m.n.) and sensory impairment appropriate to the distribution of the damaged root.



Lesions within the cord (intramedullary) produce segmental signs and symptoms.

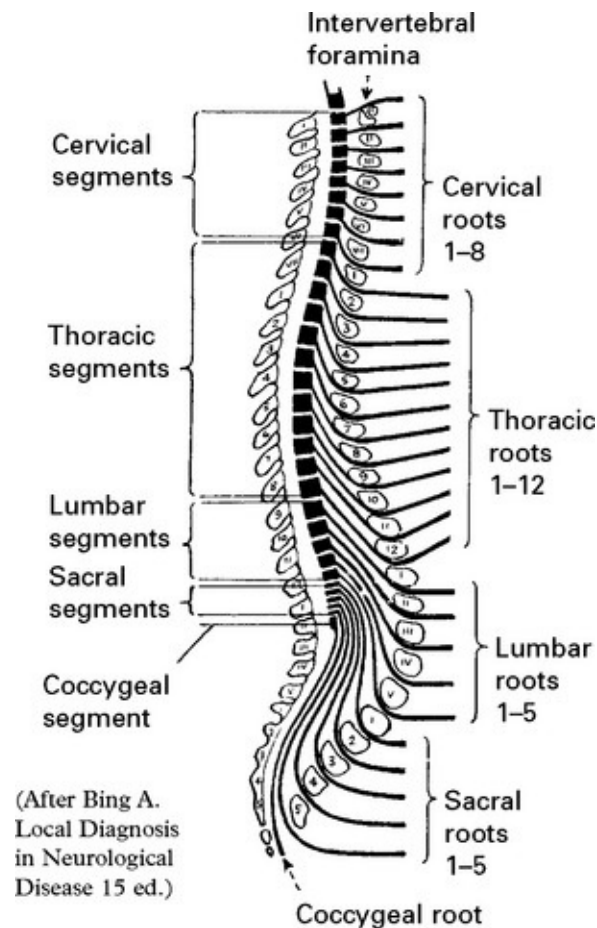
SPINAL CORD AND ROOT COMPRESSION

Level of the lesion: a lesion above the L1 vertebral body may damage both the cord and its roots. Below this, only roots are damaged.

Vascular involvement: neuronal damage from mechanical stretching

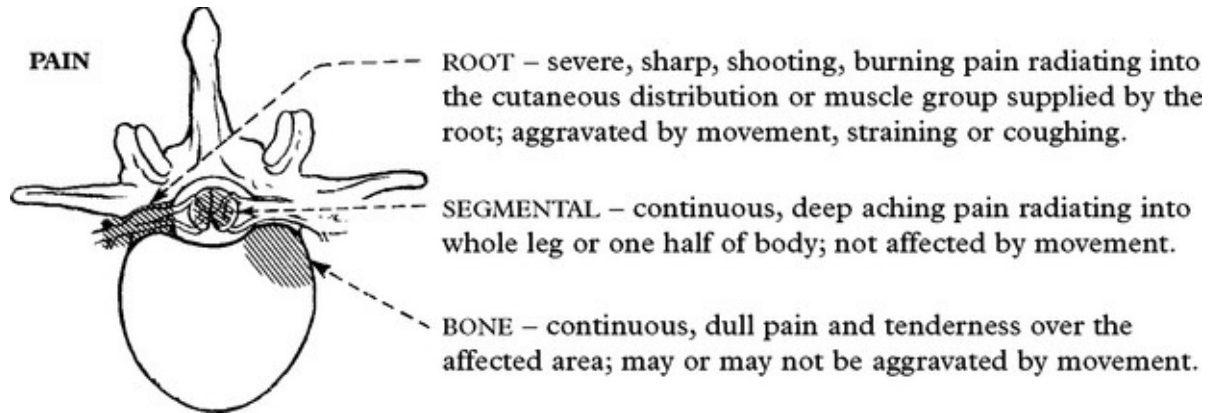
is of less importance than the vascular effects. At first venous obstruction leads to vasogenic oedema, but eventually impaired arterial flow causes irreversible spinal cord infarction. Clinical findings may suggest cord damage well beyond the level of compression, implying a distant ischaemic effect from vessel compromise at the lesion site.

Speed of onset: speed of compression effects the clinical picture. Despite producing upper motor neuron damage, a rapidly progressive cord lesion often produces a 'flaccid paralysis' with loss of reflexes and absent plantar responses. This state is akin to 'spinal shock' seen following trauma. Several days or weeks may elapse before tone returns accompanied by the expected 'upper motor neuron' signs.



Clinical features

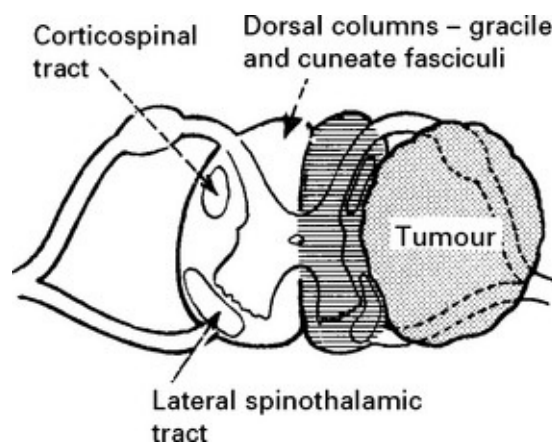
These depend on the site and level of the compressive lesion.



SPINAL CORD AND ROOT COMPRESSION – NEUROLOGICAL EFFECTS

LATERAL COMPRESSIVE LESION

Root/segmental damage

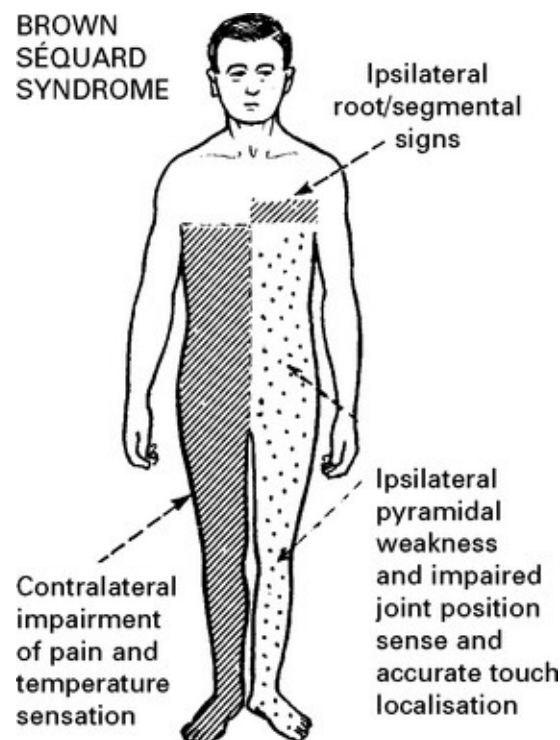


MUSCLE WEAKNESS in groups supplied by the involved root and segment with LOWER MOTOR NEURON (l.m.n.) signs: – wasting; – loss of tone; – fasciculation; – diminished or absent reflexes. N.B. motor deficit is seldom detected with root lesions above C5 and from T2 to L1.

SENSORY DEFICIT of all modalities or hyperaesthesia in area supplied by the root, but overlap from adjacent roots may prevent detection.

Long tract – signs and symptoms Partial (Unilateral) cord lesion

(Brown-Séquard syndrome)



MOTOR DEFICIT – dragging of the leg. In high cervical lesions weakness of finger and hand movements are noted on the side of the lesion.

UPPER MOTOR NEURON (u.m.n.) signs (maximal on side of lesion):

- weakness in a ‘pyramidal’ distribution, *i.e.* arms – extensors predominantly affected; legs – flexors predominantly affected.
- increased tone, clonus; – increased reflexes;

- extensor plantar response.

SENSORY DEFICIT – numbness may occur on the same side as the lesion and a burning dysaesthesia on the opposite side.

- joint position sense and accurate touch localisation (two point discrimination) impaired on side of lesion.

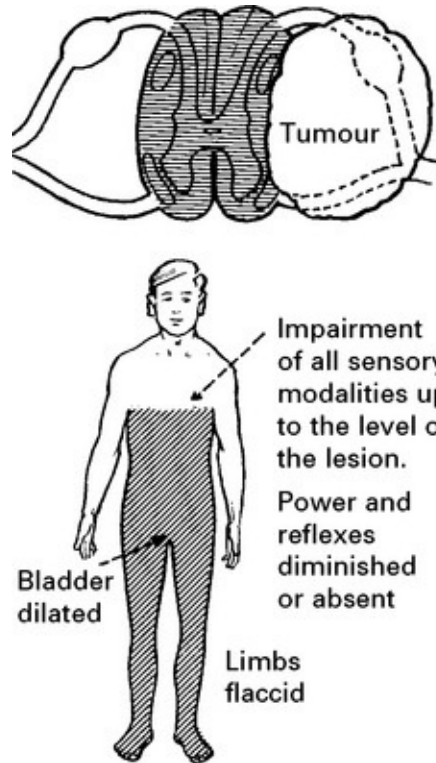
- Pinprick and temperature sensation impaired on opposite side.

In practice, cord damage is seldom restricted to one side. Usually a mixed picture occurs, with an asymmetric distribution of signs and symptoms.

Damage to sympathetic pathways in the T1 root or cervical cord causes an ipsilateral *Horner's syndrome* ([page 145](#)).

BLADDER symptoms are infrequent and only occur when cord damage is bilateral. Precipitancy or difficulty in starting micturition may precede retention.

Long tract damage – complete cord lesion

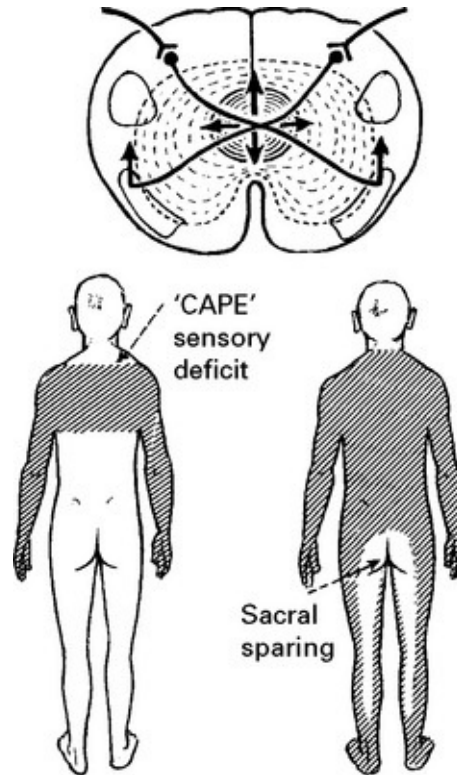


MOTOR DEFICIT: the speed of cord compression affects the clinical picture. Slowly growing lesions present with difficulty in walking; the legs may 'jump' at night. Examination reveals u.m.n. signs often with an asymmetric distribution. Rapidly progressive lesions produce 'spinal shock' – the limbs are flaccid, power and reflexes diminished or absent and plantar responses absent or extensor.

SENSORY DEFICIT: involves all modalities and occurs up to the level of the lesion.

BLADDER: patient first notices difficulty in initiating micturition. Retention follows, associated with incontinence as automatic emptying occurs. *Constipation* is only noticed after a few days. Some patients develop *priapism* (painful erection).

CENTRAL CORD LESION



Segmental damage: A central lesion initially damages the second sensory neuron crossing to the lateral spinothalamic tract; pain and temperature sensations are impaired in the distribution of the involved segment – a suspended sensory loss. As the lesion expands, anterior horn cells are also involved and a l.m.n. weakness occurs.

Long tract effects: further lesion expansion damages the spinothalamic tract and corticospinal tracts, the most medially situated fibres being involved first. With a lesion in the cervical region, the sensory deficit to pain and temperature extends downwards in a 'CAPE'-like distribution. As the sacral fibres lie peripherally in the lateral spinothalamic tract, SACRAL SPARING can occur, even with a large lesion. Involvement of the corticospinal tracts produces u.m.n. signs and symptoms in the limbs below the level of the lesion. The bladder is usually involved late.

In the cervical cord, sympathetic involvement may produce a unilateral

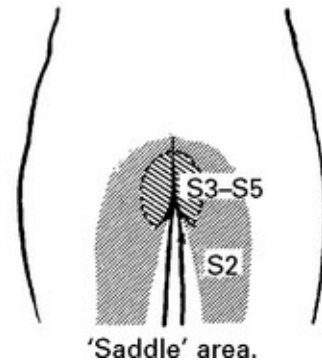
or bilateral *Horner's syndrome*.

LOWER CORD (CONUS) CAUDA EQUINA LESIONS



Root or segmental lesions may involve the upper part of the cauda equina and produce root/segmental and long tract signs as described on the previous page, e.g. an expanding proximal L4 root lesion causes weakness and wasting of the foot dorsiflexors, sensory deficit over the inner calf, an increased ankle jerk and an extensor plantar response. Bladder involvement tends to occur late.

The lower sacral roots are involved early, producing loss of motor and sensory bladder control with detrusor paralysis. Overflow incontinence ensues. Impotence and faecal incontinence may be noted. A l.m.n. weakness is found in the muscles supplied by the sacral roots (foot plantarflexors and evertors), the ankle jerks are absent or impaired and a sensory deficit occurs over the 'saddle' area.



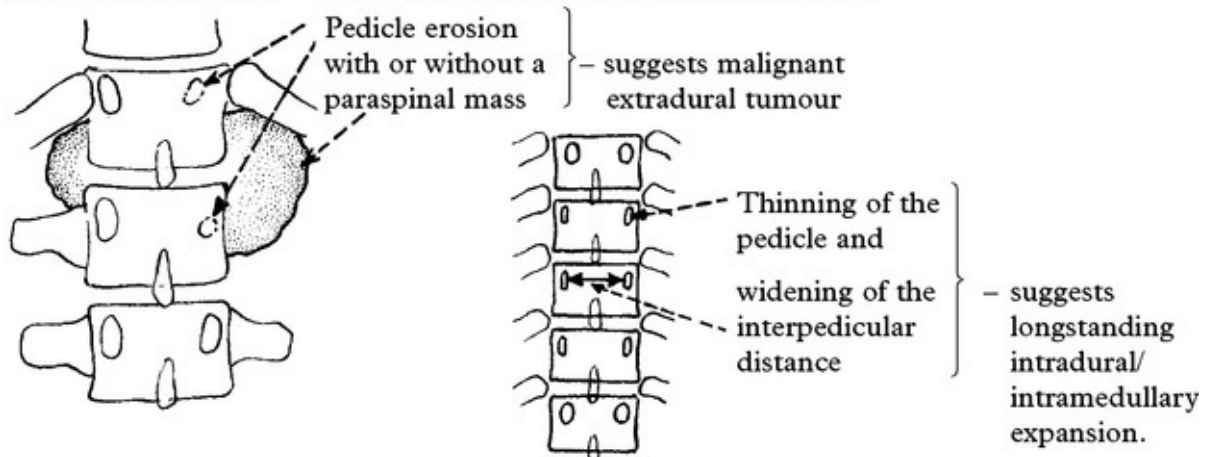
VERTEBRAL COLUMN

If a spinal cord or root lesion is suspected look for:

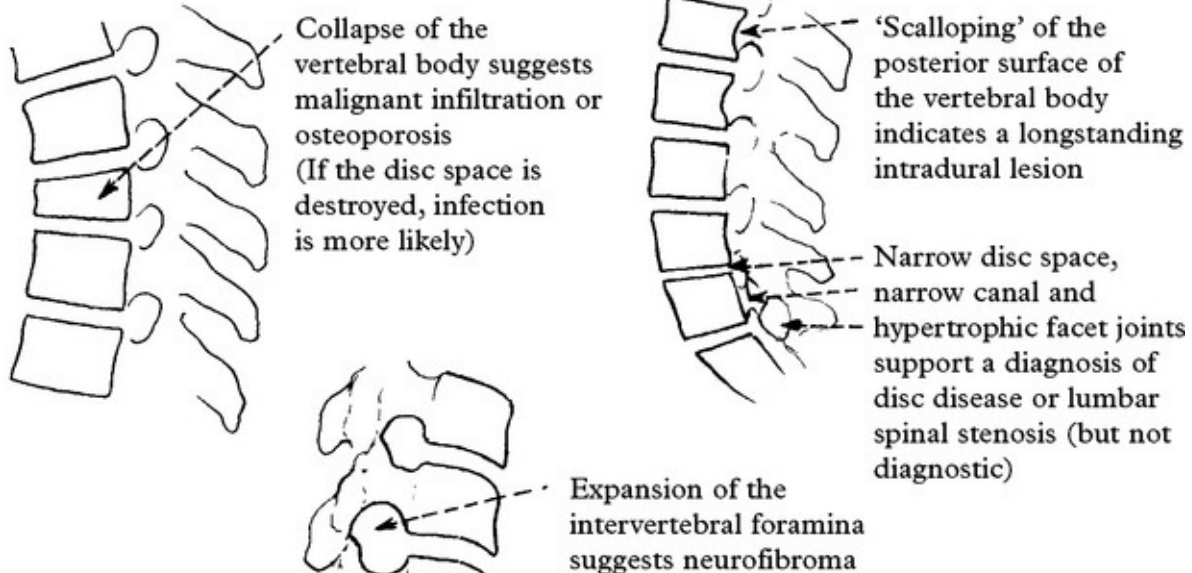
- | | | |
|---|---|---|
| - Scoliosis, loss of lordosis or limitation of straight leg raising | } | - suggests root irritation |
| - Paravertebral swelling | | |
| - Tenderness on bone percussion | } | - suggests malignant disease or infection |
| - Restricted spinal mobility | | |
| - Sacral dimple or tuft of hair | | - suggests bone, disc or root involvement |
| | | - suggests spina bifida occulta/dermoid. |

SPINAL CORD AND ROOT COMPRESSION – INVESTIGATIONS

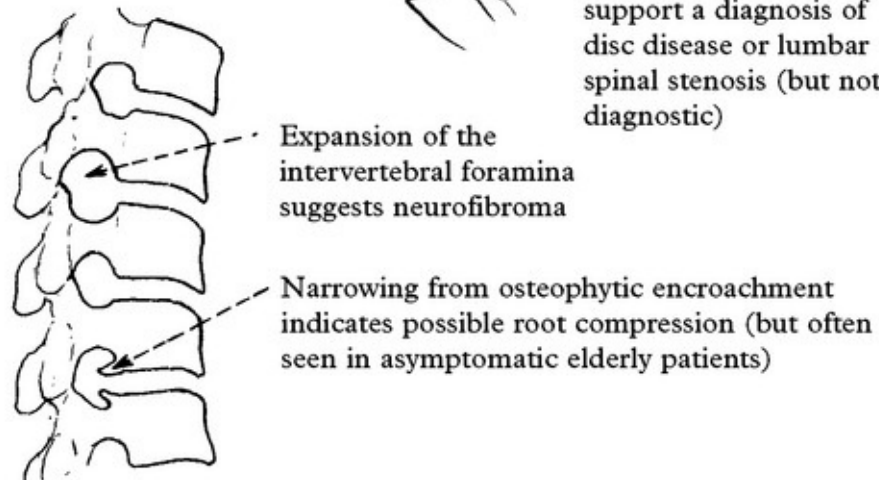
STRAIGHT X-RAY On the ANTERO-POSTERIOR views look for:



On the LATERAL view



On OBLIQUE views



MRI

This is now the investigation of choice for spinal disease, whether this lies within or outwith the dura or the spinal cord. Clinical examination

and straight X-rays may suggest the level of the lesion, but for suspected metastatic disease, a *sagittal* MRI should cover the whole spine since more than one site may be involved and the site of compression may lie many segments higher than the clinical signs indicate. The examination must involve both T1 and T2 weighted images, the former often repeated with gadolinium enhancement.



Sagittal T2 weighted MRI
showing metastatic tumour at the
4th thoracic vertebral level



Coronal T2 weighted MRI showing an intradural, extramedullary lesion (ependymoma)

On displaying an abnormality at a particular site, *coronal views* and *axial views* at selected levels may provide additional information.

MRI differentiates a syrinx ([page 401](#)) or a cystic swelling within the spinal cord from a solid intramedullary tumour ([page 400](#)).

MYELOGRAPHY

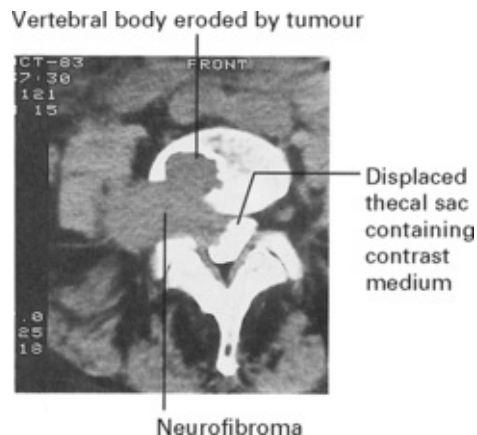
If MRI is unavailable or contraindicated *e.g.* pacemaker, myelography is used to screen the spinal cord and the cauda equina. This will identify the level of a compressive lesion and indicate its probable site *i.e.* intradural, extradural.

Even with an apparent ‘complete’ block, sufficient contrast medium may be ‘coaxed’ beyond the lesion to determine its upper extent. If not, a cervical puncture may be necessary.

Lesions in the lumbar and sacral regions require a ‘radiculogram’, outlining the lumbosacral roots.

CT SCAN/CT MYELOGRAPHY

It is impractical to use this as a screening investigation for cord compression, but if the level of interest is known, CT scanning may provide additional information.



CSF ANALYSIS

This is of limited value in cord compression. Abnormalities frequently occur, but *lumbar puncture may precipitate neurological deterioration*, presumably due to the creation of a pressure gradient.

Plain CT with axial cuts will clearly demonstrate bone erosion, osteophytic outgrowth and thickened facet joints causing narrowing of the spinal canal or intervertebral foramen. Axial cuts will also demonstrate disc herniation, the relationship of vertebral bone destruction to a paraspinal mass (e.g. metastatic tumour) and the extraspinal extent of an intraspinal lesion (e.g. neurofibroma).

CT myelography with axial cuts (CT performed either 6–12 hours after routine myelography or immediately after intrathecal injection of just a few ml of contrast) demonstrates clearly the degree of spinal cord or nerve root compression.

If cord compression is suspect then lumbar puncture and CSF analysis should await imaging.

CSF protein: often increased, especially below a complete block.

CSF cell count: a marked leucocyte count suggests an infective cause – abscess or tuberculosis.

CSF cytology may reveal tumour cells

SPINAL CORD AND ROOT COMPRESSION

TUMOURS

Incidence: The table below shows the approximate incidence of spinal tumours extracted from large series, but excludes tumours of the vertebral column and lymphoma. Spinal metastasis is by far the most common spinal tumour occurring in about 2–3% of all patients with cancer, but the incidence is an underestimate since not all undergo autopsy.

Adults		Children	
Meningioma	29%	Meningioma	10%
Schwannoma	24%	Schwannoma/Neurofibroma	24%
Ependymoma	23%	Ependymoma	10%
Astrocytoma	6%	Astrocytoma low grade	24%
Other	18%	Malignant glioma	10%
(e.g. haemangioblastoma, lipoma, ependymal cysts, metastasis)		Lipoma	10%
		Ganglioglioma	5%
Adapted from Louis et al. WHO Classification of Tumours of the CNS, IARC Press 2007		Adapted from Rickert & Paulus Child's Nervous System 2001 17:503–511	

Pathology: The pathological features of spinal tumours match those of their intracranial counterparts (see [page 303](#)).

METASTATIC TUMOUR

Occurs in 5% of all cancer patients and accounts for 50% of adult acute myelopathies.

Primary site: Usually breast, lung, prostate, kidney or myeloma (see below).

Metastatic site: Thoracic vertebrae most often involved, but metastasis may occur at any site and may be multiple.

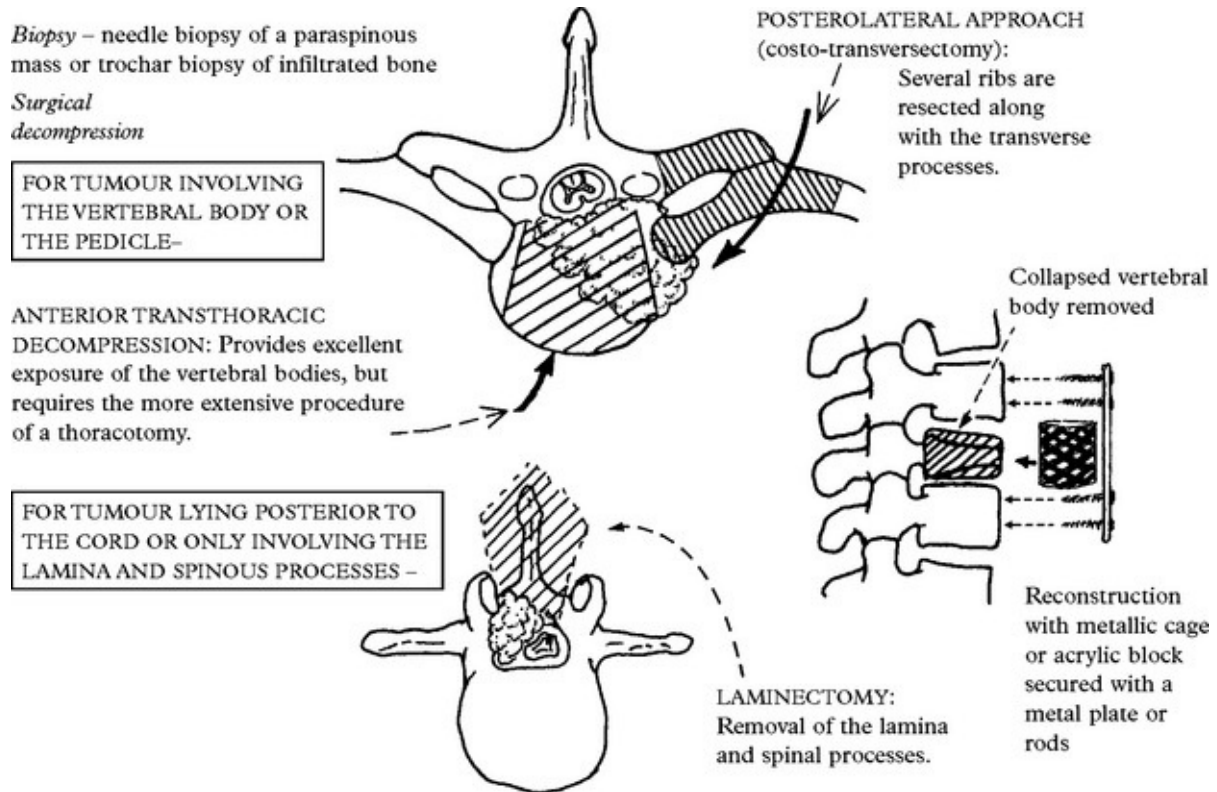
Clinical features: Bone pain and tenderness are common features usually preceding limb and autonomic dysfunction.

Investigations: Plain radiology may be diagnostic as osteolytic lesions or vertebral collapse are present in most cases. MRI will identify extradural compression and help exclude or confirm multiple level disease.

Management

In earlier years, numerous patients were subjected to a 'decompressive' laminectomy followed by radiotherapy. Since metastatic tumour usually involves the vertebral body and pedicles, removal of the spinous processes and lamina increases instability. Not surprisingly results were extremely poor and led to a swing towards radiotherapy alone. A recent randomised trial in patients with radioresistant tumours affecting one site comparing decompressive surgery plus radiotherapy against radiotherapy alone showed that surgery increased the percentage of patients who remained ambulant and who regained the ability to walk. This has led to a revival of decompressive procedures in such patients. Surgical treatment aims to establish a histological diagnosis, to decompress the spinal cord and to provide stability if instability causes pain.

Techniques



MANAGEMENT SCHEME

All patients require steroids and biopsy if no prior diagnosis exists.

Factor supporting:

Decompressive surgery (plus radiotherapy)	Radiotherapy
• Preferably ambulant, but not paraplegic > 4 hours • Radioresistant tumour • Single-level disease • Instability at the affected level • Life-expectancy > 4 months • Deterioration following previous radiotherapy.	• Radiosensitive tumour • Multi-level disease • Life-expectancy < 4 months • Stable neurological disease.

Major operative treatment is inappropriate in the elderly, when paraplegia persists for > 48 hours and in those with a dismal prognosis. In such patients, if medication fails to control pain, a palliative course of radiotherapy may help.

Prognosis: Outcome depends on the nature of the primary tumour. Median survival is 3–6 months. Early diagnosis is important to ensure that the majority of patients remain ambulant. Good prognostic factors include – ambulant before or after treatment, a radiosensitive tumour and only one level of involvement.

MYELOMA

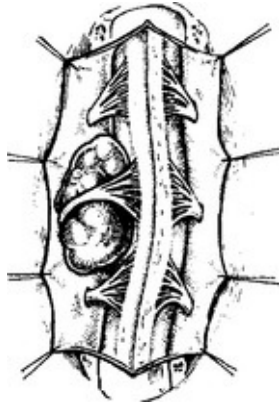
This malignant condition usually affects older age groups. It is often multifocal, involving the vertebral bodies, pelvis, ribs and skull, but solitary tumours may occur ('plasmacytoma'). Spinal cord compression occurs in 15% of patients with myeloma and rarely without vertebral body involvement due to intradural deposits. If suspect, look for characteristic changes in the plasma immunoglobulins and for Bence-Jones protein in the urine. An isotope bone scan may be less informative than a radiological skeletal survey. Bone marrow shows infiltration of plasma cells. Serum calcium levels may be high.

Management is as for metastatic tumour with additional chemotherapy. The prognosis is variable but patients may survive many years with a solitary plasmacytoma.

MENINGIOMA

Spinal meningiomas tend to occur in elderly patients and are more common in females than in males. They usually arise in the thoracic region and are almost always intradural. Slow growth often permits considerable cord flattening to occur before symptoms become evident.

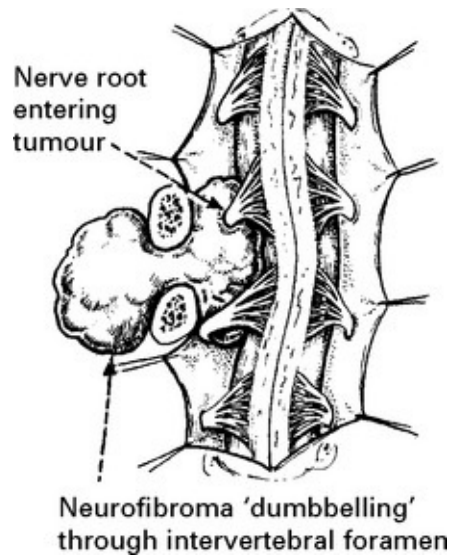
MRI or CT myelography will identify the lesion.



The operative aim is complete removal. Results are usually good, but if the tumour arises anteriorly to the cord, excision of the dural origin is difficult, if not impossible, and recurrence may result.

SCHWANNOMA/NEUROFIBROMA

Schwannomas are slowly growing benign tumours occurring at any level and arising from the posterior nerve roots. They lie either entirely within the spinal canal or 'dumbbell' through the intervertebral foramen, on occasions presenting as a mass in the thorax or posterior abdominal wall.



Neurofibromas are identical apart from their microscopic appearance ([page 304](#)) and their association with multiple neurofibromatosis (Von Recklinghausen's disease NF1 – see [page 561](#)) – look for café au lait patches in the skin.

Schwannomas tend to occur in the 30–60 age group. Typically they present with root pain. Root signs and/or signs of cord compression may follow.

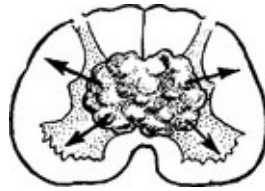
MRI or *CT myelography* identifies an intradural/extramedullary lesion. Axial views will delineate any extraspinal extension (see [page 395](#)). Complete operative removal is feasible but the nerve root of origin is inevitably sacrificed. Overlap from adjacent nerve roots usually minimises any resultant neurological deficit.

INTRAMEDULLARY TUMOURS

Intrinsic tumours of the spinal cord occur infrequently. In adults, ependymomas occur more frequently, but in children low grade astrocytomas are by far the most common. Cystic cavities may lie within the tumour or at the upper or lower pole. Benign lesions include

haemangioblastoma, lipoma, epidermoid, tuberculoma and cavernous angioma.

Clinical features

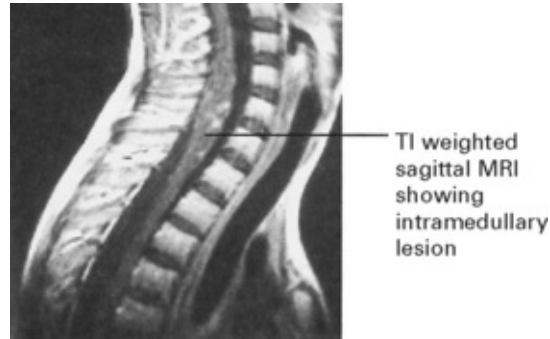


The onset is usually gradual. Segmental pain is common. Interruption of the decussating fibres of the lateral spinothalamic tract causes loss of pain and temperature sensation at the level of the involved segments.

Tumour expansion and involvement of the anterior horn cells produces a lower motor neuron weakness of the corresponding muscle groups; corticospinal tract involvement produces an upper motor neuron weakness below the level of the lesion. The sensory deficit spreads downwards bilaterally, the sacral region being the last to become involved.

Investigations

Straight X-rays occasionally show widening of the interpedicular distance or 'scalloping' of the vertebral bodies. MRI shows widening of the cord and differentiates solid tumour from syringomyelia. It also identifies the extent of the lesion and any associated cysts.



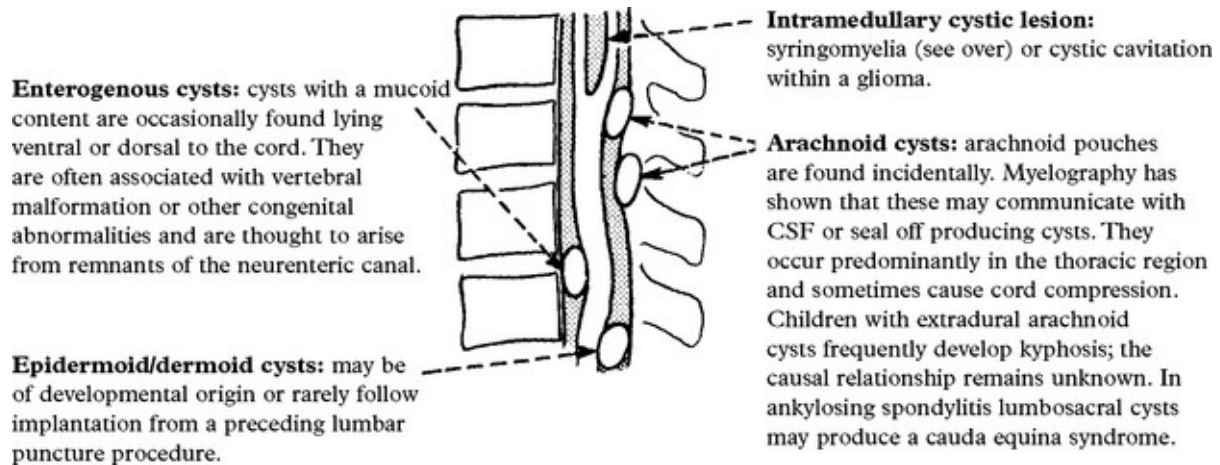
Management

When an intrinsic cord tumour is suspected, an exploratory laminectomy is required. An attempt is made to obtain a diagnosis either through a longitudinal midline cord incision or by needle biopsy. Cystic cavities within a tumour or an associated syringomyelia may benefit from aspiration. With some ependymomas and benign lesions, a plane of cleavage is evident and partial or even total removal is possible. Attempted removal of low grade astrocytomas carries less encouraging results and operation is contraindicated in malignant tumours. After tumour biopsy or removal, radiotherapy is often administered, but its value is uncertain.

EPENDYMOMA OF THE CAUDA EQUINA

Over 50% of spinal ependymomas occur around the cauda equina and present with a central cauda equina syndrome ([page 394](#)). Operative removal combined with radiotherapy usually gives good long-term results, although metastatic seeding occasionally occurs through the CSF.

SPINAL CYSTIC LESIONS



SYRINGOMYELIA

Syringomyelia is the acquired development of a cavity (syrinx) within the central spinal cord. The lower cervical segments are usually affected, but extension may occur upwards into the brain stem (syringobulbia, see [page 381](#)) or downwards as far as the filum terminale.

The cavitation appears to develop in association with obstruction:

- around the foramen magnum in conjunction with the *Chiari malformation*.
- secondarily to *trauma* or *arachnoiditis*.

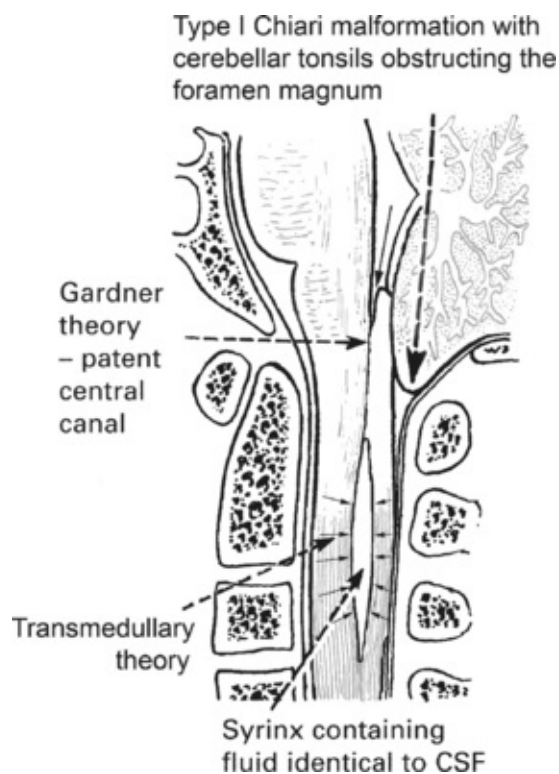
The syrinx may obliterate the central canal leaving clumps of ependymal cells in the wall. In contrast HYDROMYELIA is the congenital persistence and widening of the central canal.

Syringomyelia should be distinguished from cystic intramedullary tumours, although both pathologies may coexist.

Pathogenesis

The exact cause of this condition remains uncertain but theories abound. In 1965, Gardner proposed the '*hydrodynamic theory*', suggesting that the craniovertebral anomaly may impair CSF outflow from the 4th ventricle

to the cisterna magna. This in turn was believed to result in transmission of a CSF arterial pulse wave through a patent central canal, dilating the canal below the level of compression. This theory, however, does not explain the occurrence of syringomyelia in patients with non-patent central canals. It now seems likely that the normal free flow of CSF around the foramen magnum during the cardiac cycle becomes obstructed in patients with the Chiari malformation. In these patients downward movement of the tonsils occurs with each systole causing high CSF pressure waves which force CSF into the cord substance via the Virchow-Robin spaces (extension of the subarachnoid space around the blood vessels that penetrate the cord) *i.e.* transmedullary theory. This model does not require a patent central canal.



Clinical features



- *Dissociated sensory loss* (i.e. loss of pain and temperature sensation with retention of other senses) occurring in a cape-like distribution. Painless burns are a classic sign.
- *Wasting and weakness of the small muscles of the hand and winging of the scapula* from anterior horn cell involvement. *Scoliosis* often results.
- *Long tract signs* follow.
- *Brain stem signs* may appear, either from syringobulbia or an associated Chiari malformation.
- *Hydrocephalus* occurs in 25% but is usually asymptomatic.

Investigations

MRI is the investigation of choice (see [page 380](#)). This will demonstrate the syrinx with any associated Chiari malformation and exclude intramedullary tumour.

If MRI is unavailable – MYELOGRAPHY demonstrates widening of the spinal cord. With coexisting Chiari malformations, screening in the *supine* position will show the cerebellar tonsils descending below the foramen magnum.

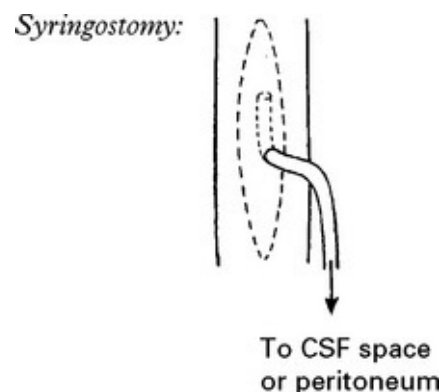
Historically introduction of air into the CSF space – AIR

MYELOGRAPHY – was used to ‘collapse’ the dilated segment thereby excluding an intrinsic cord tumour. A CT scan, six hours after injection of intrathecal contrast, may show uptake within the syrinx, but beware of misinterpreting normal contrast uptake within spinal cord tissue. Puncture of the syrinx is occasionally possible and subsequent injection of contrast shows its exact extent.

Management

The natural history is variable and operative techniques only of limited benefit. The approach depends on progression of symptoms and the presence or absence of an associated Chiari malformation.

If Chiari malformation is present – *decompression* by removing the posterior rim of the foramen magnum and posterior arch of the atlas and widening the dura with a patch, improves symptoms in most patients and should halt progression. This operation relieves the obstructed foramen magnum and alters the hydrodynamics of the syrinx. If deterioration continues, or if no associated Chiari malformation exists –



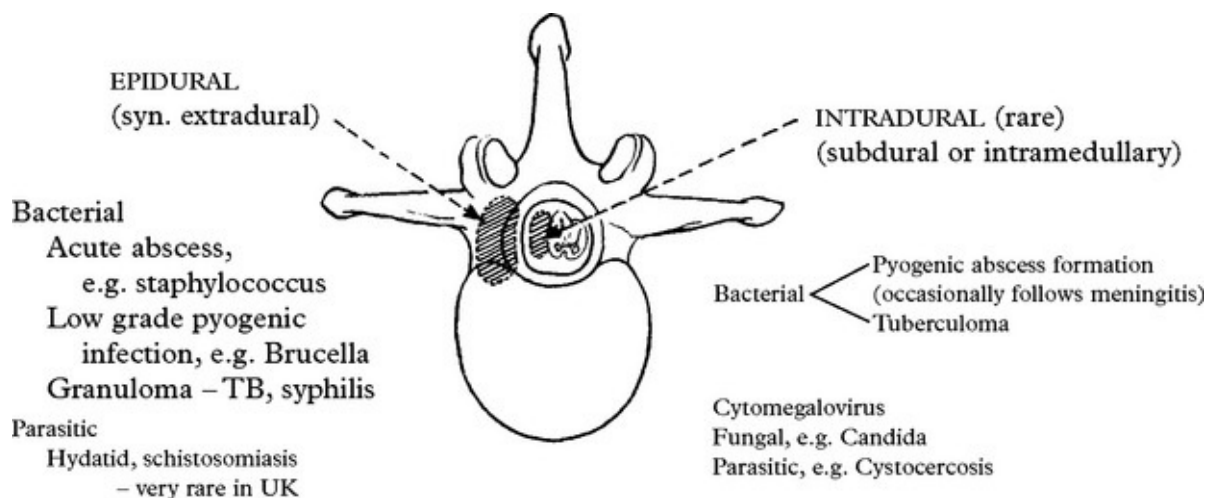
The syrinx is drained via a silastic tube into the surrounding CSF space.

Alternatively, a syringoperitoneal shunt is performed. Some patients benefit from this procedure but in others, progressive deterioration

continues.

Syringomyelia remains a difficult condition to treat. Draining the syrinx into the CSF space by syringostomy may not significantly alter the haemodynamics. A syringoperitoneal shunt may seem to be the most logical approach. Despite all efforts, about one-third of patients suffer progressive deterioration.

SPINAL INFECTION



ACUTE EPIDURAL ABSCESS

Tend to occur in debilitated patients – diabetes, malignancy, liver or renal failure, intravenous drug abuse and alcoholism.

Organism: *Staphylococcus aureus* is the most common agent (90% of cases).

Spread: Haematogenous, e.g. from a boil or furuncle, or direct from vertebral osteomyelitis.

Site: Usually thoracic, but may affect any level and be extensive. Cord damage occurs either from direct compression or secondary to a thrombophlebitis and venous infarction.

Clinical features: Develops over several days mimicking a rapidly progressive extradural tumour or haematoma with bilateral leg weakness, a sensory level and urinary retention, but distinguishing features are:

- very severe pain and tenderness over the involved site.
 - toxæmia: pyrexia, malaise, increased pulse rate.
 - rigidity of neck and spinal column, with marked resistance to flexion.
- As the abscess extends upwards, the sensory level may rise.

Investigations: *Straight X-ray* may or may not show an associated osteitis or discitis.

An *MRI* or *myelogram* confirms the site of the extradural lesion.

CSF examination, if performed shows an increased white cell count, usually polymorphonuclear, but may be normal

A *leucocytosis* is usually present in the peripheral blood and the ESR raised.

Blood cultures are often positive.

Management: Urgent decompressive laminectomy and abscess drainage combined with intravenous antibiotic therapy over some weeks provide the best chance of recovery of function. In the cervical spine, anterior collections may be drained through the disc space.

SPINAL TUBERCULOSIS (Pott's disease of the spine)

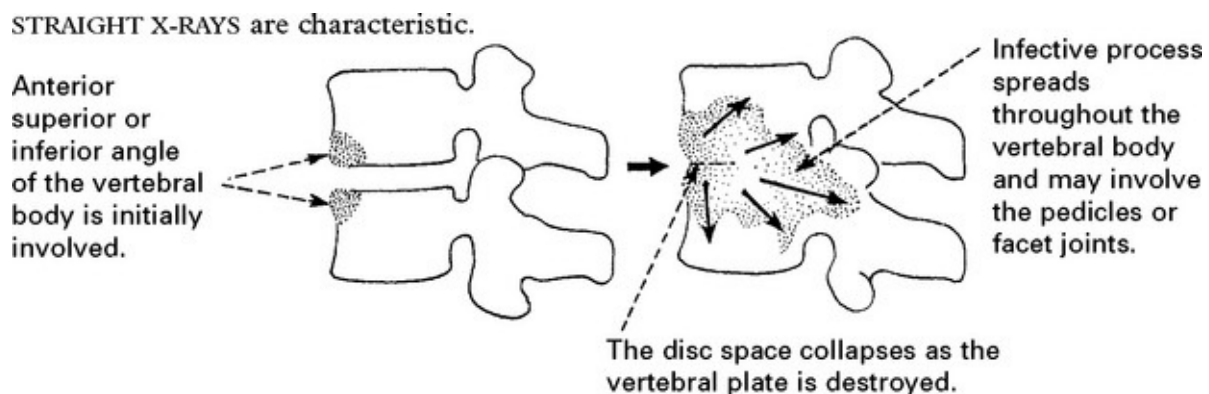
In developing countries, spinal TB is mostly a disease of childhood or adolescence. In Britain it usually affects the middle aged and is particularly prevalent in immigrant populations and in the immunocompromised. The incidence is now increasing, probably due to

the development of antibiotic resistance.

The lower thoracic spine is commonly involved and the disease initially affects the intravertebral disc and spreads to adjacent vertebral bodies.

Clinical features:

The classic systemic features of weight loss, night fever and cachexia are often absent. Pain occurs over the affected area and is made worse by weight bearing. Symptoms and signs of cord compression occur in approximately 20% of cases. The onset may be gradual as pus, caseous material or granulation tissue accumulate, or sudden as vertebral bodies collapse and a kyphosis develops.



MRI with gadolinium shows an epidural mass with paraspinal soft tissue swelling.

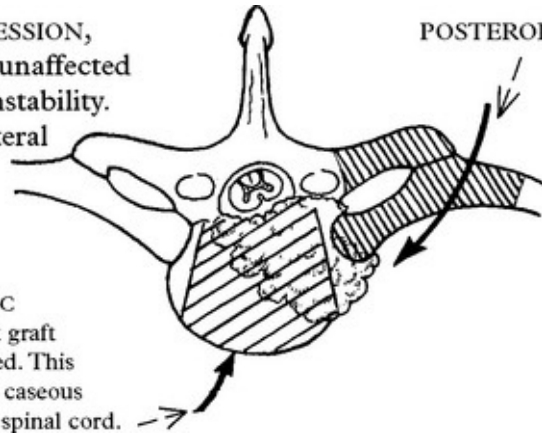
Management:

Every effort is made to establish the diagnosis. A *needle biopsy* is often sufficient, but occasionally an exploratory operation (costotransversectomy) is required. Long-term *antituberculous therapy* is commenced.

If signs of cord compression develop, decompression is necessary.

A POSTERIOR DECOMPRESSION, removing the remaining unaffected bone, is likely to cause instability. An anterior or posterolateral approach is therefore required.

ANTERIOR TRANSTHORACIC DECOMPRESSION with strut graft fusion is sometimes performed. This permits clearance of pus and caseous debris without retracting the spinal cord.

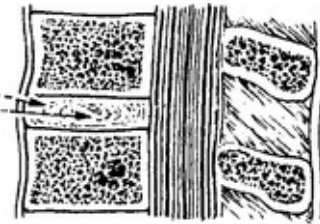


POSTEROLATERAL APPROACH (costotransversectomy): One or more ribs are resected medially, along with the transverse processes.

DISC PROLAPSE AND SPONDYLOSIS

Intervertebral discs act as shock absorbers for the bony spine.

A tough outer layer – the annulus fibrosis surrounds a softer central nucleus pulposus.



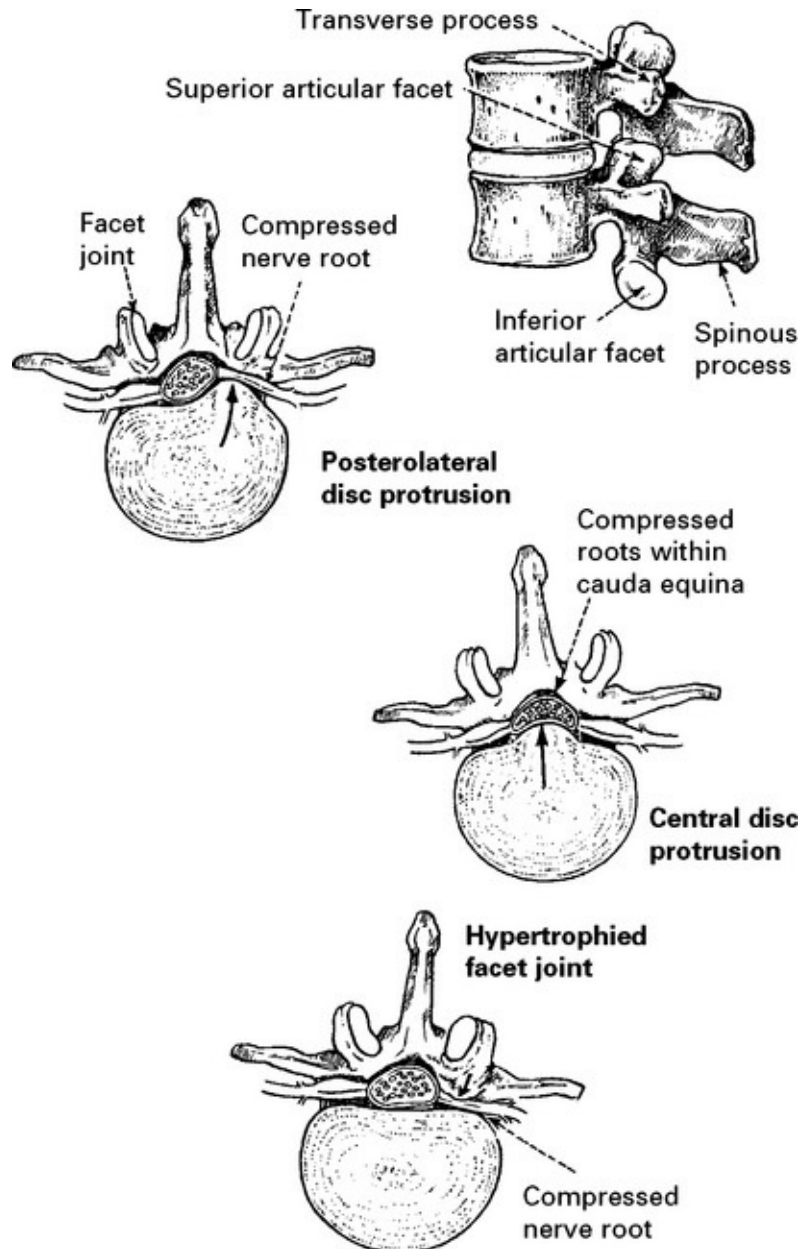
Discs degenerate with age, the fluid within the nucleus pulposus gradually drying out. Genetic factors may play a role. Disc collapse produces excessive strain on the facet joints, *i.e.* the superior and inferior articular processes of each vertebral body, and leads to degeneration and hypertrophy.

LUMBAR DISC PROLAPSE

Disc degeneration leads to a tear in the annulus fibrosis, perhaps precipitated by an injury or an excessive mechanical load. An *acute disc prolapse* occurs when the soft nucleus herniates through the annular tear causing irritation and/or compression of the adjacent nerve root. A 'free fragment' of the nucleus pulposus may extrude and lie above or below

the level of the disc space. Herniation usually occurs posterolaterally, but may occasionally occur centrally, compressing the cauda equina.

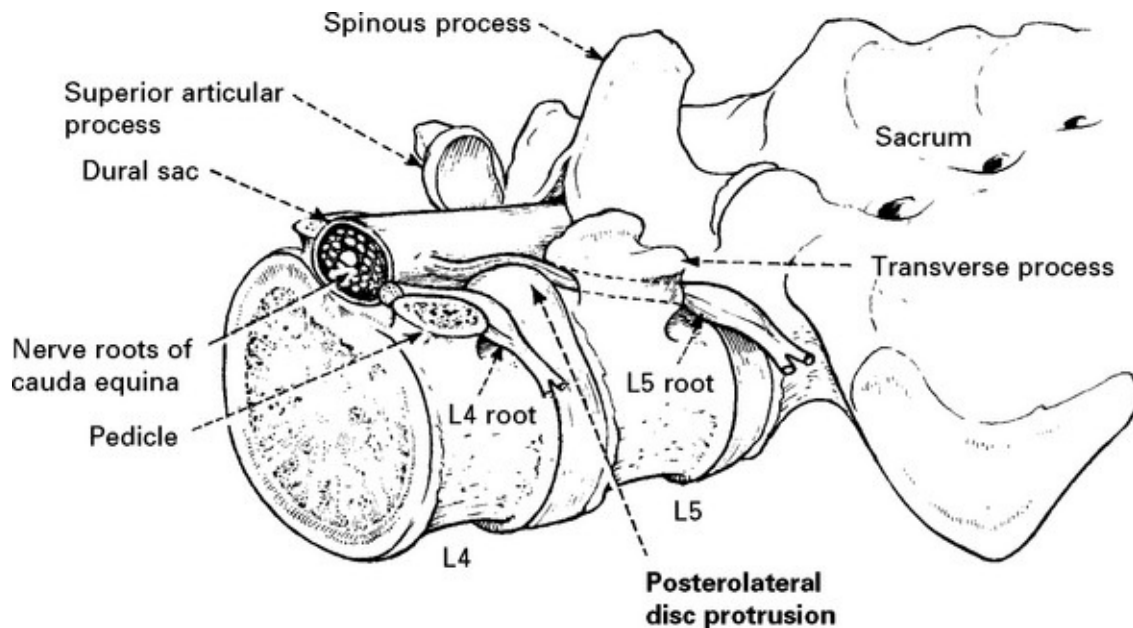
Disc degeneration may contribute to hypertrophy and degeneration of adjacent facet joints, a further source of back and leg pain and an important cause of root compression.



LUMBAR DISC PROLAPSE

A *congenitally narrowed spinal canal* increases susceptibility to the development of nerve root compression. Here the spinal canal diameter is considerably diminished and minor disc protrusion or mild joint hypertrophy may more readily compress the nerve root.

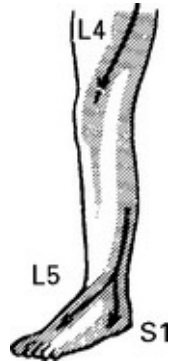
Posterolateral disc herniations usually compress the nerve root exiting through the foramen below the affected level, *e.g.* an L4/5 disc lesion will compress the L5 nerve root, but large disc protrusions or a free fragment may compress any adjacent root.



Lumbar disc lesions may occur at any level but L4/5 and L5/S1 are the commonest sites (95%).

CLINICAL FEATURES

Posterolateral disc protrusion



Injury: A history of falling, or lifting heavy weights; associated back pain often precedes the onset of leg symptoms.

Leg pain: Root irritation or compression produces pain in the distribution of the affected root and this should extend below the mid-calf. Coughing, sneezing or straining aggravates the leg pain which is usually more severe than any associated backache. If compression causes severe root damage the leg pain may disappear as neurological signs develop.

Sensory symptoms: Numbness or paraesthesia occur in the distribution of the affected root.

‘MECHANICAL’ SIGNS: Spinal movements are restricted, scoliosis is often present and is related to spasm of the erector spinae muscles, and the normal lumbar lordosis is lost.



Straight leg raising: L5 and S1 root compression causes limitation to less than 60° from the horizontal and produces pain down the back of the leg.

Dorsiflexion of the foot while the leg is elevated aggravates the pain. Elevation of the 'good' leg may produce pain in the other leg.

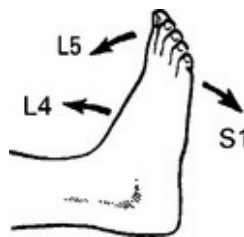
(If in doubt about the veracity of a restricted straight leg raising deficit, sit the patient up on the examination couch with the legs straight. This is equivalent to 90° straight leg raising.)

Reverse leg raising (femoral stretch)

Tests for irritation of higher nerve roots (L4 and above)



NEUROLOGICAL DEFICIT: Depends on the *predominant* root involved:



L4 – Quadriceps wasting and weakness; sensory impairment over medial calf; impaired knee jerk.

L5 – Wasting and weakness of dorsiflexors of foot, extensor digitorum longus and extensor hallucis longus; wasting of extensor digitorum brevis; sensory impairment over lateral calf and dorsum of foot.

S1 – Wasting and weakness of plantar flexors; sensory impairment over lateral aspect of foot and sole; impaired ankle jerk.

Root signs cannot reliably localise the level of disc protrusion due to variability of the anatomical distribution.

Central disc protrusion

Symptoms and signs of central disc protrusion are usually bilateral, although one side is often worse than the other.

Leg pain: Extends bilaterally down the back of the thighs. Pain may disappear with the onset of motor loss.

Paraesthesia: Occurs in the same distribution.

Sphincter paralysis: Loss of bladder and urethral sensation with intermittent or complete retention of urine occurs in most patients. Anal sensation is usually impaired and accompanies constipation.



Severe pain associated with lateral disc protrusion may inhibit micturition. In this instance, strong analgesia should allow normal micturition; the presence of normal perineal sensation excludes root

compression as the cause of the retention.

Sensory loss: Extends over all or part of the sacral area ('saddle' anaesthesia) and confirms a neurogenic cause for the sphincter disturbance.

Motor loss: Usually presents as foot drop with loss of power in the dorsiflexors and plantarflexors of both feet.

Reflex loss: The ankle jerks are usually absent on each side.

INVESTIGATION

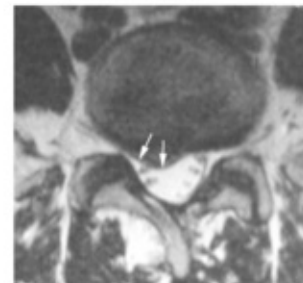
Straight X-ray of lumbosacral spine is of limited benefit in the investigation of lumbar disc disease – it may show loss of a disc space or an associated spondylolisthesis (see p. 410). Straight X-rays are *important in excluding other pathology* such as metastatic carcinoma.

MRI is the investigation of choice. Sagittal views combined with axial views at the appropriate level will demonstrate disc disease and exclude a lesion at the conus.

T1 weighted sagittal MRI

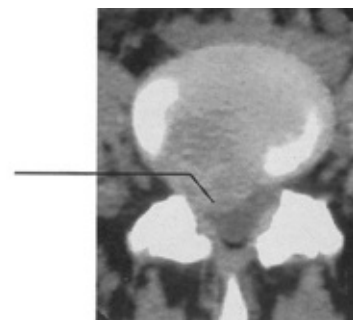


T2 weighted axial MRI



CT scanning of lowest three spaces will detect a disc protrusion and demonstrate the extent of root compression. CT scanning also clearly shows hypertrophy of the facet joints and the diameter of the spinal canal (see page 405).

NB A patient with characteristic 'root' pain in whom CT scanning is negative requires an MRI to exclude a lesion involving the conus medullaris.



Management

(a) Posterolateral disc protrusion

CONSERVATIVE: Most bouts of leg pain settle spontaneously by taking simple measures:

- *Analgesics*
- *Avoiding heavy lifting and bending.* Picking up objects from the floor should be performed by bending the knees and keeping the back straight
- *Bed rest* with a firm mattress, but *only* if pain prevents mobilisation

INDICATIONS FOR OPERATION

- Severe *unremitting leg pain* despite conservative measures.
- *Recurrent attacks* of leg pain, especially when causing repeated time loss from work.
- The development of a *neurological deficit* with unremitting pain.

PERCUTANEOUS PROCEDURES: These include a variety of techniques with the aim of decompressing the disc space by removing the nucleus pulposus by aspiration (*automated percutaneous discectomy*), laser therapy (*laser discectomy*) or radiofrequency energy (*coblation*). Although all techniques may produce some improvement in symptoms, none appears as effective as microdiscectomy. All require further evaluation.

PROSTHETIC INTERVERTBRAL DISC REPLACEMENT: Through an abdominal approach, the offending disc is removed and replaced with an artificial disc which allows a degree of movement between the adjacent vertebrae. Initial studies report good results, but as yet there is no evidence to suggest that this more extensive and more expensive procedure should replace standard microdiscectomy.

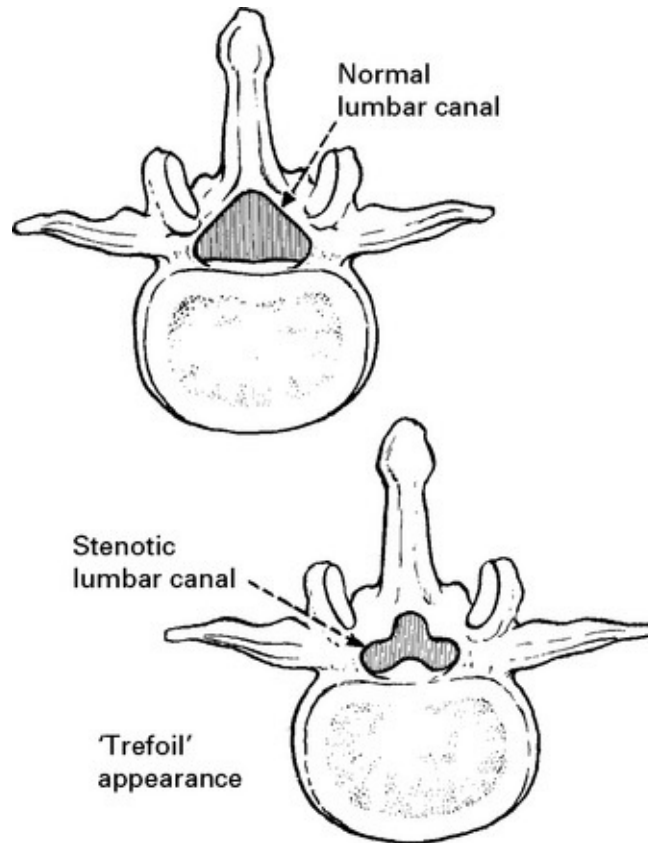
LUMBAR FUSION: This procedure has been available for many years, particularly for the treatment of low back pain. A recent randomised trial comparing lumbar fusion with an intensive rehabilitation programme found no evidence of any benefit from lumbar fusion.

After disc operation, patients are advised to avoid heavy lifting, preferably for an indefinite period. Persistence in a heavy manual job may lead to further trouble. In general, patients with clear-cut indications for operation do well, whereas those with dubious clinical or radiographic signs tend to have a high incidence of residual or recurrent problems.

(b) Central disc protrusion

Compression of the cauda equina from a central disc usually requires urgent treatment, particularly if signs and symptoms have developed within 24–48 hours. Retrospective studies suggest that the chance of recovery depends on the extent of nerve root damage at the time of the decompression, but for ethical reasons this cannot be tested by randomised trial. If symptoms have progressed to painless urinary retention with overflow incontinence, then the outcome is poor and the timing of surgery may not influence the results. In contrast to posterolateral protrusions, large central discs may require a one or two level laminectomy to minimise the risk of further root damage. After disc removal, recovery of function may continue for up to 2 years, but results are often disappointing. Although most regain bladder control, few have completely normal function and in many, disordered sexual function persists.

LUMBAR SPINAL STENOSIS



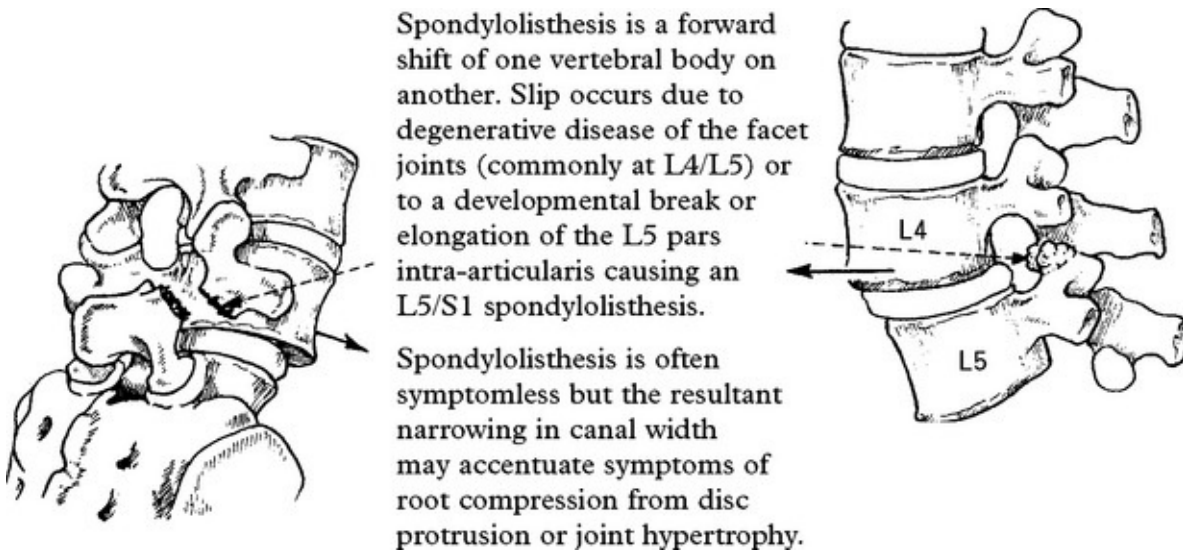
Congenital narrowing of the lumbar spinal canal, or secondary narrowing due to hypertrophic facet joints, may predispose to root compression from a herniated disc, but in addition may produce '*neurogenic claudication*'. Symptoms of root pain, paraesthesia or weakness develop after standing or walking and may be relieved by sitting, bending forwards or lying down. Straight leg raising is seldom impaired, in contrast to patients with disc protrusion. Objective neurological findings may only appear after exercise. In some patients this condition only affects one side – the '*unilateral facet syndrome*'.

Plain X-rays may show thickened joints, but MRI or CT scanning, is required to establish the diagnosis.

Treatment: Decompression of the nerve root canal either through bilateral fenestrations or via a laminectomy usually produces good

results with relief of symptoms. Implants available to distract the spinous processes at the affected level may help symptoms, but await full evaluation.

SPONDYLOLISTHESIS



Treatment: usually conservative, but if signs of root compression are present, then decompression of the root canal is necessary. Occasionally fusion is required, especially if back pain predominates.

THORACIC DISC PROLAPSE

This occurs rarely (0.2% of all disc lesions) due to the relative rigidity of the thoracic spine.

PRESENTATION

- Root pain and/or
- Progressive or fluctuating paraparesis (may lead to mistaken diagnosis).

As vascular involvement may produce damage above the level of

compression, sensory findings may be misleading.

INVESTIGATION

MRI is the investigation of choice and should clearly demonstrate the disc herniation and the extent of the associated cord compression.

CT myelography will clearly demonstrate the lesion if *MRI* is unavailable.

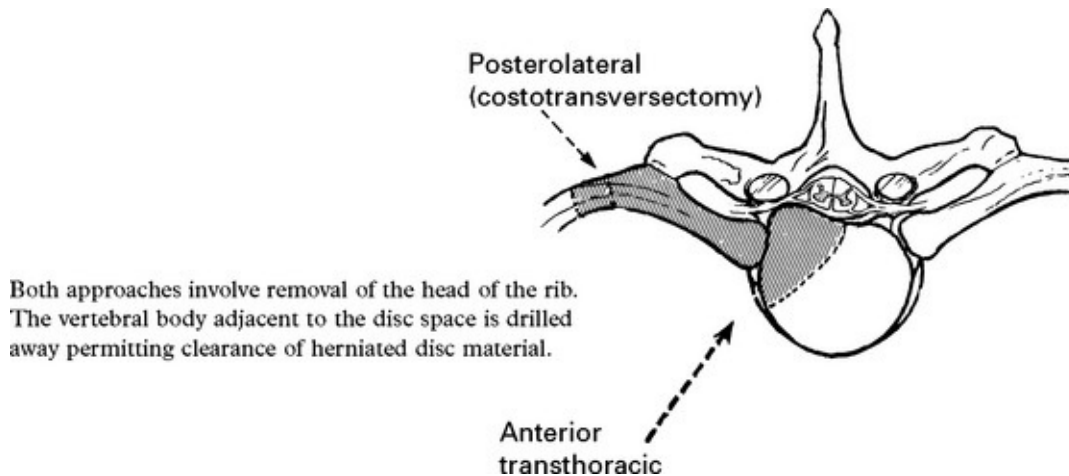


T2 weighted sagittal MRI showing mid thoracic disc compressing the spinal cord.

MANAGEMENT

Root pain – may settle with conservative treatment.

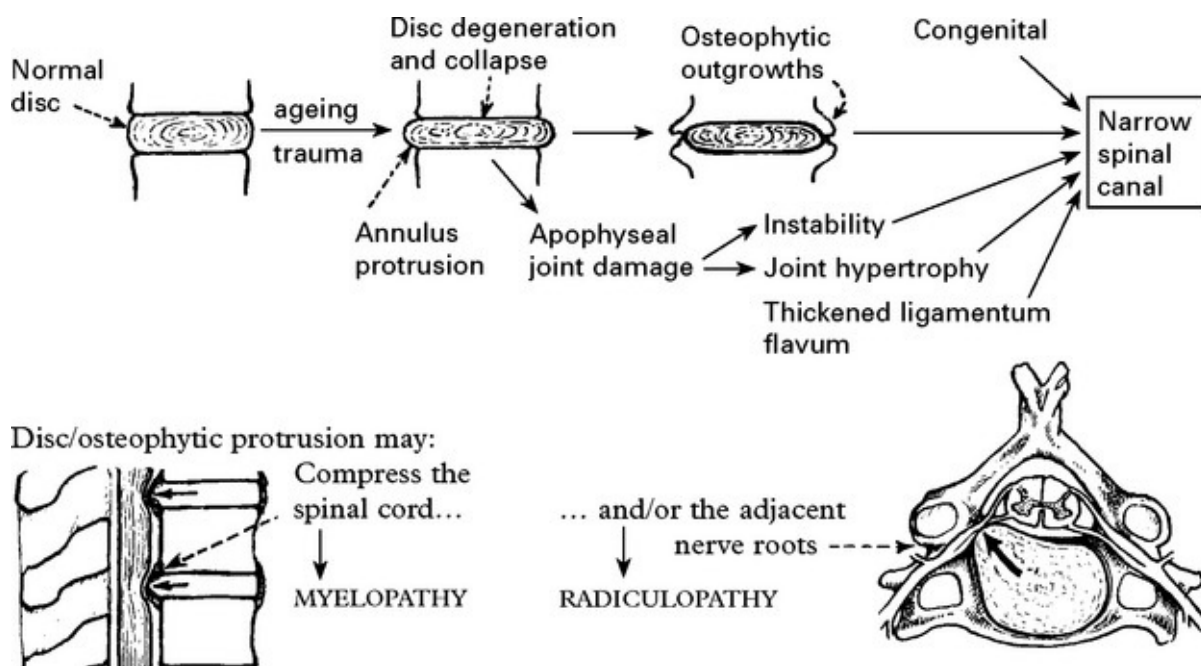
In the presence of cord compression or unremitting root pain, either a *posterolateral* or an *anterior transthoracic approach* is used to remove the disc. (A posterior approach – laminectomy – carries an unacceptably high risk of paraplegia.)



CERVICAL SPONDYLOSIS

The mobile cervical spine is particularly subject to osteoarthritic change and this occurs in more than half the population over 50 years of age; of these approximately 20% develop symptoms. Relatively few require operative treatment.

PATHOGENESIS



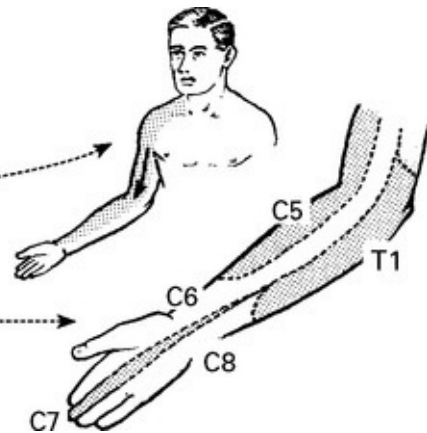
Resultant damage to the spinal cord may arise from direct pressure or may follow vascular impairment. The onset is usually gradual. Trauma may or may not predispose to the development of symptoms.

CLINICAL FEATURES

Radiculopathy

Pain: a sharp stabbing pain, worse on coughing, may be superimposed on a more constant deep ache radiating over the shoulders and down the arm.

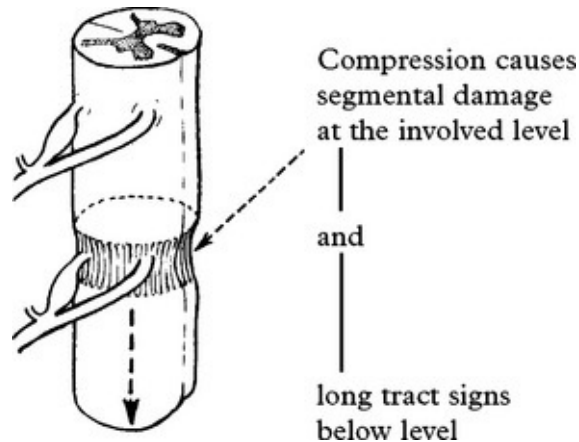
Paraesthesia: Numbness or tingling follows a nerve root distribution.



Root signs:

- *Sensory loss, i.e.* pin prick deficit in the appropriate dermatomal distribution.
- *Muscle (l.m.n.) weakness* and wasting in appropriate muscle groups, e.g. C5, C6 ... biceps, deltoid: C7 ... triceps.
- *Reflex impairment/loss, e.g.* C5, 6 ... biceps, supinator jerk: C7 ... triceps jerk.
- *Trophic change:* In long-standing root compression, skin becomes dry, scaly, inelastic, blue and cold.

Myelopathy



Arms: l.m.n. signs and symptoms, as above, at the level of the lesion
and/or

u.m.n. signs and symptoms below the level of the lesion, *e.g.* C5 lesion: deltoid and biceps weakness and wasting; reduced biceps reflex; increased finger reflex. C3/4 lesions produce syndrome of numb clumsy hands (reflecting posterior column loss).

Legs: u.m.n. signs and symptoms, *i.e.* difficulty in walking due to stiffness; 'pyramidal' distribution weakness, increased tone, clonus and extensor plantar responses; sensory symptoms and signs are variable and less prominent.

Sphincter disturbance is seldom a prominent early feature.

N.B. Involved segments may extend above or below the level of compression if the vascular supply is also impaired.

INVESTIGATION

Plain X-ray of cervical spine

Look for:

- congenital narrowing of canal, loss of lordosis.

- disc space narrowing and osteophyte protrusion (foraminal encroachment is best seen in oblique views).

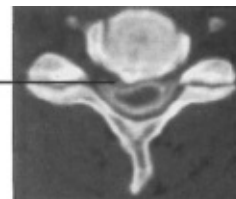
- subluxation. Flexion/extension views may be required.

MRI: the investigation of choice. *Sagittal views* clearly demonstrate cord compression at the level of the disc space. Any hyperintensity within the cord on T2 weighting reflects cord damage and may correlate with the severity of the myelopathy and outcome. *Axial views* show cord compression and the degree of foraminal narrowing.



T1 weighted MR
showing a C5/6
spondylotic bar

Myelography, particularly when combined with **CT scanning**, shows in detail the degree of spinal cord and nerve root compression from osteophytic outgrowth.



MANAGEMENT

Conservative

– Analgesics	Symptoms of radiculopathy, whether acute or chronic, usually respond to these conservative measures plus reassurance. Progression of a disabling neurological
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	deficit however demands surgical intervention.
– Cervical collar	The clinician may adopt a conservative approach when a myelopathy is mild, but undue delay in operation may reduce the chance of recovery.

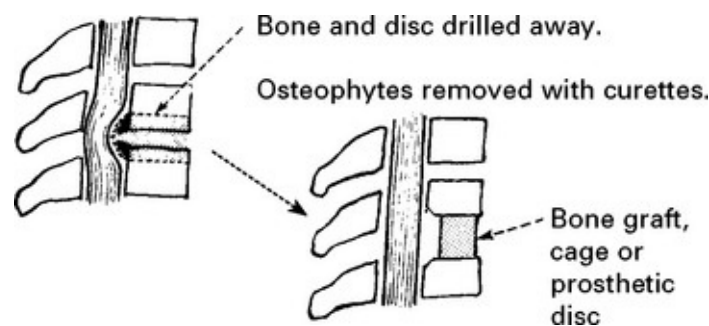
Indications for operation

1. Progressive neurological deficit – myelopathy or radiculopathy.
2. Intractable pain, when this fails to respond to conservative measures. This is rarely the sole indication for operation and usually applies to acute disc protrusion (see below) rather than chronic radiculopathy.

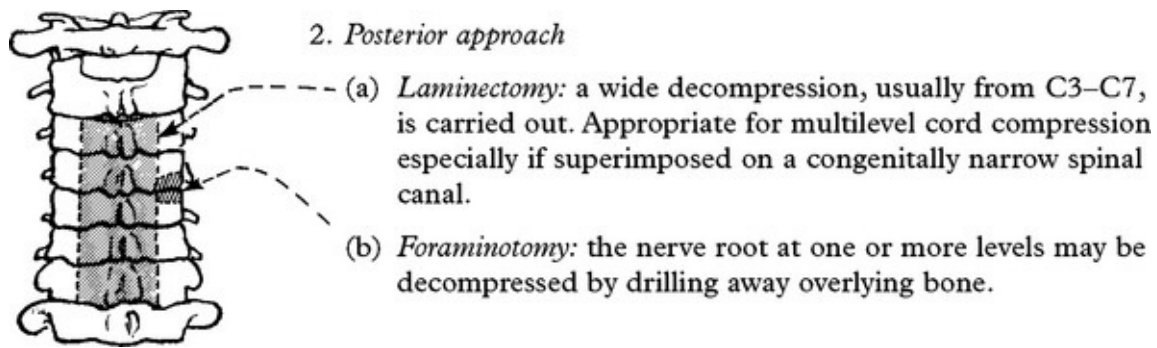
Operative techniques

1. Anterior decompression and fusion

A core of bone and disc is removed along with the osteophytic projections. Although not essential, some insert a bone graft from the iliac crest, or a metallic cage (see [page 398](#)) to promote fusion. More recently prosthetic discs have become available. There is no evidence that any one technique produces better results than another.



Most suitable for root or cord compression from an anterior protrusion at one or two levels.



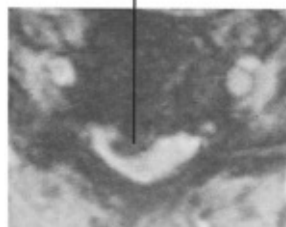
Results

Operative results vary widely in different series and probably depend on patient selection. Some improvement occurs in 50–80% of patients. Operation should be aimed at preventing progression rather than curing all symptoms.

CERVICAL DISC PROLAPSE

In contrast to cervical spondylosis, cervical ‘soft disc’ protrusion is uncommon. This tends to occur acutely in younger patients and relate to a sudden twist or injury to the neck. The protrusion usually occurs posterolaterally at the C5/C6 or C6/C7 level causing a radiculopathy rather than a myelopathy. Sagittal and axial MRI will clearly outline the disc protrusion. Operative removal through an anterior approach may be required for intractable pain or neurological deficit and gives good results

T2 weighted axial MRI showing disc protrusion



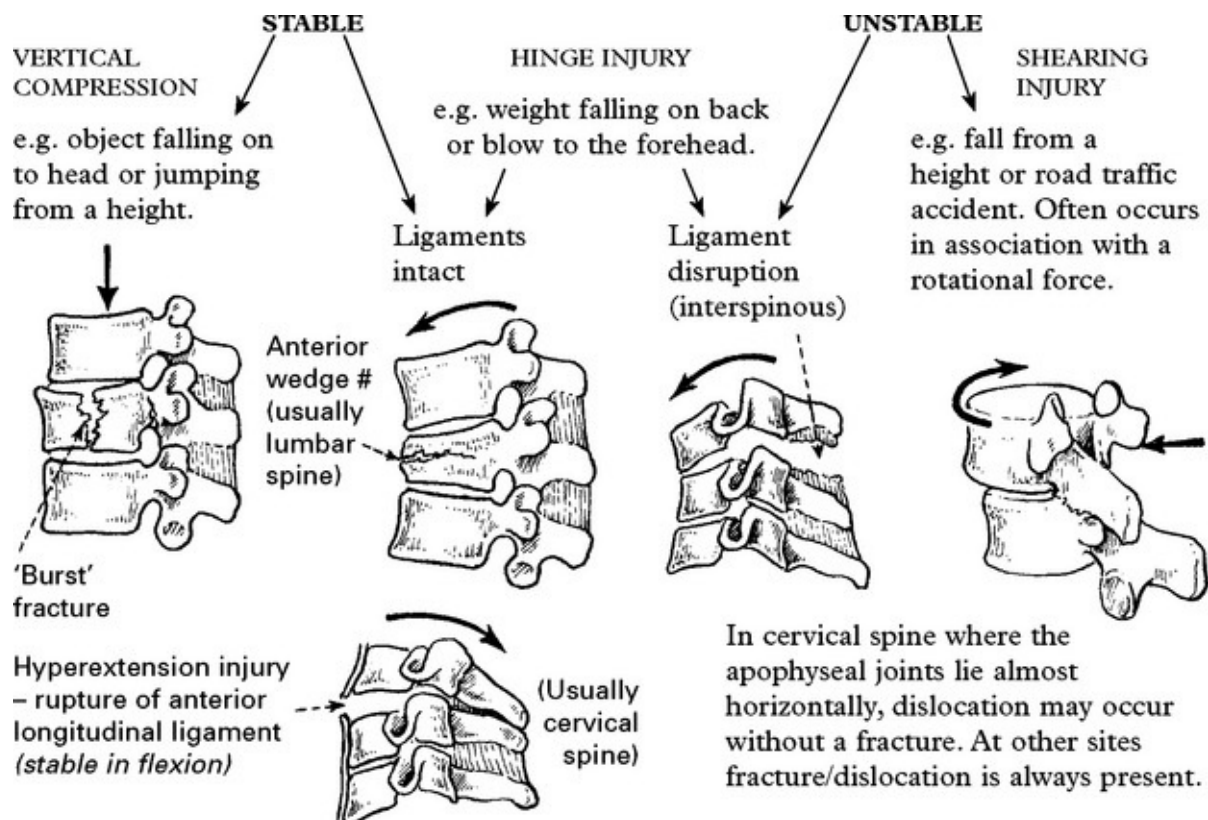
SPINAL TRAUMA

Approximately 2 per 100 000 of the population per year sustain a spinal injury. Of these, 50% involve the cervical region.

At impact, *spinal cord damage* may or may not accompany the *bony or ligamentous damage*. After impact, stability at the level of injury plays a crucial part in further management. Injudicious movement of a patient with an unstable lesion may precipitate spinal cord injury or aggravate any pre-existing damage.

MECHANISMS OF INJURY

The mechanism of injury helps determine the degree of stability:



Initial assessment

The *possibility of spinal injury must be considered* at the scene of the accident and all movements and transportation of the patient undertaken with extreme caution especially when comatose. Most spinal injuries occur in conscious patients who complain of *pain, numbness or difficulty with limb movements*.

Examination may reveal *tenderness over the spinous processes, paraspinal swelling or a gap between the spinous processes*, indicating rupture of an interspinous ligament.

Neurogenic paradoxical ventilation (indrawing of the chest on inspiration due to absent intercostal function) may occur with cervical cord damage.

Bilateral absence of limb reflexes in flaccid limbs, unresponsive to painful stimuli, indicates spinal cord damage (unless death is imminent from severe head injury.)

Painless urinary retention or priapism may also occur.

SPINAL TRAUMA – INVESTIGATIONS

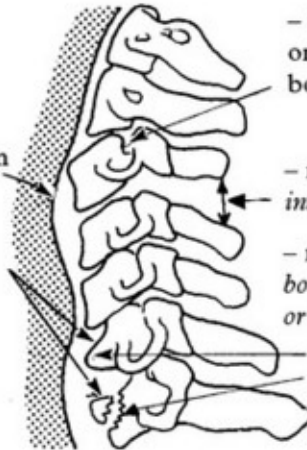
STRAIGHT X-RAYS

LATERAL VIEW

In the *cervical spine*:

– note evidence of *soft tissue swelling* between the pharynx and the vertebrae.

– ensure C6 and C7 are included in the film. If not, do a CT scan at this level.



– note any *malalignment* of the anterior or posterior margins of the vertebral body or of the lamina, i.e. *subluxation*.

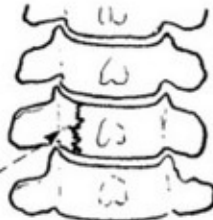
– note any undue *widening* of the interspinous distance or of the disc space.

– note *damage* to the vertebral body, apophyseal joints, lamina or spinous process, e.g. anterior wedge collapse, 'burst' fracture.

In the upper thoracic spine a CT scan or tomography may be required to demonstrate a sagittal view.

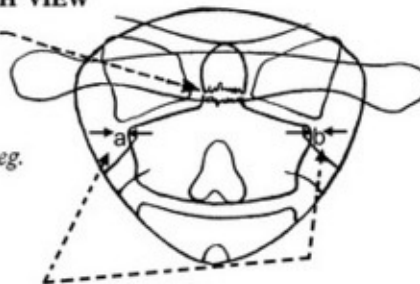
ANTERO-POSTERIOR VIEW

– note the alignment of the spinous processes and the width of the apophyseal joints and look for vertical fracture lines.



ANTERO-POSTERIOR 'OPEN MOUTH' VIEW

– may be required to demonstrate a fracture of the odontoid peg.



In fractures of C1, if the lateral masses project beyond C2 > 7 mm (i.e. $a + b$), the transverse ligament is likely to be disrupted indicating an unstable injury (Rule of Spence).

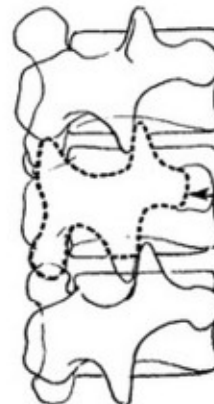
If doubt remains – take **OBLIQUE VIEWS** to demonstrate the intervertebral foramina.

CERVICAL SPINE

Disruption of the foraminal outline suggests malalignment



If in doubt about cervical stability, take **FLEXION/EXTENSION VIEWS**, but only with expert supervision.



In the **LUMBAR** spine look for the normal 'scotty dog' appearance – if misshapen, suggests a fracture/dislocation.

SPINAL TRAUMA – INVESTIGATIONS/MANAGEMENT

If straight X-rays are difficult to visualise or if clinical suspicion of cord injury persists despite normal X-rays, then a CT scan or MRI should be

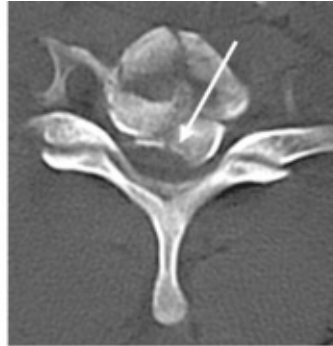
performed.

CT SCANNING

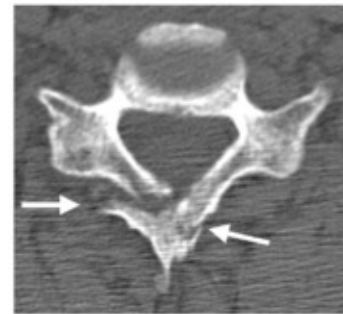
CT may demonstrate more extensive fracturing than suspected on plain X-rays and aids identification of regions not clearly shown.



Sagittal reconstruction showing dislocation of the cervico-thoracic junction



Axial views showing a burst fracture of the vertebral body with retropulsed fragments compromising the spinal canal



Axial views demonstrating fractures of the lamina and spinal process

MRI may provide additional information of soft disc prolapse or haematoma within the spinal canal, but seldom influences management.

MANAGEMENT

Management depends on the site and stability of the lesion, but basic principles apply.

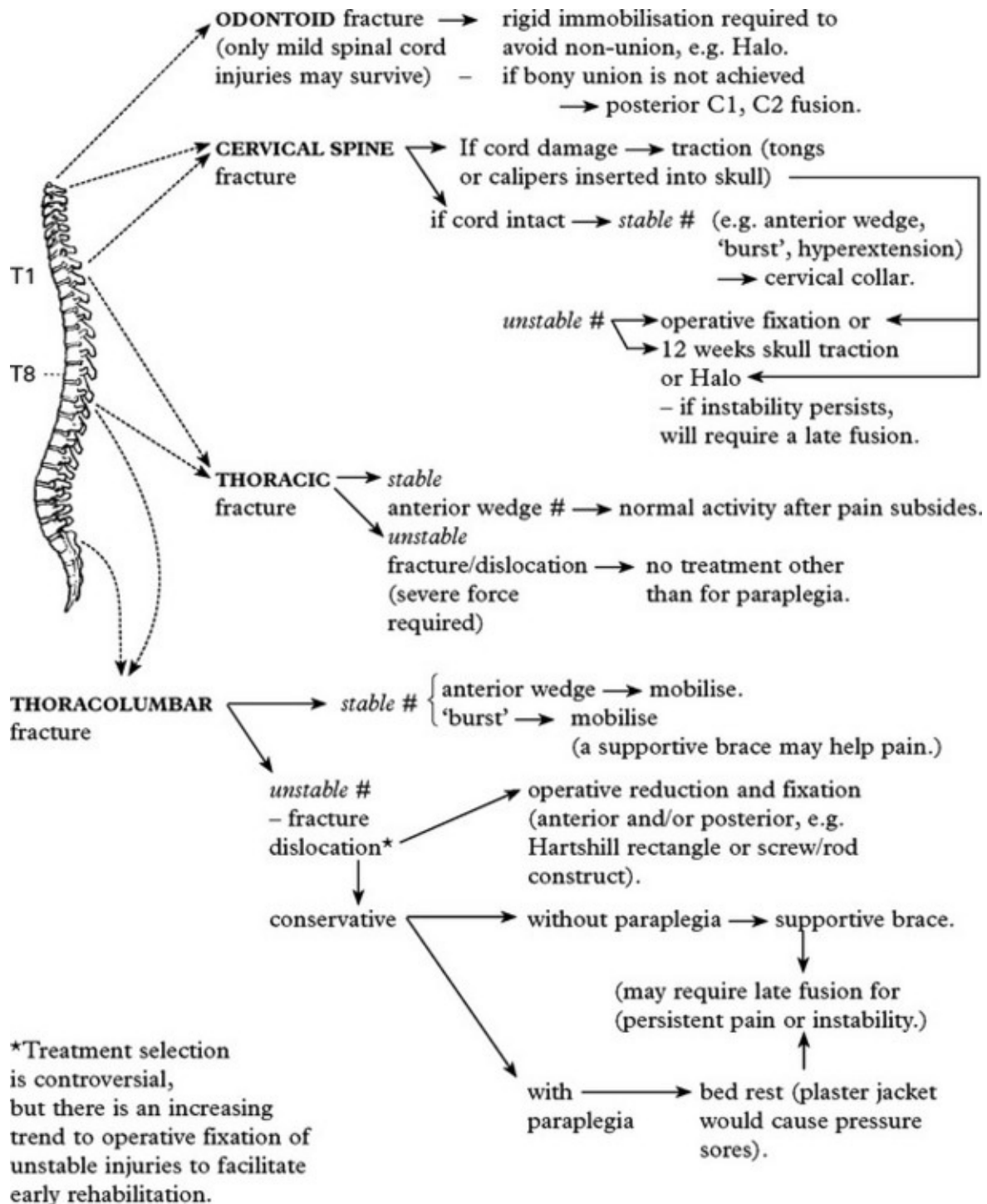
1. An *unstable* lesion risks further damage to the spinal cord and roots and requires either
 - *operative fixation* or
 - *immobilisation*, e.g. skull traction, Halo or plaster jacket.
2. There is no evidence that ‘decompressing’ the cord lesion (either anteriorly or posteriorly) improves the neurological outcome, but –
3. If a patient with normal cord function or with an incomplete cord

lesion (i.e. with some residual function) *progressively deteriorates*, then *operative decompression* is required.

Many additional therapies and techniques (e.g. steroids, cord cooling, hyperbaric oxygen) have been employed with the aim of improving neurological outcome. Although initial trials with METHYLPREDNISOLONE suggested benefit when given within 8 hours of injury, concern has been raised about the methodological techniques applied. Its use may be associated with an increased incidence of infective complications and its value in improving functional outcome remains unproven.

SPINAL TRAUMA – MANAGEMENT

Management of injury at specific sites



Management of the paraplegic patient

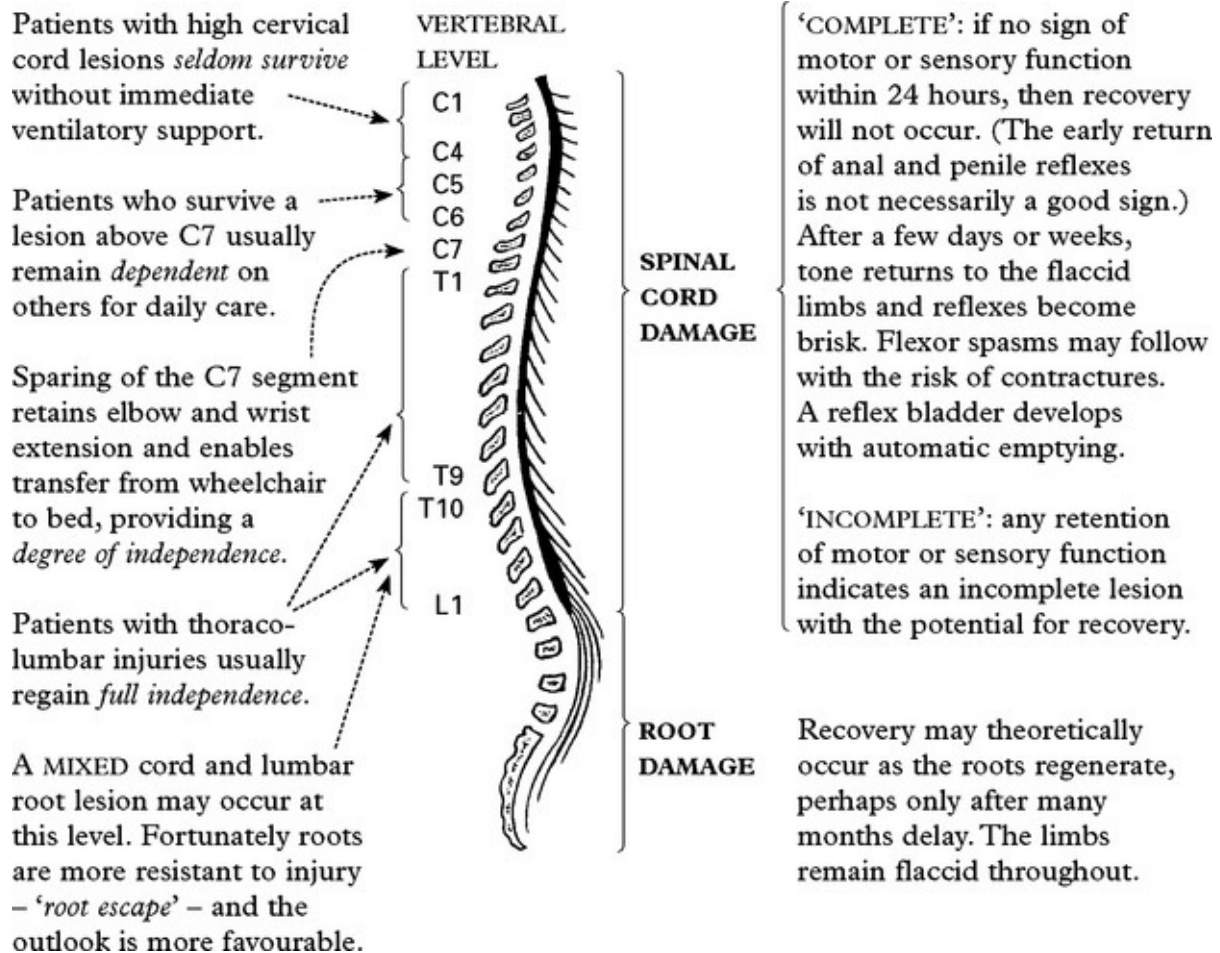
After spinal cord injury, transfer to a spinal injury centre with medical and nursing staff skilled in the management of the paraplegic patient

provides optimal daily care and rehabilitation.

Important features include:

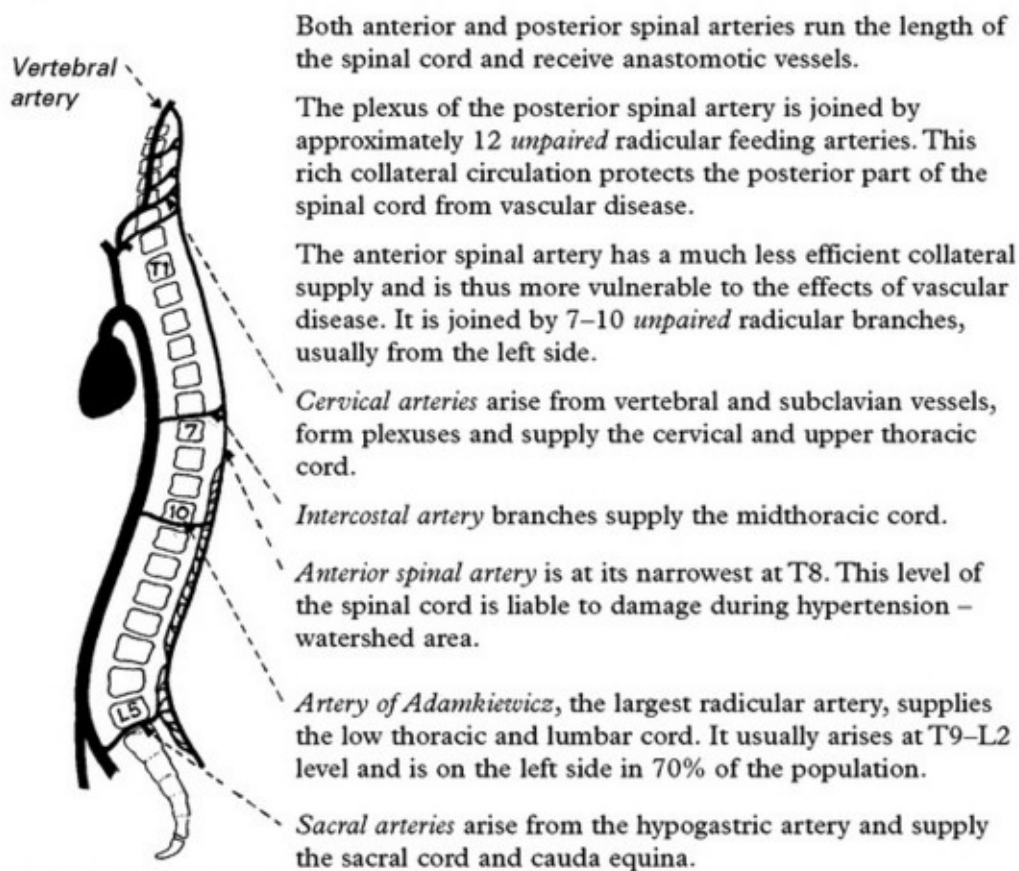
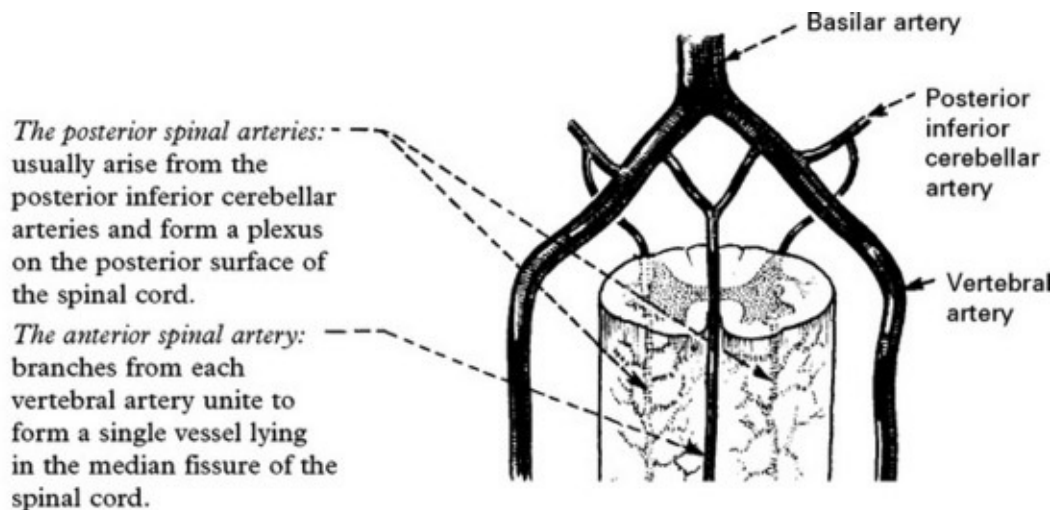
1. *Skin care* – requires meticulous attention. Two-hourly turning should prevent pressure sores. Attempt to avoid contact with bony prominences or creases in the bed sheets. Air or water beds or a sheepskin may help.
2. *Urinary tract* – long-term catheter drainage or intermittent self-catheterisation is required. Infection requires prompt treatment. Eventually, training may permit automatic reflex function (in cord lesions) or micturition by abdominal compression (in root lesions). In some, urodynamic studies may indicate possible benefit from bladder neck resection.
3. *Limbs* – intensive physiotherapy helps prevent flexion contractures (in cord injury) and plays an essential role in rehabilitation.

OUTCOME FOLLOWING SPINAL CORD OR ROOT INJURY

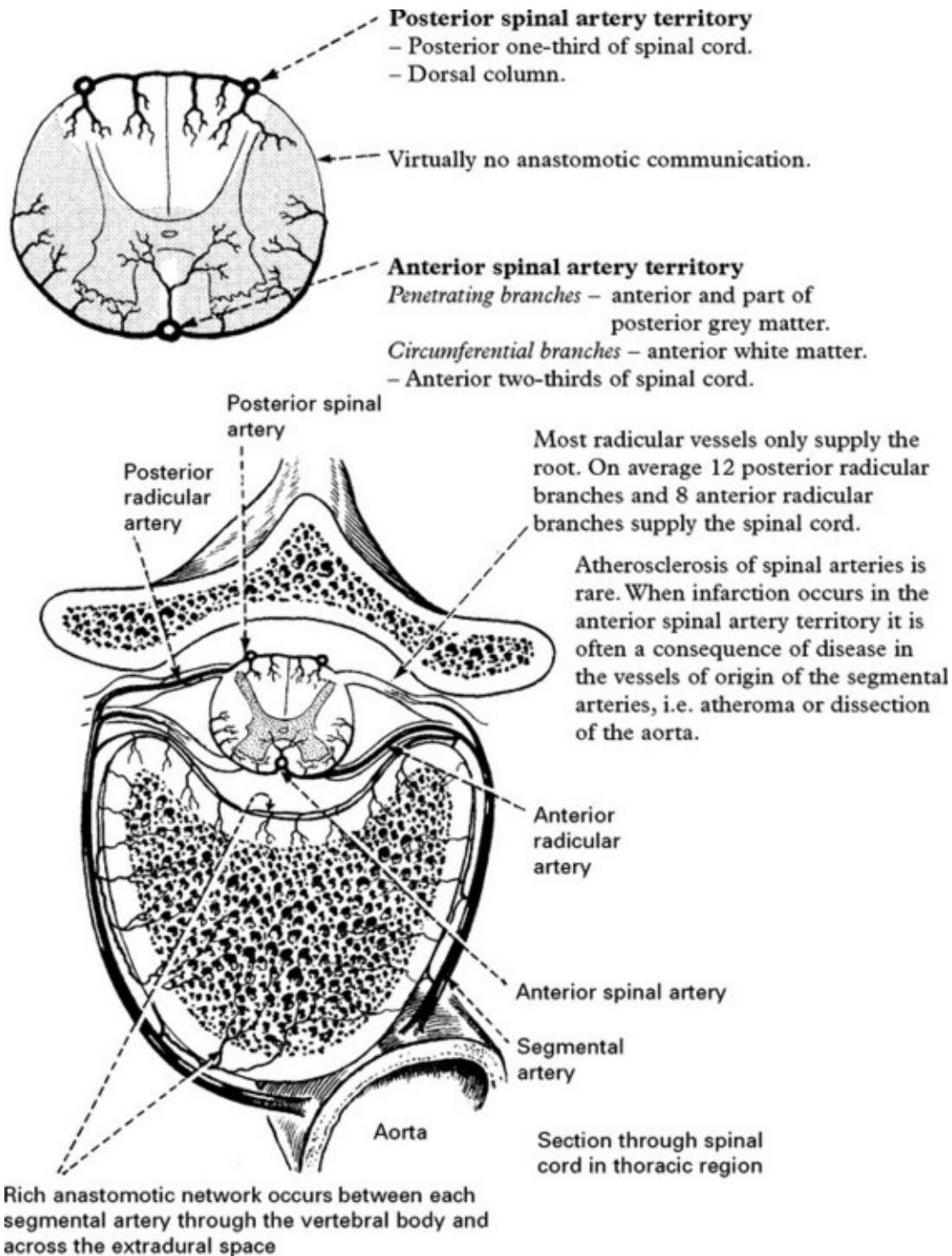


VASCULAR DISEASES OF THE SPINAL CORD

Blood supply to the spinal cord is complex; the main vessels are the anterior and posterior spinal arteries.



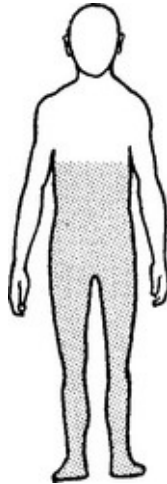
Anterior radicular branches joining anterior spinal artery



SPINAL CORD INFARCTION

Anterior spinal artery syndrome

The level at which infarction occurs determines symptoms and signs.



Characteristic features include:

- Radicular pain at onset
- Sudden para/quadruplegia
- Flaccid limbs —————> days spastic
- Areflexia —————> days hyper-reflexia and extensor plantar responses
- Sensory loss to pain and temperature up to the level of cord damage
- Preserved vibration and joint position sensation (dorsal columns supplied by the posterior spinal arteries)
- Urinary retention

When only penetrating branches are involved, long tract damage may be selective and sensory loss may be minor.

Spinal cord ischaemia due to aortic atheroma evolves slowly and preferentially affects anterior horn cells.

A pure conus syndrome ([page 394](#)) occasionally occurs.

Investigative approach

- *Exclude* other causes of acute para/quadruplegia – cord compression, Guillain-Barré syndrome – by appropriate imaging or neurophysiology
- *Confirm* spinal ischaemia by MRI (T2 weighted imaging showing hyperintense signal changes)
- *Explore* possible sources of spinal ischaemia

Small vessel diseases

diabetes – random or fasting blood glucose

vasculitis – see [pages 267–269](#)

neurosyphilis – CSF VDRL, FTA and TPI tests (see [page 499](#))

endarteritis secondary to – CSF meningeal infection or granulomatous disease

aortic (large) vessel diseases

atheromatous – vascular risk factor *e.g.* cholesterol

embolic – echocardiography, blood cultures

thrombotic – coagulation screen

dissection/aneurysm – transoesophageal echo (TOE) angiography

hypotension – ECG, cardiac enzymes

Treatment is symptomatic and the outcome variable.

Posterior spinal artery syndrome

This is rare as white matter structures are less vulnerable to ischaemia. The dorsal columns are damaged and ischaemia may extend into the posterior horns.

Clinical features:

- Loss of tendon reflexes/motor weakness
- Loss of joint position sense.

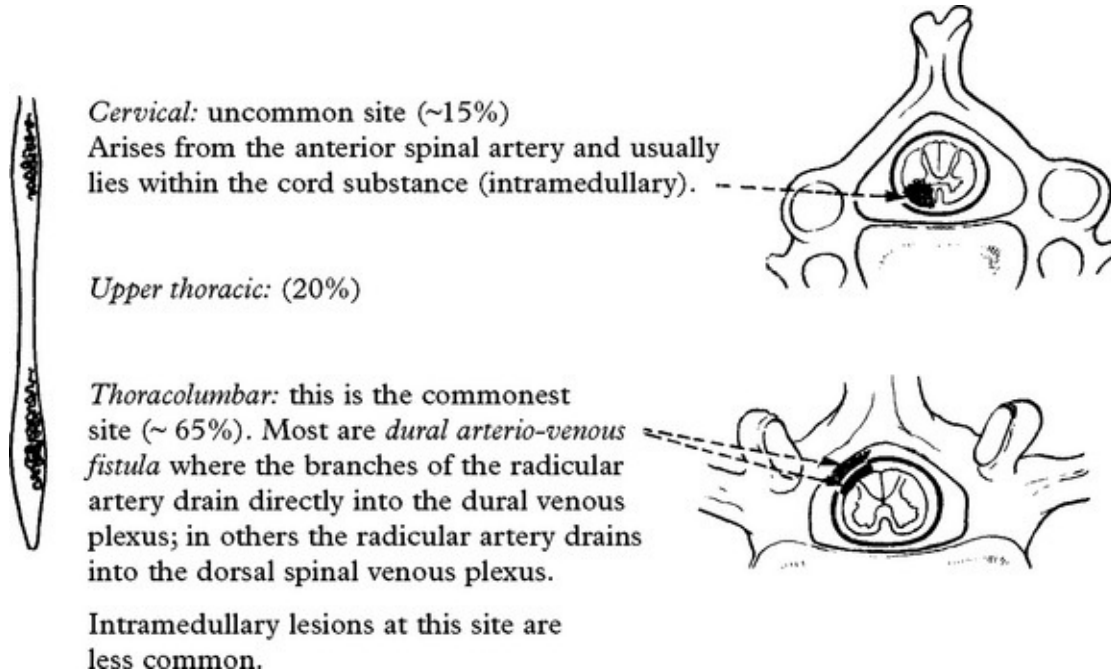
Venous infarction

A rapid 'total' cord syndrome with poor outcome often associated with pelvic sepsis.

SPINAL ARTERIOVENOUS MALFORMATION (Angiomatous malformation)

Arterio-venous malformations (AVMs) are abnormal collections of blood vessels. Arteries communicate directly with veins, bypass the capillary network and create a 'shunt'. The AVM appears as a mass of convoluted dilated vessels.

Site



Intradural AV malformations occur in younger patients at any age in either sex and are most likely congenital. Dural AV fistulae are most

common in males between 40–70 years. They are probably acquired and related to trauma.

Clinical features

SUDDEN ONSET (10–15%)

Due to

- subarachnoid haemorrhage: headache, neck stiffness, back and leg pain
- extradural haematoma
- subdural haematoma
- intramedullary haematoma (haematomyelia)

} signs of acute cord damage.

GRADUAL ONSET (85–90%)

Probably due to ↑ *venous pressure*
but other factors may play a part:

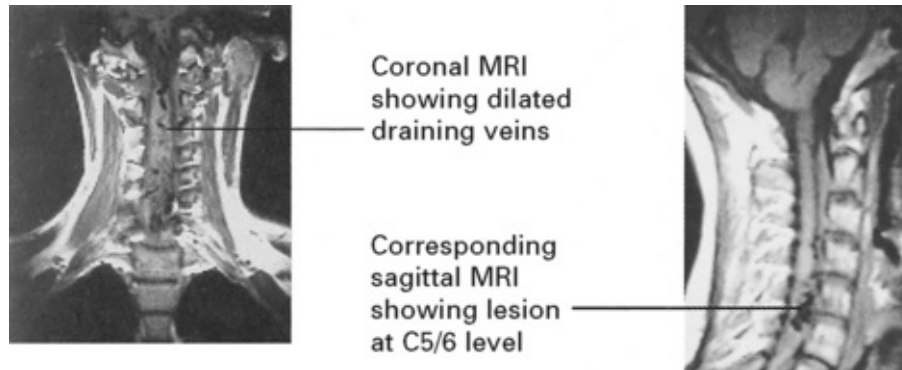
- venous thrombosis
- ‘steal’ phenomenon
- venous bulk
- arachnoiditis (if previous bleed).

} Progressive deterioration of all spinal modalities simulating cord compression. Pain is common. With thoracolumbar lesions a mixed u.m.n./l.m.n. weakness in the legs is typical. Intramedullary AVMs may cause fluctuating signs and symptoms and may mimic intermittent claudication.

A bruit may be heard overlying a spinal AVM and occasionally midline cutaneous lesions – haemangiomas, naevi or angiolipomas – are found. (Note that cutaneous angiomas are not uncommon and do not necessarily imply an underlying lesion.)

Investigation

MRI will demonstrate abnormal signal from the lesion or from draining veins. *Myelography* will also demonstrate abnormal draining veins. *Selective angiography* is required to delineate arterial feeders.



Management

Untreated, 50% of patients with gradual onset of symptoms would be unable to walk within 3 years. Treatment should prevent progression and may well improve a gait or bladder disturbance. Delay may result in irreversible cord damage.

Techniques:	<i>Embolisation</i>	– may successfully obliterate dural AVMs, particularly when fed by one or two dural arteries – may aid subsequent operative treatment – or may produce symptomatic improvement in inoperable lesions.
	<i>Surgery</i>	– It is important to identify and divide the feeding vessel and excise the shunt. Total excision of all the dilated veins is unnecessary and hazardous. Operative risk for most dural A-V fistula is low and excision provides an alternative to embolisation. In contrast, when an AVM lies within the cord substance and/or ventral to the cord, operative risks are high. Staged pre-operative embolisation may help, but in some, a conservative approach may be appropriate.

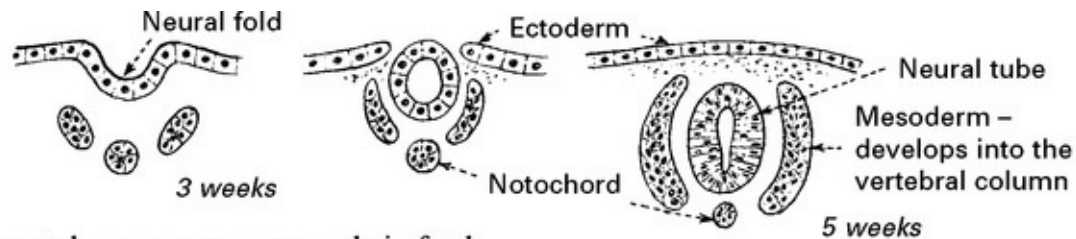
Spinal Epidural and Subdural Haematomas: These may present with a rapid onset of paraplegia. Epidural or less commonly, subdural

haematoma may occur due to rupture of a spinal AVM, after minor trauma or lumbar puncture, or spontaneously in patients with a bleeding disorder, liver disease or on anticoagulant therapy. MRI (or myelography) clearly demonstrates the lesion. Urgent decompression is required after correcting any coagulation deficit, without waiting for spinal angiography. Pathological examination of the haematoma may reveal angiomatous tissue; in other patients, there is no evident cause.

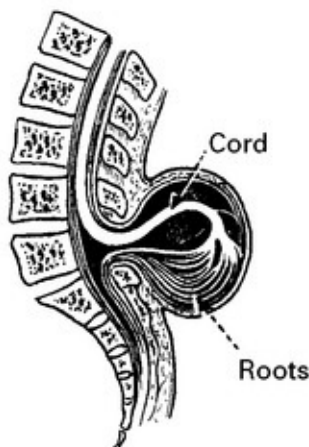
SPINAL DYSRAPHISM

SPINAL DYSRAPHISM: This term encompasses all defects (open or closed) associated with a failure of closure of the posterior neural arch.

Embryology



Developmental errors may occur early in fetal life and lead to a variety of spinal defects:



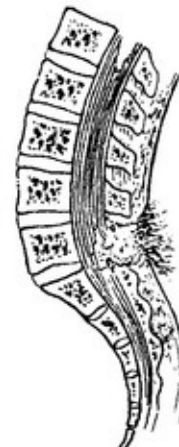
MYELOMENINGOCELE

The spinal cord and roots protrude through the bony defect and lie within a cystic cavity, lined with meninges and/or skin. In most patients, the meningeal covering ruptures and the spinal cord and roots lie exposed to the air – myelodysplasia. CSF may leak from the open lesion.



MENINGOCELE

Cystic CSF filled cavity – lined with meninges but devoid of neural tissue. The cavity communicates with the spinal canal through the bone defect (usually lumbosacral). Meningocele occurs far less frequently than myelomeningocele; they are rarely associated with other congenital anomalies.



Cutaneous defect may overlie spina bifida

SPINA BIFIDA OCCULTA

A bony deficit – present in 5–10% of the population and not clinically significant. Those who also have a lumbosacral cutaneous abnormality (*tuft of hair, dimple, sinus or 'port wine' stain*) have a high incidence of related underlying defects:

- *diastatomyelia*
- *lipoma*
- *dermoid cyst.*

These defects may cause symptoms of pain or neurological impairment after many years.

Site: 80% occur in the lumbosacral region.

Incidence: 2/1000 births in the UK, but geographical variation exists (0.2/1000 in Japan) and the incidence is declining. A familial incidence increases the risk (5% if a sibling is affected). Both genetic factors and

teratogens, e.g. sodium valproate, have a role. Folic acid before and during pregnancy provides some protection.

Associated abnormalities: *Hydrocephalus, Chiari type II, aqueduct forking.*

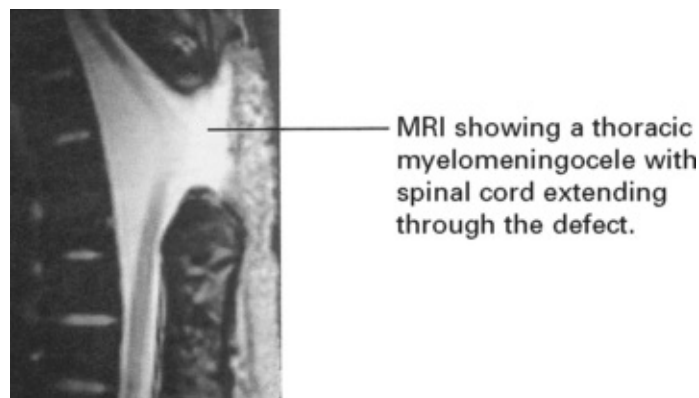
Clinical assessment

Myelomeningocele: This lesion should be carefully examined for the presence of neural elements. Transillumination of the sac may help. Observation of movement in the limbs and in specific muscle groups, occurring spontaneously and in response to pain applied both above and below the level of the lesion, helps determine the degree and level of neurological damage. Also note the presence of a dilated bladder and a patulous anal sphincter. Look for any associated congenital anomalies, e.g. hydrocephalus, scoliosis, foot deformities.

Meningocele: Patients with this lesion seldom show any neurological deficit.

Investigations

Ultrasound or *MRI* may detect neural elements extending into the sac.



Management

Myelomeningocele: Advances in both orthopaedic and urological procedures have considerably improved the long-term management of the associated disabilities in most patients. Active treatment, however, in patients with gross hydrocephalus, complete paraplegia and other multiple anomalies as well as the spinal dysraphism, may merely prolong a painful existence and in such patients, some adopt a conservative approach.

Treatment within a few days involves closure and replacement of the neural tissues into the spinal canal to prevent infection. This initial step provides time to consider the wisdom of embarking on further active management.

Meningocele: In the presence of a CSF leak, urgent excision is performed; otherwise this is deferred, perhaps indefinitely if the lesion is small.

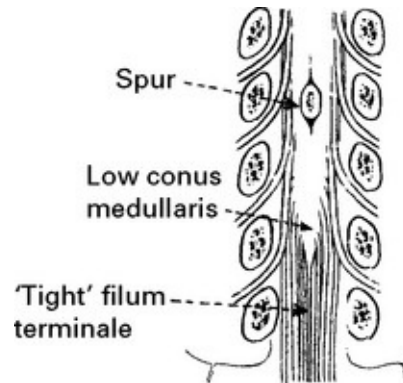
Spina bifida occulta: Treatment may not be required, although patients with a cutaneous abnormality or with neurological signs, should undergo ultrasound or MRI to exclude an intraspinal anomaly.

Antenatal diagnosis

Screening the maternal serum/amniotic fluid for alpha-fetoprotein and acetylcholinesterase, fetal ultrasonography and contrast enhanced amniography in high risk patients (e.g. with an affected sibling) provides an effective method of detecting neural tube defects and offers the possibility of therapeutic abortion. Intrauterine surgery to repair the myelomeningocele is currently under evaluation and may reduce the severity of associated defects, e.g. Chiari II.

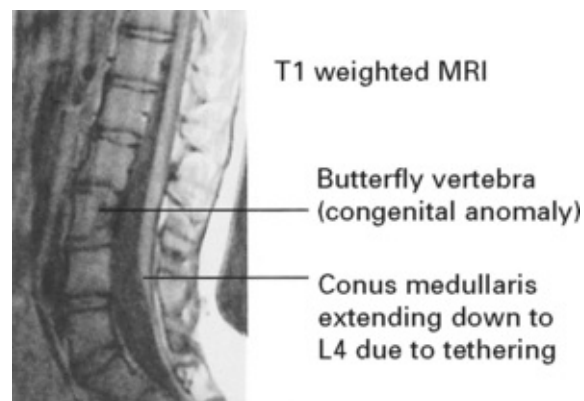
TETHERED CORD: in some patients the conus medullaries lies well below its normal level (L1), 'tethered' by the filum terminale. Since vertebral growth proceeds more rapidly than growth of the spinal cord,

tethering may produce progressive back pain or neurological impairment as the cord is stretched.



DIASTOMATOMYELIA: A congenital splitting of part of the spinal cord by a *bony, fibrous or cartilaginous* spur. This usually lies at the upper lumbar region and extends directly across the spinal canal in an antero-posterior direction. The split cord does not always reunite distal to the spur (diplomyelia).

Investigation: MRI is the investigation of choice in spinal dysraphism, but straight X-ray may reveal associated congenital abnormalities: spina bifida occulta, fused or hemivertebrae. CT scanning may help demonstrate the presence of a bony spur.



Management: Although some recommend prophylactic division of the

tethered filum terminale in the absence of neurological impairment, most reserve operative treatment for those who present with a neurological deficit, especially if there is evidence of progression, or prior to correction of any spinal deformity. In contrast, prophylactic removal of the spur in patients with diastomatomyelia is usually performed, even in the absence of neurological impairment.

LIPOMENINGOCELE

Lipomas may occur in association with spinal dysraphism and range from purely intraspinal lesions to very large masses extending along with neural tissues through the bony defect. All are adherent to the conus and closely related to the lumbosacral roots, preventing complete removal and increasing operative hazards.



CONGENITAL DERMAL SINUS TRACT/DERMOID CYST

This congenital defect results from a failure of separation of neuronal from epithelial ectoderm and may occur with other midline fusion defects, e.g. diastomatomyelia and a tethered cord. A tiny sinus in the lumbosacral region may represent the opening of a blind ending duct or may extend into the spinal canal. Dermoid cysts arise at any point along the sinus tract and often lie adjacent to the conus.

Clinical presentation varies from repeated attacks of unexplained meningitis to neurological deficits arising from the presence of an intraspinal mass. Treatment involves excision of the whole tract and any associated cyst (after treating any meningitic infection).



SECTION IV

LOCALISED NEUROLOGICAL DISEASE AND ITS MANAGEMENT C. PERIPHERAL NERVE AND MUSCLE

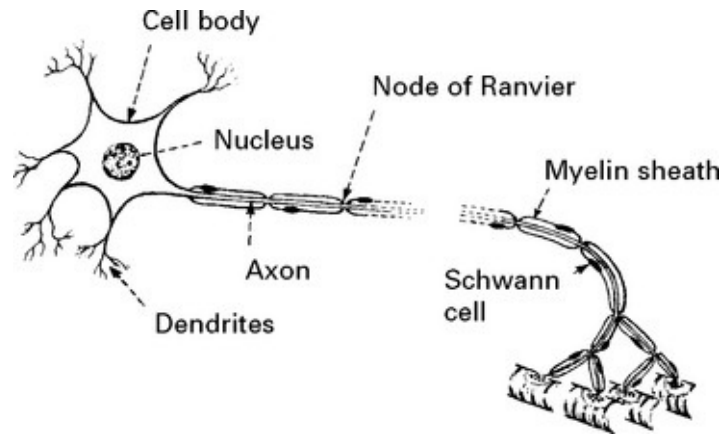
THE POLYNEUROPATHIES – FUNCTIONAL ANATOMY

The function of the peripheral nervous system is to carry impulses to and from the central nervous system. These impulses regulate motor, sensory and autonomic activities.

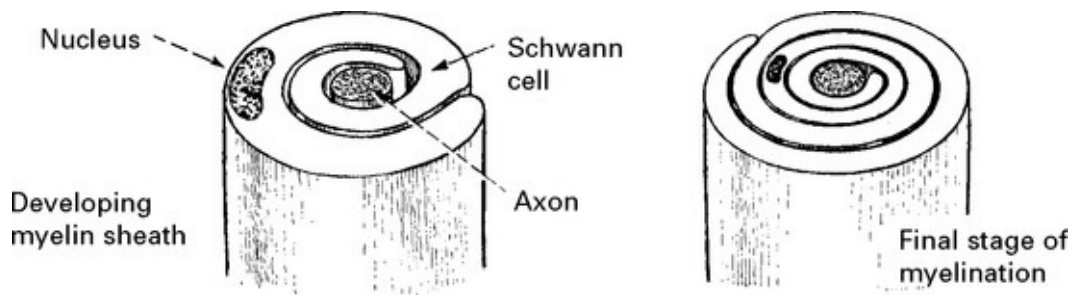
The peripheral nervous system is comprised of structures that lie outside the pial membrane of the brain stem and spinal cord and can be divided into cranial, spinal and autonomic components.

STRUCTURE OF THE NERVE CELL AND AXON

Each axon represents an elongation of the nerve cell – lying within the central nervous system, *e.g.* anterior horn cell, or in an outlying ganglion, *e.g.* dorsal root ganglion. The cell body maintains the viability of the axon, being the centre of all cellular metabolic activity.



Many axons are surrounded by an insulation of myelin, which is enveloped by the Schwann cell membrane. Myelin is a protein–lipid complex. The membrane of the Schwann cell ‘spirals’ around the axon resulting in the formation of a multilayered myelin sheath.



All axons have a cellular sheath – Schwann cell – but not all axons are myelinated.

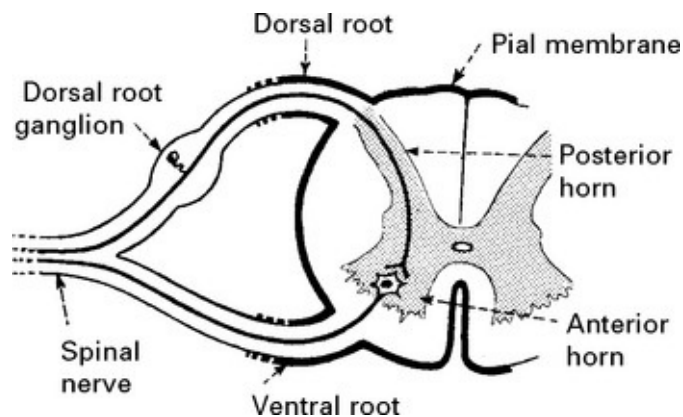
Schwann cells with associated myelin are 250–1000 μm in length and separated from each other by the node of Ranvier. The axon is bare at this node and, during conduction, impulses jump from one node to the next – *saltatory* conduction. The rate of conduction in myelinated nerves is markedly increased in comparison with unmyelinated fibres. Myelin thus facilitates fast conduction. In unmyelinated fibres conduction depends upon the diameter of the nerve fibre, this determining the rate of longitudinal current flow.

SPINAL PERIPHERAL NERVOUS SYSTEM

Entry to and exit from the central nervous system is achieved by paired spinal nerve roots (30 in all).

These dorsal and ventral roots lie in the spinal subarachnoid space and come together at the intervertebral foramen to form the spinal nerve.

The dorsal roots contains sensory fibres, arising from specialised sensory receptors in the periphery.



The dorsal root ganglia are collections of sensory cell bodies with axons extending peripherally as well as a central process which passes into the spinal cord in the region of the posterior horn of grey matter and makes appropriate central connections.

Sensation can be divided into:

- *Pain and temperature*
- *Simple touch*
- *Discriminatory sensation* – proprioception, vibration.

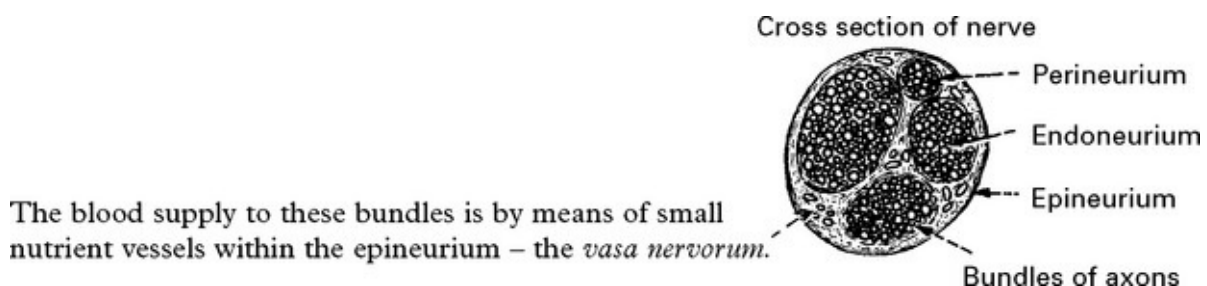
These different forms of sensation are carried from the periphery by axons with specific characteristics. The central connections and pathways vary also (see [page 200](#)).

The anterior horns of the spinal cord contain cell bodies whose axons pass to the periphery to innervate skeletal muscle – the alpha motor neurons. Smaller cell bodies also project into the anterior root and innervate the intrafusal muscle fibres of muscle spindles – the gamma motor neurons.

Each alpha motor neuron through its peripheral ramifications will innervate a number of muscle fibres. The number of fibres innervated from a single cell varies from less than 20 in the eye muscles to more than 1000 in the large limb muscles (innervation ratio). The alpha motor neuron with its complement of muscle fibres is termed the *motor unit*.

PERIPHERAL NERVES

Peripheral nerves are composed of many axons bound together by connective tissue. A ‘mixed’ nerve contains motor, sensory and autonomic axons.



Nerve fibre type

Axons within the peripheral nerve vary structurally. This is related to function. Three distinct fibre types can be distinguished:

TYPE A	2–20 μm in diameter. Myelinated.

<i>Function:</i>	Motor and sensory (vibration, proprioception). Conduction velocity: 10–70 metres/second.
TYPE B	3 μ m diameter. Thinly myelinated.
<i>Function:</i>	Mainly preganglionic autonomic, some pain and temperature. Conduction velocity: 7–5 metres/second.
TYPE C	< 1 μ m diameter. Unmyelinated.
<i>Function:</i>	Sensory – pain and temperature. Conduction velocity: < 2 metres/second.

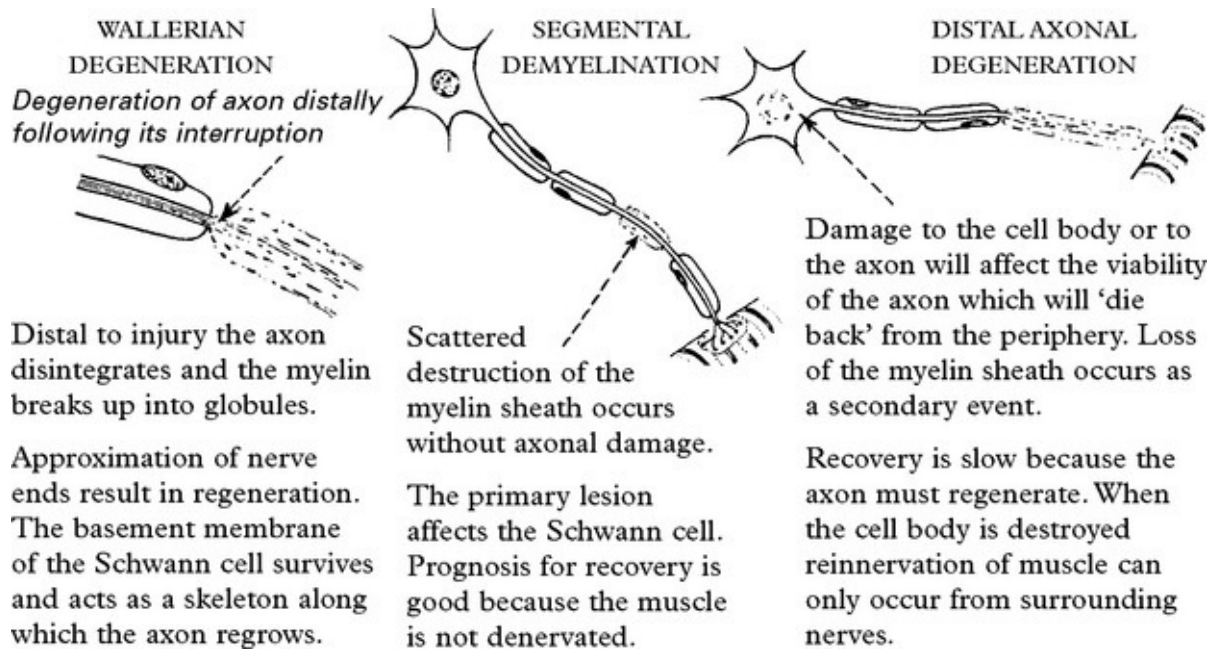
The structure of the spinal peripheral nervous system has been considered but the arrangement is also important. Spinal nerves, after emerging from the intervertebral foramen pass into the brachial plexus to supply the upper limbs and the lumbosacral plexus to supply the lower limbs.

The thoracic nerves supply skeletal muscles and subserve sensation of the thorax and abdomen.

The Autonomic Nervous System is described on [page 457](#).

PATTERNS OF INJURY

Damage may occur to: axon, myelin sheath, cell body, supporting connective tissue and nutrient blood supply to nerves. Three basic pathological processes occur.



THE POLYNEUROPATHIES – SYMPTOMS

Sensory

Negative phenomena – loss of sensation.

Disease of *large myelinated fibres* produces loss of touch and joint position perception.

Patients complain of difficulty in discriminating textures. Their hands and feet feel like cotton wool. Gait is unsteady, especially when in darkness where vision cannot compensate for loss of joint position sensation (proprioception).

Disease of *small unmyelinated fibres* produces loss of pain and temperature appreciation as a consequence of which painless burns/trauma result. Damage to joints without pain results in a 'neuropathic' joint (Charcot's joint) in which traumatic deformity is totally painless.

Positive phenomena

Disease of *large myelinated fibres* produces paraesthesia – a ‘pins and needles’ sensation with a peripheral distal distribution.

Disease of *small unmyelinated fibres* produces painful positive phenomena:

International Association for the Study of Pain (IASP) definitions has clarified the following.

– analgesia	absent sensitivity to a painful stimulus
– hyperalgesia	increased sensitivity to a painful stimulus
– hypoalgesia	reduced sensitivity to painful stimulus
– hyperaesthesia	increased sensitivity to any stimulus
– hypoaesthesia	reduced sensitivity to any stimulus
– hyperpathia	increased sensitivity with increasing threshold to repetitive stimulation
– allodynia	pain provoked by a non-painful stimulus

Complex regional pain syndromes (CRPS) were previously termed ‘reflex sympathetic dystrophy’ and ‘causalgia’. These may follow a simple soft tissue injury (CRPS-1) or injury to a large peripheral nerve (CRPS-2). Allodynia and hyperalgesia are associated with local changes in temperature and skin appearance (oedema and discoloration). The pain has a burning, shooting quality. Motor manifestations (weakness or involuntary movements) are common and the pathophysiologic mechanism unknown.

Motor

The patient notices weakness:

- When distal, *e.g.* difficulty in clearing the kerb when walking
- When proximal, *e.g.* difficulty in climbing stairs or combing hair
- Cramps may be troublesome
- Twitching of muscles (fasciculation) may be felt.

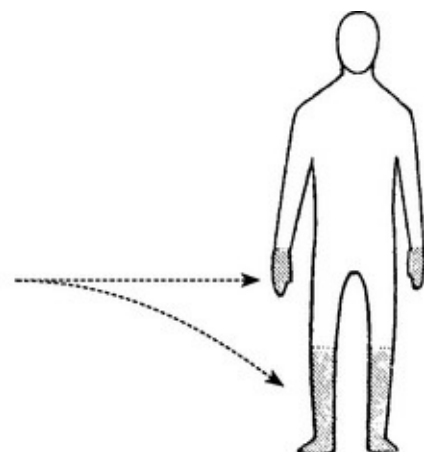
THE POLYNEUROPATHIES – SIGNS

SENSORY EXAMINATION

All modalities are tested	} Functions of large myelinated sensory fibres.
Light touch	
Two point discrimination	
Vibration sensation	
Joint position perception	} Functions of small unmyelinated and thinly myelinated sensory fibres.
Temperature perception	
Pain perception.	

Initially the area of total sensory loss is defined. The test object, *e.g.* a pin, should be moved from anaesthetic to normal area; it is more accurate to state when an object is felt rather than when it disappears.

In polyneuropathies, sensory loss is symmetrical and follows a characteristic stocking and glove distribution.



Examination of gait is important; with joint position impairment, sensory ataxia is evident. Romberg's test is positive (see [page 28](#)).

Neuropathic burns/ulcers or joints may be present.

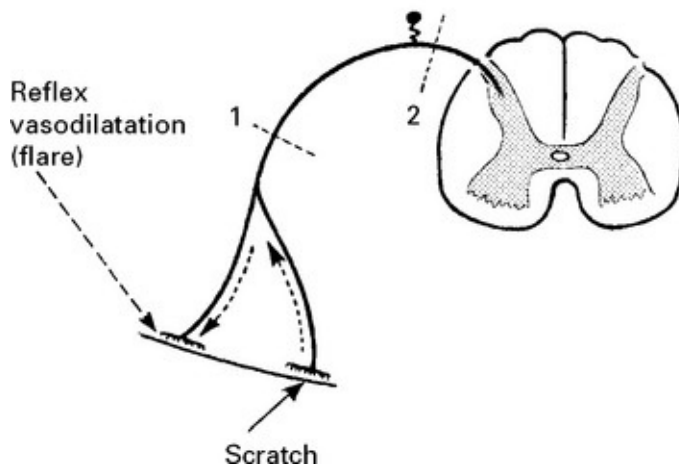
Trophic changes

- Cold blue extremities.
- Cutaneous hair loss.
- Brittle finger/toe nails occasionally occur.

The AXON REFLEX can be used to 'place' lesions in the sensory pathway.

Normally:

the skin is scratched – local vasoconstriction (white reaction) } due to local
next – local oedema (red reaction) } histamine release.
and finally – surrounding vasodilation or flare, dependent on antidromic
impulses from the dorsal root ganglion along an intact sensory
neuron.



1. A distal sensory lesion will result in an absent flare response.
2. A proximal root lesion will not impair the response.

MOTOR EXAMINATION

Muscle wasting. Evident in axonal but absent in demyelinating neuropathies. Oedema of immobile limbs may mask wasting. The 1st dorsal interosseus muscle in the upper limbs and extensor digitorum brevis in the lower limbs are muscles that commonly first show wasting in the neuropathies, but examine all muscle groups. Look for *fasciculations* – irregular twitches of groups of muscle fibres due to

diseased anterior horn cells, these may be induced by exercise or muscle percussion.

Muscle weakness. Weakness is proportional to the number of affected motor neurons. It develops suddenly or slowly and is generally symmetrical, usually starting distally in the lower limbs and spreading to upper limbs in a similar manner before ascending into proximal muscle groups. This pattern of progression is supposedly due to the ‘dying back’ of axons towards their nerve cells – the longest being the most vulnerable.

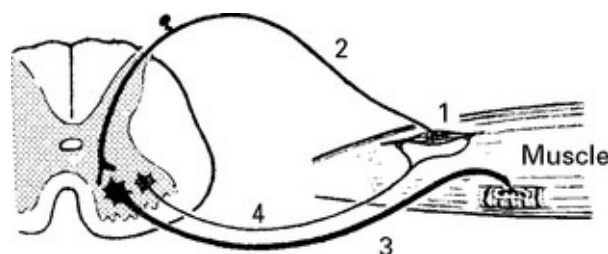
Some neuropathies, *e.g.* Guillain-Barré, chronic inflammatory demyelinating polyneuropathy, may affect proximal muscle groups first.

In severe neuropathies, truncal and respiratory muscle involvement occurs. Respiratory muscle weakness may result in death.

Tendon reflexes

The tendon reflex depends on:

- stretch of the muscle spindle (1),
- activation of spindle afferent fibres (2),
- monosynaptic projections to the alpha motoneurons (3).



The gamma motoneuron fibres, projecting to the spindle (4), ‘modulate’ activity in the reflex loop.

Reflexes commonly tested:

Deltoid	– C5,6 – Circumflex nerve	Triceps	– C6,7,8 – Radial nerve
Biceps	– C5,6 – Musculocutaneous nerve	Knee	– L2,3,4 – Femoral nerve
Supinator	– C6,7 – Radial nerve	Ankle	– S1,2 – Sciatic nerve

The tendon reflexes are lost when any component of the reflex response is affected by disease. Reflexes are lost early in peripheral neuropathies when power and muscle bulk appear normal. Distal reflexes are generally lost before proximal ones.

THE POLYNEUROPATHIES – CLASSIFICATION

There are several approaches to classification:

by MODE OF ONSET – acute, subacute, chronic

by FUNCTIONAL DISTURBANCE – motor, sensory, autonomic, mixed

by PATHOLOGICAL PROCESS – axonal, demyelinating

by CAUSATION – *e.g.* infections; carcinomatous, diabetic, inflammatory, vascular

by DISTRIBUTION – *e.g.* symmetrical, asymmetrical; proximal, distal.

The following table based primarily on mode of onset is for reference. Certain neuropathies will be dealt with separately (see [pages 439–444](#)).

CAUSE	FUNCTIONAL DISTURBANCE	PATHOLOGY
ACUTE: days to 4 weeks		
Inflammatory (Guillain Barré syndrome)	Predominantly motor	Demyelinative with perivascular lymphocytic infiltration
	Distal or proximal	

	Autonomic disturbance	
Diphtheria ——	Cranial nerve onset	Demyelinative. No inflammatory infiltration.
	Mixed motor/sensory	
Porphyria ——	Motor (may begin in arm).	Axonal
	Autonomic disturbance	
	Minimal sensory loss.	

SUBACUTE – occasionally CHRONIC: months and years		
ASYMETRICAL and MULTIFOCAL		
Infections		
Leprosy	Sensory neuropathy, often multifocal; associated depigmentation	Spectrum from paucibacillary (few organisms with intense inflammation) to multibacillary (many organisms with little inflammation)
HIV	Range of associated neuropathies	
Vasculitic disorders		
Polyarteritis nodosa;	Usually presents with mononeuritis multiplex or an asymmetrical sensorimotor neuropathy.	Vasculitis with Wallerian degeneration in distal nerves
Wegner's granulomatosis;		
Churg-Strauss syndrome		
	Often painful	
Non-systemic vasculitis	As above without systemic features	

SUBACUTE and CHRONIC: months and years		
SYMMETRICAL		
Metabolic and endocrine disorders		
Diabetes	Most commonly distal sensorimotor	
	But wide range of other forms (see later)	
Uremia	Distal sensorimotor	Axonal degeneration
Hypothyroidism	Distal sensorimotor	
Acromegaly	Distal sensorimotor	

CAUSE	FUNCTIONAL DISTURBANCE	PATHOLOGY
Nutritional deficiencies		
Vitamin B1 (thiamine)	Predominantly sensory, with burning feet.	Axonal degeneration with segmental demyelination
Includes alcoholic neuropathy	Weakness may develop.	
	Autonomic involvement common but mild	
B12 deficiency	Predominantly sensory; may be associated spinal cord involvement	
Malignant disease		
Paraneoplastic	Sensory or sensorimotor	Axonal; may be associated antibodies (anti-Hu)
Infiltrative	Multifocal, often a polyradiculopathy	More common with lymphoma
Paraprotein associated		
Monoclonal gammopathy (IgG, IgA, IgM)	Sensorimotor neuropathy	Axonal with segmental demyelination
Chronic inflammatory demyelinating polyneuropathy (CIDP) (see later)		
Amyloid		

Primary, secondary or familial	Sensorimotor neuropathy often with autonomic involvement	Thickened nerves with amyloid deposition and small fibre neuropathy
	May present as multiple mononeuropathies	
Inherited neuropathies		
Charcot-Marie-Tooth disease (see below)		
Refsum's disease	Phytanic acid storage disorder. Sensorimotor neuropathy with ichthyosis, retinitis pigmentosa and deafness	
Drug induced		
Wide range of drugs induce neuropathies including:		
Antibiotics	Metronidazole; ethambutol; isoniazid; nitrofurantoin; dapsone	
Oncology drugs	Adriamycin; cisplatin; taxanes; vincristine	
HIV drugs	Didanosine; stavudine; zalcitabine	
Others	Amiodarone; gold; phenytoin	
Toxin induced		
Solvents		
Heavy metals	Lead; arsenic; thallium	

Chronic idiopathic axonal neuropathy:

In patients about 20% of patients with a chronic neuropathy no cause is identified. Follow up of cohorts of such patients has found that while their symptoms slowly progress they do not develop significant disability.

INVESTIGATION OF NEUROPATHY

Investigation of a neuropathy will be led by the history and the pattern of the neuropathy. In many patients the diagnosis will be relatively straightforward, for example a typical distal symmetrical neuropathy in a patient with diabetes or with a history of alcoholism. Where the

aetiology is known and the neuropathy mild and typical there is often no need for further investigation. However, in many patients the diagnosis is not clear and then the investigations will be led by the pattern of the neuropathy. Unlike the situation for chronic neuropathies (see previous page) the *cause of acute or subacute neuropathy* can usually be defined.

For a patient with a *distal symmetrical sensorimotor neuropathy*:

Initial investigations:

Glucose, HbA1C, urea and electrolytes, liver function tests, thyroid, protein electrophoresis, ESR or plasma viscosity, vitamin B12 and folate, chest X-ray and nerve conduction studies (see below).

Further investigations (depending on clinical history):

Glucose tolerance test

Autoantibodies (including extractable antibodies (anti-ro and anti-la), rheumatoid factor, ANCA, tissue transglutaminase antibodies. Angiotensin converting enzyme (ACE) levels.

Anti-neuronal antibodies (anti-Hu or anti-Yo).

Anti-ganglioside antibodies, anti-myelin associated glycoprotein (anti-MAG) antibodies.

Genetic studies (see later) – including clinical examination of relatives

Porphyria screen

HIV, lyme

CSF studies (cells, protein glucose)

Screen for primary tumour (CT scanning or PET scanning)

Asymmetrical or multifocal neuropathies are much less common and

there are usually clues in the history to direct investigation towards what is most frequently an underlying inflammatory disease, for example vasculitis or a specific inflammatory neuropathy. Inflammatory markers and autoantibodies may be helpful. In such patients nerve conduction studies and nerve biopsy may lead to diagnosis.

SPECIFIC INVESTIGATIONS

Nerve conduction studies (see [pages 60–61](#)) are useful to confirm there is a neuropathy and to determine the type and distribution of the pathology.

Nerve conduction studies will distinguish axonal from demyelinating neuropathies. They may be able to demonstrate asymmetrical involvement, pointing to a multifocal pathology. They may demonstrate conduction block, an area of focal demyelination, indicative of acquired demyelinating neuropathies.

Nerve biopsy

A biopsy is most likely to aid diagnosis in asymmetric multiple mononeuropathies (vasculitis, amyloidosis, sarcoidosis, etc.). The sural nerve is usually chosen, provided it is involved clinically and neurophysiologically.

THE POLYNEUROPATHIES – SPECIFIC TYPES

GUILLAIN BARRÉ SYNDROME (ACUTE INFLAMMATORY DEMYELINATING POLYNEUROPATHY)

Incidence: 2 per 100 000 population per year. Characteristically it occurs 1–3 weeks after a viral or other infection or immunisation.

Aetiology/pathology

The condition may follow viral infection, *e.g.* varicella-zoster, mumps and *cytomegalovirus*. It is also associated with *Mycoplasma*, *Campylobacter*, infections, immunisations with both live and dead vaccines, antitoxins, trauma, surgery and, rarely, malignant disease and immunodeficiency.

Both antibody and cell-mediated reactions to peripheral nerve myelin are involved. Some patients produce antibodies to myelin glycoproteins or gangliosides, others develop a T cell-mediated assault on myelin basic protein.

Segmental demyelination results with secondary axonal damage if the process is severe. Perivascular infiltration with lymphocytes occurs within peripheral nerves and nerve roots. Lymphocytes and macrophages release cytotoxic substances (cytokines) which damage Schwann cell/myelin.

When axon damage and nerve cell death occur, regeneration cannot take place.

Clinical features

Sensory symptoms predominate at the beginning with paraesthesia of the feet, then hands. Pain, especially back pain, is an occasional initial symptom. Weakness next develops – this may be generalised, proximal in distribution or commence distally and ascend. Tendon reflexes are absent or depressed. In severe cases, respiratory and bulbar involvement occurs. Weakness is maximal three weeks after the onset. Tracheostomy/ventilation is required in 20% of cases. Facial weakness is present to some extent in 50% of cases. Papilloedema may occur when CSF protein is markedly elevated (blocked arachnoid villi?). Autonomic involvement – tachycardia, fluctuating blood pressure, retention of urine – develops in some cases.

Variants are common (20% of cases).

- acute motor axonal neuropathy (AMAN) – often after campylobacter infection
- acute motor, sensory axonal neuropathy (AMSAN)

Investigations

CSF protein is elevated in most patients but often not until the second or third week of illness. Cells are usually absent but in 20% up to 50 cells/mm³ may be found.

Nerve conduction studies

When carried out early in the illness, these may be normal. Findings of multifocal demyelination soon develops with slowing of motor conduction, conduction block and prolonged distal motor latencies.

Ancillary investigations

Performed to identify any precipitating infection: e.g. viral and bacterial studies. Electrolytes are checked for inappropriate secretion of antidiuretic hormone and immune complex glomerulonephritis.

Diagnosis is based on clinical history supported by CSF and neurophysiological investigation and exclusion of acute spinal cord disease, porphyria and myasthenia gravis.

Some antibodies have been identified as being associated with some sub-types including:

AMAN: anti-GD1a and GM1

Acute sensory neuropathy: anti-GD1b

Treatment

Supportive care in HDU/ITU with prevention of respiratory and autonomic complications provides the best chance of a favourable outcome. Signs of impending respiratory failure – forced vital capacity (FVC) below 18 ml/kg, arterial $PaCO_2 > 6.5$ kPa and $PaO_2 < 8$ kPa on oxygen – indicate elective intubation for ventilation. When respiratory assistance is likely to exceed 2 weeks, tracheotomy should be performed.

Subcutaneous low molecular weight heparin with support stockings must be given where the degree of immobility makes thromboembolism a possible complication.

Both plasma exchange (PE) and intravenous immune globulin (IVIG), 0.4 g/kg daily for 5 days – are equally effective at speeding recovery and improving outcome. IVIG is the preferred treatment because of ease of administration but is not without side effects (flu-like symptoms, vasomotor instability, congestive cardiac failure, thrombotic complications – strokes and myocardial ischaemia, transient renal failure and anaphylaxis. There is a very small risk of infection, including theoretically variant CJD)

Treatment is generally given to those who can no longer walk and is deferred in milder cases.

Steroids are not indicated, two trials showing no benefit.

Outcome

Mortality – 2%. Of those progressing to respiratory failure, 20% are left severely disabled and 10% moderately disabled. In milder cases the outcome is excellent. Recurrence – 3%.

Miller Fisher variant of Guillain Barré

The Miller Fisher syndrome consists of ophthalmoplegia, areflexia and ataxia without significant limb weakness. Serum IgG antibodies to a

specific ganglioside are characteristic (anti-GQ1B antibodies). Management is that of Guillain Barré.

CHRONIC INFLAMMATORY DEMYELINATING POLYNEUROPATHY (CIDP)

Similar to Guillain Barré but with a progressive or fluctuating course over weeks or months and rarely involving cranial nerves or respiratory function.

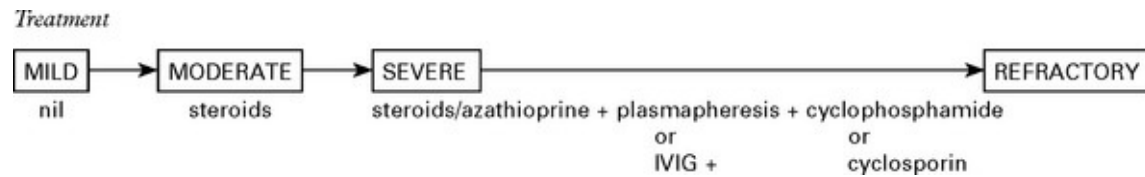
Pathology: Segmental demyelination with remyelination (onion bulb formation) and sparse mononuclear inflammatory change.

Prevalence – 3% of all neuropathies

Incidence – 5 per million

Age of onset: mean 35 yrs (fluctuating course – younger, progressive – older)

<i>Diagnosis:</i>	Electrophysiology	– conduction velocity < 70% of normal – conduction block (outwith entrapment sites) – prolonged distal latencies
	Distinguish from	– hereditary neuropathy (CMT type 1, page 444) – paraprotein and lymphoma associated neuropathy (page 443) – multifocal motor neuropathy with conduction block (page 443) – HIV neuropathy (page 516)



About two thirds of patients respond to steroids, PE or IVIG. In moderate/severe cases steroids should be used initially (cost and ease of use) followed, if response unsatisfactory, by IVIG and finally PE. Despite little evidence, immunosuppressive drugs (azathioprine, cyclophosphamide or cyclosporin) are deployed in resistant cases.

Outcome with treatment – 30% symptom free – 45% mild disability – 25% severe disability

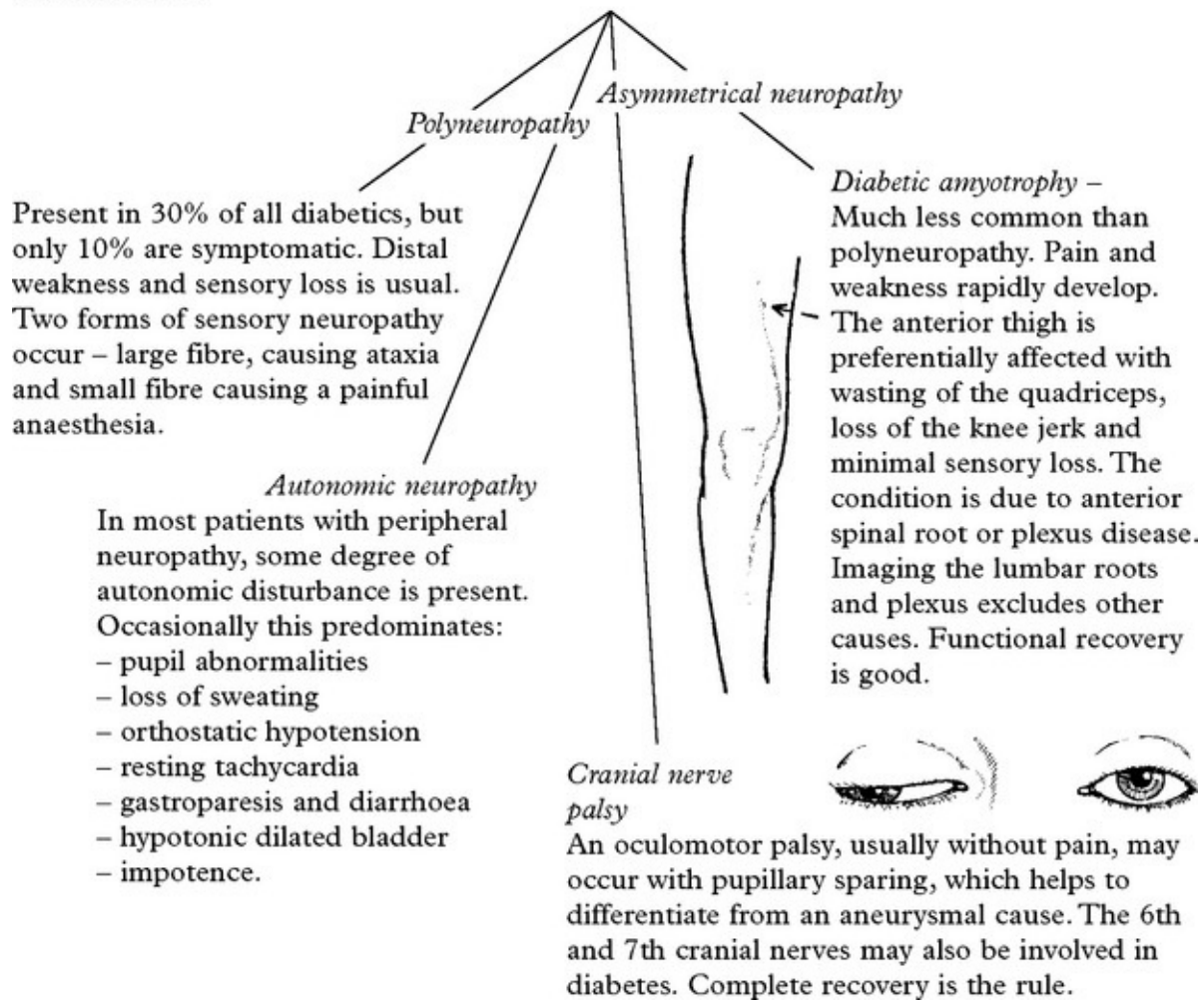
DIABETIC NEUROPATHY

This condition is uncommon in childhood and increases with age.

Peripheral nerve damage is related to poor control of diabetes. This is more common in insulin-dependent patients. Damage results from either metabolic disturbance with sorbitol and fructose accumulation in axons and Schwann cells or an occlusion of the nutrient vessels supplying nerves (vasa vasorum). The frequent occurrence of neuropathy with other vascular complications – retinopathy and nephropathy – suggests that the latter is the more usual mechanism. Neurological complications correlate with levels of glycosylated haemoglobin A1C, an indicator of the long-term control of hyperglycaemia.

Classification

Classification



Treatment

Improved control of diabetes is essential.

Carbamazepine, gabapentin, pregabalin, tricyclic antidepressants or α -adrenergic blockers, e.g. phenoxybenzene, help control pain.

Drugs which reduce aldose reductase and halt accumulation of sorbitol and fructose in nerves are being evaluated.

Management of autonomic neuropathy – see [page 460](#).

Asymmetrical neuropathies usually spontaneously recover, whereas prognosis for symmetric neuropathies is less certain.

CARCINOMATOUS POLYNEUROPATHY

Sensory or mixed 'sensorimotor' neuropathy is often associated with malignant disease, particularly small cell carcinoma of the lung. Neuropathy may also occur with Hodgkin's disease and lymphomas. The neuropathy is characterised by the presence of antibodies (anti Hu) that are detected in serum. Such antibodies not only recognise antigen in tumours but also bind to peripheral nervous system neurons.

Pathology

The sensory type is characterised by degeneration and inflammatory changes in the dorsal root ganglion. The ventral roots and peripheral nerve motor fibres are spared. In the sensorimotor type, degeneration of the dorsal root ganglion is less marked and axonal and demyelinating changes affect motor and sensory fibres equally.

Clinical features

Symptoms and signs may predate the appearance of causal malignant disease by months or even years.

Sensory neuropathy: Progressive sensory loss often commencing in upper limbs is associated with paraesthesia, unpleasant 'burning' dysaesthesia and sensory ataxia.

Sensorimotor neuropathy: The onset is gradual with distal sensory loss and mild motor weakness. Occasionally a more acute, severe neuropathy resembling Guillain-Barré syndrome occurs.

Detection and treatment of the underlying malignancy may lead to recovery of the neuropathy. Alternatively the use of immunosuppressive agents, plasma exchange or intravenous gammaglobulin (IVIG) may help.

MULTIFOCAL MOTOR NEUROPATHY WITH CONDUCTION BLOCK

This presents with asymmetric lower motor neuron weakness and may be mistaken for motor neuron disease. Neurophysiological investigation shows 'conduction block' at sites distant from possible entrapment. Antibodies to gangliosides (Anti GM1) are found in serum. Immunosuppressive treatment (cyclophosphamide) or intravenous immunoglobulin (IVIG) when indicated, result in clinical improvement.

NEUROPATHIES ASSOCIATED WITH PARAPROTEINS

Approximately 10% of patients with late onset chronic peripheral neuropathy have a circulating monoclonal paraprotein in the serum. If myeloma, lymphoma, amyloidosis and Waldenström's macroglobinaemia are excluded, the condition is referred to as a 'monoclonal gammopathy of uncertain significance' (MGUS). IgM is reactive, in 50% of cases, against myelin associated glycoprotein, anti-MAG antibodies, which can be demonstrated to bind myelin. Neuropathies may be axonal, demyelinating or mixed and show a variable response to immunotherapy.

PORPHYRIA

Acute intermittent porphyria is a rare autosomal dominant disorder in which symptoms of abdominal pain, psychosis, convulsions and peripheral neuropathy occur.

The metabolic fault occurs in the liver. An increased production of porphobilinogen is reflected by its increased urinary excretion. δ -amino laevulic acid, a porphyrin precursor, is also increased.

Clinical features

The onset is acute and predominantly motor with upper limb and occasional cranial nerve involvement. Respiratory failure occurs in severe cases. Autonomic involvement with tachycardia, blood pressure changes, abdominal pain and vomiting often develop. The neuropathy must be distinguished from Guillain-Barré.

Clinical course is variable. Spontaneous recovery occurs over several weeks. Respiratory failure will require ventilation and carries a poor prognosis. During an attack, a high carbohydrate diet and prevention and treatment of electrolyte disturbances are essential. Chelating agents (EDTA or Penicillamine) are used in severe cases. Recurrent attacks may be anticipated. Certain drugs may precipitate these attacks and must be avoided, e.g. sulphonamides, barbiturates, phenytoin, griseofulvin.

THE POLYNEUROPATHIES – SPECIFIC TYPES – INHERITED NEUROPATHIES

CHARCOT-MARIE-TOOTH DISEASE (previously called hereditary motor and sensory neuropathy (HMSN))

A heterogeneous group of disorders with a prevalence of 1:2500 – the largest category of genetic neurological disease. The characteristic appearance is that of distal wasting, the lower limbs having an ‘inverted wine bottle’ appearance.

Classification	Clinical features	Pathology	Neuro-physiology	Inheritance
CMT type I	Age of onset < 30 yrs. Wasting and weakness of intrinsic foot muscles, peroneal and tibial groups. Distal upper limb involvement. Pes cavus/hammer toes – 75%. Palpable peripheral nerves – 25%. Associated ataxia and tremor – 10%.	Demyelination with thickened 'onion bulb' areas of remyelination.	Motor conduction velocities slowed < 38 m/sec in common peroneal nerve.	<i>Autosomal dominant</i> 70% of cases are due to duplication of 17p 11.2 in PMP (peripheral myelin protein) 22 gene. X linked cases result from point mutations in the Connexin 32 gene.
CMT type II	Age of onset > 30 yrs. Wasting and weakness as type I. Foot deformities absent. Peripheral nerves not palpable.	Axonal loss.	Motor conduction velocities normal or marginally slowed.	<i>Autosomal dominant</i> – mechanism 20% have mitofusin gene mutations.
Dejerine-Sottas disease	Age of onset: childhood. Wasting and weakness may be proximal. Peripheral nerves and spinal roots thickened. CSF protein elevated.	Demyelination with 'onion bulb' formation.	Motor conduction velocities profoundly slowed – 5–10 m/sec	<i>Autosomal recessive</i> – point mutation chromosome 1 or 17 or sporadic



Many complex forms of hereditary neuropathies occur and the above classification is far from complete with the genetic basis for many now determined. Some pedigrees show additional features such as – optic atrophy, retinopathy, deafness, ataxia, spasticity and cardiomyopathy. Such 'extra' features complicate a simple classification. Treatment is symptomatic with provision of appropriate footwear, splints or orthopaedic procedures to maintain mobility. In adult onset disease, the rate of progression is exceedingly slow. The demonstration of genetic markers and the application of nerve conduction studies allows early and correct diagnosis in those at risk. Nerve biopsy is of no diagnostic value.

Other rare forms of hereditary neuropathy

– Hereditary sensory and autonomic neuropathies	– Autosomal recessive Childhood onset
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	Characterised by insensitivity to pain and disordered sweating
– Hereditary neuropathy with liability to pressure palsies (HNPP)	– Autosomal dominant (deletion in PMP 22 gene) Adult onset Characterised by recurrent entrapment neuropathies e.g. carpal tunnel syndrome
– Hereditary neuropathy with spinocerebellar degeneration	– e.g. Friedreich's ataxia (pages 552–3)
– Hereditary neuropathy with metabolic defect	– e.g. Familial amyloid neuropathy – mutation of transthyretin gene Porphyria – abnormality of hepatic haem biosynthesis Refsum's disease – abnormality of phytanic acid metabolism.

PLEXUS SYNDROMES AND MONONEUROPATHIES

Disease of a single peripheral or cranial nerve is termed *mononeuropathy*. When many single nerves are damaged one by one, this is described as *mononeuritis multiplex*. Damage to the brachial or lumbosacral plexus may produce widespread limb weakness which does not conform to the distribution of any one peripheral nerve. A knowledge of the anatomy and muscle innervation of the plexuses and peripheral nerves is essential to localise the site of the lesion and thus deduce the possible causes.

Certain systemic illnesses are associated with the development of mononeuropathy or mononeuritis multiplex:

- diabetes mellitus
- sarcoidosis

- vasculitis
- leprosy (worldwide commonest cause)

Entrapment mononeuropathies result from damage to a nerve where it passes through a tight space such as the median nerve under the flexor retinaculum of the wrist. These are often related to conditions such as acromegaly, myxoedema and pregnancy, in which soft tissue swelling occurs. A familial tendency to entrapment neuropathy has been described.

Cranial nerve mononeuropathies have been dealt with separately.

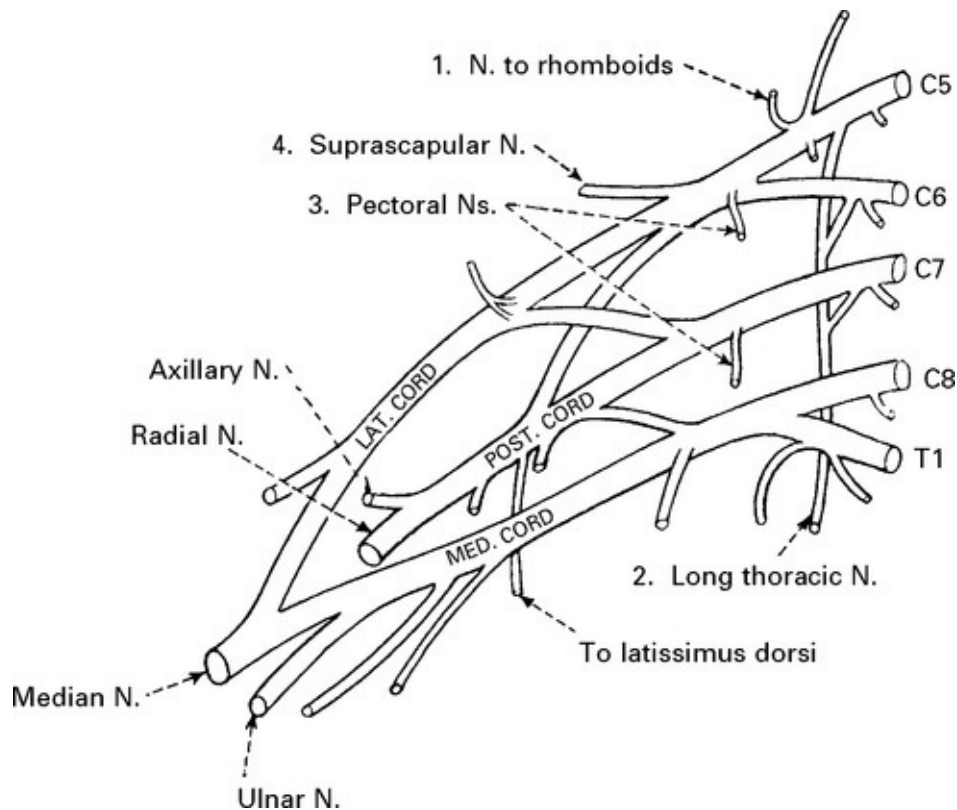
BRACHIAL PLEXUS

The plexus lies in the posterior triangle of the neck between the muscles scalenius anterior and scalenius medius.

At the root of the neck the plexus lies behind the clavicle.

The plexus itself gives off several important motor branches:

1. Nerve to rhomboids
2. Long thoracic nerve – to serratus ant.
3. Pectoral nerves – to pectoralis major
4. Suprascapular nerve – to supraspinatus infraspinatus

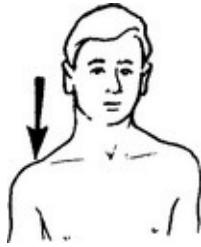


BRACHIAL PLEXUS SYNDROMES

UPPER PLEXUS LESION (C5C6)

Traction on the arm at birth (Erb-Duchenne paralysis) or falling on the shoulder (especially motor cyclists) may damage the upper part (C5C6) of the plexus.

Deltoid	}	paralysed.
Supraspinatus		
Infraspinatus		
Biceps	}	elbow flexors – also paralysed.
Brachialis		
Adductors of shoulder – mildly affected.		



When damage to C5C6 is more proximal, nerve to rhomboids and long thoracic nerve may be affected.

POSTERIOR CORD LESION (C5C6C7C8)

Deltoid	} paralysed
Extensors of elbow (triceps)	
Extensors of wrist (extensor carpi radialis longus and brevis, extensor carpi ulnaris)	
Extensors of fingers (extensor digitorum)	

LOWER PLEXUS LESION (C8T1)

Forced abduction of the arm at birth (Klumpke's paralysis) or trauma may produce damage to the lower plexus. This results in paralysis of the intrinsic hand muscles producing a claw hand, C8T1 sensory loss and a Horner's syndrome ([page 145](#)) if the T1 root is involved.

N.B. A combined ulnar and median nerve lesion will produce a similar picture in the hand but with involvement also of flexor carpi ulnaris and pronator teres.



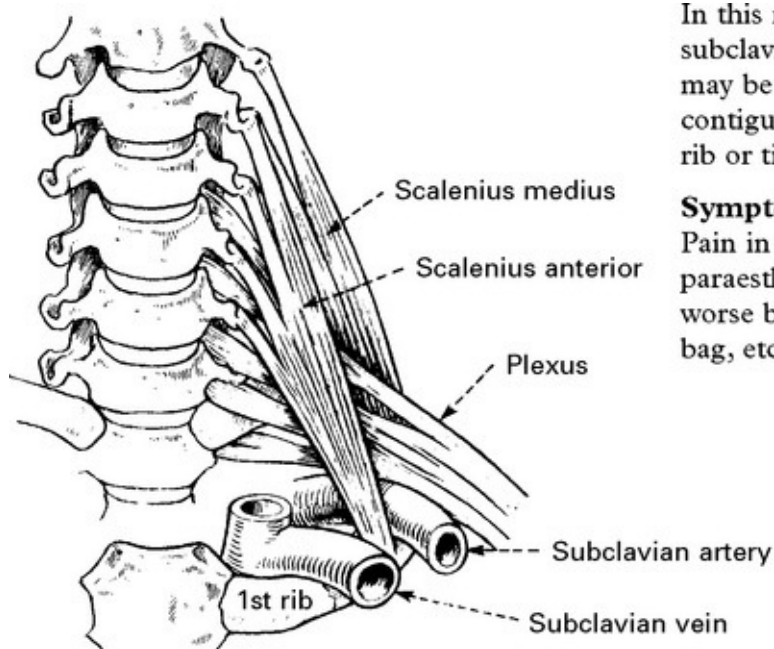
TOTAL BRACHIAL LESION

This results in complete flaccid paralysis and anaesthesia of the arm.

The presence of a Horner's syndrome indicates proximal T1 nerve root involvement.

N.B. When trauma is the cause of brachial paralysis, early referral to a specialist unit with experience in the surgical repair of plexus injuries is advised.

THORACIC OUTLET SYNDROME



In this rare disorder the brachial plexus, subclavian artery and subclavian vein may be compressed in the neck by contiguous structures such as a cervical rib or tight fibrous band.

Symptoms

Pain in the neck and shoulder with paraesthesia in the forearm, made worse by carrying a suitcase, shopping bag, etc.

Signs

Sensory loss in a T1 distribution.

Wasting and weakness of thenar and occasionally interosseous muscles.

Signs of vascular compression:

- Unilateral Raynaud's phenomenon.
- Pallor of limb on elevation.

- Brittle trophic finger nails.
- Loss of radial pulse in arm on abduction and external rotation at the shoulder or on bracing the shoulders – ADSON's sign.
- Subclavian venous thrombosis may occur, especially after excessive usage of arm.

Investigation

Coronal MRI is the definitive investigation.

Plain radiology of the thoracic outlet may reveal a cervical rib or prolonged transverse process. Nerve conduction/electromyography will distinguish this from other peripheral nerve lesions. Arteriography or venography is occasionally necessary if there are obvious vascular problems.

Treatment

In middle-aged people with poor posture and no evidence of abnormality on plain radiology, neck and postural exercises are helpful.

In younger patients with clinical and electrophysiological changes supporting the radiological abnormalities, exploration and removal of a fibrous band or rib may afford relief.

In this rare disorder the brachial plexus, subclavian artery and subclavian vein may be compressed in the neck by contiguous structures such as a cervical rib or tight fibrous band.

Symptoms

Pain in the neck and shoulder with paraesthesia in the forearm, made worse by carrying a suitcase, shopping bag, *etc.*

Ensure orthopaedic mimics, rotator cuff injury or shoulder joint

contractures, are considered prior to assessment.

BRACHIAL NEURITIS (Neuralgic amyotrophy)

Brachial neuritis is a relatively common disorder sometimes associated with:

- Viral infection (infectious mononucleosis, cytomegalovirus)
- Vaccination (tetanus toxoid, influenza)
- Strenuous exercise

In most cases it develops without any evident precipitating cause.

Clinical features

- Acute onset with preceding shoulder pain.
- Weakness is usually proximal, though the whole arm may be affected.
- Occasionally both arms are affected simultaneously.
- Sensory findings are minor (loss over the outer aspect of the shoulder) and occur in 50%.
- Reflex loss occurs.
- Wasting is apparent after 3–6 weeks.
- Recurrent episodes can occur, especially in the presence of a family history.

Differential diagnosis	Investigation
A painful weak arm. Consider:	
– Cervical spondylosis.	CSF may show a mild protein rise and a pleocytosis.

– Cervical disc lesion. – Brachialgia due to local bursitis. – Polymyalgia rheumatica.	<i>Nerve conduction studies</i> will show slowing in affected nerves after 7–10 days.
--	--

Treatment

Narcotic analgesics may be required if pain is extreme. Corticosteroids are normally given though the value of immunotherapy is uncertain. By 2 yrs – 75% have fully recovered. Brachial neuritis may be familial. Recurrent attacks occur in Hereditary Neuropathy with liability to Pressure Palsies – HNPP ([page 444](#)). Autosomal dominant forms of Hereditary Neuralgic Amyotrophy (HNA), both acute and chronic are described, some linked to chromosome 17q.

PANCOAST'S TUMOUR

Involvement of the plexus by apical lung tumour (usually squamous cell carcinoma). The lower cervical and upper thoracic roots are involved.

Clinical features

- Severe pain around the shoulder and down the inside of the arm.
- Weak wasted hand muscles.
- Sensory loss (C8T1).
- Horner's syndrome (invasion of sympathetic chain and stellate ganglion).
- Rarely medial extension can involve the recurrent laryngeal nerve (hoarseness & bovine cough).

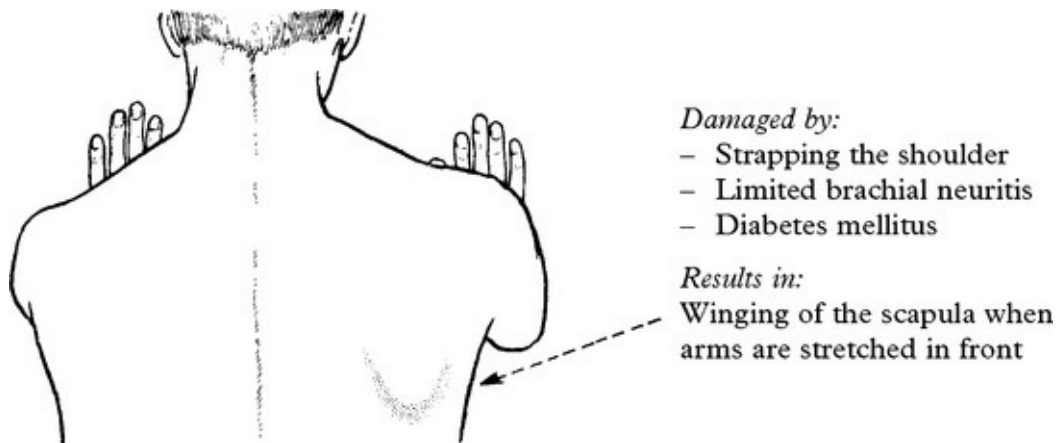
RADIATION PLEXOPATHY

Now infrequently seen after irradiation of axillary nodes in breast Ca. Onset usually 2–4 yrs after exposure to high radiation dose (> 44–50 Gy). Thickening of the vascular endothelium causes ischaemia of the plexus. Symptoms start with paraesthesia in the hand and progress slowly to involve all lower plexus structures with wasting, weakness, reflex & sensory loss.

UPPER LIMB MONONEUROPATHIES

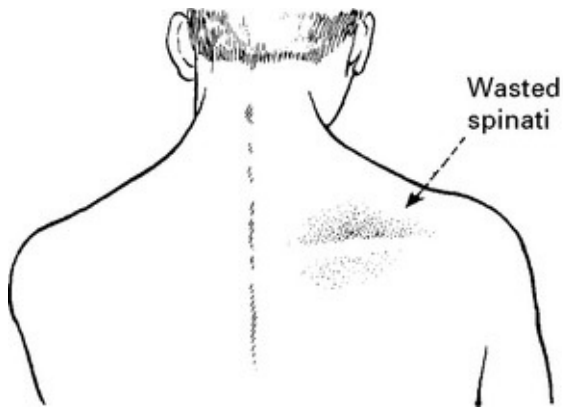
LONG THORACIC NERVE (C5C6C7)

Supplies: Serratus anterior muscle



SUPRASCAPULAR NERVE (C5C6)

Supplies: Supraspinatus and infraspinatus muscles.



Damaged by:

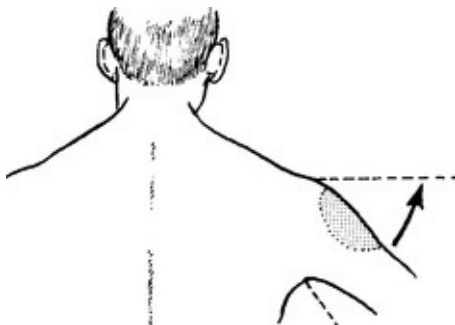
- [as for Long thoracic nerve (above)]
- Carrying heavy objects over shoulder (rucksack or pitchfork)

Results in:

- Weakness of abduction of arm (supraspinatus)
- Weakness of external rotation of arm (infraspinatus).

AXILLARY NERVE (posterior cord) (C5C6)

Supplies: Deltoid and teres minor muscles.



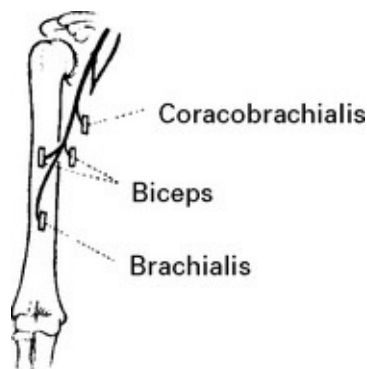
Damaged by:

- Shoulder dislocation.
- Limited brachial neuritis.

Results in:

- Weakness of abduction of shoulder between 15–90° and sensory loss over the outer aspect of the shoulder.

MUSCULOCUTANEOUS NERVE (Lateral cord) (C5C6)



Sensory supply: Lateral border of the arm.

Damaged by:

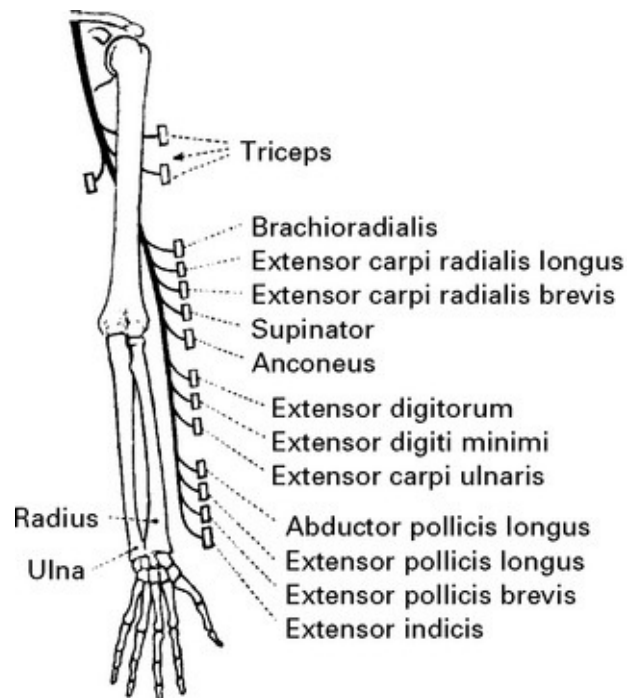
- Fracture of the humerus.
- Systemic causes.

Results in:

- Weakness of elbow flexion and forearm supination with characteristic sensory loss and absent biceps reflex.



RADIAL NERVE (Posterior cord) (C6C7C8)



Sensory supply: Dorsum of hand. The nerve descends from the axilla,

winding posteriorly around the humerus. The deep branch – the posterior interosseous nerve – lies in the posterior compartment of the forearm behind the interosseous membrane.



Damaged by:

- Fractures of the humerus.
- Prolonged pressure (Saturday night palsy).
- Intramuscular injection.
- Lipoma, fibroma or neuroma.
- Systemic causes.



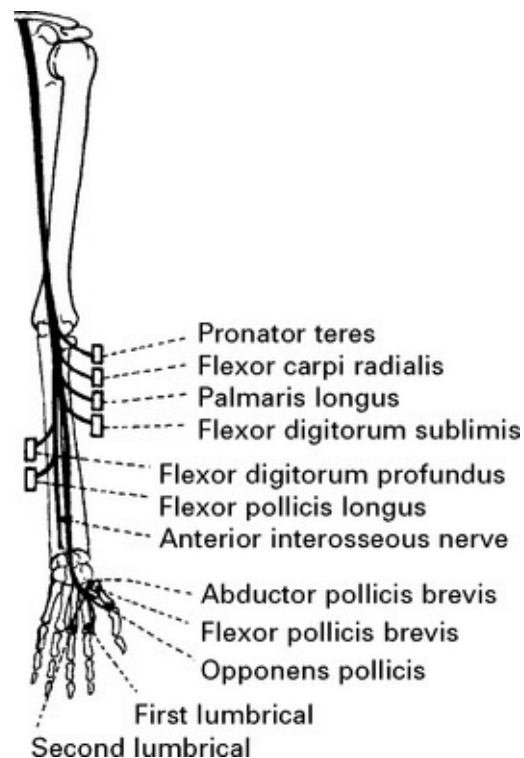
Results in:

- Weakness and wasting of muscles supplied, characterised by wrist drop with flexed fingers (weak extensors). Sensory loss on dorsum of hand and forearm. Loss of triceps reflex (when lesion lies in the axilla) and supinator reflex.

The *posterior interosseous branch* of the radial nerve can be compressed at its point of entry into the supinator muscle. The clinical picture is similar to a radial nerve palsy, only brachioradialis and wrist extensors are spared. Examination shows weakness of finger extension with little or no wrist drop.



MEDIAN NERVE (Lateral and medial cords) (C7C8)

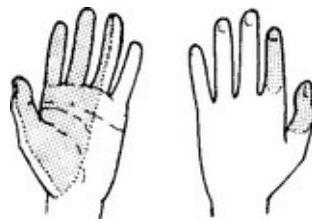


Sensory supply:

Palmar surfaces of the radial border of the hand.

The nerve lies close to the brachial artery in the upper arm. It passes under the transverse carpal ligament as it approaches the palm of the

hand.



Damaged by:

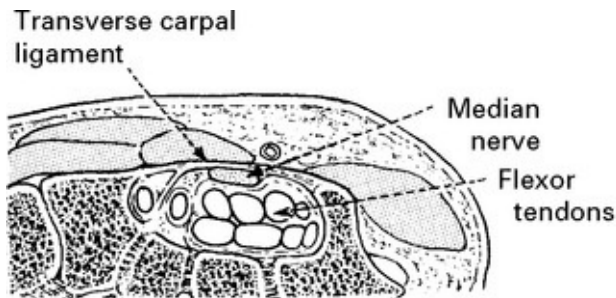
- Injury in axilla, e.g. dislocation of shoulder, compression in the forearm – anterior interosseous branch, compression at the wrist (carpal tunnel syndrome).

Results in:

- Weakness of abduction and apposition of thumb.
- Weakness of pronation of the forearm.
- Deviation of wrist to ulnar side on wrist flexion.
- Weakness of flexion of distal phalanx of thumb and index finger.
- Wasting of thenar muscles is evident.
- Sensory loss is variable but most marked on index and middle fingers

Carpal tunnel syndrome

The most common entrapment neuropathy, more frequent in women, results from median nerve entrapment under the transverse carpal ligament at the wrist.



- Causes:* – Connective tissue thickening, e.g.
- Rheumatoid arthritis
 - Acromegaly
 - Hypothyroidism.
 - Infiltration of ligament, e.g. amyloid disease.
 - Fluid retention, e.g. in pregnancy, weight gain.

<i>Causes:</i>	– Connective tissue thickening, e.g.	
	–	Rheumatoid arthritis
	–	Acromegaly
	–	Hypothyroidism.
	–	Infiltration of ligament, e.g. amyloid disease.
	–	Fluid retention, e.g. in pregnancy, weight gain.

Symptoms:

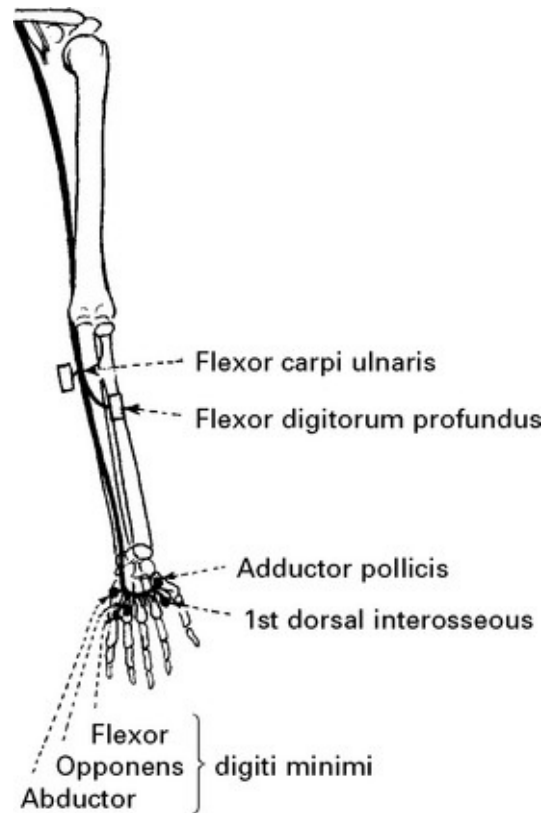
Pain, especially at night, and paraesthesia, eased by shaking the hand or dangling it out of the bed.

Objective findings may follow with cutaneous sensory loss and wasting and weakness of thenar muscles (abductor and opponens pollicis). Percussion on the nerve at the wrist produces heightened paraesthesia (Tinel's sign).

Nerve conduction studies are helpful in confirming diagnosis by showing slowing of conduction over the wrist.

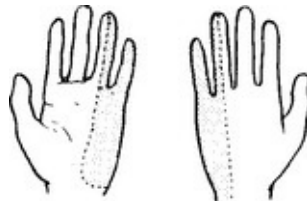
Treatment: of the cause, weight loss and diuretics. Surgical division of the transverse ligament if symptoms fail to improve produces excellent results (90% symptom free).

ULNAR NERVE (Medial cord) (C7C8)



Sensory supply:

Both palmar and dorsal surfaces of the ulnar border of the hand.

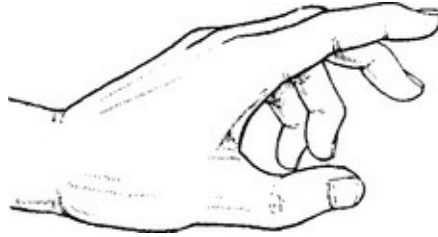


In the upper arm the nerve is closely related to the brachial artery and the median nerve, and passes behind the medial epicondyle of the humerus into the forearm.

In the hand, close to the hamate bone, it divides into deep and superficial branches.

Damaged by:

- Injury at elbow, e.g. dislocation.
- Entrapment at elbow or distal to the medial epicondyle.
- Pressure on the nerve in the palm of the hand damages the deep branch resulting in wasting and weakness without sensory loss.



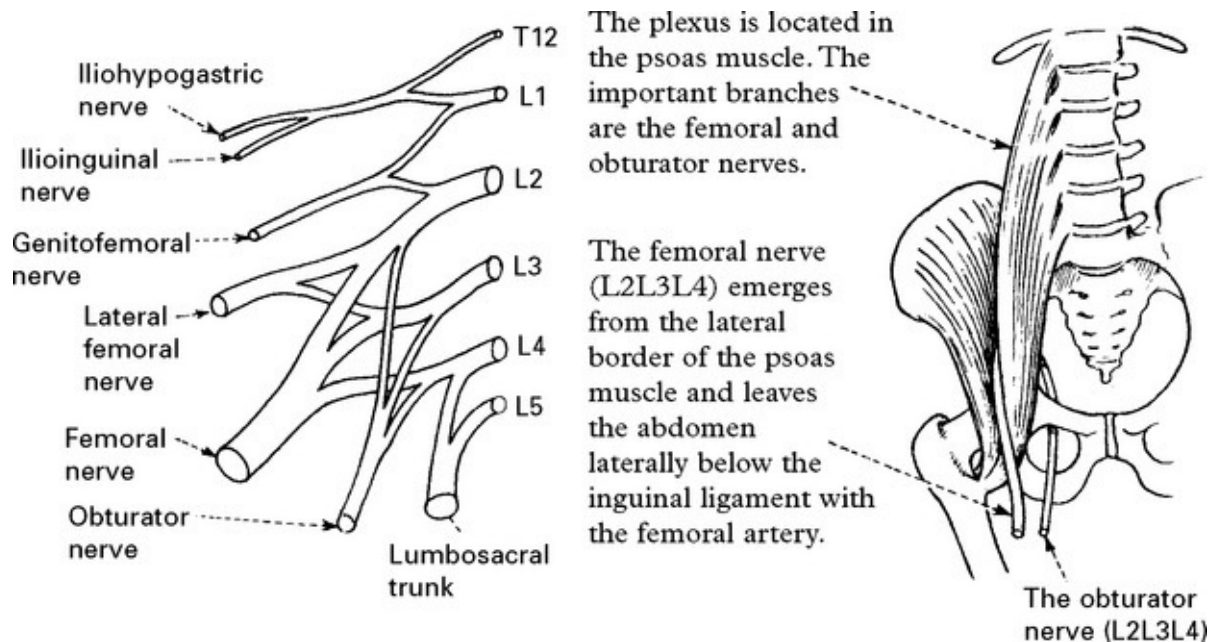
Results in:

- Weakness and wasting of muscles supplied, with a characteristic posture of the hand – *ulnar claw hand* – as well as sensory loss. The level of the lesion dictates the extent of the motor paralysis. Nerve conduction studies are helpful in confirming entrapment at the elbow.

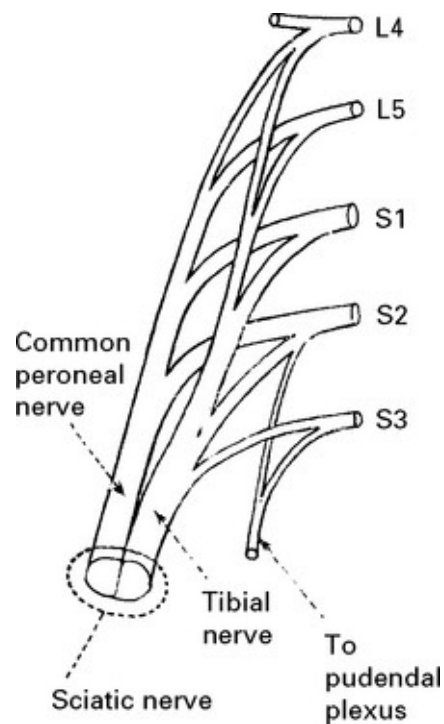
Surgical transposition may be necessary in such cases.

LUMBOSACRAL PLEXUS

LUMBAR PLEXUS



SACRAL PLEXUS



The plexus is located on the posterior wall of the pelvis. The five roots of the plexus divide into anterior and posterior divisions. The L4L5S1S2

divisions form the common peroneal nerve.

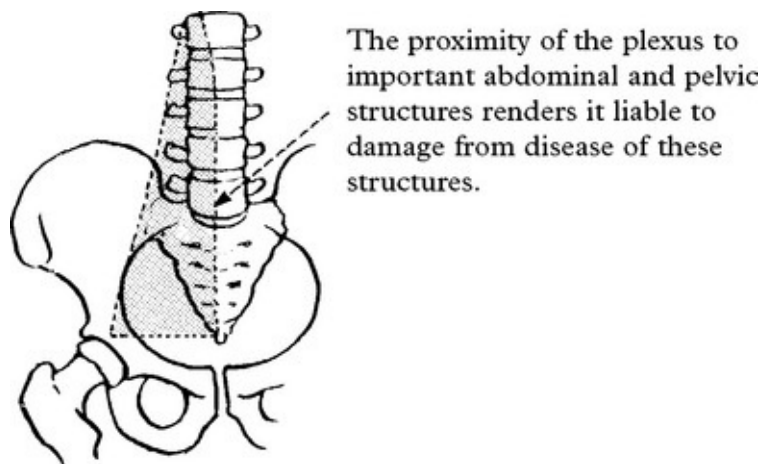
The L4L5S1S2S3 anterior divisions form the tibial nerve. Both these nerves fuse to form the sciatic nerve.



The posterior divisions S2S3 pass to the pudendal plexus.

The common peroneal and tibial nerves (sciatic nerve) leave the pelvis by the greater sciatic foramen. In the popliteal fossa the sciatic nerve splits into its constituent nerves.

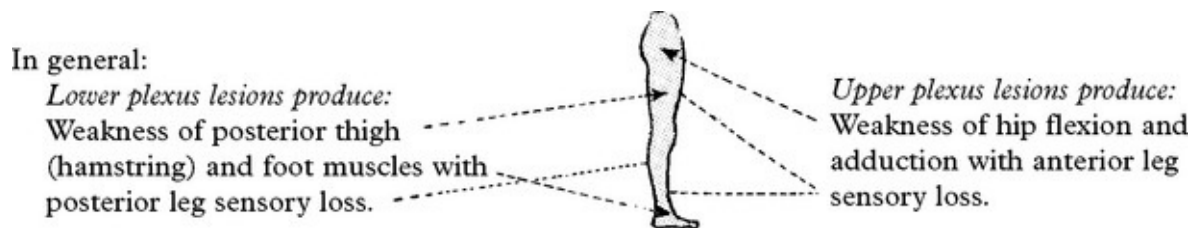
LUMBOSACRAL PLEXUS SYNDROMES



Trauma following surgery, *e.g.* hysterectomy, lumbar sympathectomy

or during labour. Compression from an abdominal mass, *e.g.* aortic aneurysm. Infiltration from pelvic tumour. Radiotherapy.

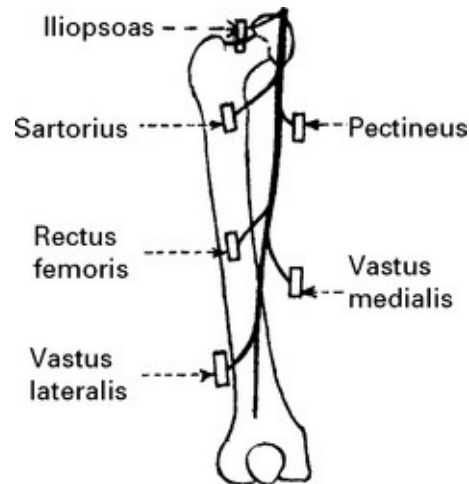
Symptoms may be unilateral or bilateral, depending upon causation. Weakness, sensory loss and reflex changes are dictated by the location and extent of plexus damage. Pain of a severe burning quality may be present; it may be worsened by coughing, sneezing, *etc.*



The lumbosacral plexus may be affected in the same way as the brachial plexus in brachial neuritis – lumbosacral neuritis – the association with infection, *etc.*, being similar. Recovery is usually good. Recurrent and familial cases occur. Plexus lesions also occur in *diabetes mellitus* and *vasculitis*. In both, the symptoms and signs may be bilateral. Investigate with CT/MR and neurophysiology.

LOWER LIMB MONONEUROPATHIES

FEMORAL NERVE (L1L2L3)



Damaged by:

- Fractures of the upper femur
- Congenital dislocation of the hip, hip surgery
- Neoplastic infiltration
- Psoas muscle abscess
- Haematoma into iliopsoas muscle (haemophilia, anticoagulants)
- Systemic causes of mononeuropathy, *e.g.* diabetes.

Results in:

- Weakness of hip flexion
- Weakness of knee extension with wasting of thigh muscles
- Sensory loss over the anterior and medial aspects of the thigh
- The knee jerk is lost.

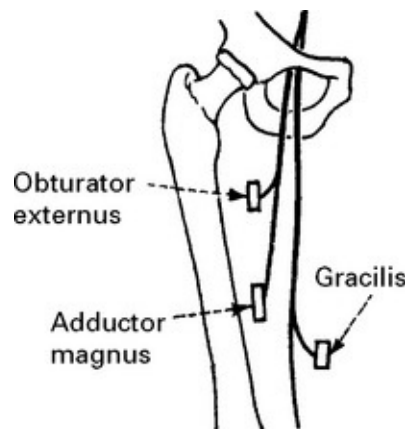
OBTURATOR NERVE (L2L3L4)

Damaged by:

- Same process as the femoral nerve.
- During labour and occasionally as a consequence of compression by hernia in the obturator canal.

Results in:

- Weakness of hip external rotation and adduction.
- The patient may complain of inability to cross the affected leg on the other.
- Sensory loss is confined to the innermost aspect of the thigh.
- The adductor reflex is absent (adductor response to striking the medial epicondyle).



SCIATIC NERVE (L4L5S1S2)

The nerve descends between the ischial tuberosity and the greater trochanter of the femur. In the thigh it innervates the hamstring muscles (semitendinosus, semimembranosus and biceps).

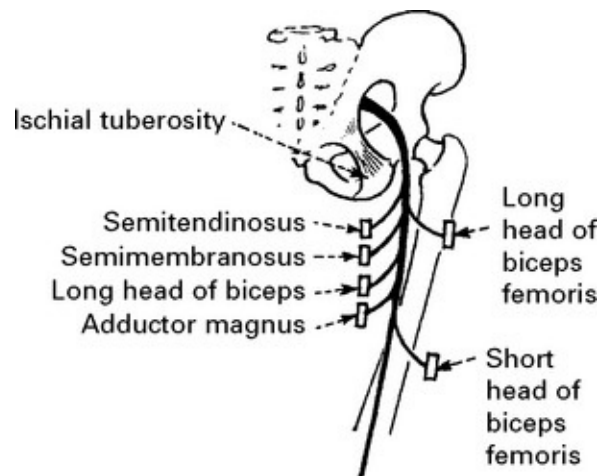
Damaged by:

- Congenital or traumatic hip dislocation.

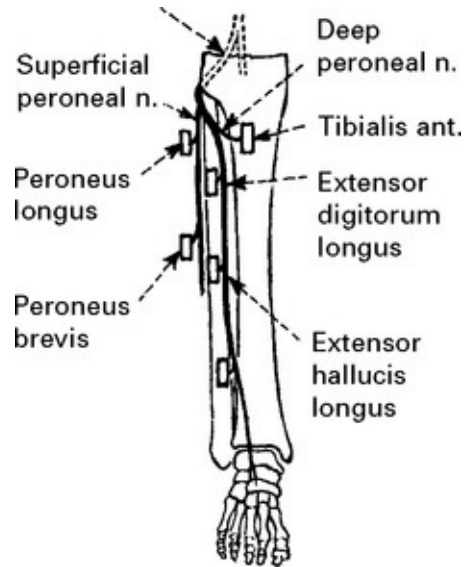
- Penetrating injuries.
- Accidental damage from ‘misplaced’ intramuscular injection.
- Entrapment at sciatic notch.
- Systemic causes of mononeuropathy

Results in:

- Weakness of hamstring muscles with loss of knee flexion.
- Distal foot and leg muscles are also affected.
- Sensory loss involves the outer aspect of the leg.
- The ankle reflex is absent.



COMMON PERONEAL NERVE (L4L5)



The nerve arises from the division of the sciatic nerve in the popliteal fossa. It bears a close relationship with the head of the fibula as it winds anteriorly. It divides into superficial and deep branches as well as giving off a purely sensory branch which, with sensory twigs from the tibial nerve, forms the sural nerve, mediating sensation from the dorsum and lateral aspect of the foot.

Damaged by:

- Trauma to the head of the fibula; pressure here from kneeling, crossing legs.
- Systemic causes of mononeuropathy, e.g. diabetes.

Results in: Weakness of dorsiflexion and eversion of the foot. The patient walks with a ‘foot drop’. Sensory loss involves the dorsum and outer aspect of the foot. Partial common peroneal nerve palsies are common with very selective muscle weakness.

POSTERIOR TIBIAL NERVE (S1S2)

This nerve also arises from the division of the sciatic nerve in the popliteal fossa and descends behind the tibia, terminating in the medial

and lateral plantar nerves which innervate the small muscles of the foot. The sensory branch contributes to the *sural nerve*.

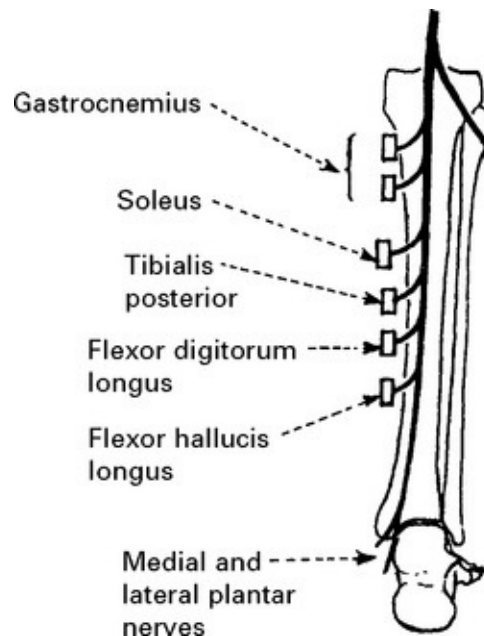
Damaged by:

Trauma in the popliteal fossa.

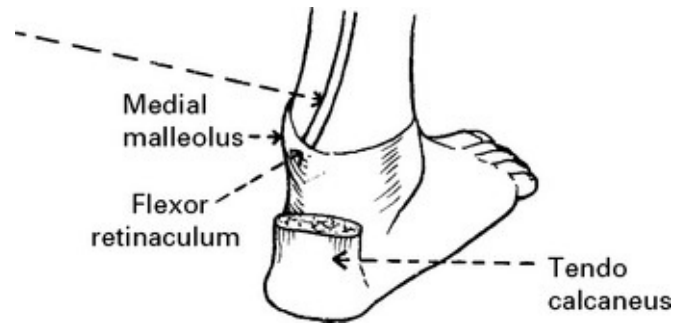
- Fracture of the tibia.
- Systemic causes of mononeuropathy.

Results in:

- Weakness of plantar flexion and inversion of the foot.
- The patient cannot stand on toes.
- Sensory loss involves the sole of the foot.
- The ankle reflex is lost.

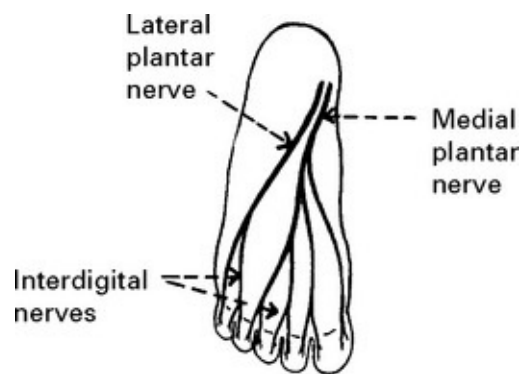


Tarsal tunnel syndrome



PLANTAR AND SMALL INTERDIGITAL NERVES

Compression of these nerves at the sole of the foot produces localised burning pain. Involvement of interdigital nerves produces pain and analgesia in adjacent halves of neighbouring toes.



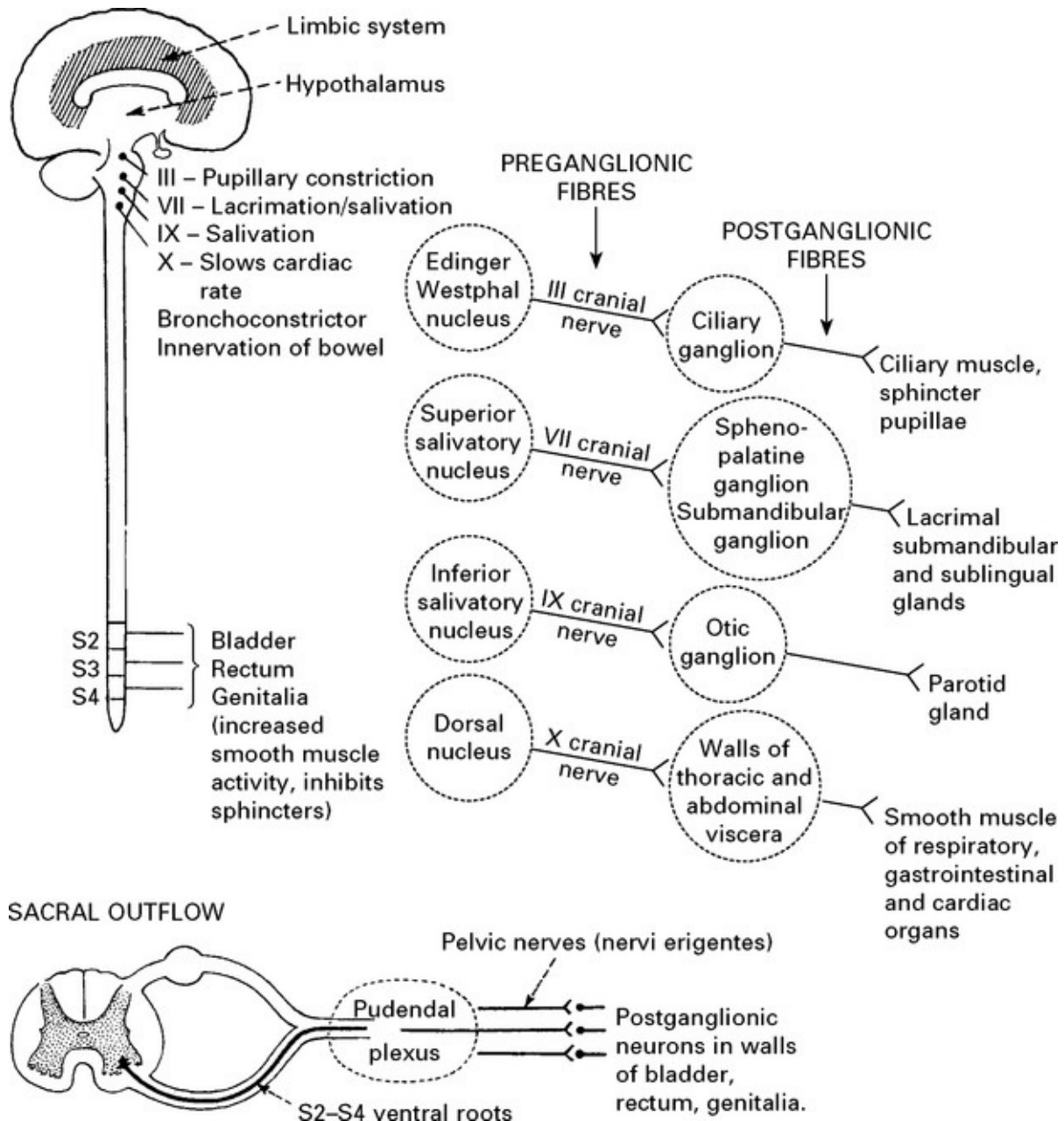
AUTONOMIC NERVOUS SYSTEM

The autonomic nervous system maintains the visceral and homeostatic functions essential to life. It is divided into SYMPATHETIC and PARASYMPATHETIC components and contains both motor (efferent) and sensory (afferent) pathways.

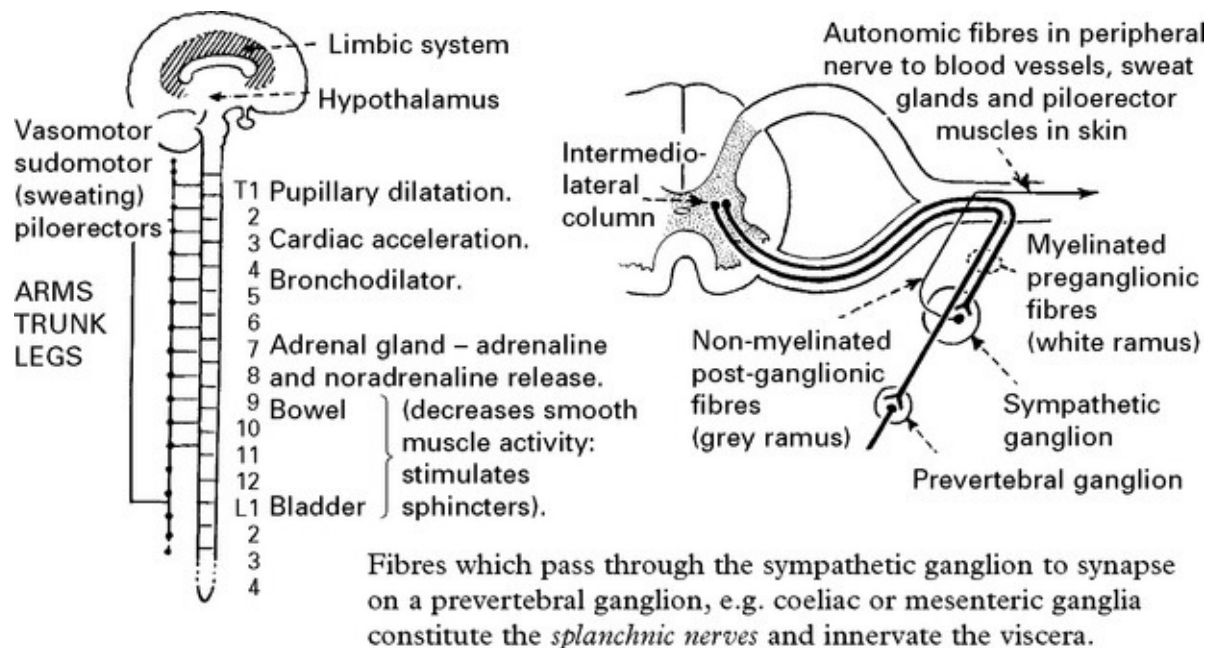
Both sympathetic and parasympathetic systems are regulated by the *limbic system*, *hypothalamus* and *reticular formation*. Fibres from these structures descend to synapse with preganglionic neurons in the intermediolateral column T1–T12 (sympathetic) and in the III, VII, IX

and X cranial nerve nuclei and S2–S4 segments of the cord (parasympathetic).

PARASYMPATHETIC OUTFLOW



SYMPATHETIC OUTFLOW



AFFERENT AUTONOMIC NERVOUS SYSTEM

Sympathetic

Terminate in spinal cord in intermediate zone of grey matter – in relation to preganglionic neurons.

Function: Important in the appreciation of visceral pain.

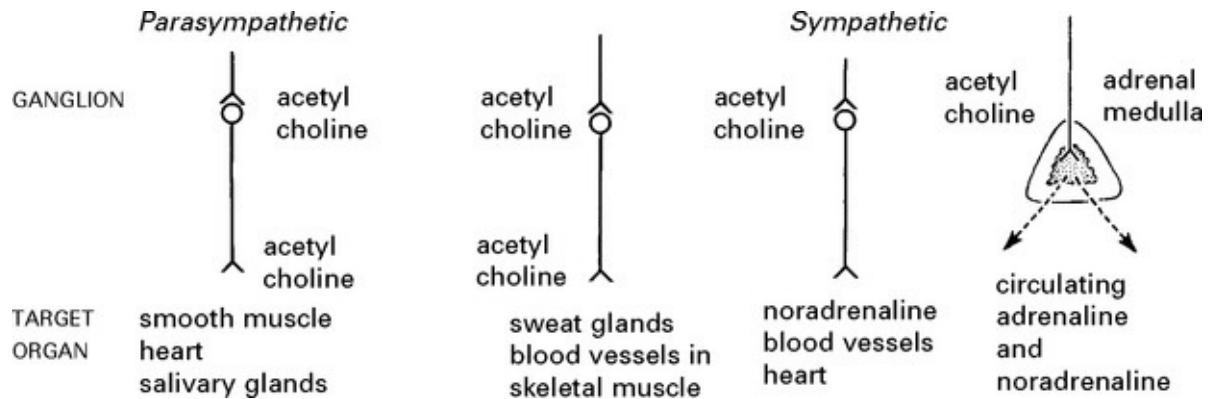
Parasympathetic

Afferent fibres from the mouth and pharynx, and respiratory, cardiac and gastrointestinal systems, travelling in the VII, IX and X cranial nerves, terminate in the nucleus of tractus solitarius.

Function: Important in maintaining visceral reflexes.

The sacral afferents end in the S2–S4 region in relation to preganglionic neurons.

NEUROTRANSMITTER SUBSTANCES



TESTS OF AUTONOMIC FUNCTION

BLOOD PRESSURE CONTROL

1. Maintenance of blood pressure with alteration in posture – is normally dependent upon reflex baroreceptor function. A fall in BP occurs with efferent or afferent lesions – postural (orthostatic) hypotension.

2. Exposure to cold induces vasoconstriction and a rise in BP – cold pressor test. Stress will produce a similar pressor response, e.g. ask patient to do mental arithmetic.

Both central and peripheral lesions affect these tests.

3. Valsalva manoeuvre:

The patient exhales against a closed glottis, increases intrathoracic pressure and thus reduces venous return and systemic BP. The heart rate accelerates to maintain BP. On opening the glottis, venous return increases and an overshoot of BP with cardiac slowing occurs. An impaired response occurs with afferent or efferent autonomic lesions.

4. Noradrenaline infusion test:

A postganglionic sympathetic lesion results in ‘supersensitivity’ of denervated smooth muscle to adrenaline, with a marked rise in BP

following infusion.

HEART RATE

1. Massage of the carotid sinus should stimulate the baroreceptors, increase vagal parasympathetic discharge and slow the heart rate. Either efferent or afferent lesions abolish this response.

2. Atropine test:

Intravenous atropine 'blocks' vagal action and with intact sympathetic innervation results in an increase in heart rate.

SWEATING

A rise in body temperature causes increased sweating, detectable on the skin surface with starch-iodide paper. Any lesion from the central to the postganglionic sympathetic system impairs sweating.

SKIN TEMPERATURE

Skin temperature is a function of the sympathetic supply to blood vessels. With pre-or postganglionic lesions the skin becomes warm and red. With chronic postganglionic lesions the skin may become cold and blue (denervation hypersensitivity) compare the temperature of various regions.

PUPILLARY FUNCTION

Check the response to light and accommodation.

Pharmacological tests are important:

1. Atropine – blocks parasympathetic system – dilates pupil.

2. Cocaine – stimulates adrenergic receptors – dilates pupil.

In highly specialised units detailed neurophysiology (e.g. thermal

threshold measurements) and plasma concentrations of neurotransmitters and hormones at rest and in response to baroreceptor stimulation are employed to characterize the site and selectivity of the autonomic lesion.

AUTONOMIC NERVOUS SYSTEM – SPECIFIC DISEASES

Symptoms of autonomic dysfunction occur in many common conditions which affect both the parasympathetic and sympathetic pathways e.g. cerebrovascular disease.

The following are less common disorders which primarily may affect the autonomic nervous system –

IDIOPATHIC ORTHOSTATIC HYPOTENSION

Two types of this condition are recognised:

1. Due to degeneration of sympathetic postganglionic neurons.
2. Due to degeneration of sympathetic preganglionic neurons of the intermediolateral column T1–T12 – Multiple system atrophy (MSA).

In the latter disorder, features of extrapyramidal system involvement are also found.

Both disorders are characterised by: postural hypotension: anhidrosis (absent sweating): impotence: sphincter disturbance: pupillary abnormalities.

The disorders may be separated pharmacologically; the postganglionic disorders shows hypersensitivity (denervation hypersensitivity) to noradrenaline infusion.

Treatment

Drugs such as fludrocortisone increase blood volume and may prevent

postural hypotension.

DIABETIC AUTONOMIC NEUROPATHY

Symptoms of autonomic dysfunction are common in long-standing insulin-dependent diabetics:

Impotence/retrograde ejaculation.

Bladder dysfunction – decreased detrusor muscle action – resulting in increased residual volume.

Nocturnal diarrhoea.

GI dysfunction – vomiting from gastroparesis.

Orthostatic hypotension.

These problems arise from damage to both sympathetic and parasympathetic postganglionic neurons.

Treatment

Improve diabetic control and treat symptoms *e.g.* fludrocortisone for BP control.

GUILLAIN-BARRÉ SYNDROME – Guillain-Barré syndrome (see previous chapter). Autonomic involvement is common and may present major problems in patient management. The lesion may involve the afferent or efferent limbs of the cardiovascular reflexes (baroreceptor reflexes) resulting in postural hypotension, episodes of hypertension and cardiac dysrhythmias.

Occasionally the postinfectious neuropathy is purely autonomic.

HEREDITARY SENSORY & AUTONOMIC NEUROPATHY (HSAN)

This group of rare, generally recessively inherited disorders are characterised by insensitivity to pain, anhidrosis (absence of sweating), orthostatic hypotension and unexplained fevers from birth. Riley-Day syndrome is typical of these though its gene mutation is confined to Ashkenazi Jewish families.

PRIMARY AMYLOIDOSIS

Autonomic involvement with orthostatic hypotension, impotence, diarrhoea and bladder involvement may accompany sensorimotor neuropathy in the primary and hereditary forms. Amyloid infiltration affects autonomic ganglia.

ADIE'S SYNDROME

A tonic pupil ([page 144](#)) associated with areflexia and occasionally widespread autonomic dysfunction, e.g. segmental hypohidrosis (absent sweating) and diarrhoea.

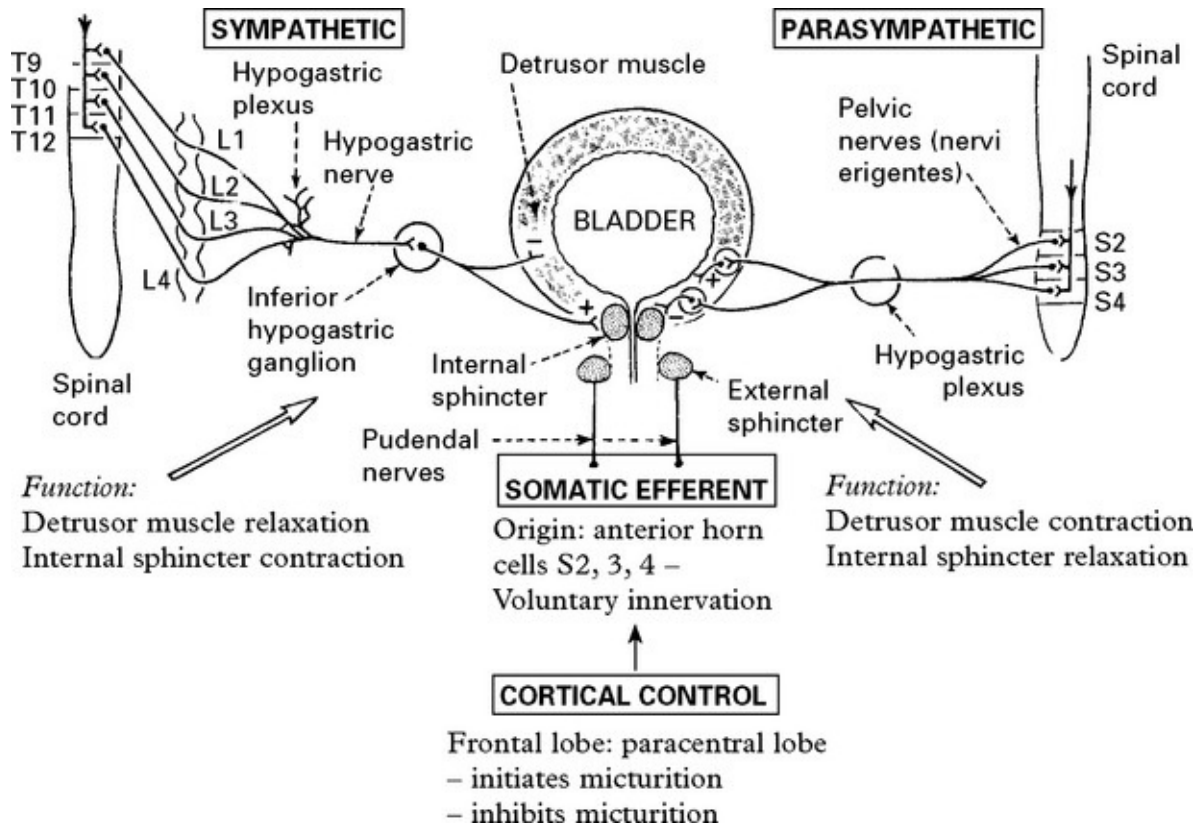
AUTONOMIC DYSFUNCTION IN QUADRIPLÉGIA (autonomic dysreflexia)

A high cervical lesion which completely severs the spinal cord, e.g. traumatic cervical fracture/dislocation will isolate all but the cranial parasympathetic outflow. As a result, disturbed autonomic function is inevitable but variable.

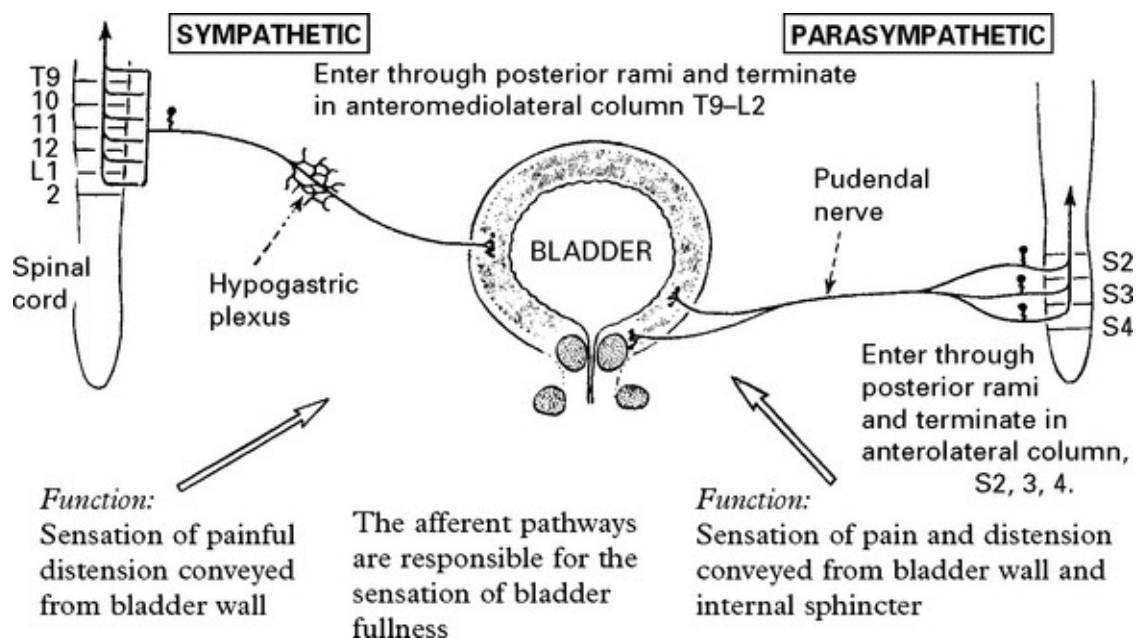
Autonomic reflexes are retained – Passive movement or tactile stimulation of limbs may result in blood pressure rise, bradycardia, sweating, reflex penile erection (priapism).

AUTONOMIC NERVOUS SYSTEM – BLADDER INNERVATION

Efferent innervation



Afferent innervation



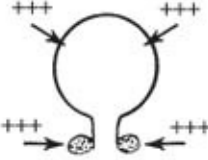
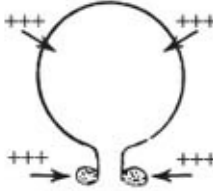
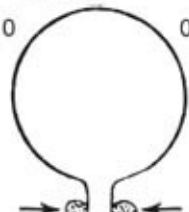
MICTURITION

PROCESS OF MICTURITION

1. Cortical centre – removal of conscious inhibition of micturition.
2. Voiding – wave-like detrusor muscle contractions with relaxation of internal and external sphincters.
3. Voiding completed – detrusor muscle relaxation contraction of internal sphincter contraction of external sphincter.
4. Voiding may be voluntarily interrupted before complete bladder emptying by forced voluntary contraction of the external sphincter.

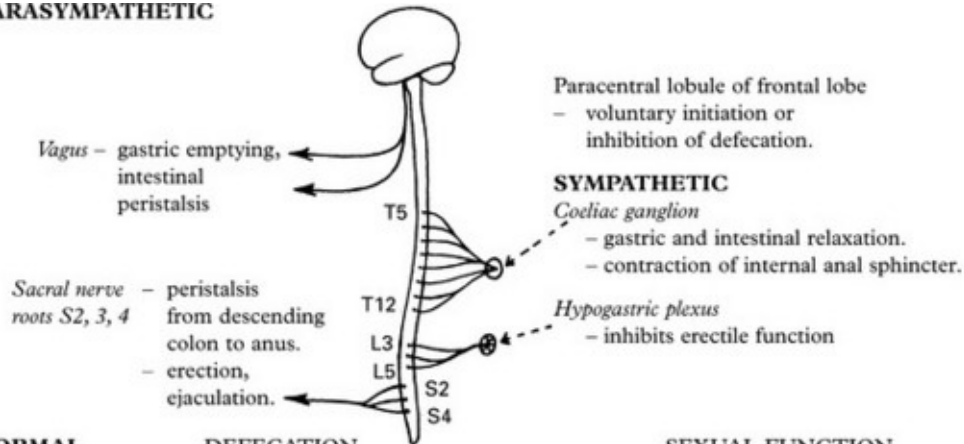
DISORDERS OF MICTURITION

Urinary and, less commonly, associated faecal incontinence occurs in women following traumatic childbirth with injury to the innervation of striated pelvic floor musculature.

	Partial incomplete upper motor neuron bladder	Complete/late partial upper motor neuron bladder	Lower motor neuron bladder
Pathophysiology	Loss of cortical inhibition leads to overactive bladder Loss of co-ordination of bladder contraction and sphincter relaxation	Partial lesion (as on left) leads to additional bladder dilatation and incomplete emptying	Cauda equina lesion causes parasympathetic lesion (S2–4) leading to loss of bladder tone and to dilatation Sphincter intact – sympathetic (T9–12)
			
Symptoms	Urgency Frequency	Urgency Frequency Retention Incomplete emptying Dribbling overflow incontinence	Retention or dribbling overflow incontinence
Associations	May be bilateral upper motor neuron signs in legs	May be bilateral upper motor neuron signs in legs	Loss of perianal sensation Reduced anal tone
Treatments	Anticholinergics, e.g. oxybutinin, tolterodine	Intermittent self catheterisation (ISC) or indwelling catheter Anticholinergics (if severe intravesical botulinum toxin)	ISC or indwelling catheter

BOWEL AND SEXUAL FUNCTION

PARASYMPATHETIC



NORMAL PROCESS

DEFECATION

1. Faeces arrive at rectosigmoid junction:
 - cortical awareness of urge to defecate
 - release of sympathetic tone.
2. Relaxation of pelvic floor muscles and internal anal sphincter. Lowering of anorectum.
3. Voluntary opening of external anal sphincter.
4. Parasympathetic peristalsis and Valsalva manoeuvre empty the rectum.

SEXUAL FUNCTION

Parasympathetic:

- penile/clitoral erection.
- Reflex – in response to tactile stimulation of erogenous zones.
- Psychogenic – sexual thoughts or visual erotic stimulation – orgasm, ejaculation

Sympathetic:

- mainly anti-erectile action.

COMPLETE OR PARTIAL CORD LESION



Bowel atony for up to 1 week. → Faecal retention with impaction and faecal fluid overflow (spurious diarrhoea). Impaired/absent external sphincter tone becomes spastic after days or weeks.
→ Regular bowel emptying reflexly in response to digital stimulation or suppositories achieves continence.

CONUS LESION



→ Mixed upper and lower neuron pattern

CAUDA EQUINA LESION



Flaccid external sphincter. Faecal retention with impaction and faecal fluid overflow.
Regular clearance of constipated stool by manual evacuation or Valsalva manoeuvre achieves continence.



Prolonged reflex erection (priapism) may occur for 2–3 days, then:

- Erections and ejaculation lost for weeks or months, then:
- Reflex erections (only tactile) appear but reflex ejaculation seldom returns.

Fertility is impaired or lost



Vaginal sensation and lubrication are lost. Fertility is retained.

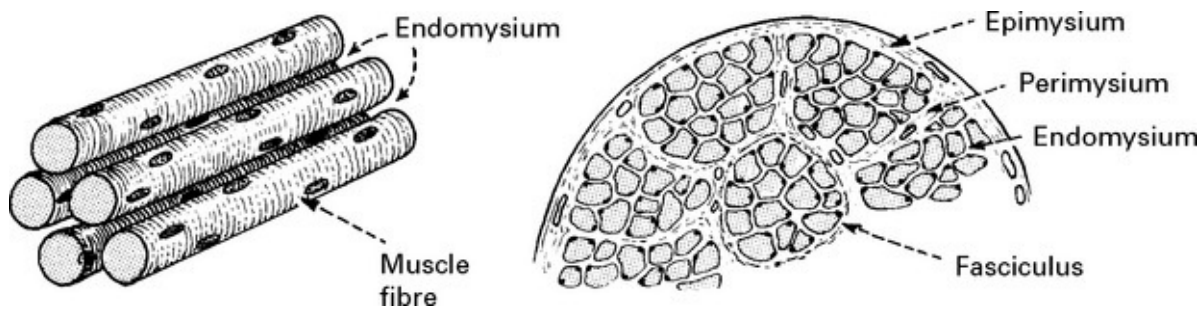
Loss of genital sensation.

Loss of reflex erections and ejaculation (psychogenic erection may be retained). Male infertile; female fertility retained. Male erection may be achieved using phosphodiesterase-5 inhibitors such as sildenafil, tadalafil or cardenafil or by injection of prostaglandins into the corpora (Caverject).

DISEASES OF SKELETAL (VOLUNTARY) MUSCLE

Normal skeletal muscle morphology

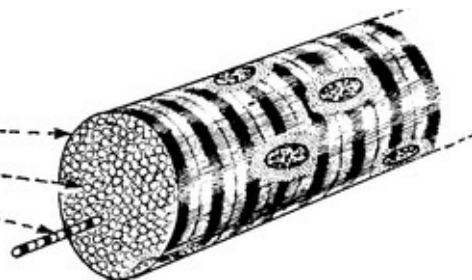
A skeletal muscle is composed of a large number of muscle fibres separated by connective tissue (endomysium) and arranged in bundles (fasciculi) in which the individual fibres are parallel to each other. Each fasciculus has a connective tissue sheath (perimysium) and the muscle itself is composed of a number of fasciculi bound together and surrounded by a connective tissue sheath (epimysium).



The three envelopes (sheaths) are made up of connective tissue richly endowed with blood vessels and fat cells (lipocytes).

The muscle fibre

This is a large multinucleated cell with an outer membrane – SARCOLEMMA and a cytoplasm – SARCOPLASM within which lie the MYOFIBRILS



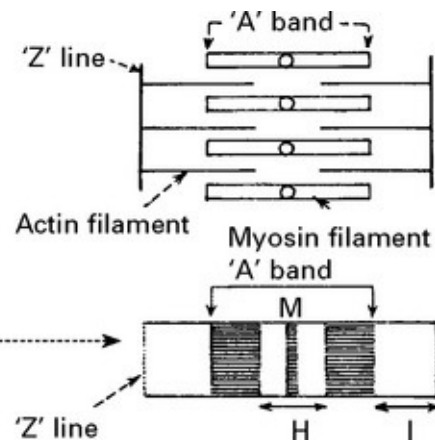
Each muscle fibre has its own endplate approximately half way along its length.

The cell also contains mitochondria, endoplasmic reticulum and microsomes – the usual cellular constituents.

Fats, glycogen, enzymes and myoglobin lie within the sarcoplasm and related structures.

The **MYOFIBRILS** are the contractile components of muscle.

Each myofibril is 1 μm in diameter and contains filaments of *myosin* and *actin* interdigitating with each other between each Z line. When muscle contracts or relaxes these filaments slide over each other producing shortening and lengthening of the muscle fibre. The striated appearance of skeletal muscle is a consequence of differing concentrations of actin and myosin. These resultant bands are designated as shown.



MUSCLE MORPHOLOGY AND FUNCTION

Fibre type

It is possible to distinguish two main types of muscle fibre on pathological and physiological study:

Type I: Slow twitch, fatigue resistant.

Type II: Fast twitch, fatigue dependent.

Characteristics:	Type I	Type II
	ATPase stain: Light	ATPase stain: Dark
	Oxidative metabolism	Glycolytic metabolism
	Abundant mitochondria	High glucogen content

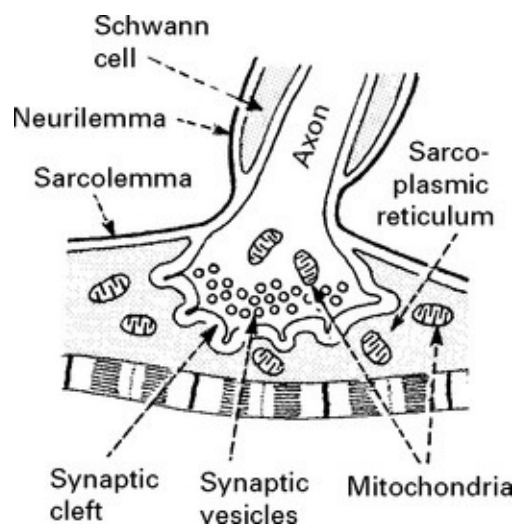
The muscle fibre type is influenced by its innervation that further determines its pattern of use; all muscle fibres innervated by a single motor neuron (the motor unit) have identical physiological and pathological parameters. The distribution of muscle fibre types differ in specific muscles within the body according to function – the muscles of the erector spinae are rich in oxidative, fatigue resistant fibres while the

converse is true in triceps.

Neuromuscular junction

Each muscle fibre receives a nerve branch from the motor cell body in the anterior horn of the spinal cord or cranial nerve motor nuclei.

When a nerve fibre reaches the muscle it loses its myelin sheath and its neurilemma then merges with the sarcolemma under which the axon spreads out to form the motor endplate. The axon fibre with its endings and muscle fibres it supplies is called the MOTOR UNIT. The number of muscle fibres in a motor unit varies: in the eye muscles it is small (5–10), whereas in the limb muscles the number is large (in the gastrocnemius about 1800). Each motor unit contains only one type of muscle fibre, *i.e.* type I or type II. The neuromuscular junction is the point at which neuromuscular transmission is effected. The motor endplate is separate from the sarcoplasm by the synaptic cleft.



Physiology

Muscle contraction results from the following:

- 1) A depolarisation wave arrives at the axon terminus and opens voltage

sensitive Ca^{2+} channels

2) Ca^{2+} entry triggers fusion of synaptic vesicles with the axon membrane which then release acetylcholine into the synaptic cleft

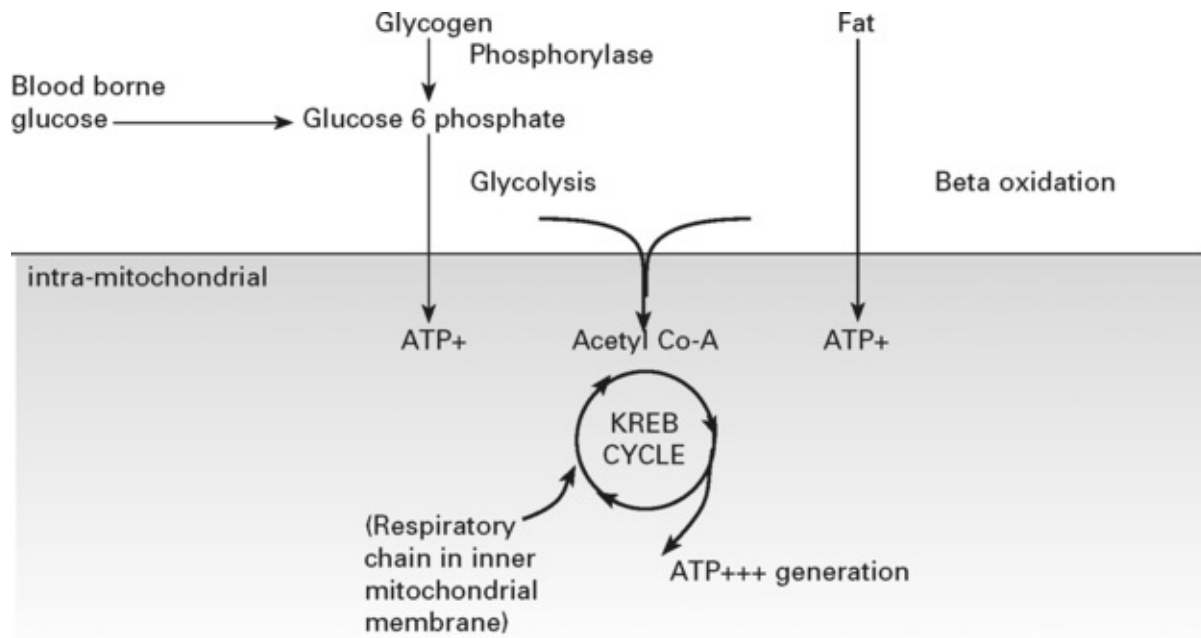
3) Acetylcholine attaches to endplate receptors with Na^{+} entry into muscle. Post synaptic depolarisation initiates an action potential that spreads along the sarcolemmal membrane.

4) Release of Ca^{2+} from the sarcoplasmic reticulum and the interaction of actin and myosin result in muscle contraction.

The enzyme cholinesterase, found in high concentration at motor endplates, destroys acetylcholine so that normally a single nerve impulse only gives rise to a single muscle contraction.

Biochemistry

Muscle contraction requires adenosine triphosphate (ATP). This may be generated either by carbohydrate breakdown (glycogenolysis and glycolysis,) or lipid breakdown (beta-oxidation). These non-oxygen requiring processes produce only a limited amount of ATP but also generate Acetyl-Co-A which, in the presence of oxygen, is further metabolised through the Krebs cycle within the mitochondria. This process yields even greater quantities of ATP.



The biochemical pathways yielding energy from the Krebs cycle reactions are dependent on proteins coded for by both the nuclear and mitochondrial genome. Mitochondrial DNA (mtDNA) is present in many copies per mitochondrion, with many mitochondria per cell. The usual state is that all an individual's mtDNA has the same sequence – homoplasmy – but in the mitochondrial disorders mutations are frequently present in only a proportion of the mtDNA – heteroplasmy. The distribution of these populations is not homogeneous across tissues and these features make the diagnosis of disorders associated with abnormalities of mtDNA difficult when the mutation may not be detected in blood but may be present in varying amounts in muscle or other affected tissues (see [page 481](#)).

MUSCLE DISEASE – HISTORY, EXAMINATION AND INVESTIGATIONS

History taking

This must include

- *A family history* paying particular attention to additional features that can be associated with muscle disease (e.g. deafness if a mitochondrial disorder is suspected or diabetes and cataracts with Myotonic Dystrophy).
- *Age at onset.* Parents describe delay in early motor milestones or a history of poor athletic abilities. Old photographs show long-standing facial weakness or ptosis.
- *A full drug and alcohol history.*
- *Terminology.* Patients should be given the opportunity to expand on terms such as ‘weakness’, ‘cramp’ or ‘fatigue’. These are often used to describe symptoms distinct from their strict medical definitions.
- *Pattern of weakness.* Proximal weakness will produce difficulty in descending stairs or rising from a low chair or drying hair. Distal weakness causes difficulty with latch keys, ascending stairs and scuffing toes.
- *Pain and cramp.* Their relationship to exercise should be noted. In disorders of glycolysis a cramp develops in the exercising muscle after a minute or so whereas in Carnitine-palmityl transferase deficiency cramp and rhabdomyolysis follows some hours later.
- *Fatigability.* This occurs in neuromuscular transmission disorders and mitochondrial disease.

Examination

This must assess

- *Walking* – here a waddling or foot drop gait is noted or other neurological problems such as Parkinsonism identified.

- *The distribution of weakness and wasting* will distinguish proximal, distal and generalised myopathies. Involvement of anatomically adjacent muscles is a feature of the muscular dystrophies. The face must be carefully examined for minor bilateral facial weakness; mild ptosis and limitation of extraocular movements. Muscle weakness should be graded using a standard scale (Medical Research Council scale – [page 19](#)).
- *The presence of pseudohypertrophy and contractures* (easily missed at hips, ankles and elbows) should be noted.

Investigations

- *Creatine kinase (CK)*: this sarcoplasmic enzyme is released from the damaged muscle membrane. High levels are associated with Muscular Dystrophies and Rhabdomyolysis but normal values do not exclude milder muscle disease (benign recessive dystrophies, mitochondrial and some metabolic disorders).
- *Neurophysiology*: may differentiate neurogenic from myopathic weakness and provide evidence of muscle membrane damage (e.g. inflammatory myopathies), but normal studies do not exclude muscle disease.
- *Muscle biopsy*: Routine staining of frozen material identifies some disorders but immunohistochemical analysis and appropriate mutation studies are needed for the diagnosis of others (e.g. Muscular Dystrophies). The choice between needle and open biopsy is difficult – the former is simpler but no less painful; the latter may be preferable to avoid sampling error.

INHERITED MUSCLE DISORDERS

The muscular dystrophies (MD) are genetically determined progressive

disorders of muscle characterised by cycles of muscle fibre necrosis, regeneration, eventual fibrosis and replacement with fatty tissue. Originally defined and described on patterns of weakness (e.g. Facioscapulo-humeral muscular dystrophy) they are now defined on the basis of known gene loci and protein product. This is not yet possible in all dystrophies but a continuing reclassification is taking place. Many disorders are associated with abnormalities in the dystrophin associated glycoprotein complex. Congenital myopathies are associated with morphological muscle abnormalities without necrosis and with a more benign prognosis. The metabolic myopathies present with pain, weakness or fatigue.

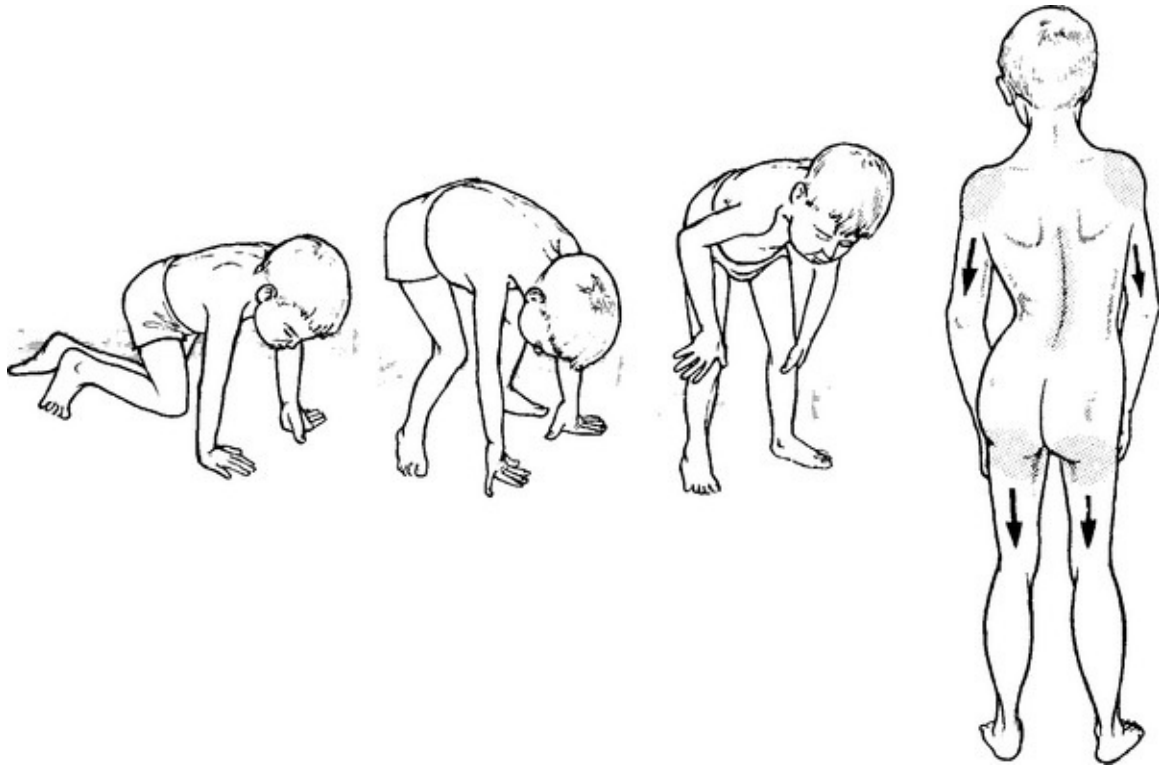
Xp2.1 DYSTROPHIES (DUCHENNE & BECKER MUSCULAR DYSTROPHY)

The gene for dystrophin is located at Xp2.1. Point mutations and deletions affecting the terminal domains are more often associated with the severe clinical phenotype of Duchenne, while deletions within the central rod domain are associated with the milder Becker Dystrophy.

DUCHENNE DYSTROPHY

Clinical features

Duchenne MD has an incidence of 1:3500 male births. It is characterised by delayed early motor development usually noted between ages 1 and 3 years, followed by scoliosis, contractures and eventual loss of ambulation at around 12 years of age. Pseudohypertrophy of muscle, in particular the calf, is a characteristic (occurring in 80%) but not a pathognomonic feature.



The child cannot climb stairs or rise from a low chair and when attempting to rise from the ground will 'climb up him' – Gower's sign (not diagnostic of the condition, but indicative of pelvic muscle weakness).

Investigation

Gene testing on serum may establish the diagnosis. The Dystrophin gene is large and many protocols only screen a part of it. A 'negative test' therefore does not rule it out and muscle biopsy with immunological testing is necessary. This demonstrates the absence of dystrophin. Female carriers can be detected by PCR.

Creatine kinase (CK) – substantially elevated (several thousand times). The enzyme is raised at birth and elevated in female carriers (in earlier times this formed the basis for counselling).

Electrocardiogram – 80% show conduction disorders, tall precordial R

waves and deep left precordial Q waves. Echocardiography should be repeated occasionally to detect developing cardiomyopathy.

Electromyography – shows severe myopathic change.

Life expectancy has risen from late teens to late 20s or early 30s with the use of surgery to correct scoliosis, active control of contractures and non-invasive ventilation. Corticosteroids slow progression and delay onset of disability, though the optimum regimen is still uncertain. Death occurs from respiratory insufficiency and infection or is 'sudden' and presumed to be related to cardiac disease. Long term care of affected individuals should be co-ordinated with anticipation rather than reaction to the evolution of disease.

BECKER DYSTROPHY

Abnormalities within the dystrophin gene may be associated with a spectrum of presentations from Duchenne to the milder condition described by Becker. Becker MD is rarer than Duchenne MD – incidence 1:35 000, presenting at a later age usually with limb-girdle involvement and pseudohypertrophy. These later milder presentations may also occur in some female carriers of the mutation. Cardiac involvement may be symptomatic in up to 10% of affected individuals and female carriers and is not related to the mutation or the severity of limb muscle disease.

The diagnosis is established in up to 80% of cases with serum DNA analysis. In the rest a combination of immunohistochemical demonstration of the relative absence of dystrophin, elevated CK, the clinical pattern and pedigree analysis make the diagnosis.

MUSCULAR DYSTROPHIES

DYSTROPHIES WITH PARTICULAR PATTERNS OF WEAKNESS

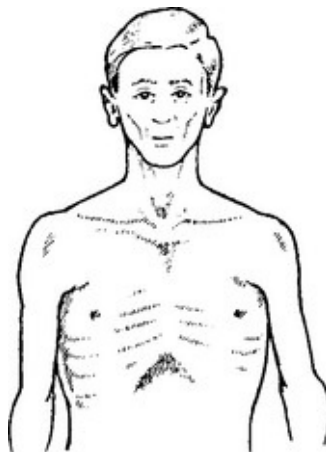
Facioscapulohumeral (FSH)

An autosomal dominant disorder, variable in severity and associated with a contraction of a series of 3.3 kB repeats at locus 4q35. Incidence 1–2:100 000. The mechanism by which this mutation causes disease is not known.

The clinical features include

- Facial weakness (which may be mild or asymmetrical)
- Periscapular weakness producing winging of the scapula and rising up of the scapulae on attempted abduction
 - Weakness of the humeral muscles
- A predominantly proximal lower limb pattern of weakness giving a dromedary or camel-backed gait

Pseudohypertrophy is not a feature.



Severity is variable, ranging from severe childhood forms to later onset disease that may be asymptomatic. CK levels may only be raised to 1.5–2 upper limit or normal. EMG and muscle biopsy will show myopathic abnormalities but have no specific features; although secondary inflammatory change on biopsy may lead to an erroneous diagnosis of

Polymyositis. Cardiac involvement is not a feature. High frequency sensorineural hearing loss and exudative retinal telangiectases complicate some early onset cases (Coat's syndrome). Prognosis is dependent on the degree of respiratory muscle involvement. Some may benefit from ventilatory support.

Scapuloperoneal

A dominant or recessive disorder that involves proximal upper and distal lower limb muscles. Onset is in adulthood with foot drop followed by weakness in scapular deltoid, triceps and biceps muscle groups. Differentiation from spinal muscular atrophy and inflammatory muscle disease is difficult.

Distal

Distal weakness due to primary dystrophies is rare with the exception of Myotonic Dystrophy. Both autosomal dominant and recessive patterns are described and may involve upper or lower limb muscles at onset. Some are associated with vacuolation of muscle fibres.

Emery-Dreifuss

Rare but important because of its cardiac complications. Both X-linked and dominant forms reported (the dominant form is now classified as LGMD type 2). Contractures of the spine produce an appearance of hyperextension. Contractures of elbows and ankles occur early. Weakness may be in a scapuloperoneal distribution. Life threatening cardiac condition defects are virtually universal and ventricular tachyarrhythmias occur in a proportion. Patients will require pacing and some have implanted defibrillators. Respiratory muscle weakness may occur.

Oculopharyngeal

This is another very rare pattern of weakness associated with a small GCG trinucleotide expansion in the PABP2 gene on chromosome 14. Inheritance is autosomal dominant. Occurs with a mean age of onset of 50 years with a combination of ptosis, ophthalmoparesis and dysphagia. Limb weakness may occur. Muscle biopsy shows rimmed vacuoles and filamentous intranuclear inclusions.

Limb girdle syndromes and limb girdle muscular dystrophy (LGMD)

Slowly progressing proximal weakness is a common presentation of both primary and secondary myopathies. A large number of proteins with differing functions produce a similar LGMD phenotype. Recessive forms are more common than dominant ones. The differential diagnosis of limb girdle distribution weakness is wide (see table).

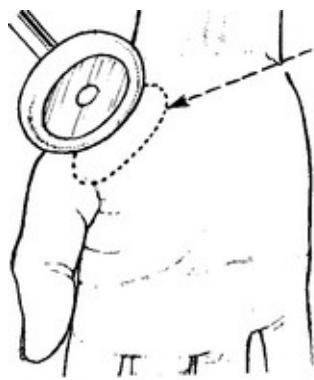
CATEGORIES	EXAMPLES	SUGGESTIVE FEATURES
Non-dystrophic genetic myopathies	Desmin myopathy, congenital structural myopathies (nemaline etc.)	Early onset, presence of contractures, often very thin muscles yet only mild weakness
Metabolic myopathies	Acid Maltase deficiency, McArdles disease, mitochondrial disorders	Pain, variability, exercise intolerance
Endocrine	Hypo-and hyperthyroidism, osteomalacic myopathy, Cushing's syndrome	Diffuse pattern of weakness, endocrine features may not be prominent
Toxic/metabolic	Steroid therapy, alcohol, statins	Should be apparent from history
Inflammatory	Polymyositis	See discussion below
Limb Girdle Muscular Dystrophy (LGMD)	At least 3 dominant and 9 recessive forms. Precise diagnosis requires specialised investigation	Symmetry, focal involvement of individual muscles, cardiac conduction defects, contractures from early stages

MYOTONIC DYSTROPHY (MyD)

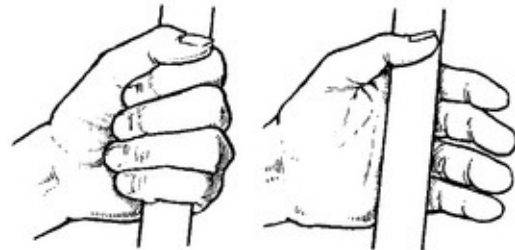
Myotonic Dystrophy is an autosomal dominant multisystem disorder caused by an unstable trinucleotide repeat expansion in a non-coding sequence at position 19q13.3. This expansion is thought to be pathogenic because of indirect effects on adjacent gene(s). It may present at any age with an incidence of 5 per 100 000.

Whilst neuromuscular features may not be prominent, the condition is usually characterised by the presence of MYOTONIA – failure of immediate muscle relaxation after contraction has ceased.

It can be demonstrated by:



1. Striking a muscle with the tendon hammer and watching the resultant 'dimple' persist for a while before filling up.
2. Asking the patient to grip an object then suddenly release it. The slow relaxation and opening of the hand grip will make the object appear 'stuck' to the fingers.



Clinical features

The facial appearance is typical:



Frontal baldness
Myopathic face with ptosis
Jaw hanging and wasting of muscles of mastication resulting in hollowing of temporal fossae and cheeks
Wasting of neck and shoulder girdle muscles also is evident

In the limbs – weakness and wasting are distal though the hands are spared until late.

- Cataracts
- Disorders of smooth muscle; gut motility disorders, constipation, poor bladder emptying.
- Cardiac disease; dilated cardiomyopathy and atrio-ventricular block requiring cardiac pacing
- Respiratory failure; due to intercostals and diaphragmatic weakness, impaired swallowing with risk of aspiration and central sleep apnoea (many patients benefit from nocturnal respiratory support).
- Diabetes; due to insulin resistance
- Testicular atrophy and subfertility

Diagnosis

In classic adult-onset cases, clinical diagnosis is straightforward with demonstration of progressive distal and bulbar dystrophy in the presence of myotonia, with frontal balding, and cataracts. Clinical diagnosis can be more difficult in mild cases, where cataracts may be the only manifestation. Direct analysis, by Southern blotting on peripheral leucocytes, of the size of the CTG repeat permits DNA diagnosis. Normal individuals have 5 to 37 CTG repeats, whereas patients have 50 to several thousand CTG repeats.

The importance of recognition of the disorder lies in the management of complications and genetic advice. The gene defect instability (number of repeats) between generations accounts for the wide clinical variability (phenotype) of MyD. Females are at risk of delivering a severely affected child who, due to respiratory failure, may not survive the neonatal period. Occasionally persons first present, either spontaneously or following anaesthesia, with unexpected respiratory failure or sudden

death.

When molecular tests are negative but clinical features suggestive two rare alternative disorders are considered –

1. DYSTROPHIA MYOTONICA 2 (MyD 2)

Genetically distinct form of myotonic dystrophy. Affected family members show remarkable clinical similarity to classic MyD (myotonia, proximal and distal limb weakness, frontal balding, cataracts, and cardiac arrhythmias). Disease locus maps to a 10 cM region of 3q.

2. PROXIMAL MYOTONIC MYOPATHY (PROMM)

This dominant disorder presents with myotonia in 30s–40s and mild proximal weakness in the fifth to seventh decades of life. Muscle biopsy demonstrates a non-specific mild myopathy with hypertrophy of type 2 fibres. Cataracts identical to those found in MyD occur in 15 to 30% of patients. Cardiac symptoms (arrhythmias) are infrequent. The gene causing PROMM is also located on 3q, suggesting that PROMM and MyD 2 are either allelic disorders or caused by closely linked genes.

DYSTROPHIES: GENERAL PRINCIPLES

It may not be possible to diagnose or exclude a specific type of dystrophy but practical issues apply to all:

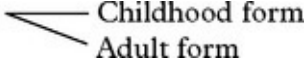
- *Genetics.* The implications for the family of differing modes of inheritance are clear. Help should be sought from a clinical geneticist to discuss these even if no molecular diagnosis has been reached but an inherited disorder suspected. Isolated cases of LGMD may represent a new dominant mutation and its phenotype is extremely variable. Both patients and their partners should be made aware of such issues.
- *Cardiac disease.* This is critically important in the Emery-Dreifuss

syndrome where life-threatening conduction defects are inevitable but also occur in Xp2.1 related dystrophies and Polymyositis. In the absence of a proven diagnosis, ECGs should be performed at 12 monthly intervals and echocardiography also if symptoms suggestive of cardiac failure develop.

– *Respiratory failure* related to diaphragmatic weakness, a prominent feature of Xp2.1, MD, LGMD, other forms of MD and inflammatory muscle disease. Late deterioration in some of the congenital myopathies may also lead to sleep disordered breathing. It is important to be aware of this, as non-invasive nocturnal ventilatory support is frequently beneficial to such patients.

INFLAMMATORY MYOPATHY

Primary inflammatory myopathies are clinically, pathologically and therapeutically distinct entities. Inflammatory changes are probably due to an immune mediated process rather than directly pathogenic. These are acquired as opposed to the inherited dystrophies and are classified as follows:

Polymyositis 

Dermatomyositis

Inclusion body myositis

Inflammatory myopathy associated with malignant disease

Inflammatory myopathy associated with collagen vascular disorders – e.g. lupus erythematosus, systemic sclerosis, rheumatoid arthritis.

Infective – *Viral* e.g. coxsackie, echo. *Parasitic*, e.g. cysticercosis,

trichinosis, taenia solium, toxoplasma, toxocara.

Sarcoid myopathy – some with this multisystem disease have granulomas in skeletal muscle.

POLYMYOSITIS/DERMATOMYOSITIS

There are two principal forms of inflammatory myopathy – polymyositis and dermatomyositis – which are separated clinically by the dermatological findings in the latter. All age groups are affected. Annual incidence is 8 per 100 000. These disorders are sporadic though familial cases are described.

An autoimmune basis for these disorders is supported by:

- response to immunosuppressive therapy.
- association with other known immunological disorders, *e.g.* collagen vascular disorders.
- elevated IgG in blood and presence of circulating autoantibodies, *e.g.* antinuclear antibody in some cases.
- an increased incidence of certain histocompatibility antigens (HLA antigens) – B₈, DR3.
- the reproduction of a similar disorder in laboratory animals by injection of muscle extract with Freund's adjuvant.

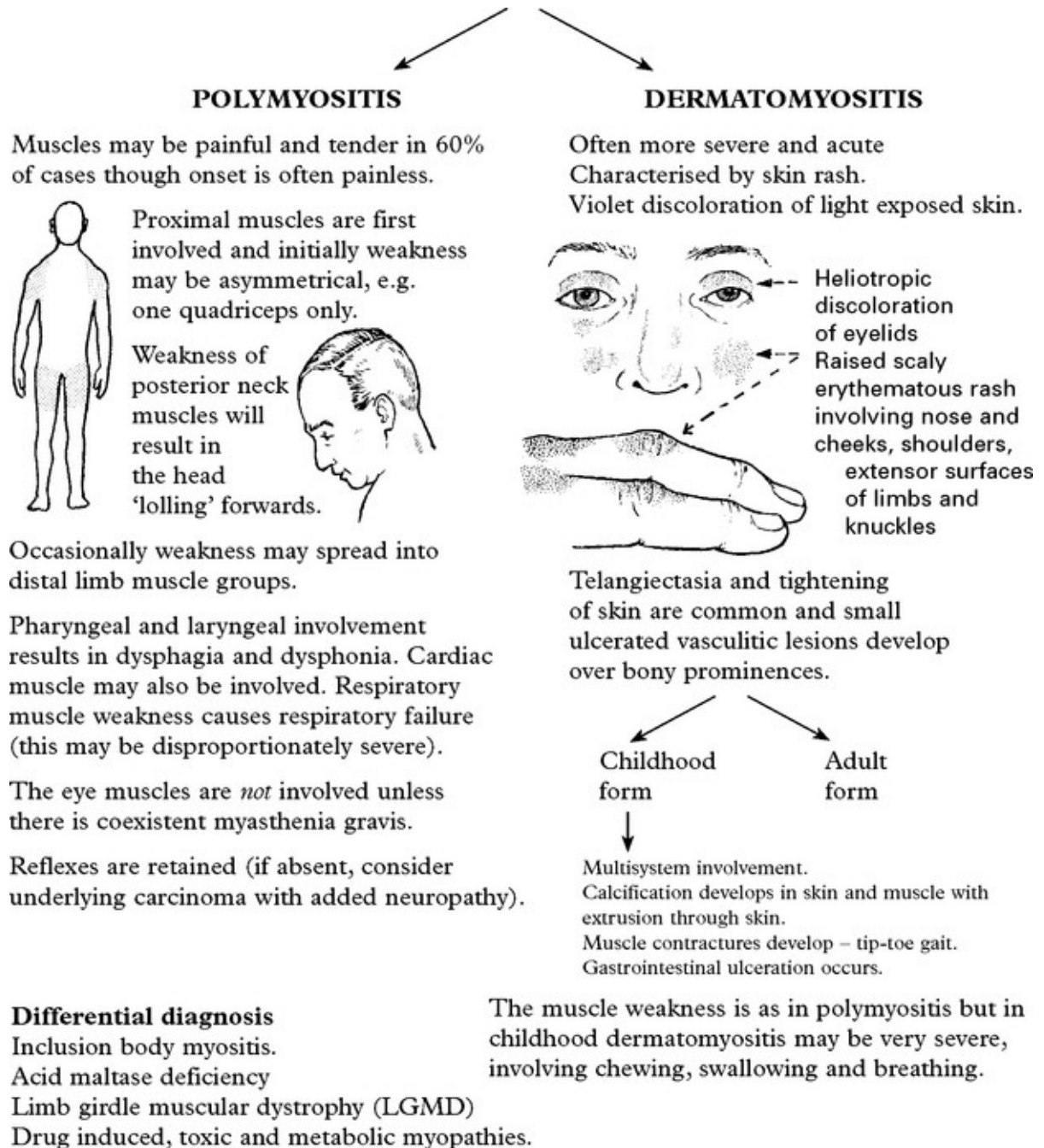
Humoral and cell mediated immune mechanisms seem responsible for these disorders but the trigger factor(s) remain unknown.

Clinical presentation

Onset is acute or subacute over a period of several weeks and may follow systemic infection.

Systemic symptoms prevail at onset, *e.g.* lassitude, and are then

followed by muscle weakness. Extensive oedema of skin and subcutaneous tissues is common (especially in the periorbital region).



INCLUSION BODY MYOSITIS (IBM)

Recognised now as the commonest inflammatory muscle disorder in the

middle aged and elderly (women are less commonly affected and more likely to be younger). Unlike the other inflammatory myopathies symmetrical weakness is painless and distal including foot extensors and finger flexors. May be associated neuropathy. Most patients have a protracted course unaffected by immunosuppressive therapies. Occasionally it is associated with connective tissue disorders such as Sjögren's syndrome.

Investigations

Diagnosis is supported by the following investigations:

<i>Muscle enzymes</i>	<i>Circulating antibodies</i>
Creatine kinase (CK) is elevated.	<i>e.g.</i> rheumatoid factor, antinuclear factor. Present in 40%.
Released from necrotic muscle, it is an indicator of disease activity and severity	
<i>Electromyography</i>	<i>Erythrocyte sedimentation rate (ESR)</i>
Shows a typical myopathic pattern.	Elevated in most patients.

Muscle biopsy shows necrosis of muscle fibres with inflammatory cells – lymphocytes, plasma cells, leucocytes.

The distinguishing features of the common inflammatory myopathies and responses to treatment are summarised as follows:

	Dermatomyositis	Polymyositis	Inclusion body myositis
Clinical features	Proximal weakness	Proximal weakness	Axial and asymmetric distal weakness
Neurophysiology	Myopathic	Myopathic	Mixed neurogenic/ myopathic
Pathology	Necrosis, secondary inflammatory infiltrate often perivascular, perifascicular atrophy of muscle fibres. B cell mediated.	Necrosis, inflammatory infiltrate, T cell mediated necrosis; invasion of healthy muscle fibres	Necrosis, inflammatory cell infiltrate. Vacuolation with inclusion bodies and paired helical filaments at EM
Therapy	Steroids, intravenous immunoglobulin	Steroids; usually with azathioprine	None proven to benefit
Associations	Paraneoplastic in adults	Weakly paraneoplastic	Sjögren's etc.

OUTCOME OF INFLAMMATORY MYOPATHIES

The natural history of these conditions is uncertain; mortality is low though perhaps only a minority recover completely. Inclusion body myositis is slowly and steadily progressive. Polymyositis and dermatomyositis respond in varying degrees to treatment and eventually become inactive. Safe monitoring of treatments and protection against side effects (e.g. steroid induced bone disease) is critical.

POLYMYOSITIS AND DERMATOMYOSITIS ASSOCIATED WITH MALIGNANT DISEASES

Approximately 10% of adults with inflammatory myopathy have underlying neoplasia usually carcinoma. In dermatomyositis, of those over 40 years of age as many as 60% harbour neoplasia. Neoplasia may present before or after the development of inflammatory myopathy.

POLYMYOSITIS AND DERMATOMYOSITIS ASSOCIATED WITH COLLAGEN VASCULAR DISEASES

Approximately 15% of adults with inflammatory myopathy have symptoms and signs of an associated collagen vascular disorder.

In 5–10% of persons with these disorders (systemic lupus erythematosus etc.), inflammatory myopathy develops at some stage in their illness.

In the ‘overlap’ syndromes (mixed collagen vascular diseases) muscle involvement is more common.

ENDOCRINE MYOPATHIES

Unlike inflammatory myopathy the weakness in these conditions is more chronic and is unassociated pathologically with inflammation. Serum CK is usually normal and EMG and biopsy (if performed) show non-specific myopathic changes. Correction of the underlying endocrine disturbance results in recovery. Usually the other features of endocrine dysfunction are more problematical and myopathy is of secondary importance.

Pituitary

Acromegaly

Proximal weakness with fatigue. Entrapment neuropathies, *e.g.* carpal tunnel syndrome may complicate the clinical picture of myopathy. Other features of growth hormone excess are evident.

Parathyroid

Hyperparathyroidism and osteomalacia.

Weakness of a proximal distribution with muscle tenderness occurs in 50% of patients with osteomalacia but is less common in primary hyperparathyroidism. The legs are mainly affected and a waddling gait results. Pathogenesis of hyperparathyroid myopathy is uncertain;

hypercalcaemia, Vitamin D deficiency or chronic phosphate deficiency is implicated, tetany results and the CK may be elevated but weakness is uncommon.

Adrenal

Hyperadrenalism and hypoadrenalism

These may both be associated with proximal myopathy. Muscle weakness, fatigue and cramping are frequent in *Addison's disease* with attacks of severe episodic hypokalaemic weakness (periodic paralysis) requiring glucocorticoid and mineral corticoid replacement. Hypokalaemic periodic paralysis is also frequent in *hyperaldosteronism*. *Cushing's syndrome* and exposure to excessive exogenous glucocorticoids commonly results in insidious proximal weakness. Reduction of steroid dosage results in improvement.

Thyroid

Hyperthyroidism

Weakness occurs in 20% of thyrotoxic patients. Shoulder girdle weakness is more marked than pelvic. Reflexes are brisk, fasciculation and atrophy may be present. *Distinction must be made from motor neuron disease*. There is always clinical evidence of thyrotoxicosis in these patients. Diagnosis is confirmed by thyroid function studies.

Hypothyroidism

Hypothyroidism impairs muscle glycolysis and mitochondrial oxidative capacity. Proximal weakness involves pelvic girdle more than shoulder. Painful cramps and muscle stiffness are common.

Muscle enlargement in limbs and tongue often occur (Hoffman's syndrome). There is always clinical evidence of hypothyroidism in these

patients. Diagnosis is confirmed by thyroid function tests and response to thyroid hormone therapy is excellent.

In chronic proximal weakness, careful clinical history taking, examination and appropriate investigation will separate the various endocrine causes.

CHANNELOPATHIES: PERIODIC PARALYSES AND MYOTONIA

Periodic paralyses and congenital myotonias are associated with defects in ion channels and are grouped together as channelopathies. The *primary periodic paralyses* are classified into two categories: *hypokalaemic* and *hyperkalaemic* (or potassium-sensitive). Hypokalaemic periodic paralysis shows the clearest relationship between episodic weakness and alterations in potassium. Hyperkalaemic periodic paralysis is more accurately a ‘potassium sensitive’ periodic paralysis as weakness can be provoked by potassium administration, whilst serum potassium may rise only marginally during spontaneous attacks. *Paramyotonia* can be associated with either hypo or hyperkalaemic periodic paralysis.

Importantly episodes of weakness, with alterations in serum potassium, are most commonly *secondary* to drugs (e.g. diuretics and corticosteroids) or disorders such as alcoholism, renal and endocrine disease. (See [page 478](#).)

Hypokalaemic periodic paralysis	Hyperkalaemic periodic paralysis	Paramyotonia congenita
Autosomal dominant. The gene has been mapped on chromosome 1, mutations	Autosomal dominant or recessive.	Autosomal dominant. Same gene as in hyperkalaemic form. Na ⁺ channel defect. Onset in infancy. Precipitated by: rest

resulting in upset of the dihydropyridine receptor, a voltage-gated calcium channel. Onset in second decade. Precipitated by: exercise, carbohydrate load. Commences in proximal lower limb muscles and rapidly becomes generalised. Onset usually in morning on waking. Attacks last from 4 to 24 hours. Bulbar muscles/respiration unaffected. K^+ falls as low as 1.5 meq/l. <i>Treatment:</i> Acute – oral KCl. Prophylactic – acetazolamide; low carbohydrate, high K^+ diet. With age, attacks become progressively less frequent.	Chromosome 17 location. Na^+ channel gene defect Onset in infancy/childhood. Precipitated by: rest after activity or by cold. Commences in lower limbs and evolves rapidly. Attacks are of short duration (less than 60 min). Myotonia is evident in some patients. K^+ rises only slightly. <i>Treatment:</i> Acute – intravenous calcium gluconate or sodium chloride. Prophylactic – Acetazolamide is effective	after exercise, fasting and cooling. Commences in proximal muscles. Repetitive muscle contractions produce increasing stiffness. EMG findings are specific with marked spontaneous activity in limb cooling. <i>Treatment.</i> Na^+ channel blockers, Tocainide or Mexiletine.
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	prophylaxis.	
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Normokalaemic periodic paralysis	Thyrotoxic periodic paralysis	Congenital myotonia
There are patients with episodic weakness in whom no alteration in serum potassium can be found. Many are sensitive to the administration of oral potassium salts. Treatment is the same as for the hyperkalaemic form but there is no response to acetazolamide. Muscle biopsy in these patients showed occasional vacuoles and prominent tubular aggregates.	Attacks of paralysis are associated with hypokalaemia and are clinically similar to those of the hypokalaemic form. Mainly occurs in Asians and rarely in non-Asians. The majority of patients experience their first attack in their 30s. There is a marked (20 to 1) male to female predominance.	Dominant form (Thomsen's disease) and recessive (Becker's) are both caused by mutation in the chloride channel gene. Myotonia can be triggered by cold, improving with exercise. May have muscle hypertrophy. Treatment with quinine, phenytoin or mexilitene reduces myotonia.

METABOLIC AND TOXIC MYOPATHIES

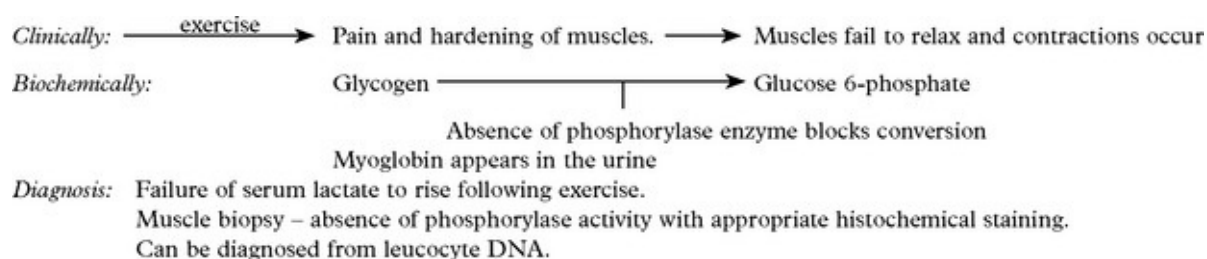
Metabolic myopathies

A group of genetically determined biochemical disorders of muscle characterised by myalgia, cramps, weakness and fatigue. These are divided into conditions with *reduced exercise tolerance* and those of *static weakness*. These complex disorders of muscle carbohydrate and lipid metabolism require specialist evaluation. Diagnosis requires detailed muscle staining to demonstrate enzyme loss critical to specific metabolic pathways.

The following disorders are representative but not comprehensive.

McArdle's disease – disorder of carbohydrate metabolism – block in glycolytic pathway (phosphorylase deficiency). Muscle phosphorylase

deficiency is a phenotypically heterogeneous autosomal recessive disorder. In some patients phosphorylase is absent whilst in others present but defective. The gene defect localises to chromosome 10.



<i>Diagnosis:</i>	Failure of serum lactate to rise following exercise. Muscle biopsy – absence of phosphorylase activity with appropriate histochemical staining. Can be diagnosed from leucocyte DNA.
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Treatment with oral fructose may help.

Carnitine palmitoyltransferase deficiency – Carnitine palmitoyltransferase (CPT) enzymes transfer fatty acids across the muscle mitochondrial membrane. CPT 1 attaches and CPT 2 detaches these fatty acids.

Infrequent episodes of myalgia and myoglobinuria following fasting or strenuous exercise. Onset is in adolescence, occasionally in adulthood. Though an autosomal recessive disorder, males are more commonly symptomatic. Neurological examination is normal. Serum CK, EMG and muscle biopsy (including histochemistry) are normal between attacks. Patients are advised to take a low fat/high protein and carbohydrate diet and to avoid prolonged exercise or fasting.

Acid maltase deficiency (ADM) – A lysosomal glycogen storage

disease with infantile, childhood, and adult types. The casual gene localises to chromosome 17 with different mutations accounting for ages of onset. Treatment is supportive, genetic counselling essential.

Infantile AMD (Pompe's disease) – progressive muscle weakness, cardiomegaly with congestive heart failure. Death occurs before 1 year. Glycogen accumulates in cardiac, skeletal muscle and in the CNS.

Childhood AMD – slower clinical course, with respiratory muscle weakness developing between 5 and 20 years. Histologically, muscle contains glycogen-filled vacuoles.

Adult AMD – proximal weakness in 3rd or 4th decade mimicking limb-girdle muscular dystrophy or polymyositis. Respiratory muscles are severely affected with risk of death from respiratory failure. Muscle biopsy again shows glycogen-filled vacuoles. Liver, heart and central nervous system are spared.

Carnitine deficiency – Carnitine transports long-chain fatty acids into the mitochondria. Deficiency results in systemic or myopathic features.

Systemic carnitine deficiency – childhood onset weakness with hypoglycaemic encephalopathy, precipitated by fasting and resembling Reye's syndrome ([page 508](#)). Serum and muscle carnitine levels are low. Biopsy shows an excessive number of lipid droplets in type 1 fibres. The liver, kidney, and heart contain excessive lipid. Cardiomyopathy is fatal.

Myopathic carnitine deficiency – muscle weakness, exertional myalgias and myoglobinuria. Onset of symptoms is usually in childhood but can be delayed until adulthood. Some cases are complicated by cardiomyopathy. Muscle biopsy shows excessive lipid droplets, especially in type 1 fibres. Muscle and serum carnitine levels are low.

Toxic myopathies

Necrotising myopathy is the pathological consequence of toxic muscle insult characterised by muscle weakness, pain, and tenderness. Investigations – elevated serum CK, myoglobinuria, myopathic motor units and fibrillation on EMG. Muscle biopsy – necrosis and regeneration. Numerous drugs have been incriminated. Statins are the most common culprits. Other examples include clofibrate in renal failure or hypoalbuminaemia. Epsilon-aminocaproic acid, procainamide, zidovudine (AZT) and phencyclidine. Focal muscle necrosis can be caused by intramuscular injections.

MITOCHONDRIAL DISORDERS

The DNA of the mitochondria (mtDNA) is circular, and while mitochondria themselves reproduce by binary fission, mtDNA replication is controlled by the eukaryotic genome. Mitochondrial disorders are transmitted through the maternal line and not by affected males (mtDNA transmits through the ovum not sperm). The relative proportion of normal to abnormal mtDNA determines the degree of expression (phenotype) for the mutation. As well as muscle involvement, characterised by ragged red muscle fibres on biopsy, many other clinical features are associated with mtDNA mutation syndromes and include:

- Seizures, respiratory insufficiency, weakness and vomiting and failure to thrive in the neonate.
- Developmental delay, ataxia, optic atrophy, progressive external ophthalmoplegia, sensorineural deafness, stroke-like episodes, dementia, exercise intolerance, and short stature.
- Renal failure, diabetes mellitus, cataract and cardiomyopathy.

Certain specific syndromes are recognised though overlap and diversity of phenotype is common.

CPEO (Chronic progressive external ophthalmoplegia)	MERRF (Myoclonic epilepsy with ragged red fibres)
Adolescence/adult onset of ptosis and ophthalmoplegia (without diplopia) often associated with mild proximal myopathy. When associated with heart block and retinopathy – Kearns Sayre syndrome (KSS). <i>Differentiate</i> from ocular myasthenia (page 152). <i>Investigations</i> Large deletion in mt DNA in skeletal muscle biopsy and serum. <i>Prognosis</i> good with very slow progression. Heart block may require pacing.	Adult onset of myoclonus, seizures and ataxia occasionally associated with respiratory failure. Disease expression is variable. <i>Differentiate</i> from other types of myoclonic epilepsy. <i>Investigations</i> Elevated lactate i.e. serum and CSF. ‘Ragged red’ fibres in muscle biopsy and point mutation in tRNA Lys gene of mt DNA in skeletal muscle biopsy and serum. <i>Prognosis</i> is poor in fully expressed disease – death from seizures or respiratory failure.
MELAS (Mitochondrial encephalopathy, lactic acidosis and stroke-like syndrome)	LHON (Leber’s hereditary optic neuropathy)
Adult onset of stroke-like episodes (posterior hemisphere) associated with focal seizures and vascular headache. ‘Strokes’ are not in vascular territories and are due to failure to utilise substrates rather than to a lack of them. <i>Differentiate</i> from other causes of ‘young’ stroke. <i>Investigations</i> CT/MRT shows posteriorly placed ischaemic changes, elevated lactate in serum and CSF. ‘Ragged red’ fibres on muscle biopsy and two-	Adult subacute onset of loss of central vision, initially unilateral but bilateral in all patients after 1 year. Visual acuity may be reduced to hand movements only due to marked optic atrophy. Male/female ratio: 3:1. <i>Differentiate</i> from optic neuritis, alcohol/tobacco amblyopia and anterior ischaemic optic neuropathy. <i>Investigations</i> Several mt DNA

point mutation in tRNA Leu gene of mt DNA. <i>Prognosis</i> is variable. Seizures and headache followed by 'strokes' and eventual dementia.	mutations have been detected in serum. <i>Prognosis</i> for visual recovery varies and depends on the specific mutation, as do other accompanying neurological features. In the majority visual loss is irreversible.
NARP (Neuropathy, ataxia and retinitis pigmentosa)	Leigh's syndrome
Adult onset of sensory/motor neuropathy, ataxia and chronic visual impairment. The rarest mitochondrial syndrome. In some, shares a similar molecular basis as Leigh's syndrome and can demonstrate maternal, autosomal recessive or X linked inheritance. <i>Differentiate</i> from other causes of ataxic neuropathy <i>e.g.</i> Freidreich's. <i>Investigations</i> Point mutations mt DNA (ATPase) detected in serum. <i>Prognosis</i> is uncertain, dementia occurs in time.	Infant or childhood onset of subacute necrotising encephalomyelopathy characterised by psychomotor retardation, ataxia, optic atrophy and ophthalmoplegia. <i>Differentiate</i> from other causes of progressive encephalopathy of childhood <i>e.g.</i> inborn errors of metabolism. <i>Investigations</i> CT/MRI brain stem changes, elevated lactate and pyruvate dehydrogenase complex in CSF and serum and various mutations at Xp 22.1 and mt DNA (ATPase). <i>Prognosis</i> is poor with early death.

There is no proven therapy for these conditions. Co-morbid conditions such as infection, cardiac involvement and diabetes mellitus should be treated conventionally. Pharmacologic therapies that may bypass biochemical defects are worth using *e.g.* L. Carnitine, Ubiquinone, riboflavin, thiamine and free radical scavengers (Vits C and E).

MYASTHENIA GRAVIS

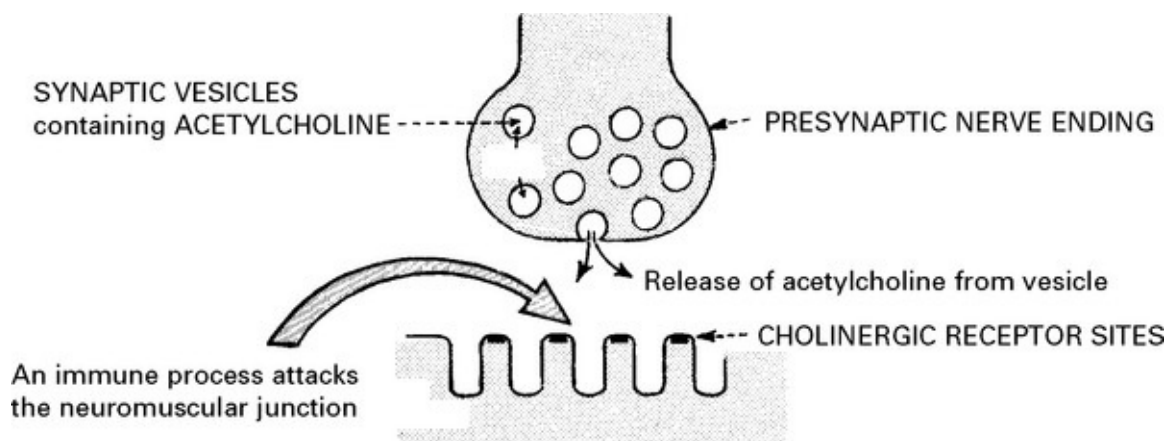
Myasthenia gravis is a disorder of neuromuscular transmission characterised by:

- Weakness and fatiguing of some or all muscle groups.
- Weakness worsening on sustained or repeated exertion, or towards the end of the day, relieved by rest.

This condition is a consequence of an autoimmune destruction of the NICOTINIC POSTSYNAPTIC RECEPTORS FOR ACETYLCHOLINE.

Myasthenia gravis is rare, with a prevalence of 5 per 100 000. The increased incidence of autoimmune disorders in patients and first degree relatives and the association of the disease with certain histocompatibility antigens (HLA) – B₇, B₈ and DR₂ – suggests an IMMUNOLOGICAL BASIS.

AETIOLOGY



Antibodies bind to the receptor sites resulting in their destruction (complement mediated). These antibodies are referred to as ACETYLCHOLINE RECEPTOR ANTIBODIES (AChR antibodies) and are demonstrated by radioimmunoassay in the serum of 90% of patients.

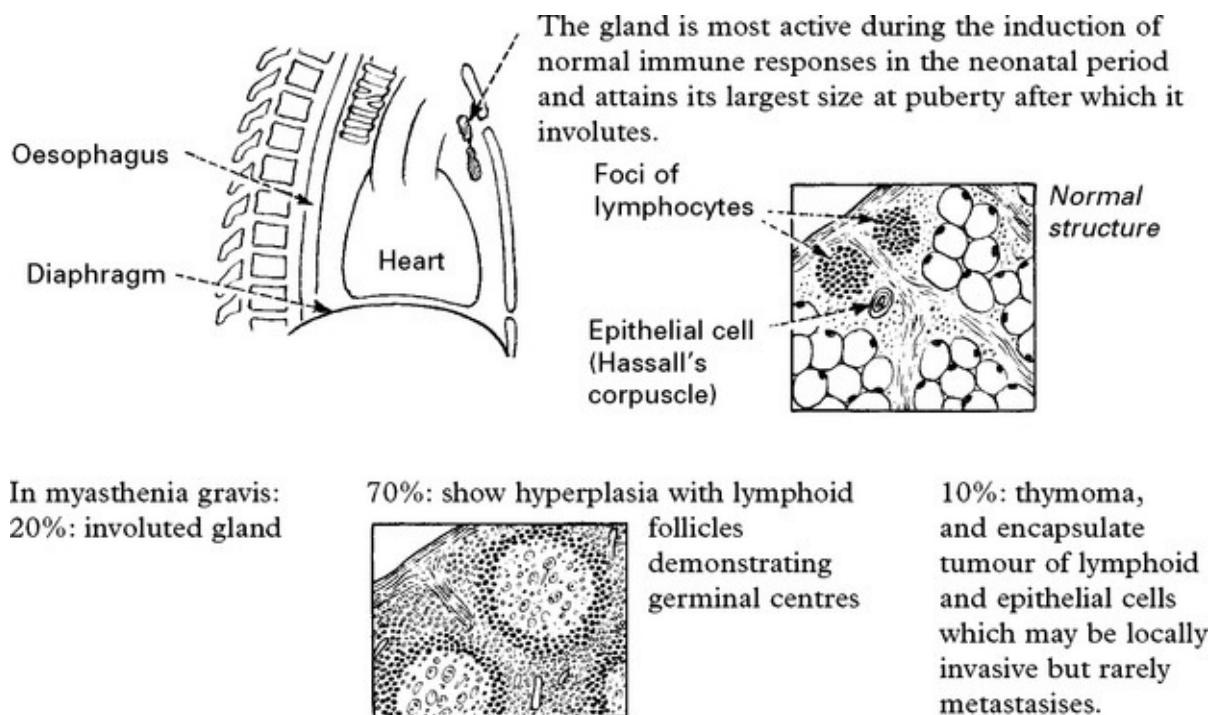
Human purified IgG (containing AChR antibodies) injected into mice induces myasthenia-like disease in these recipient animals.

In human myasthenia gravis a reduction of acetylcholine receptor sites has been demonstrated in the postsynaptic folds. Reduced receptor synthesis and increased receptor destruction, as well as the blocking of receptor response to acetylcholine, all seem responsible for the disorder.

The rôle of the thymus: Thymic abnormalities occur in 80% of patients. The main function of the thymus is to affect the production of T-cell lymphocytes, which participate in immune responses. Thymus dysfunction is noted in a large number of disorders which may be associated with myasthenia gravis, e.g. systemic lupus erythematosus.

MYASTHENIA GRAVIS – PATHOLOGY

Changes are found in the thymus gland and in muscle.



Muscle biopsy may show abnormalities:

- Lymphocytic infiltration associated with small necrotic foci of muscle fibre damage.
- Muscle fibre atrophy (type I and II or type III alone).
- Diffuse muscle necrosis with inflammatory infiltration (when associated with thymoma).

Motor point biopsy may show abnormal motor endplates. Supravital methylene blue staining reveals abnormally long and irregular terminal nerve branching.

Light and electron microscopy show destruction of ACh receptors with simplification of the secondary folds of the postsynaptic surface.

CLINICAL FEATURES

Up to 90% of patients present in early adult life (<40 years of age). Female:male ratio 2:1. The disorder may be selective, involving specific groups of muscles.

Several clinical subdivisions are recognised:

Class 1 – ocular muscles only – 20%	} 80%
Class 2 – Mild generalised weakness	
Class 3 – Moderate generalised and mild to moderate ocular-bulbar weakness	
Class 4 – Severe generalised and ocular-bulbar weakness	
Class 5 – Myasthenic crises	

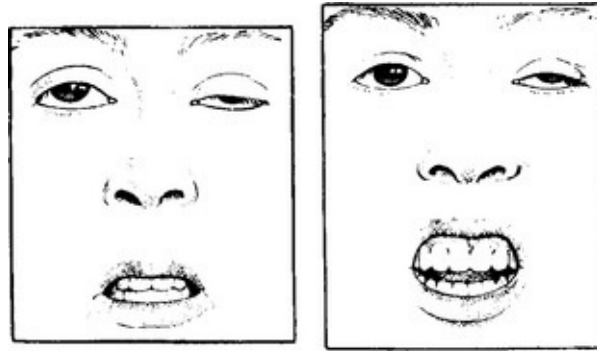
Approximately 40% of class I will eventually become widespread. The rest remain purely ocular throughout the illness.

Respiratory muscle involvement accompanies severe illness.

Cranial nerve signs and symptoms

- Ocular involvement produces ptosis and muscle paresis.

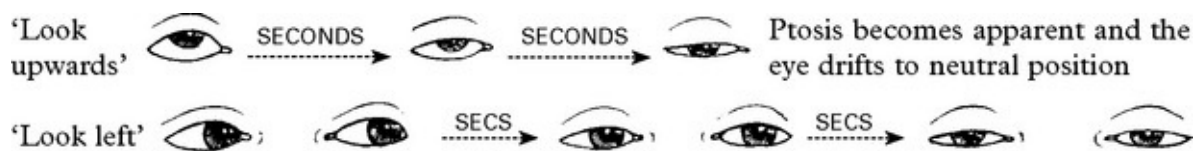
- Weakness of jaw muscles allows the mouth to hang open.
- Weakness of facial muscles results in expressionless appearance.
- On smiling, buccinator weakness produces a characteristic smile (myasthenic snarl).



Bulbar involvement may result in: – dysarthric dysphonic speech and dysphagia. – nasal regurgitation of fluids – nasal quality to speech.



The demonstration of *fatiguing* is important in reaching diagnosis and in monitoring the response to treatment:



Fatiguing of other bulbar muscles may be demonstrated by:

- blowing out cheeks against pressure.
- counting as far as possible in one breath, *etc.*

The tongue occasionally shows the characteristic triple grooved appearance with two lateral and one central furrow.

Limb and trunk signs and symptoms

Weakness of neck muscles may result in lolling of the head. Proximal limb muscles are preferentially affected. Fatigue may be demonstrated by movement against a constant resistance.

Limb reflexes are often hyperactive and fatigue on repeated testing.

Muscle wasting occurs in 15% of cases.

Stress, infection and pregnancy and drugs that alter neuromuscular transmission all exacerbate the weakness

Natural history: (Before treatment became available)	10% of patients entered a period of remission of long duration.
	20% experienced short periods of remission (1 to several months).
	30% progressed to death.
	The remainder showed varying degrees of disability accentuated by exercise.

MYASTHENIA GRAVIS – DIFFERENTIAL DIAGNOSIS

Distinguish from:

- The patient who complains of fatiguing easily – general weakness/debility (e.g. chronic fatigue syndrome) & functional weakness.
- The patient with progressive ophthalmoplegia, e.g. mitochondrial myopathy, oculopharyngeal dystrophy.

- The patient with multiple sclerosis – diplopia, dysarthria and fatigue with a relapsing and remitting course.
- The patient with the Lambert-Eaton myasthenic syndrome (see [page 549](#)).

INVESTIGATION

PHARMACOLOGICAL

Anticholinesterase drugs are used to confirm diagnosis.

Tensilon (edrophonium) – short action, 2–4 minutes, given i.v. 2–10 mg slowly, with atropine pretreatment to counter muscarinic side effects (nausea and bradycardia – resuscitation facilities need to be available). This is positive when clear improvement in weakness occurs on objective testing. A control injection of saline and blinded observer can be useful. The Tensilon test may be negative in ocular myasthenia and give a false positive in the Lambert-Eaton syndrome.

SEROLOGICAL

Acetylcholine receptor antibodies (anti-AchR) are detected in 90% of patients and are virtually specific to this disease. In ocular myasthenia, only 60% show antibodies. Magnitude of titres correlates with disease severity. Anti-Muscle specific Kinase (anti-MUSK) antibodies are found in a proportion of anti-AchR negative patients.

Other antibodies *e.g.* microsomal, colloid, rheumatoid factor, gastric parietal cell antibody – are occasionally found, reflecting the overlap with other autoimmune disorders.

Anti striated muscle antibodies are found in 30% of all patients and in 90% of those with thymoma.

ELECTROPHYSIOLOGICAL

Reduction of the amplitude of the compound muscle action potential evoked by repetitive

supramaximal nerve stimulation – ‘the decrementing response’.

Various rates of stimulation; even as low as 3/second may produce a decrementing response.



Single fibre electromyography – measure of ‘Jitter’ – the time interval variability of action potentials from two single muscle fibres of the same motor unit – is a more sensitive index of neuromuscular function and is increased (95% of mild cases are abnormal).

ADDITIONAL

Chest X ray will show a large mediastinal mass but will not exclude a small thymoma. CT of chest should be performed in all newly diagnosed cases.

MYASTHENIA GRAVIS – TREATMENT

In severely ill patients, the first priority is to protect respiration by intubation and, if necessary, ventilation.

Anticholinesterase drugs

This is the longest established form of treatment (1930s).

Anticholinesterase drugs inhibit **cholinesterase**, the enzyme responsible for the breakdown of acetylcholine, allowing enhanced receptor stimulation. As a result, more acetylcholine is available to effect neuromuscular transmission.

ANTICHOLINESTERASES	DURATION OF ACTION	METHOD OF ADMINISTRATION
Edrophonium	4 min	– Intravenous
Neostigmine	2 hours	– Intravenous, intramuscular, oral
Pyridostigmine	4 hours	– Oral

A muscarinic inhibitor, atropine, may be required to counter side effects (nausea, vomiting, diarrhoea, muscle fasciculations and increasing weakness). Anticholinesterases rarely give complete symptomatic relief and large doses can result in a *cholinergic crisis*

- worsening weakness
- increased sweating, saliva and bronchial secretions
- small pupils (miosis)
- eventual respiratory failure.

Atropine may mask early warning symptoms of this potential life-threatening state.

Steroids

Because this disorder is immune-mediated steroids are a logical choice in generalised and occasionally severe ocular disease. Prednisone 60 mg/day is initially used. Deterioration may briefly occur before improvement. Because of this low-dose regimes are often preferred, increasingly slowly from prednisone 25 mg alternate days. Once a response occurs, dosage is reduced.

Immunosuppressants other than steroids

These drugs (azathioprine and cyclosporine) are considered in patients who do not respond to steroids or who require an unacceptably high steroid maintenance dose.

Thymectomy

There are two indications for this:

1. When thymoma is present

2. When myasthenia is generalised and benefits of surgery outweigh risks.

Trans-sternal is preferred to supra-sternal approach giving better chance of total clearance. Within 5 yrs of surgery 70% of patients are in remission.

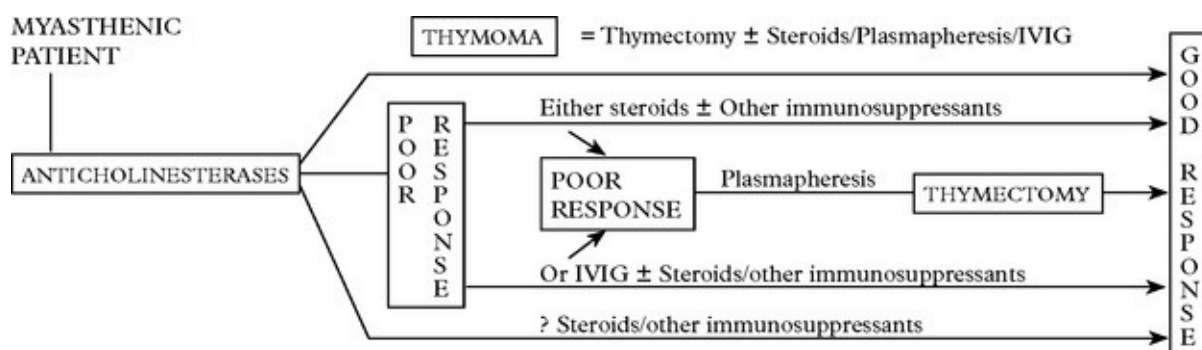
Plasmapheresis

Plasma filtration removes antibodies and other circulating factors and has short term benefit (4–6 weeks). A plasma volume of 1.5–2 litres is exchanged 3–5 times over a 6–8 day period. The technique is expensive and carries risks (hypotension, metabolic disturbance and thromboembolism). It is used to stabilise refractory cases and prior to thymectomy in severe disease.

Immunoglobulin (IVIG)

May be used in place of plasmapheresis at a dose of 400 mg per kg intravenously daily for 5 days. Mechanism may act by blocking ACh receptors. A positive response (75% of patients) lasts for 2–3 months. Treatment is expensive and long term effects and complications unknown.

SUMMARY OF TREATMENT



Anticholinesterases should not be required throughout the whole

illness. When immunological control of the disease is obtained, these drugs may be stopped.

EMERGENCY TREATMENT – MYASTHENIC/CHOLINERGIC CRISES

- Identify and treat precipitating cause, e.g. infection, drug interaction or overdose
- Sit patient at 45°, clear airway, give nasal O₂ and if overt respiratory failure – intubate and ventilate for as long as required.

<i>Myasthenic crisis</i>	<i>Cholinergic crisis</i>
– IV neostigmine 8–12 mg/24 hrs – sc. Atropine 0.5 mg tds – Prednisolone 100 mg daily – Consider plasmapheresis or IVIG – Change IV to oral anticholinesterases when able to swallow	– Withdraw all anticholinesterases – Monitor respiratory function (vital capacity) – Wean from ventilation when appropriate – Re-introduce oral anticholinesterases in low dose and gradually increase

NEONATAL form of myasthenia gravis: this develops in a number of infants of myasthenic mothers.

- Suggested by poor crying/sucking and floppy limbs.
- Presents within 48 hours of birth and may persist until the end of 3rd month.
- Caused by passive transplacental passage of IgG (acetylcholine receptor antibodies).

– Treatment with anticholinesterases is required until spontaneous recovery occurs. Remission occurs following exchange transfusion. This disorder may occur in infants even when their mother has been in remission for many years.

CONGENITAL MYASTHENIAS

These non-immunologic disorders are due to pre, post and mixed synaptic defects. They generally present in infancy though onset can be delayed into adult life. Characteristically fatiguing weakness affects limb (with associated skeletal abnormalities when early age of onset), ocular, bulbar and respiratory muscle groups. AChR antibodies are absent, electrophysiological assessment complex and treatments supportive though some respond to anticholinesterases or 3,4-diaminopyridine.

SECTION V

MULTIFOCAL NEUROLOGICAL DISEASE AND ITS MANAGEMENT

BACTERIAL INFECTIONS – MENINGITIS

ACUTE BACTERIAL MENINGITIS

In most cases the infection causing meningitis arises in the nasopharynx; intravascular invasion (bacteraemia) and penetration of the blood–brain barrier follow mucosal involvement with entry into the CSF. Bacteria may invade the subarachnoid space directly by spread from contiguous structures, *e.g.* sinuses and fractures. Specific characteristics of the capsule determine whether meninges are breached. Humoral defences against bacteria are absent in the CSF offering little resistance to infection.

Causative organisms

In neonates	– Gram –ve bacilli, <i>e.g.</i> <i>E. coli</i> , <i>Klebsiella</i> .
	<i>Haemophilus influenzae</i> .
In children	– <i>Haemophilus influenzae</i> . Pneumococcus (<i>Strep. pneumoniae</i>).
	Meningococcus. (<i>Neisseria meningitidis</i>).
In adults	– Pneumococcus. Meningococcus.

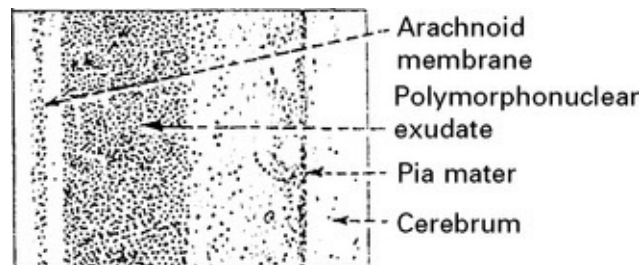
Other bacteria – *Listeria monocytogenes*, *Streptococcus pyogenes* and *Staphylococcus aureus* – are occasionally responsible.

Host factors (congenital or acquired immune deficiency, hyposplenism and alcoholism) predispose to infection, as do environmental factors (overcrowding and poverty).

Infections of mixed aetiology (two or more bacteria) may occur following head injury, mastoiditis or iatrogenically after lumbar puncture.

Pathology

The presence of the blood–brain barrier limits host defence mechanisms and enables multiplication of organisms.



A purulent exudate most evident in the basal cisterns extends throughout the subarachnoid space.

The underlying brain, although not invaded by bacteria, becomes congested, oedematous and ischaemic.

The integrity of the pia mater normally protects against brain abscess formation.

The cytokines, interleukin, tumour necrosis factor, and prostaglandin E2 are released as part of an acute inflammatory response. They increase vascular permeability, cause a loss of cerebrovascular autoregulation and exacerbate neuronal injury.

The inflammatory exudate may also affect vascular structures crossing the subarachnoid space producing an *arteritis* or *venous thrombophlebitis*

with resultant *infarction*. Similarly, cranial nerves may suffer direct damage.

Hydrocephalus can result from CSF obstruction.

Clinical

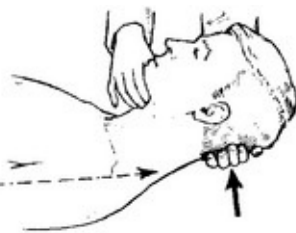
The classical clinical triad is fever, headache and neck stiffness.

<i>Prodromal features (variable)</i>	<i>Meningitic symptoms</i>
A respiratory infection otitis media or pneumonia associated with muscle pain	Severe frontal/occipital headache
	Stiff neck
	Photophobia.

Systemic signs: – High fever. Transient purpuric or petechial skin rash in meningococcal meningitis.

Meningitic signs:

Neck stiffness – gentle flexion of the neck is met with boardlike stiffness



Kernig's sign – stretching the lumbar roots produces pain

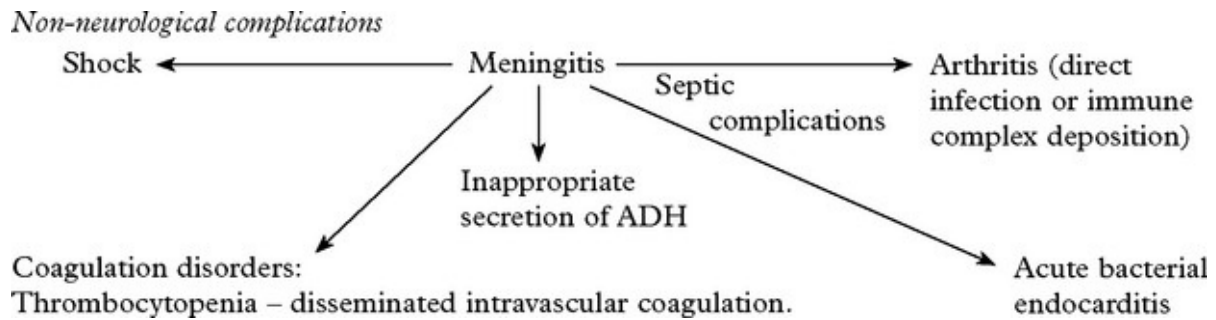


Associated neurological signs

- Impaired conscious level
- Focal or generalised seizures are frequent.
- Cranial nerve signs occur in 15% of patients.
- Sensorineural deafness (not due to concurrent otitis media but to direct cochlear involvement) – 20%

- Focal neurological signs – hemiparesis, dysphasia, hemianopia, papilloedema – occur in 10%.

Non-neurological complications



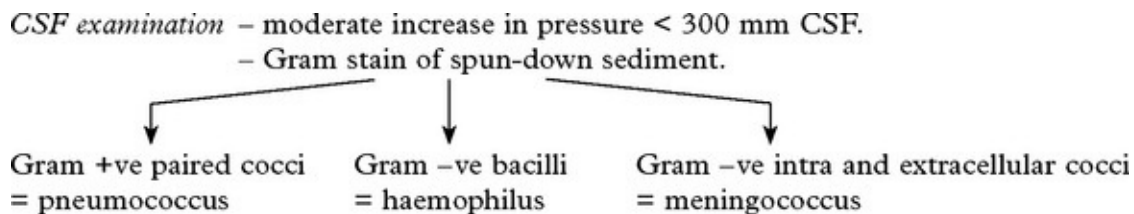
Features specific to causative bacteria

<i>Haemophilus meningitis</i>	<i>Meningococcal meningitis</i>	<i>Pneumococcal meningitis</i>
Generally occurs in small children. Preceding upper respiratory tract infection. Onset abrupt with a brief prodrome.	Often occurs in epidemics where the organism is carried in the nasopharynx. Septicaemia can occur with arthralgia; purpuric skin rash. When overwhelming, confluent haemorrhages appear in the skin due to disseminated intravascular coagulation.	Predominantly an adult disorder. Often associated with debilitation, e.g. alcoholism. May result from pneumonia, middle ear, sinus infection or follow splenectomy. Onset may be explosive, progressing to death within a few hours.
Outcome		
Generally good	Gradual onset – good prognosis. Sudden onset with septicaemia – poor	Mortality – 20%. Poor prognostic signs –

Less than 5% mortality.	outcome. Overall mortality – 10%.	coma, seizures, increased protein in CSF.
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Investigations

1. If patient has altered consciousness, focal signs, papilloedema, a recent seizure or is immunocompromised a CT brain should be done before LP. However, do not delay treatment – take blood cultures and commence antibiotics (see below) *prior* to scanning.
2. If above signs are absent or CT scan excludes a mass lesion → *confirm diagnosis with a lumbar puncture and identify the organism.*



- cell count is elevated, 100–10 000 cells/mm³(80–90% polymorphonuclear leucocytes).
- glucose is depressed.
- enzyme lactic dehydrogenase is elevated.
- culture CSF

Serological/immunological tests

The latex particle agglutination (LA) test, for the detection of bacteria antigen in CSF, has a sensitivity 80% for haemophilus and pneumococcus and 50% for meningococcus (100% specificity). The polymerase chain reaction (PCR), for the detection of bacteria nucleic

acid in CSF, is available for all the suspected organisms. The specificity and sensitivity of PCR is unknown and the delay (3 to 5 days) to process results, makes the test less helpful than the combination of Gram's stain, culture, and the LA test.

Blood cultures

- Organism isolated in 80% of cases of Haemophilus meningitis.
- Pneumococcus and meningococcus in less than 50% of patients.

3. Check serum electrolytes.

- important in view of the frequency of inappropriate antidiuretic hormone secretion.

4. Detect the source of infection.

– Chest X-ray – pneumonia – Sinus X-ray – sinusitis	– Skull X-ray – fracture – Petrous views – mastoiditis
---	--

Treatment

Once meningitis is suspected, treatment must commence immediately, often before identification of the causative organism. Antibiotics must penetrate CSF, be in appropriate bactericidal dosage and be sensitive to causal organism once identified.

Initial therapy (before organism identification)

Neonates (above 1 month)	– ampicillin, + aminoglycoside and cephalosporin
Children (under 5 years)	– vancomycin + 3rd generation cephalosporin
Adults	– vancomycin + 3rd generation cephalosporin

Immunocompromised patient

– vancomycin + ampicillin + cephalosporin

Steroids

A four-day regimen of dexamethasone, starting before or with the first dose of antibiotics, is now recommended in children with haemophilus and adults with bacterial meningitis likely to be pneumococcal. Meta-analysis found a risk reduction of neurological sequelae and mortality of about 30% in pneumococcal meningitis, with no clear difference with other organisms.

Therapy after organism identification

ORGANISM	ANTIBIOTIC	ALTERNATIVE THERAPY
<i>Haemophilus</i>	Ampicillin or 3rd generation cephalosporin according to sensitivities	Chloramphenicol Fluoroquinolone Cefepime
Pneumococcus	Benzylpenicillin or 3rd generation cephalosporin according to sensitivities	Chloramphenicol Fluoroquinolone Meropenem
Meningococcus	Benzylpenicillin or 3rd generation cephalosporin according to sensitivities	Chloramphenicol Fluoroquinolone Meropenem
<i>E. coli</i>	3rd generation cephalosporin	Aztreonam, fluoroquinolone, meropenem, ampicillin
<i>Listeria</i>	Ampicillin ± gentamicin	Chloramphenicol Cotrimoxazole

Duration

Meningococcus } continue for at least 1 week after afebrile.
Haemophilus }

Pneumococcus – continue for 10–14 days after afebrile

Monitoring

In a deteriorating patient, CT scan will exclude the development of hydrocephalus, abscess or subdural empyema. In suspected sinus thrombosis MR venography may be required.

Remove any source of infection, *e.g.* mastoidectomy or sinus clearance.

In meningococcal meningitis the risk to household contacts is increased (500–800x) and chemoprophylaxis should be offered – rifampicin 600 mg b.d. for 48 hours. Vaccines are also available.

Meningitis/CSF shunts

Meningitis infection may follow CSF drainage operations for hydrocephalus. This may occur in the immediate postoperative period or be delayed for weeks or months. Clinical features of raised intracranial pressure may coexist due to shunt blockage. Bacteraemia is inevitable and blood cultures identify the responsible organism – usually *Staphylococcus albus*. The infection seldom resolves with antibiotic therapy alone and shunt removal is usually required.

BACTERIAL INFECTIONS – CNS TUBERCULOSIS

Tuberculosis is an infection caused in man by one of two mycobacteria – *Mycobacterium tuberculosis* and *Mycobacterium bovis*. The disease involves the nervous system in 10% of patients.

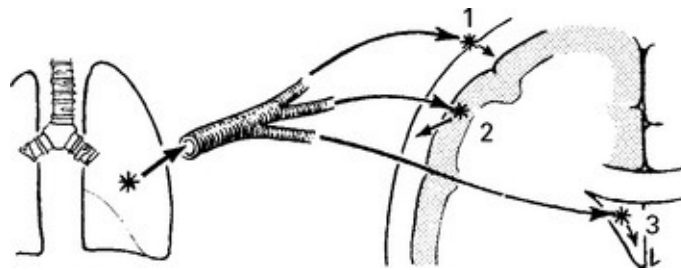
MENINGITIS

This is the commonest manifestation of tuberculous infection of the nervous system. *In children*, it usually results from bacteraemia following the initial phase of primary pulmonary tuberculosis.

In adults, it may occur many years after the primary infection.

Following bacteraemia, metastatic foci of infection lodge in:

1. Meninges
2. Cerebral or spinal tissue
3. Choroid plexus



Rupture of these encapsulated foci results in spread of infection into the subarachnoid space. In adults, reactivity of metastatic foci may occur spontaneously or result from impaired immunity (e.g. recent measles, alcohol abuse, administration of steroids).

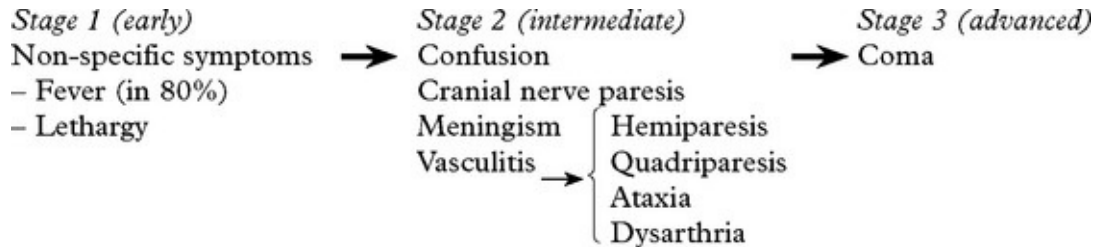
The clinical features of tuberculous meningitis (TBM) result from:

– Infection.	
– Exudation	– which may obstruct the basal cisterns and result in hydrocephalus.
– Vasculitis	– secondary to inflammation around vessels, resulting in infarction of brain and spinal cord.

The basal meninges are generally most severely affected.

Clinical features

The majority of patients are adults; childhood TBM is now rare. Nonspecific prodromal symptoms develop over 2–8 weeks.



Staging is useful for predicting outcome.

Seizures may occur at the onset. Involuntary movements (chorea, myoclonus) occur in 10%.

Atypically the illness may develop slowly over months presenting with dementia or rapidly like pyogenic (bacterial) meningitis. Occasionally cerebral features prevail rather than signs of meningitis.

Untreated, the illness may progress from phase 1 to death over a 3-week period.

Arachnoiditis inflammatory exudate may result in hydrocephalus/dementia/blindness.

TUBERCULOUS MENINGITIS

Investigations

- *General*: Anaemia, leucocytosis. Hyponatraemia (if inappropriate ADH secretion occurs).
- *Cerebrospinal fluid*

- Cell count, differential count, cytology (50–4000/mm³ – predominantly lymphocytes)
 - Glucose, with a simultaneous blood sugar (< 50% blood glucose)
 - Protein (> 1g/l)
 - Acid-fast stain, Gram stain, appropriate bacteriologic culture and sensitivity, India ink (all causes of lymphocytic meningitis)
 - Cryptococcal antigen, herpes antigen (other causes of lymphocytic meningitis)
 - Culture for *M. tuberculosis* (50–80% positive)
 - Polymerase chain reaction (PCR) to detect Mycobacterium DNA – specificity and sensitivity 100% and 70%.
- *Tuberculin skin test*: Positive in 50% of cases. (Negative if recent steroids or acquired primary infection.)
 - *Chest x-ray*: – hilar lymphadenopathy/infiltrate/cavitations/effusion/scar.
 - *CT scan and MRI* – hydrocephalus, basal meningeal thickening, infarcts, oedema, tuberculomas and obliteration of the subarachnoid space.



*CT scan with
i.v. contrast
Thickened,
enhancing
basal
meninges*

Diagnosis

Diagnosis is based on the clinical presentation with characteristic CSF findings.

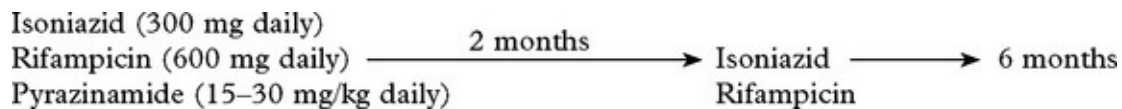
DIFFERENTIAL DIAGNOSIS – Viral meningoencephalitis – Subacute/chronic meningitis (see [pages 517–8](#)).

Treatment

If suspect, commence antituberculous treatment.

Recommended treatment programme:

Normal regime:



Drug resistance suspected due to previous antituberculous therapy, *e.g.*

- Developing countries
- History of previous infection.
 - Add a fourth drug – streptomycin (1 g daily) or ethambutal (25 mg/kg daily).

Isoniazid and pyrazinamide penetrate meninges well; other drugs penetrate less well especially when the inflammation begins to settle.

Side effects:

- Isoniazid may produce peripheral neuropathy – protect with pyridoxine 50 mg daily.
- Ethambutol may produce optic atrophy – check colour vision.
- Streptomycin may cause 8th cranial nerve damage (vertigo and deafness).
- Nausea, vomiting, abnormal liver function and skin rashes may occur with all antituberculous drugs.

Evidence concerning the duration of antituberculous treatment is conflicting. Conventionally therapy is given for 6–9 months, although some still recommend it for 24 months.

Intrathecal therapy: Since CSF penetration, especially with streptomycin, is poor, some recommend intrathecal treatment. Streptomycin 50 mg may be given daily or more frequently in seriously ill patients.

When obstructive hydrocephalus occurs, combined intraventricular (through the shunt reservoir or drainage catheter) and lumbar intrathecal treatment injections may be administered.

Steroid therapy: A recent Cochrane review reported that adjunctive steroids reduce neurological sequelae, hearing loss and mortality in patients with TBM without HIV. Insufficient data are available to recommend the use of steroids in HIV positive TBM.

Hydrocephalus

Progressive dilatation of the ventricles impairing conscious level requires CSF drainage – either temporarily with a ventricular catheter (permitting intraventricular drug administration) or permanently with a ventriculoperitoneal/atrial shunt. Surgery may also be considered for coexistent tuberculomas and tuberculous abscesses though these often resolve with drug therapy.

The course of treated tuberculous meningitis

Outcome is influenced by the patient's age, general state of health, timing of initiation of treatment and the development of arachnoiditis and vascular complications.

Treatment in early stages is associated with a 10% mortality, in later stages with a 50% mortality. Of those who survive, neurological

sequelae persist in 30% – hemiplegia, hypothalamic/pituitary dysfunction, blindness, deafness, dementia and epilepsy.

With treatment, CSF sugar quickly returns to normal; the cellular reaction gradually diminishes over 3–4 months; the protein level may take a similar time to return to normal.

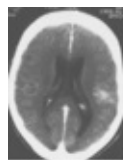
Tuberculous meningitis in AIDS

Atypical mycobacteria such as *M. avium* and *fortuitum* should be considered. Response to treatment is generally good. TBM tends to occur in the earlier phases of immunodeficiency with CD4 T cell count, at < 400 per mm³.

OTHER FORMS OF CNS TUBERCULOUS INFECTION

TUBERCULOMAS OF THE BRAIN

Tuberculomata may occur in cerebral hemispheres, cerebellum or brain stem with or without tuberculous meningitis, and may produce a space-occupying effect. They consist of caseating granulomas made up of epithelioid cells and macrophages containing mycobacteria. Lesions may be single or multiple. CT and MRI demonstrate lesions but appearances are not pathognomonic. Most resolve over a few weeks with antituberculous therapy.



CT scan with i.v. contrast
Enhancing tuberculoma at
grey/white matter junction

POTT'S DISEASE

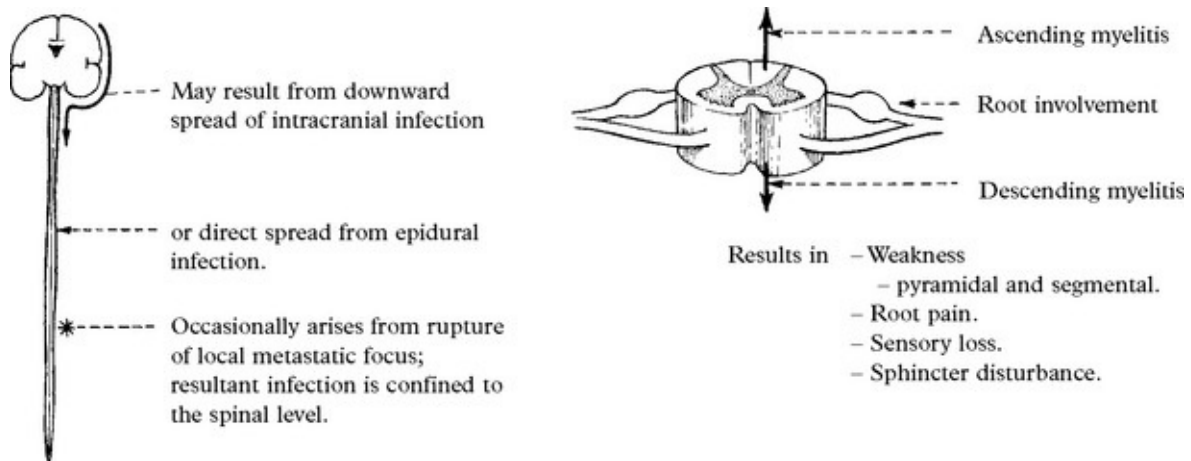
Chronic epidural infection follows tuberculous osteomyelitis of the vertebral bodies. This arises in the lower thoracic region, can extend

over several segments and may spread through the intervertebral foramen into pleura, peritoneum or psoas muscle (psoas abscess).

TUBERCULOUS MENINGOMYELITIS

Infection of the leptomeninges results in an exudate that encases the spinal cord and nerve roots. This produces back pain, paraesthesia, lower limb weakness and loss of bowel and bladder control. Imaging may be normal while CSF shows high protein, lymphocytes and rarely acid fast bacilli. This disorder is now more frequent in AIDS patients. Differential diagnosis includes cytomegalovirus, cryptococcus, syphilis and lymphoma. Laminectomy and meningeal biopsy may be required to establish diagnosis. When suspected, empirical therapy with antituberculous drugs is appropriate.

Clinical features:



SPIROCHAETAL INFECTIONS OF THE NERVOUS SYSTEM

SYPHILIS

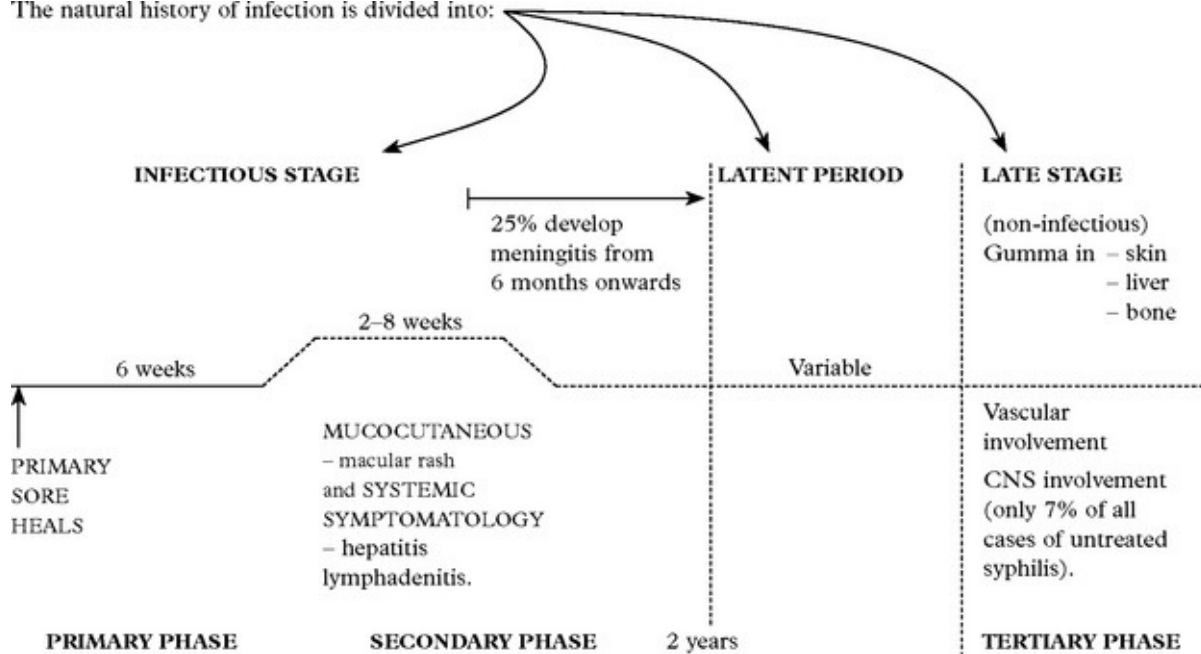
This infectious disease is caused by the spirochaete *Treponema pallidum*. Entry is by:

- inoculation through skin or mucous membrane (sexually transmitted)
- acquired syphilis.
- transmission in utero – congenital syphilis.

In the last 30 years, there has been a steady decline in incidence regardless of race and ethnicity. Despite this, it still remains an important health problem in certain geographic areas.

Up to 10% of patients with HIV will test positive for syphilis. All patients with neurosyphilis should be tested for this.

The natural history of infection is divided into:



The *chancre* or *primary sore* on skin or mucous membrane represents the local tissue response to inoculation and is the first clinical event in acquired syphilis.

The organism, although present in all lesions, is more easily demonstrated in the primary and secondary phases.

In congenital syphilis fetal involvement can occur even though many years may elapse between the mother's primary infection and

conception.

Widespread recognition and efficient treatment of the primary infection have greatly reduced the late or tertiary consequences.

Not all patients untreated in the secondary phase progress to the tertiary phase.

In HIV patients the neurological complications occur earlier and advance more quickly.

Investigations

Spirochaetes can be demonstrated microscopically by dark field examination in primary and secondary phase lesions.

Serological diagnosis depends on detection of antibodies.

1. Nonspecific (Reagin) antibodies (IgG and IgM).

Reagin tests involve complement fixation.

The Venereal Disease Research Laboratory (VDRL) test is the commonest and when strongly positive indicates active disease (may be negative in HIV).

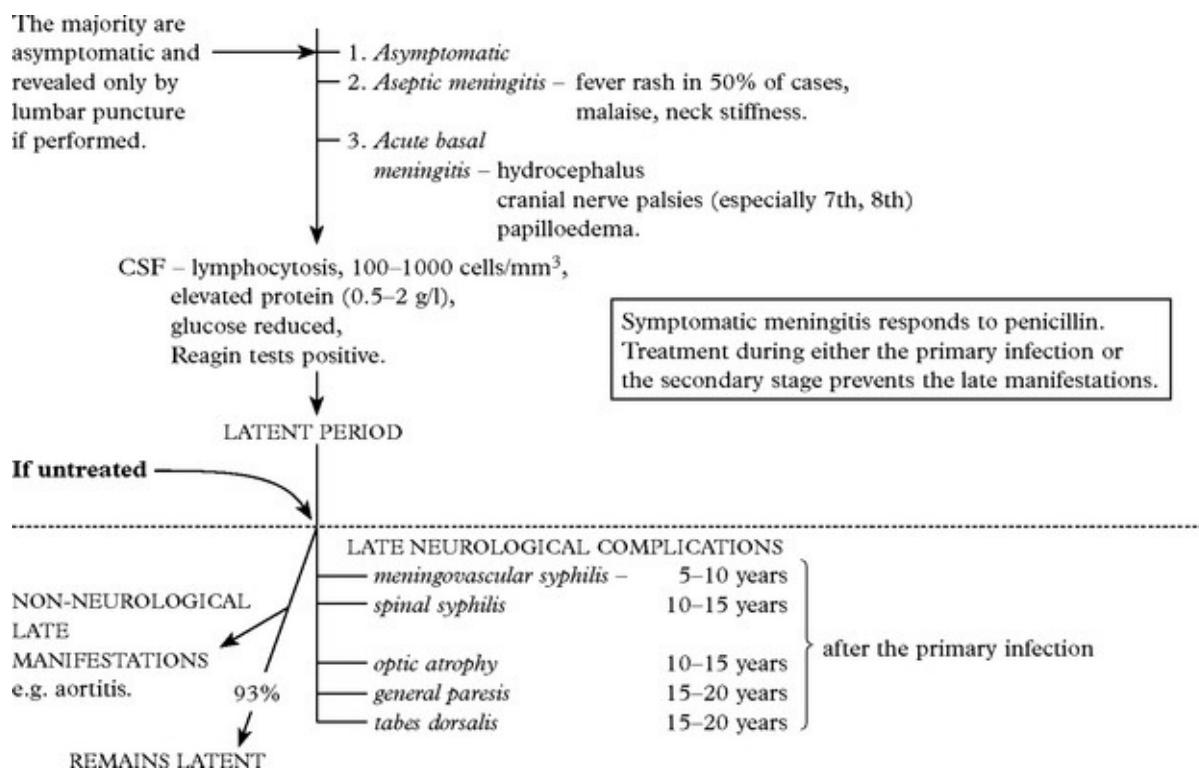
2. Specific treponemal antibodies (do not differentiate between past and present infection). Fluorescent treponemal antibody absorption (FTA) test and Treponema immobilisation (TPI) test.

3. *Treponema pallidum* DNA can be detected in the CSF of patients by PCR (sensitivity 60%).

SPIROCHAETAL INFECTION – NEUROSYPHILIS

The initial event in neurosyphilis is meningitis. Of all untreated patients 25% develop an acute symptomatic syphilitic meningitis within 2 years of the primary infection.

ACUTE SYPHILITIC MENINGITIS: Three clinical forms are recognised:



Late neurological complications occur in only 7% of **untreated** cases.

These forms are exceptionally rare and the clinical syndromes mentioned above seldom occur in a ‘pure’ form.

MENINGOVASCULAR SYPHILIS

‘Early’ late manifestation resulting in an obliterative endarteritis and periarteritis.

Presents as a ‘stroke’ in a young person – hemisphere, brain stem or spinal. Granulations around the base of the brain may produce cranial nerve palsies or even hydrocephalus.

CSF – lymphocytes 100/mm³, protein ↑, gammaglobulin ↑, positive serology. Penicillin arrests progression.

SPINAL SYPHILIS

Chronic meningitis with subpial damage to the spinal cord.

Presents as a progressive paraplegia, occasionally with radicular pain and wasting in upper limbs – ERB's PARAPLEGIA. CSF – as meningovascular syphilis. Penicillin arrests progression.

OCULAR MANIFESTATIONS

Meningitis around optic nerve with subpial necrosis may be the only manifestation of late syphilis.

Presents as a constriction of the visual fields with a progressive pallor of the optic disc:

- if both eyes are affected, the vision is rarely saved.
- if only one eye is involved, treatment with penicillin will save the other.

Neuroretinitis, uveitis and chorioretinitis occur, especially in HIV patients.

GENERAL PARESIS

Characterised by dementia – with memory impairment, disordered judgement and disturbed affect – manic behaviour, delusions of grandeur (rare).

There are two phases:

1. Preparalytic – with progressive dementia.
2. Paralytic – when corticospinal and extrapyramidal symptoms and signs develop associated with involuntary movements (myoclonus).

Argyll Robertson pupils may be present (see [page 146](#)).

At autopsy, meningeal thickening, brain atrophy and perivascular infiltration with plasma cells and lymphocytes are evident; culture from the cortex may reveal an occasional treponema.

CSF – lymphocytes 50/mm³, protein ↑ 0.5–2 g/l, gammaglobulin ↑.

Reagin tests in CSF positive in the majority.

Treatment in the preparalytic phase will halt progression in 40%.

TABES DORSALIS

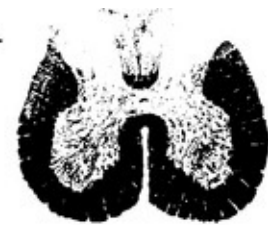
Posterior spinal root and posterior column dysfunction account for symptoms.

Posterior spinal root and posterior column dysfunction ----->

Pupillary abnormality (Argyll Robertson) and optic atrophy occur. Peripheral reflexes are lost and joint position and vibration sensation is impaired. A positive Romberg's test (page 28) indicates a sensory ataxia.

Pain loss results in trophic lesions and occasionally a Charcot joint may develop. ----->

Urinary incontinence, impotence and constipation also occur. 'Lightning pains', visceral crises (abdominal pain/diarrhoea) and rectal crises (tenesmus) are frequent.



Repeated trauma to an insensitive joint results in 'painless' osteoarthritis and joint destruction.

The CSF is more normal than in general paresis. The Reagin test may be negative in 30 per cent. Treatment may produce some improvement; it will not reverse joint destruction.

SYPHILITIC GUMMA presenting as an intracranial mass is extremely rare.

TREATMENT OF NEUROSYPHILIS

Penicillin G.	2–4 megaunits i.v.	(When patient sensitive to penicillin

or	4-hourly for 10 days.	↓
Procaine Penicillin	600 000 units i.m. daily for 15 days.	erythromycin or tetracycline may be given orally over 30 days.)
Benzathine Penicillin	2–4 megaunits i.m. weekly × 3.	

To prevent congenital syphilis penicillin should be given to all neonates and infected mothers during the first 4 months of pregnancy.

The Jarisch-Herxheimer reaction – tachycardia/fever – occurs in one-third of patients within a few hours of commencing treatment; it is believed to be due to endotoxin release from killed organisms. Steroids should counter the reaction, especially in tertiary syphilis.

CSF follow up: CSF is checked initially and at 6 monthly intervals until normal.

Cell count and degree of positivity of VDRL are the best indicators of persistent infection.

Failure of treatment is common in HIV positive patients and more frequent retesting of blood and CSF is necessary.

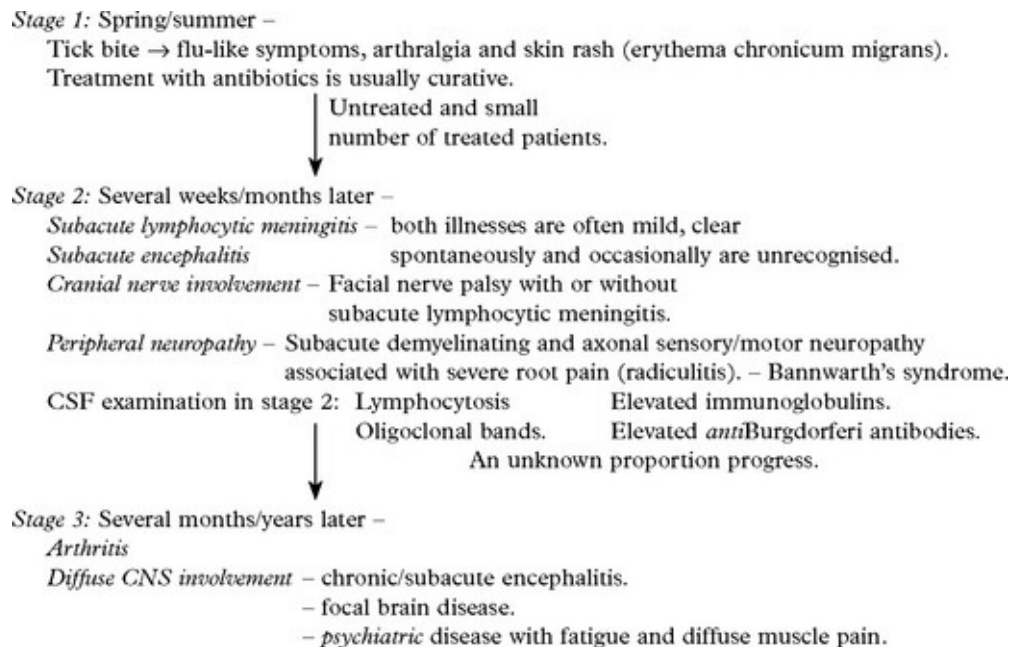
SPIROCHAETAL INFECTION

LYME DISEASE (NEUROBORRELIOSIS)

Originally described in the community of Old Lyme, this is a disorder, caused by the spirochaete *Borrelia burgdorferi*, characterised by relapsing and remitting arthralgia associated with a characteristic skin rash (*erythema chronicum migrans*) and neurological features. The organism, related to the treponemes, is prevalent throughout Europe and North America and is carried by ixodes ticks.

Clinical features

Only a minority of persons bitten by an infected tick develop the disease. Spirochaetocidal activity in normal serum and the immune response normally provide protection. It rarely occurs in HIV patients.



Diagnosis

Antibody tests

- Immunofluorescence assay (IFA)
 - Enzyme-linked immunoabsorbent assay (ELISA).
- } in serum and CSF.

In endemic areas up to 5% of the population are positive, although with lower titres than symptomatic patients.

In patients from endemic areas:

- | | | | |
|--|---|---|--|
| with meningitis/CN palsy
encephalitis/radiculitis
+ CSF profile
+ positive serology | } | diagnosis is definite, but
in stage 3 this is often
uncertain and blind trials
of therapy are given. | PCR if available gives the definitive answer.

MRI is abnormal in 25% with subcortical
(T2) white matter lesions. |
|--|---|---|--|

Treatment

Stage 1 – Oral antibiotics: penicillin, erythromycin or tetracycline.

Stage 2 – I.V. penicillin G. 20 million units for 10 days (or ceftriaxone).

Stage 3 – as stage 2.

If symptoms persist – wrong diagnosis with misleading titres, or –

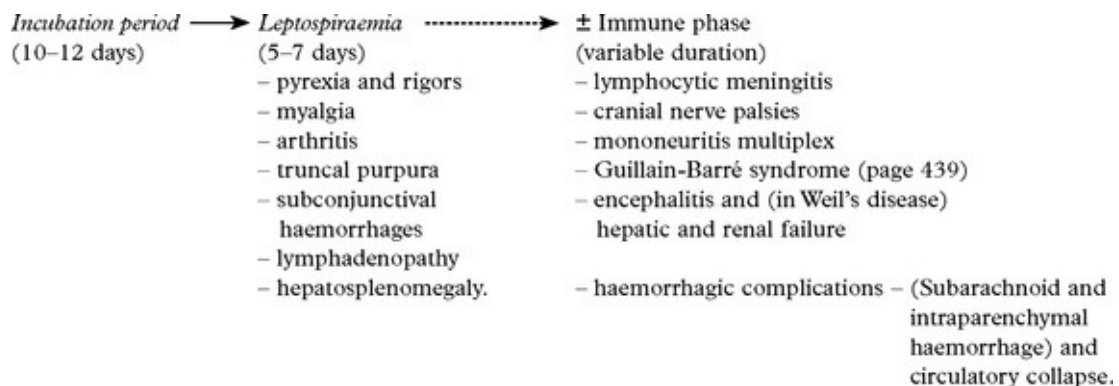
immune mediated damage.

Steroids can be used in late stages when symptoms have not responded to antibiotics.

LEPTOSPIROSIS

Leptospira interrogans is transmitted to man in the infected urine of wild and domestic animal carriers. Subclinical infection commonly occurs in high-risk occupations, e.g. sewer workers. Symptomatic illness is usually mild and only 10% of patients develop jaundice and haemorrhagic complications (Weil's disease).

Clinical features



Diagnosis

A combination of abnormal liver and renal function with elevated creatine kinase suggest the diagnosis. Leptospirae can be isolated from blood and CSF (in the immune phase) but diagnosis is usually confirmed by demonstrating agglutinating antibodies (ELISA detected IgM).

Treatment

The disease is usually self limiting and therapy unnecessary. Early treatment in the leptospiraemic phase with Penicillin G 12 million units daily and tetracycline 500 mg four times per day may minimize the

immune-mediated complications. Support of hepatic/renal failure and management of haemorrhagic complications may be life-saving.

PARASITIC INFECTIONS OF THE NERVOUS SYSTEM – PROTOZOA

TOXOPLASMOSIS

A worldwide parasitic infection affecting many species, including man.

Organism: An anaerobic intracellular protozoan, *Toxoplasma gondii*.

The majority of infections in man are asymptomatic (30% of the population have specific antibodies indicating previous exposure).

In the host



Transmission: Eating uncooked meat or contact with faeces of an infected dog or cat (definitive hosts).

There are two forms of toxoplasmosis:

CONGENITAL – when a previously unaffected woman contracts infection during pregnancy (subclinical infection); transplacental spread results in fetal infection. Premature delivery occurs in 25%.	ACQUIRED – symptomatic infection is uncommon and may be associated with underlying systemic disease or immunosuppression (e.g. AIDS). Fever and fatigue with muscle weakness and lymphadenopathy result. Abnormal lymphocytes in peripheral blood leads to confusion with <i>infectious mononucleosis</i>. The neurological features are those of a meningoencephalitis with focal signs and depressed conscious level. Choroidoretinitis occasionally occurs.
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Neurological complications:

- hydrocephalus,
- aqueduct stenosis,
- microcephaly.

Non-neurological features:

- skin rash, jaundice, hepatosplenomegaly, choroidoretinitis.

Skull X-ray shows:

- curvilinear calcification (basal ganglion and periventricular regions).

Varying degrees of organ involvement may occur. The only manifestation may be choroidoretinitis in an otherwise healthy child.



Areas of atrophic choroid, exposing the white sclera.

Retinal pigment epithelium becomes hyperplastic – densely pigmented areas result.

Diagnosis:

Organisms are seldom identified.

IgG antibodies indicate previous exposure, positive IgM and high or rising IgG confirm active infection.

Serological tests may be negative in AIDS.

In *acquired infection* CT shows characteristic ring shaped contrast enhancement. MRI is even more sensitive. Brain biopsy is necessary for exclusion of CNS lymphoma and for definitive diagnosis.

N.B. Rubella, cytomegalovirus and herpes simplex can also spread transplacentally and cause jaundice and hepatosplenomegaly. Cytomegalovirus may also produce choroidoretinitis and intracranial calcification.

Treatment

Sulphadiazine and pyrimethamine (Dapaprim) with folinic acid for 6 weeks. In AIDS newer drugs, such as clarithromycin and azithromycin, have also been used with some success. In this patient group recurrence after discontinuation of therapy mandates life long treatment. Give steroids when choroidoretinitis is present.

MALARIA

Plasmodium falciparum, the agent of malignant tertiary malaria, is responsible for *cerebral malaria*. Infected red blood cells adhere to vascular endothelium and block the microcirculation. Endothelial damage produces cerebral oedema. Confusion, focal signs, convulsions and coma occur. Diagnosis depends on demonstrating parasites in peripheral blood. Parenteral anti malaria treatment (chloroquine), exchange transfusion and supportive therapy may be life saving. Overall mortality is 10%. Complete recovery without sequelae is expected in survivors.

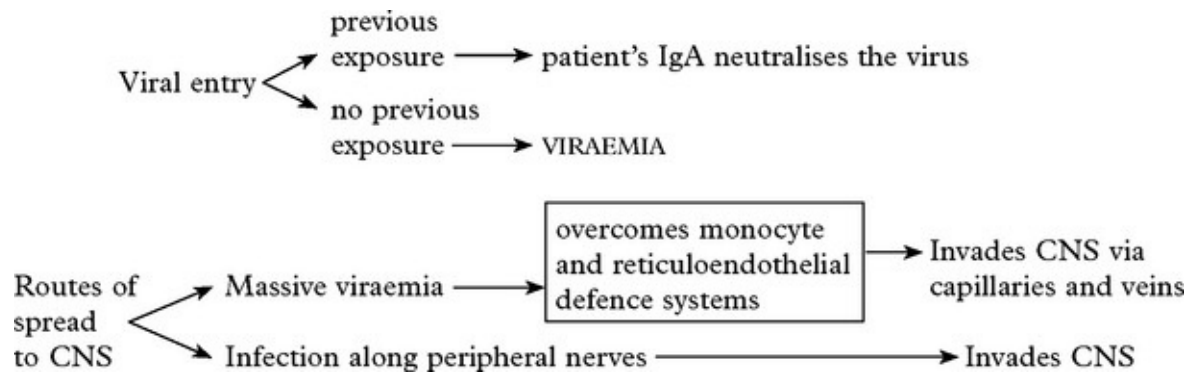
VIRAL INFECTIONS

General principles

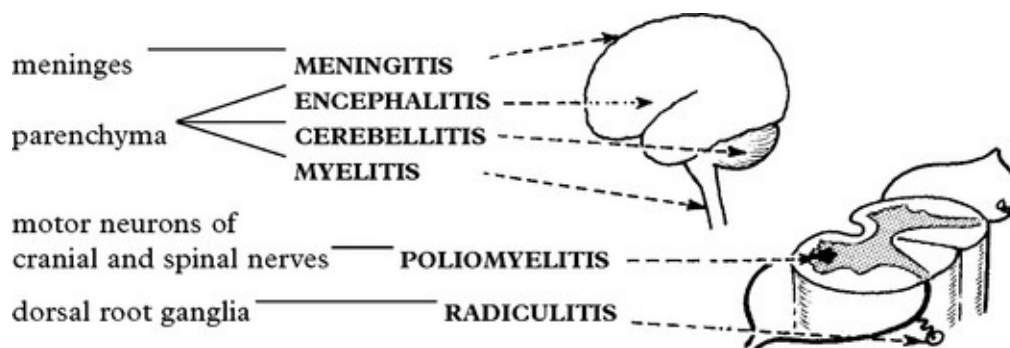
Invasion of the nervous system may occur as part of a generalised viral infection.

Occasionally nervous system involvement is disproportionately severe and symptoms of generalised infection are slight.

Viruses enter the body through the: *respiratory tract, gastrointestinal tract, genitourinary tract or by inoculation through the skin.*



After CNS penetration, the clinical picture depends upon the particular virus and the cells of the nervous system which show a specific susceptibility.



Some viruses cause a chronic, progressive infection, others remain dormant for many years within the nervous system before becoming symptomatic.

MENINGITIS

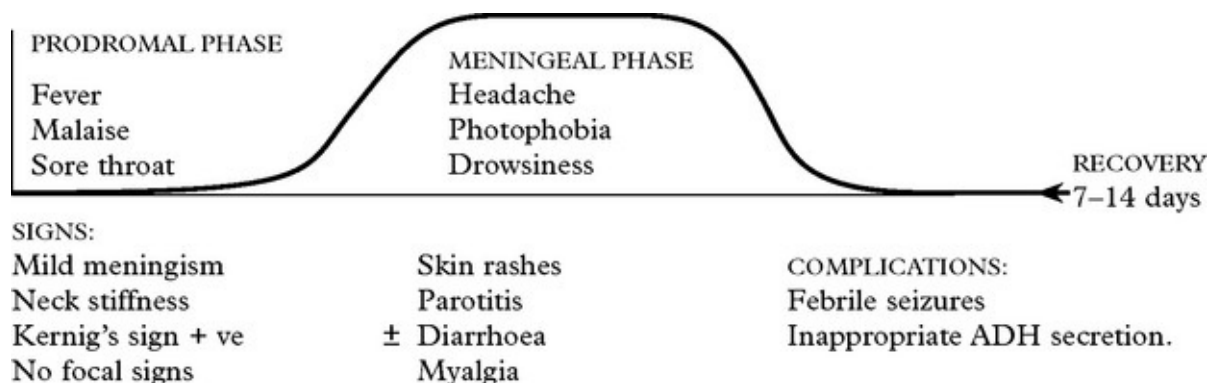
Meningitis is the commonest type of viral infection of the central nervous system. The term *aseptic meningitis* includes viral meningitis as well as other forms of meningitis where routine culture reveals no other organisms.

Common causal viruses –	Rare causal viruses –

ENTEROVIRUSES	LYMPHOCYTIC CHORIOMENINGITIS
MUMPS VIRUS	HUMAN IMMUNODEFICIENCY VIRUS (HIV)
HERPES SIMPLEX (subtype 2)	WEST NILE VIRUS
EPSTEIN-BARR VIRUS (EBV)	

VIRAL INFECTIONS – MENINGITIS

Clinical features of acute aseptic meningitis



Enterovirus infection e.g. Coxsackie or echo viruses – affects children/young adults and occurs seasonally in late summer.

Spread is by the faecal/oral route.

Mumps – affects children/young adults. Winter/spring incidence.

Herpes simplex (type 2) – accounts for 5% of viral meningitis. Develops in 25% of patients with primary genital infection (suspect in sexually active adults). Can cause a recurrent meningitis (Mollaret's meningitis).

Lymphocytic choriomeningitis – affects any age and is a consequence of airborne spread from rodent droppings.

Human Immunodeficiency Virus (HIV) – suspect in high risk groups ([page 515](#)). HIV antibodies are often absent and develop 1–3 months later during convalescence.

Investigations

The CSF cell count is elevated (lymphocytes or monocytes) with a normal glucose and protein. PCR detection of viral DNA/RNA in CSF though diagnostic, is rarely thought necessary. Virus may be cultured from throat swabs or stool. Serological tests on serum in acute and convalescent phases are especially valuable in detecting mumps and herpes simplex (type 2).

Differential diagnosis

From other causes of an aseptic meningitis which are usually subacute or chronic in onset:

- *Tuberculous or fungal meningitis*
- *Leptospirosis*
- *Sarcoidosis*
- *Carcinomatous meningitis*
- Partially treated *bacterial meningitis*
- *Parameningeal* chronic infection which evokes a meningeal response, e.g. mastoiditis.

The self-limiting and mild nature of viral meningitis should not lead to confusion with these more serious disorders.

Prognosis is excellent and **treatment** symptomatic.

VIRAL INFECTIONS – PARENCHYMAL

Viruses may act:

directly → acute viral encephalitis or meningoencephalitis, or *indirectly via the immune system* → allergic or postinfectious encephalomyelitis and

postvaccinial encephalomyelitis.

Also, a ‘toxic’ encephalopathy may develop during the course of a viral illness in which inflammation is not a pathological feature – REYE’S SYDNROME.

ACUTE VIRAL ENCEPHALITIS

Viral infection causes neuronal and glial damage with associated inflammation and oedema.

Viral encephalitis is a worldwide disorder with the highest incidence in the tropics.

Common causal viruses:

Worldwide:	Rare forms in specific areas:	
– Mumps	St Louis	– Arthropod-borne – USA
– Herpes simplex	West Nile	– Arthropod-borne – Africa/India
– Varicella zoster	Russian spring/summer	– Arthropod-borne – eastern Europe
– Epstein-Barr		
– Arboviruses		

Encephalitis following childhood infections – measles, varicella, rubella – is presumed to be *postinfectious* and not due to direct viral invasion, though the measles virus has occasionally been isolated from the brain.

Clinical features:

Signs and symptoms:

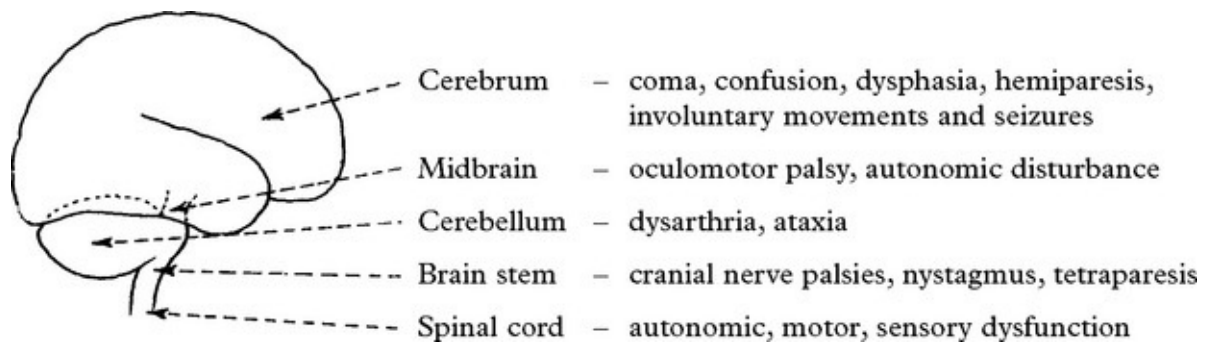
General: pyrexia, myalgia, *etc.*

Specific to causative virus, *e.g.* features of infectious mononucleosis (Epstein-Barr).

Meningeal involvement (slight) → neck stiffness, cellular response in

CSF.

Signs and symptoms of parenchymal involvement – focal and/or diffuse.



In general, the illness lasts for some weeks.

Prognosis is uncertain and depends on the causal virus as do neurological sequelae.

Herpes Simplex (HSV) and Varicella-Zoster (VZV) commonly cause disease in humans.

HERPES SIMPLEX ENCEPHALITIS

HSV-1 is the commonest cause of sporadic encephalitis.

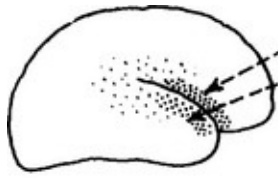
One third occur due to primary infection; two thirds have pre-existing antibodies (*reactivation*).

Clinical features

A worldwide disorder occurring during all seasons and affecting all ages.

Incidence: 1/250 000

General symptoms at onset – headache, fever – with evolution over several days to seizures and impaired conscious level.



Inferior frontal and temporal lobes are selectively involved and signs and symptoms reflect this – olfactory or gustatory hallucinations, behavioural disturbance, complex partial seizures, dysphasia (dominant hemisphere) and hemiparesis. Cerebral oedema may result in tentorial herniation.

Investigations

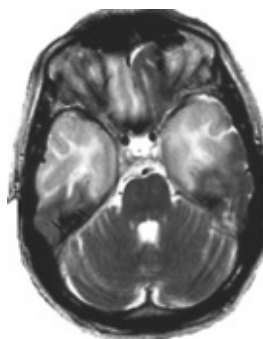
MR imaging: T2 weighted MRI showing temporal and orbitofrontal hyperintensities typical of herpes simplex encephalitis.

CSF examination reveals 5–500 lymphocytes. The protein is mildly elevated and the glucose is normal.

EEG examination shows generalised slowing with bursts of ‘periodic’ high voltage slow wave complexes over the involved temporal lobe.

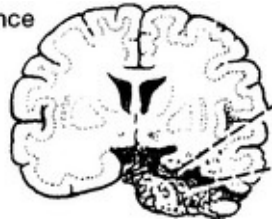
Polymerase chain reaction (PCR) on CSF may be negative in the first 48 hours. The quantity of HSV DNA then increases and, if initially negative and the clinical course is suggestive, the examination should be repeated. Also paired sera and CSF should be sent for HSV antibody (CSF HSV-specific antibody can still be detected up to 30 days).

Brain biopsy seldom required in view of the above new diagnostic techniques.



MR imaging:
T2 weighted
MRI showing
temporal and
orbitofrontal
hyperintensities
typical of
herpes simplex
encephalitis

This shows evidence of a necrotising encephalitis with intranuclear eosinophilic inclusion bodies.



Demonstrate herpes simplex antigen by immunofluorescence.

Isolate virus by culture (positive in 48 hours).

Differential diagnosis

Consider:

- other forms of encephalitis
- cerebral abscess
- brain tumour.

Treatment: Acyclovir inhibits DNA synthesis, is relatively non-toxic and significantly reduces morbidity and mortality. When the diagnosis is considered, treatment must start without delay.

Varicella-Zoster Virus (VZV) encephalitis may complicate chicken pox, or a cutaneous zoster eruption. CSF shows a mild lymphocytosis (<100 cells/mm³), a slight increase in the protein and a normal glucose. PCR detects VZV DNA. The virus can be grown from CSF and antibodies detected. Treatment with acyclovir or famciclovir is effective. Vasculitis may complicate.

VIRAL INFECTIONS – REYE'S SYNDROME

REYE'S SYNDROME

This rare encephalopathy, associated with fatty changes in the liver and other viscera, is almost exclusively confined to children. It is due to aspirin useage in infection with Influenza A, Influenza B or varicella-zoster viruses.

Incidence

Since 1980 the incidence of this condition has dropped dramatically, in part due to avoidance of salicylates in children.

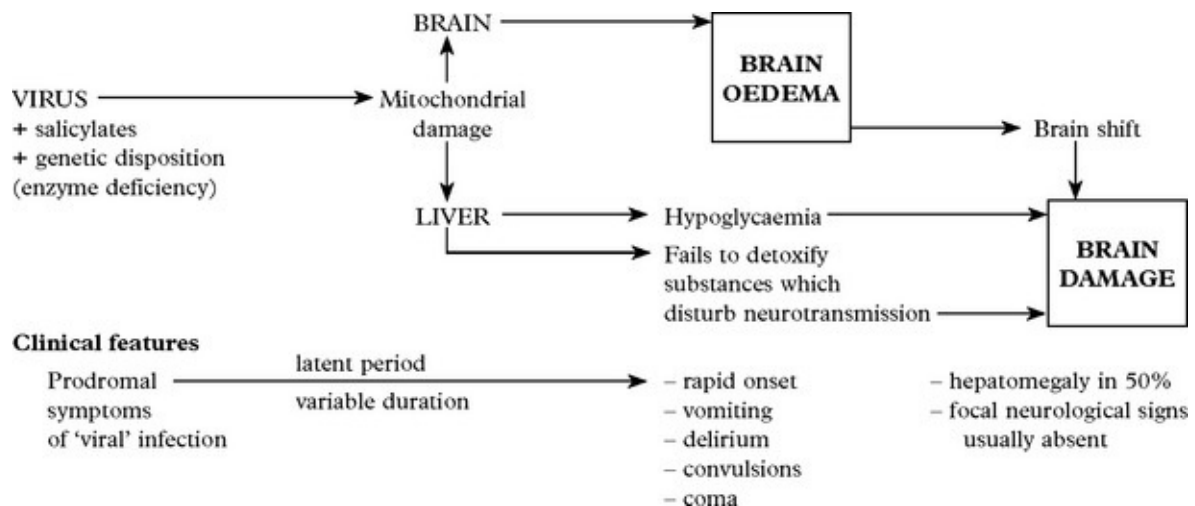
Pathology

Neurons and glial cells are swollen; the liver, heart and kidney show fatty infiltration.

Pathogenesis

Viral synergism with an environmental factor, *e.g.* salicylates, may be responsible.

Morphological changes in mitochondria indicate a central role.



Death results from raised intracranial pressure.

Investigations

– Raised liver enzymes (ALT & AST) – Elevated serum ammonia	– Hypoglycaemia (in infants) – Prolonged prothrombin time	– Increase in serum fatty acids – Aminoaciduria
--	--	--

CT/MRI show appearances of diffuse cerebral oedema

Differential diagnosis

Consider other causes of raised intracranial pressure in childhood, especially

- lead encephalopathy,
- lateral sinus thrombosis, e.g. following mastoiditis.

Treatment

Treatment aims at lowering intracranial pressure with the aid of intracranial pressure monitoring (see [page 52](#)). In addition, blood glucose must be maintained and any associated coagulopathy treated. Reduction of ammonia may be achieved by peritoneal dialysis or exchange transfusion.

Prognosis

Early diagnosis and supportive treatment has reduced the mortality from 80% to 30%.

When raised intracranial pressure is present, mortality increases to 50% and a high proportion of survivors have cognitive disorders.

A condition similar to Reye's syndrome occurs in some children with family history of 'sudden infant death'. A deficiency of medium chain acetyl-CoA dehydrogenase (an enzyme essential for fatty acid metabolism) is found. Carnitine deficiency results as a consequence of 'alternative pathway' fatty acid metabolism. *Siblings of children with Reye's syndrome should be screened for this disorder.*

VIRAL INFECTIONS – CHRONIC DISORDERS

In these disorders the infection results in a chronic progressive neurological condition.

The evidence of a viral etiology is:

Direct – finding of inclusion bodies, demonstration of viral particles or isolation of virus.

Indirect – relationship of onset of symptoms to a preceding viral illness, transmission of illness from one host to the next

N.B. Not all these features are present in any one illness

SUBACUTE SCLEROSING PANENCEPHALITIS (SSPE)

Caused by measles-like paramyxovirus – isolated from brain biopsy.

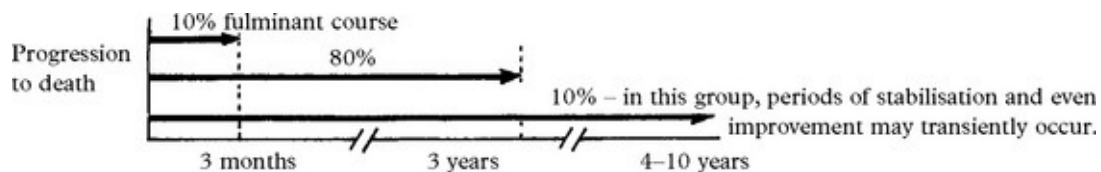
Less common with the availability of widespread primary measles vaccination.

Clinical features: A worldwide disorder. Incidence: 1 per million per year. Onset: between ages 7–10 years.

Stage 1: Behavioural problem, declining school performance, progression → dementia

Stage 2: Chorioretinitis, myoclonic jerks, seizures, ataxia, dystonia

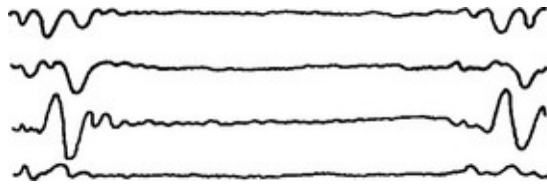
Stage 3: Lapses into rigid comatose state



The illness may occur after measles vaccination or following clinical infection at an early age (under 2 years).

Accompanying features of infection, *i.e.* pyrexia, leucocytosis, are absent.

Investigations



CSF examination shows elevated γ globulin with IgG oligoclonal bands; elevated measles antibodies (75% of total CSF IgG).

Blood examination shows elevated serum measles antibodies.

EEG – shows periodic high voltage slow wave complexes on a low voltage background trace.

Pathology

Changes involve both white and grey matter, especially in the posterior hemispheres. Brain stem, cerebellum and spinal cord are also affected. Oligodendrocytes contain eosinophilic inclusion bodies. Marked gliosis occurs with perivascular lymphocyte and plasma cell cuffing.

Treatment: There is no effective treatment. Since the introduction of measles vaccination there has been a marked reduction of SSPE.

Subacute measles encephalitis may follow measles infection in children on *immunosuppressive drug treatment* or with *hypogammaglobulinaemia*. The clinical course is different however from SSPE and EEG and CSF findings are less specific.

PROGRESSIVE RUBELLA PANENCEPHALITIS

Similar to SSPE with a fatal outcome, caused by rubella virus.

Presents at a later age (10–15 years)	CSF shows high γ globulin.
Progressive dementia.	Antibodies elevated in serum and CSF to rubella.
Ataxia. Spasticity. Myoclonus.	<i>Biopsy</i> does not show inclusion bodies.
Treatment: No effective treatment	

PRION DISEASES

Fatal conditions characterised by the accumulation of a modified cell

membrane protein – Prion protein or PrP (proteinaceous infectious particle) within the central nervous system.

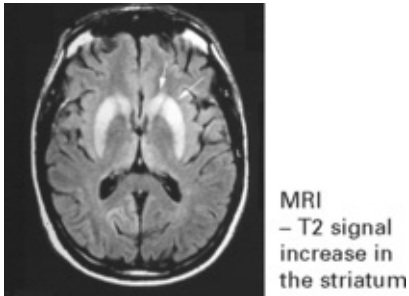
Clinical features are dependent on site and rate of deposition of PrP. A similar disorder in cattle, bovine spongiform encephalopathy (BSE) may be a source of infection in man.

The Prion theory

Experimental and epidemiological evidence supports transmissibility. Physical properties of the infective agent – heat and radiation resistance and absence of nucleic acid – suggests it is comprised solely of protein. This infectious protein when inoculated modifies normal cell membrane protein which acts as a template for further conversion to abnormal protein. This host-encoded protein accumulates without any inflammatory or immune response. In familial cases a point mutation in the prion gene explains disease susceptibility.

Creutzfeldt-Jakob disease (CJD)

A worldwide disorder with incidence 1:1 000 000. Approximately 90% of cases are sporadic and 10% familial caused by mutations in the prion protein (PRNP) gene on chromosome 20. Age of onset 50–60 years. Non specific symptoms at onset (anxiety and depression) are rapidly followed by myoclonus, ataxia, akinetic rigid state, dementia. Death within 12 months is usual. Iatrogenic disease occurs following corneal or dural grafts, depth electrodes and cadaveric derived human growth hormone treatment.



Investigation

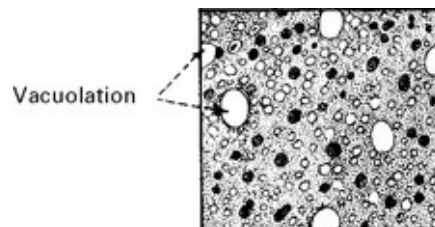
EEG – 1–2 Hz triphasic sharp waves with periodic complexes

CSF – increase in protein 14–3–3 (a protein kinase inhibitor)

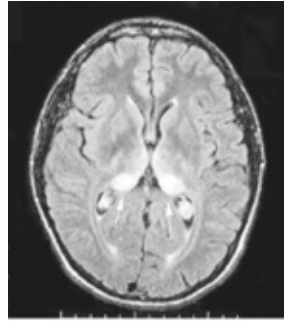
[The combined EEG and CSF findings, where positive, have diagnostic sensitivity/specificity of 97%]

Pathology

– Neuronal degeneration occurs with marked astrocytic proliferation and amyloid plaque formation. Vacuolation of glial cells results in a characteristic spongiform appearance.



Treatment – supportive.



MRI – T2 signal increase in the thalamus (pulvinar sign)

New Variant CJD (vCJD)

Generally affects younger age group. Psychiatric symptoms of depression, anxiety, or withdrawal are common early manifestations. Neurological symptoms appear approximately 6 months later, with paraesthesias an early feature. Eventually sufferers exhibit ataxia, progressive dementia and involuntary movements (myoclonus, chorea, or dystonia).

Only 50% of patients have protein 14–3–3 proteins in CSF. EEG reveals nonspecific slowing (the periodic complexes of sporadic CJD are absent).

PRNP gene mutations are found with patients homozygous for methionine at the 129 codon. Neuropathological changes – ‘florid’ plaques in the cerebral and cerebellar cortex, severe thalamic gliosis, and spongiform change with diffuse accumulation of prion proteins.

Gerstmann Straussler syndrome (GSS)

A similar disorder condition to CJD. Cases are familial and characterised by specific pathology of spongiform changes associated with amyloid plaques containing PrP immunoreactive proteins. Clinical features are nonspecific – ataxia, Parkinsonism, dementia. Death occurs within 5 years of onset.

Kuru

An extensively studied disorder of Papua New Guinea. It is of interest in view of man to man spread from cannibalism.

VIRAL INFECTIONS – MYELITIS AND POLIOMYELITIS

MYELITIS

Acute viral transverse myelitis is rare. It can occur in association with measles, mumps, Epstein-Barr, herpes zoster/simplex, enterovirus infections HIV, HTLV-1 and 2 and smallpox. Fever, back and limb pain precede paralysis, sensory loss and bladder disturbance. Initially paralysed limbs are flaccid, but over 1–2 weeks spasticity and extensor plantar responses develop. Good recovery occurs in 30%. Death from respiratory failure is rare (5%).

Investigations

Myelography when performed is normal. MRI may demonstrate focal cord signal changes. CSF shows elevated protein with a neutrophil or lymphocytic response. Serological tests will occasionally identify the causal virus. Electrophysiology distinguishes from Guillain-Barré syndrome.

Treatment

Supportive; the place of steroids remains unproven.

It is not clear whether the pathological effects (perivenous demyelination) result from direct or delayed (immunological) reactions to the virus.

POLIOMYELITIS

An acute viral infection in which the anterior horn cells of the spinal cord and motor nuclei of the brain stem are selectively involved. A

major cause of paralysis and death 30 yrs ago, now rare with the introduction of effective vaccines and improved sanitation.

Causative viruses:

The poliovirus is a picornavirus (RNA virus).

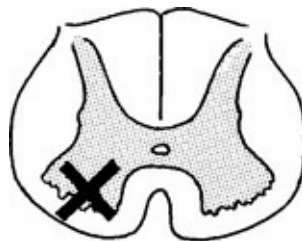
Three immunological distinct strains have been isolated. Immunity to one does not result in immunity to the other two.

Coxsackie and echoviruses (also picornaviruses), may produce a clinically identical disorder. West Nile virus can produce a polio-like flaccid paralysis.

Pathology

Initially – inflammatory meningeal changes, followed by – inflammatory cell infiltration (polymorphs and lymphocytes) around the brain stem nuclei and anterior horn cells. Neurons may undergo necrosis or central chromatolysis.

Microglial proliferation follows.



Mode of spread

Spread by faecal/oral route. Once ingested the virus multiplies in the nasopharynx and gastrointestinal tract.



Penetration of GI tract
results in viraemia

but CNS involvement occurs in only a very small
proportion. Most infected patients are asymptomatic.
Virus excretion continues in the faeces for as long as
three months after the initial infection – carrier state.

Epidemiology

A highly communicable disease which may result in epidemics.

Seasonal incidence – late summer/autumn.

Worldwide distribution, although more frequent in northern temperate climates.

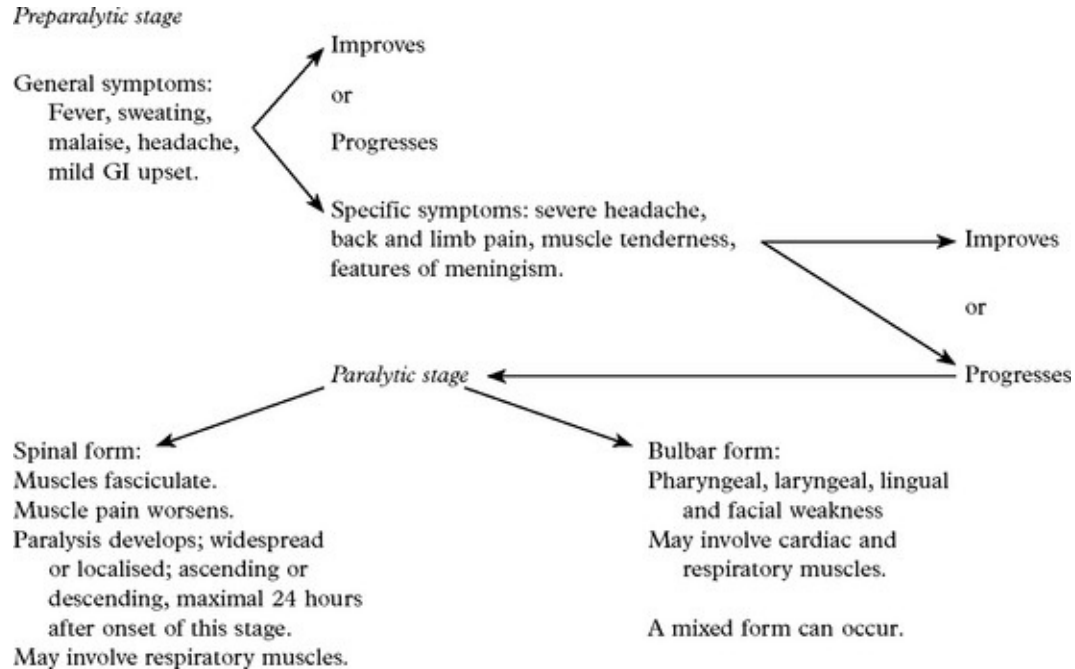
Prophylactic vaccination has produced a dramatic reduction in incidence in the last 25 years. In developing countries without a vaccination programme, the disease remains a problem.

Clinical features

Infection may result in:

- Subclinical course + resultant immunity (majority)
- Mild nonspecific symptoms of viraemia + resultant immunity
- Meningism without paralysis (PREPARALYTIC) + resultant immunity
- Meningism followed by paralysis (PARALYTIC) + resultant immunity.

VIRAL INFECTIONS – POLIOMYELITIS



Diagnosis

During the meningeal phase, consider other causes of acute meningitis.

Once the paralytic phase ensues, distinguish from the Guillain-Barré syndrome and transverse myelitis.

The clinical picture + CSF examination (polymorphs and lymphocytes increased; protein elevated with normal glucose) are sufficient to reach the diagnosis.

Poliovirus RNA can be detected in faeces or CSF by PCR.

Prognosis

In epidemics, a mortality of 25% results from respiratory paralysis. Improvement in muscle power usually commences one week after the onset of paralysis and continues for up to a year.

Only a proportion of muscles remain permanently paralysed; in these, fasciculation may persist. In affected limbs, bone growth becomes retarded with shortening as well as thinning.

Treatment

The patient is kept on bed rest and fluid balance carefully maintained.

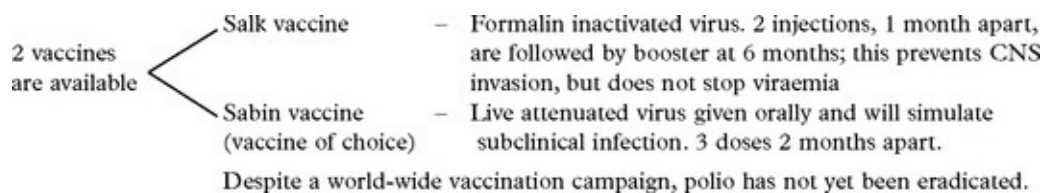
Respiratory failure may require ventilation.

Avoid the development of deformities in affected limbs with physiotherapy and splinting.

Post-Polio Syndrome

A significant proportion of polio patients develop late sequelae often 30 yrs after initial illness – fatigue, myalgia and progressive muscle atrophy with weakness are characteristic.

Prophylaxis



VIRAL INFECTIONS – VARICELLA-ZOSTER INFECTION

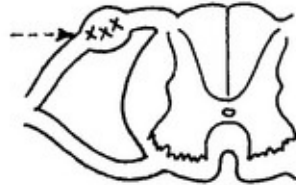
Varicella (chickenpox) and herpes zoster (shingles) are different clinical manifestations of infection by the same virus – *Varicella-Zoster*, a DNA human herpes virus.

Conditions caused:	– an acute encephalitis – viral meningitis – myelitis	– postinfectious encephalomyelitis – postinfectious polyneuropathy (Guillain-Barré syndrome).
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SHINGLES

This occurs after virus reactivation, dormant after the primary infection (chickenpox).

Pathology: The virus involves the *dorsal root* (sensory) ganglion of the spinal cord or the *cranial nerve* sensory ganglion – trigeminal or geniculate.



The inflammatory process may spread into the spinal cord and involve posterior and anterior horns. Similarly inflammatory changes may occur in the brain stem.



Clinical features

Patients are usually over 50 years of age. Sexes are affected equally. Recurrence is rare. Often occurs in immunocompromised patients e.g. lymphoma. Also associated with spinal/nerve root trauma.

Initial feature:

A vesicular skin rash associated with a burning, painful sensation. Vesicles contain clear fluid and conform to a dermatome distribution. After 1–3 weeks, the vesicles crust over and leave irregular skin depigmentation with scarring.



Motor weakness occurs in 20% due to damage of the anterior horn cell. More widespread spinal (myelitis) or encephalic involvement occurs in the immunodeficient. In these patients extensive cutaneous lesions are common (disseminated zoster).

Cranial nerve ganglia involvement:

– *Trigeminal:* usually ophthalmic division with vesicles above the eye and associated corneal ulceration – HERPES ZOSTER OPHTHALMICUS.

– *Geniculate*: vesicles within the external auditory meatus and ear drum. Ipsilateral deafness and facial weakness result. – RAMSAY HUNT SYNDROME.

Diagnosis: Based on clinical features. Virus DNA can be detected in vesicular fluid by PCR.

Treatment

This depends on the severity and location of skin lesions. Mild disease requires symptomatic treatment only. Severe disease, involvement in immunocompromised patients, or ophthalmic vesicles require acyclovir either orally or intravenously.

POST HERPETIC NEURALGIA

This is a condition which occurs in 10% of all patients. The incidence rises with age. A chronic, uncomfortable, burning pain presents in the territory of the involved dermatome. The pathogenesis is unknown.

Treatment with antidepressants, anticonvulsants, *e.g.* carbamazepine, transcutaneous stimulation (TCS) or sympathetic ganglion block may help, but results are unpredictable.

Varicella and Herpes Zoster CNS involvement

Patients with AIDS and other immunocompromising disorders risk severe, life-threatening CNS involvement – encephalitis, cerebral vasculitis, myelitis or brain stem encephalitis. Herpes Zoster ophthalmicus can be associated, in the middle aged, with delayed major cerebral artery territory infarction. This presents 4–6 weeks after infection. Stroke also occurs as a remote complication of childhood varicella, usually within 12 weeks of clinical chickenpox. These are both due to virus-induced damage to cerebral arteries (vasculitis). The role of anti-viral drugs and steroids is uncertain.

OPPORTUNISTIC INFECTIONS

These infections occur in immunocompromised patients. Certain types of immunological deficiency tend to be associated with specific forms of infection.

<i>T cell/macrophage deficiency</i> Causes: e.g. AIDS Lymphoreticular tumours Immunosuppressant drugs Organisms: <i>Viruses</i> – Cytomegalovirus Herpes simplex/zoster JC virus <i>Bacteria</i> – <i>Listeria</i> <i>Nocardia</i> <i>Mycobacterium</i> , etc. <i>Fungi</i> – <i>Aspergillus</i> <i>Candida</i> <i>Mucoraceae</i> <i>Parasites</i> – <i>Toxoplasmosis</i>	<i>B cell immunoglobulin deficiency</i> Chronic lymphatic leukaemia Primary hypogammaglobulinaemia Splenectomy Measles Enteroviruses <i>Streptococcus pneumoniae</i> <i>Haemophilus influenzae</i> <i>Pseudomonas aeruginosa</i> , etc.	<i>Granulocyte deficiency</i> Marrow infiltration Aplastic anaemia Chemotherapy/ radiotherapy Enterobacteria <i>Staphylococcus aureus</i> <i>P. aeruginosa</i> , etc. <i>Aspergillus</i> <i>Candida</i> <i>Mucoraceae</i>
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CLINICAL SYNDROMES, DIAGNOSIS AND TREATMENT

Clinical Syndrome	Diagnosis	Treatment
CEREBRAL ABSCESS	CT scan	→ Common bacteria
	Blood culture	→ <i>Listeria</i>
	Drainage or biopsy and culture	→ <i>Nocardia</i>
		→ <i>Candida</i>
OTHER INTRA-CRANIAL MASS LESIONS	CT/MR scan	→ <i>Aspergillus</i>
	Serum antibodies	→ Surgical removal (if indicated)
	Biopsy	→ Amphotericin B & 5-fluorocytosine
		→ Pyrimethamine, sulphadiazine & folinic acid
ENCEPHAL-OPATHY	EEG	
	CT/MR scan	→ <i>Toxoplasmosis</i>
	Serum antibodies	→ Cytomegalovirus
	Viral isolation (biopsy)	→ JC virus
	Intracellular inclusion bodies	(progressive multifocal encephalopathy)
		→ Pyrimethamine, sulphadiazine & folinic acid
CRANIAL NERVE PALSIES	Skull X-ray	
	CT/MR scan	→ <i>Candida</i>
	CSF/blood culture	→ <i>Mucoraceae</i>
	Serum antibodies	→ Amphotericin B & 5-fluorocytosine
	Nasal swab	→ Radical sinus and orbital surgery and amphotericin B
MENINGITIS	CT/MR scan	→ Common bacteria
	CSF/blood culture	→ <i>Mycobacterium</i>
	Serum antibodies	→ <i>Cryptococcus</i>
	Antigen agglutination tests	→ <i>Listeria</i>
		→ <i>Nocardia</i>
		→ <i>Candida</i>
RETINITIS	Serum antibodies	→ <i>Toxoplasmosis</i>
	Viral isolation	→ Cytomegalovirus

ACQUIRED IMMUNODEFICIENCY SYNDROME (AIDS)

Human immunodeficiency virus (HIV-I) has *lymotropic* (CD4 lymphocytes) and *neurotropic* (microglial) properties. Neurological features develop in 80% of infected individuals manifesting as either direct effects of the HIV virus or infections, tumours and associated vascular disorders due to immunodeficiency. AIDS is the end stage of chronic infection.

Prevalence of AIDS and HIV infection

Certain individuals are 'at risk' of infection:

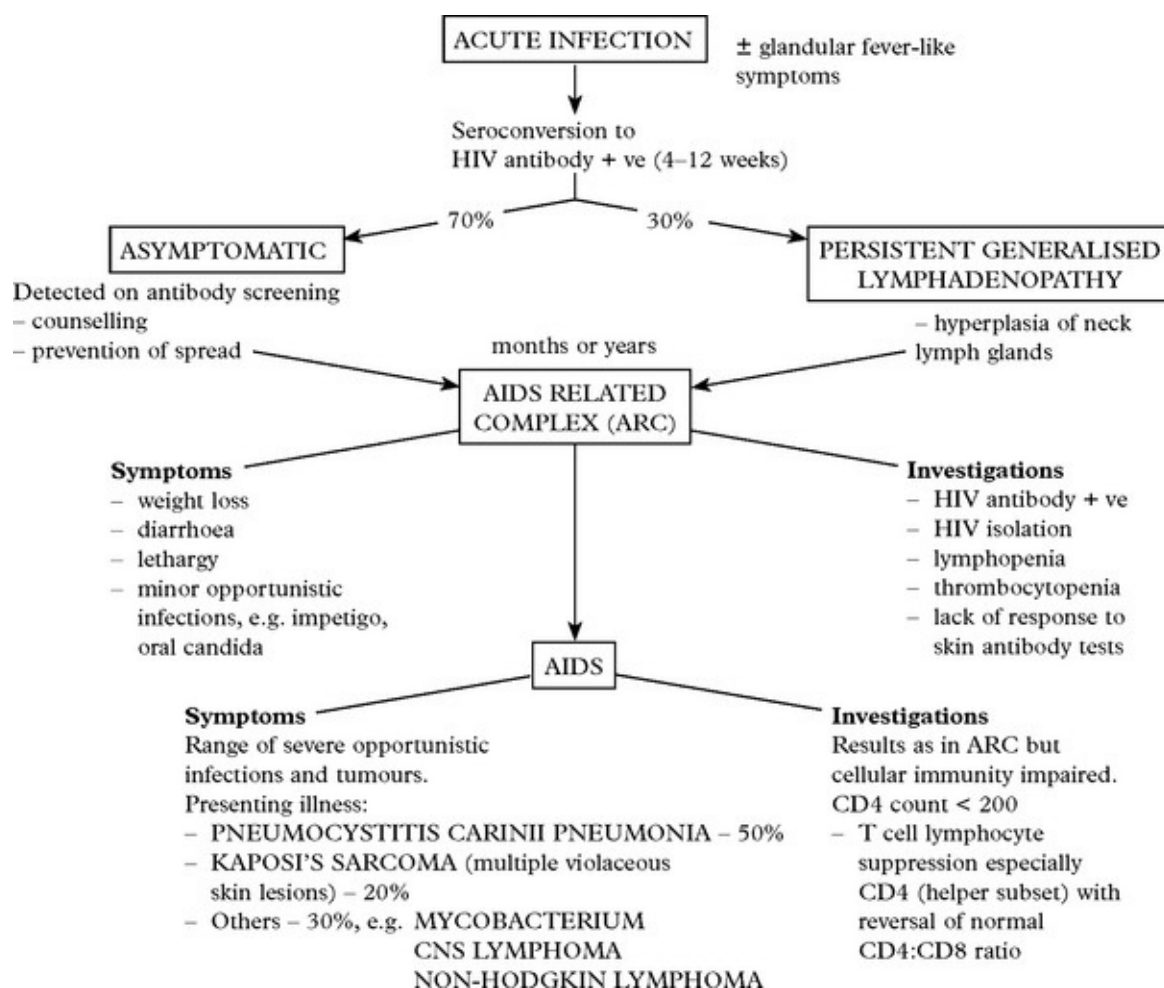
- homosexual males } and heterosexual partners
- I.V. drug users }
- Babies born to infected individuals
- Recipients of blood products, e.g. haemophiliacs.

The incidence of HIV infection in 'at risk' groups varies considerably.

Sex education, supply of clean needles to addicts, active drug-dependence programmes and specific precautions in the preparation of blood products are necessary to limit its spread.

Current prevalence of HIV – USA 450/100,000

CLINICAL COURSE OF HIV INFECTION



NEUROLOGICAL PRESENTATIONS OF HIV INFECTION

Cerebral tumours

Primary cerebral lymphoma
Metastatic systemic lymphoma
Metastatic Kaposi's sarcoma

CT/MR scan

Infections

Encephalitis

Cytomegalovirus
Herpes zoster/simplex
Toxoplasmosis
Progressive multifocal
leukoencephalopathy

CT/MR scan
Antibody tests
Biopsy

Cerebral abscess

E. coli
Aspergillus –
Candida
Nocardia

CT/MR scan
Aspiration/culture
Antibody tests

Meningitis

HIV-1
Mycobacterium
Listeria –
Syphilis
Aspergillus

CSF exam and culture
Antibody tests

Peripheral neuropathy

Herpes zoster radiculopathy
Cauda equina syndrome (cytomegalovirus)
Acute reversible demyelination
Chronic demyelination

Nerve conduction studies
Antibody tests

AIDS dementia (in 15%)

Direct HIV infection with
demyelination and perivascular
inflammatory changes. Intellectual
decline of subcortical type
(page 126).

CT/MR scan
Psychometry

Retinopathy

Cytomegalovirus
Toxoplasmosis

Fundal exam.
Antibody tests

Myelopathy

Acute reversible
Compression – abscess
– systemic lymphoma.
Ascending – cytomegalovirus
herpes zoster/simplex

Spinal CT/MRI
Antibody tests

Vascular disorders

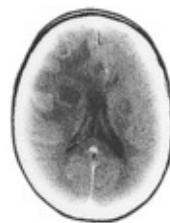
Intracranial haemorrhage
Cerebral infarction (septic
emboli or thrombosis)

CT/MR scan

Treatment

Opportunistic infection – treatment of specific infection (see [page 514](#)).

With known HIV + ve patients, invasive procedures such as biopsy are often avoided and trials of therapy are administered, e.g. cerebral toxoplasmosis – trial of pyrimethamine and sulphadiazine, monitored with CT/MRI. If lesions do not resolve → biopsy (? lymphoma).



Cerebral
toxoplasmosis

Highly active antiretroviral therapy (HAART) with effective treatment

for infections and neoplastic complications has significantly improved outcome. Mean survival time for HIV-infected persons currently exceeds 10 years. The prolonged survival of HIV-infected persons increases their risk of developing PML or CNS lymphoma (these responding poorly to treatment).

HAART management comprises two nucleoside reverse transcriptase inhibitors (e.g. *zidovudine* and *didanosine*) and a protease inhibitor (e.g. *ritonavir* or *indinavir*), or a nonnucleoside reverse transcriptase inhibitor (*nevirapine*). This combination is given to HIV-infected individuals with detectable viral loads or immunologic dysfunction (less than 500 CD4+ cells/mm³). HAART results in immunological and neurocognitive improvement, even when HIV is advanced. Treatment aims at reducing the viral load to undetectable levels, PCR having a central role in monitoring therapy and identifying drug resistance.

SUBACUTE/CHRONIC MENINGITIS

This entity is characterised by symptoms and signs of meningeal irritation which persist and progress over weeks without improvement. Unlike acute meningitis, the onset is insidious; cranial nerve signs and focal deficits such as hemiparesis, dementia and gradual deterioration of conscious level may predominate. Predisposing factors include immunosuppression or immunocompromised host. The outcome depends upon aetiology and the early instigation of appropriate treatment.

Chronic meningitis is associated with certain CSF findings.

- Lymphocytosis + low glucose (relative to serum level)
- Lymphocytosis + normal glucose.

Diagnosis depends upon CSF examination.

Lumbar puncture should be performed in suspicious cases as soon as

CT scan has ruled out a mass lesion.

SUBACUTE/CHRONIC MENINGITIS WITH A LOW CSF GLUCOSE

Western herbs shown to suppress or inhibit pro-inflammatory cytokine production

Causes	Diagnosis	Specific features	Treatment
<i>M. tuberculosis</i>	See page 494		
Fungi <i>Cryptococcus</i> <i>Nocardia</i> <i>Candida</i> <i>Aspergillus</i>	Diagnosis suggested by chest X-ray – pulmonary infiltrations, CT/MR evidence of meningeal enhancement and associated hydrocephalus CSF abnormalities lymphocytosis, low glucose and high protein with appropriate staining, culture and agglutination/complement-fixation tests.	Clinical features similar to tuberculous meningitis.	Amphotericin B + fluorocytosine or fluconazole
Carcinomatous meningitis – lung/breast/gastrointestinal tract Leukaemia/ lymphoma Glioma Medulloblastoma Melanoma	Evidence of primary neoplasm. <i>CT</i> or <i>MR</i> evidence of meningeal enhancement <i>CSF</i> Malignant cells seen in fresh centrifuged filtered sample. Tumour markers: – carcinoembryonic antigen (CEA) – β -microglobulin Meningeal biopsy rarely necessary but diagnostic	Back pain/radicular involvement common. Hydrocephalus in 30%	Consider irradiation followed by intrathecal methotrexate or monoclonal targeting (see page 314). Leukaemia/lymphoma requires specialist advice

Causes	Diagnosis	Specific features	Treatment
Parameningeal infections Cerebral abscess Epidural abscess Sinusitis Mastoiditis	Evidence of primary infected source <i>X-rays</i> Sinuses, mastoids <i>CT/MR scan</i> – cerebral or cerebellar abscess <i>CSF</i> microscopy/culture <i>Blood</i> cultures	Prodromal sinus or middle ear infection	Appropriate antibiotic therapy and, if indicated, surgical drainage of loculated parameningeal infection
Bacteria: <i>Treponema</i> <i>Brucella</i> <i>Leptospira</i> <i>Listeria</i> <i>Borrelia burgdorferi</i>	<i>CSF</i> Isolate organism (if possible) Serological tests Serum Serological tests	<i>Treponema</i> – page 498 Sexual contact <i>Brucella</i> Contact with infected cattle <i>Leptospira</i> – page 502 Contact with contaminated rat, dog or cattle urine <i>Listeria</i> – page 514 Contaminated foods <i>Borrelia burgdorferi</i> – page 501 Tick bite	Appropriate antibiotic
Miscellaneous Parasites, e.g. toxoplasma – see page 503 Sarcoidosis – see page 360 Behçet's disease Whipple's disease Systemic lupus erythematosus – see page 266			
Chemical – leakage from epidermoid dermoid cyst or craniopharyngioma – Intrathecal drugs and contrast material			

Despite extensive investigation, a group of patients with chronic meningitis exists in whom no cause is found.

DEMYELINATING DISEASE

Demyelinating disorders of the central nervous system affect *myelin*

and/or *oligodendroglia* with relative sparing of *axons*.

The central nervous system is composed of *neurons* with *neuroectodermal* and *mesodermal* supporting cells.

The neuroectodermal cells comprise:

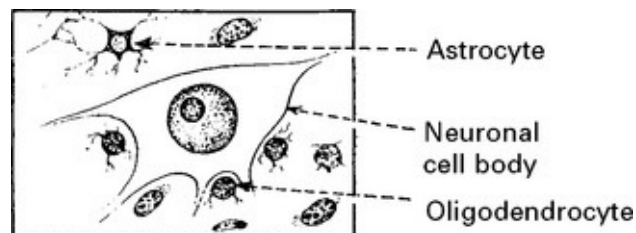
astrocytes

ependymal cells

oligodendrocytes.

The oligodendrocytes, like Schwann cells in the peripheral nervous system, are responsible for the formation of *myelin* around central nervous system *axons*.

One Schwann cell myelinates one axon but one oligodendrocyte may myelinate several contiguous axons, and the close proximity of cell to axon may not be obvious by light microscopy.



Oligodendrocytes are present in grey matter near neuronal cell bodies and in white matter near axons.

Myelin is composed of *protein* and *lipids*. Protein accounts for 20% of total content. The lipid fraction may be divided into:

cholesterol

glycophosphatides (lecithins)

sphingolipids (sphingomyelins).

The laying down of myelin in the central nervous system commences at the fourth month of fetal life in the median longitudinal bundle, then in frontal and parietal lobes at birth. Most of the cerebrum is myelinated by the end of the 2nd year. Myelination continues until the 10th year of life.

Myelin disorders may be classified as diseases in which:

1. Myelin is inherently abnormal or was never properly formed – these disorders generally present in infancy and early childhood and have a biochemical basis, *e.g.* leukodystrophy.
2. Myelin which was normal when formed breaks down as a consequence of pathological insult, *e.g.* *multiple sclerosis*.

MULTIPLE SCLEROSIS

Multiple sclerosis (MS) is a common demyelinating disease, normally characterised by focal disturbance of function and a relapsing and remitting course.

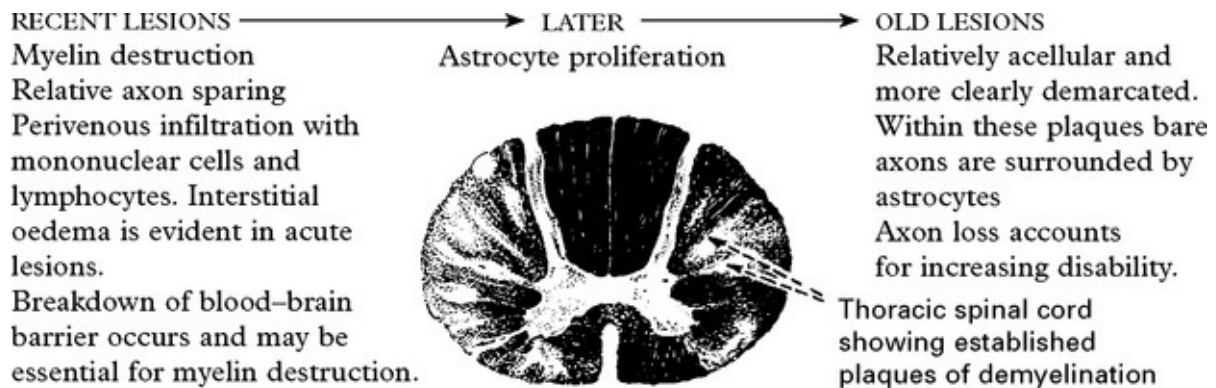
The disease occurs most commonly in temperate climates and prevalence differs at various latitudes:

	<i>Latitude (°N)</i>	<i>Rate/100 000 (adjusted)</i>
Orkneys and Shetland.....	60	309
England (Cornwall).....	51	63
Italy (Bari).....	41	13

The disease usually occurs in young adults with a peak age incidence of 20–40 years. Slightly more females than males are affected. There is a 3% risk of disease if a sibling or parent is affected.

PATHOLOGY

Scattered lesions with a greyish colour, 1 mm to several cm in size, are present in the white matter of the brain and spinal cord and are referred to as *plaques*. The lesions lie in close relationship to veins (postcapillary venules) – perivenous distribution.



PATHOGENESIS

Immune deficiency has been suggested. This might explain the possible persistence of a latent virus and variations in immune status could be the basis of ‘relapses and remissions’. T lymphocytes and macrophages found in plaques may be sensitized to myelin antigens.

Hereditary/genetic factors appear significant. There is an increased familial incidence of multiple sclerosis. This has led to the study of *histocompatibility antigens* (HL-A). An association between A3, B7, B18 and DW2/DRW2 and multiple sclerosis has been demonstrated. Concordance rate in monozygotic twins is 30% and in dizygotes 5%. Affected women transmit MS to offspring more frequently than affected men suggesting that mitochondrial genes contribute to inheritance.

Viruses may be important in the development of multiple sclerosis, infection perhaps occurring in a genetically/immunologically susceptible

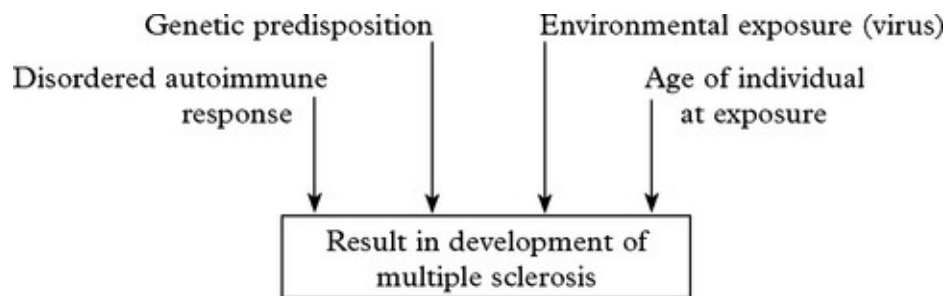
host.

Elevated serum and CSF antibody titres have been found to:

– varicella zoster, measles, rubella and herpes simplex during relapse.

Biochemical: No biochemical effect has been demonstrated – myelin appears normal before breakdown and the proposed excess of dietary fats or malabsorption of unsaturated fatty acids is unproven.

In summary – the causation is probably multifactorial.



CLINICAL FEATURES

Peak age of onset	– 20–25 years
Childhood onset rare	– 2%
Patients presenting > 50 years	– 5%
Patients presenting > 60 years	– 1%

Multiple sclerosis is usually characterised by:

- Signs and symptoms of widespread white matter disease.
- A relapsing and remitting or progressive course.

Symptoms at onset

1. *Vague symptoms*: lack of energy, headache, depression, aches in limbs – may result in diagnosis of psychoneurosis. These symptoms are eventually associated with:

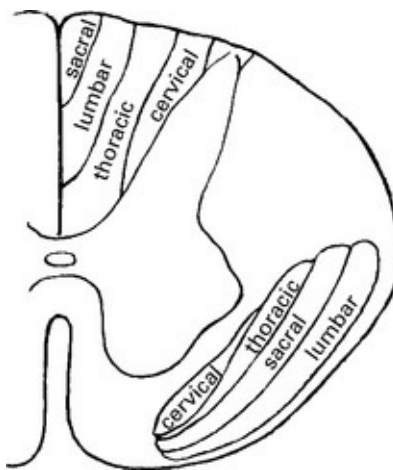
2.

2. <i>Precise symptoms</i> : (initial symptom of multiple sclerosis expressed as a %)	Sensory disturbance	– 40%
	Retrobulbar neuritis	– 17%
	Limb weakness	– 12%
	Diplopia	– 11%
	Vertigo	} – 20%
	Ataxia	
	Sphincter disturbance	

Trigeminal neuralgia may be an early symptom of multiple sclerosis, and this should be considered in the young patient with paroxysmal facial pain.

Sensory symptoms

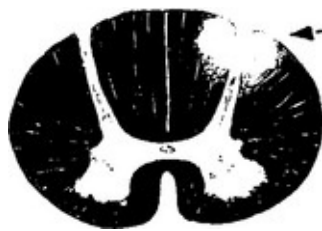
Numbness and paraesthesia are common and often minor and transient. Paraesthesia is more often due to posterior column demyelination than to spinothalamic tract involvement.



Posterior column lesions result in impaired vibration sensation and joint position sensation. In such cases a limb may be rendered ‘useless’ by the absence of positional awareness.

Lhermitte’s sign: with cervical posterior column involvement sudden neck flexion will evoke a ‘shock-like’ sensation in the limbs.

Spinothalamic lesions result in dysaesthesia – an unpleasant feeling of burning, coldness or warmth, with associated sensory loss to pain and temperature contralateral to the lesion.



A plaque at the posterior root entry zone will result in loss of *all* sensory modalities in that particular root distribution.

Motor symptoms

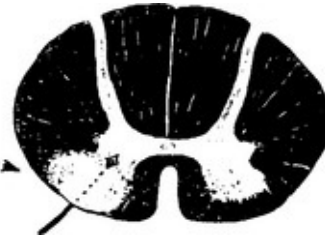
Monoparesis and paraparesis are the most common motor symptoms. Hemiparesis and quadriparesis occur less commonly.

Paraparesis is the result of spinal demyelination, usually in the cervical region.

Signs: – Increased tone

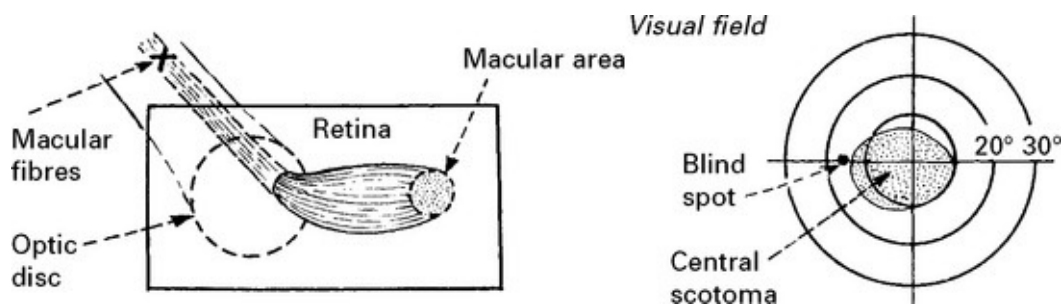
- Hyperactive tendon reflexes, extensor plantar response and absent abdominal reflexes
- Pyramidal distribution weakness.

N.B. A plaque at the anterior root exit zone may result in lower motor neuron signs (reflex loss and segmental wasting)

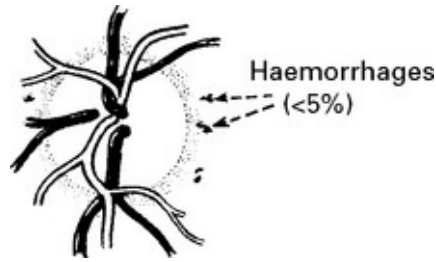


Loss of vision

Acute optic neuritis (Retrobulbar neuritis): Visual loss associated usually with a central scotoma and recovery over some weeks. This commonly occurs in young adults. The visual loss develops over several days and is often associated with pain on ocular movement (irritation of the dural membrane around the optic nerve). In milder forms, only colour vision is affected. Typically only one eye is affected, although occasionally both eyes simultaneously or consecutively are involved.



On examination: Disturbance of visual function ranges from a small central scotoma to complete loss. Fundal examination reveals swelling – papillitis – in up to 50% of patients, depending upon the proximity of the plaque to the optic nerve head.



‘Sheathing’ from an inflammatory exudate around peripheral retinal venules is common. Reduced visual acuity distinguishes papillitis from papilloedema.

Investigation: Visual evoked responses (VERs) show delay. High resolution CT or MRI of the optic nerve excludes tumour. MR confirms the presence of plaque.

Treatment: The optic neuritis study group showed that IV or oral steroids compared with placebo accelerated recovery though at 2 years there was no significant difference in eventual visual function. Oral steroids were associated with a higher risk of recurrent optic neuritis. Intravenous steroids appeared (within the next 2 years) to reduce the risk of subsequent MS.

Outcome: 90% of patients recover most vision, although symptoms may transiently return following a hot bath or physical exercise – Uhthoff’s phenomenon. Following recovery the optic disc develops an atrophic appearance with a pale ‘punched out’ temporal margin.

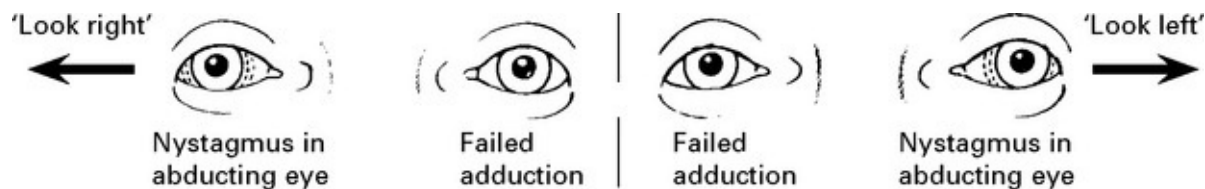
Subsequent course: The optic neuritis study group reported 12% of cases had developed clinically definite MS within 2 years (4% with a normal and 30% with an abnormal cranial MRI). Thereafter the risk is 5–6% per annum.

Acute bilateral optic neuritis: less common than unilateral disease and progression to MS not as likely. Occasionally followed by a transverse myelitis (Neuromyelitis optica, [page 529](#)). Examination of mitochondrial

DNA distinguishes from Leber's hereditary optic neuropathy ([page 551](#)).

Disturbance of ocular movement

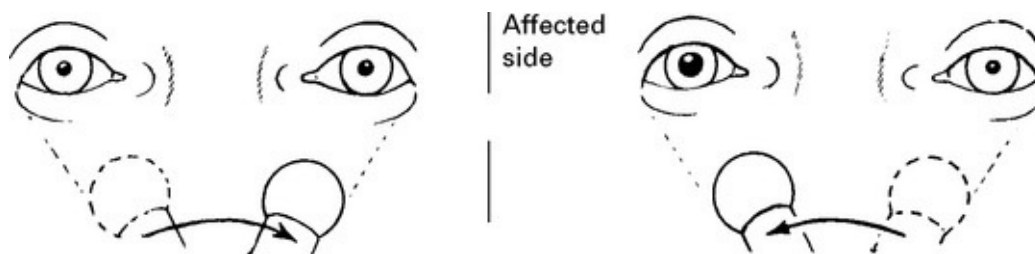
Diplopia may result from demyelination affecting the brain stem pathway of the III, IV or VI cranial nerves. Abnormality of eye movements with or without diplopia occurs when supranuclear or internuclear connection are involved. The latter results from a lesion in the medial longitudinal fasciculus – *internuclear ophthalmoplegia* (I.N.O.) – and in young persons is pathognomonic of MS.



Nystagmus may be an incidental finding on neurological examination. It is unusual in multiple sclerosis when the eyes are in the primary position, and is commonly seen on lateral gaze. Pupillary *abnormalities* may occur from:

- sympathetic involvement in the brain stem (Horner's syndrome)
- III nerve involvement, or
- II nerve involvement.

The swinging light test is a sensitive test of impaired afferent conduction in the II nerve. Alternating the light from one eye to the other results eventually in 'pupillary escape' – the pupil dilates despite the presence of direct light.



OTHER FEATURES

Vestibular symptoms: Vertigo of central type may be a presenting problem or it may develop during the course of the illness. Hearing loss is rare.

Ataxia of gait and limb inco-ordination are frequently present. The gait ataxia may be cerebellar or sensory type (see Romberg's test). Limb inco-ordination, intention tremor and dysarthria indicate cerebellar involvement.

Sphincter disturbance with urgency or precipitancy of micturition and eventual incontinence occurs. Conversely urinary retention in a young person may be the first symptom of disease. On direct questioning, impotence is frequently found.

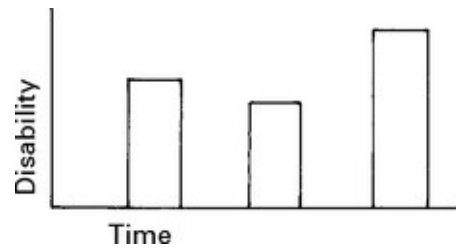
Mental changes: Mood change – euphoria or depression occur. Cognitive impairment develops in advanced cases. Generalised fatigue is common.

Emotional lability: Uncontrolled outbursts of crying or laughing, result from involvement of pseudobulbar pathways.

Paroxysmal (symptoms occurring momentarily throughout any stage of the disease): paraesthesia, dysarthria, ataxia, pain, e.g. trigeminal neuralgia, photopsia (visual scintillations), epilepsy.

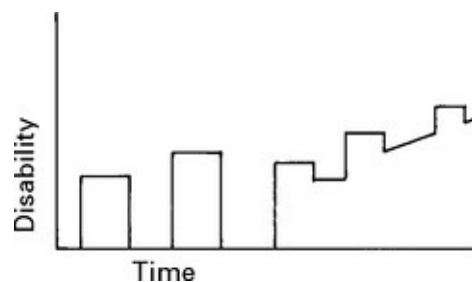
CLINICAL COURSE

The pattern of illness in individual sufferers cannot be predicted. Several different rates of disease activity and progression have been defined.

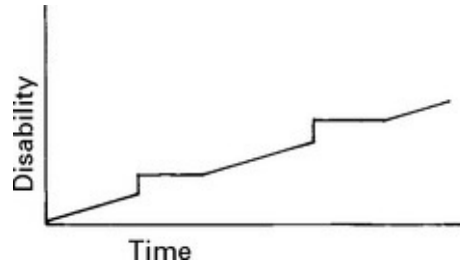


1. *Relapsing and remitting* Of all patients with MS, 70% pass through this stage. With each attack recovery is virtually complete. This phase of illness may persist for many years. The explanation of why relapses take place is unknown.

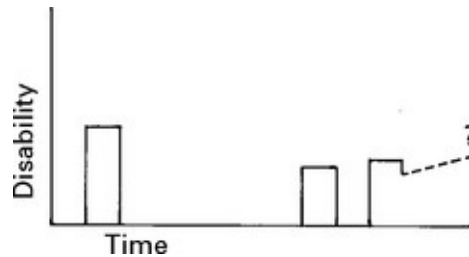
2. *Secondary progressive and relapsing/remitting secondary progressive* After a period of time, relapsing and remitting MS attacks are followed by incomplete recovery and cumulative loss of function and disability. At any one time, the chronic progressive stage accounts for 20% of all sufferers. Converting from relapsing and remitting to secondary progressive occurs on average 6–10 years after the initial symptoms.



3. *Primary progressive* This form is common in late onset MS (>45 yrs) and accounts for 15% of all patients. Symptoms and signs are usually spinal and relapses absent in the context of insidious progression.



4. *Benign* This is defined as low disability (EDSS <3) 10 years after onset. The true incidence of such cases is difficult to define and patients may still progress in time to major disability. Some support for a benign form comes from the occasional incidental autopsy finding of MS.



Recognition of different phases of MS is essential in selecting patients for new disease-modifying treatments. The degree of disability can be recorded using specific scales such as the Kurtzke score or the Extended Disability Status Score (EDSS).

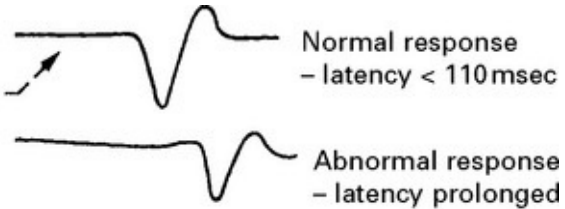
[This is a 10 point non-linear scale where 1 = no symptoms or signs, 6 = a walking aid to achieve a short distance, 8 = restricted to bed/wheelchair and 10 = death due to MS.]

INVESTIGATIONS

The development of imaging and laboratory testing has advanced diagnostic accuracy.

Neurophysiological: may detect a second asymptomatic lesion (see [page 54](#)).

1. *Visual evoked potential (VEP)*. In optic nerve involvement the latency of the large positive wave (p.102) is delayed beyond 110 msec. The amplitude of the waves may also be reduced.



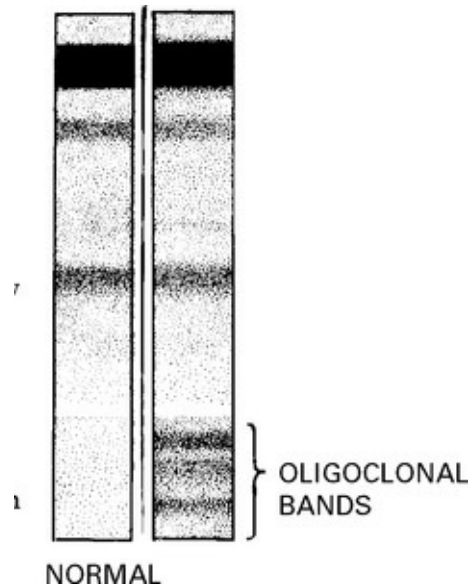
2. *Somatosensory evoked response (SSEP)* may detect central sensory pathway lesions.

3. *Brain stem auditory evoked potential (BAEP)* may detect brain stem lesions.

Cerebrospinal fluid examination by lumbar puncture

A mild pleocytosis (25 cells/mm³), mainly lymphocytes, is occasionally found. The total protein may be elevated, although this rarely exceeds 100 mg/l. An increase in gammaglobulin occurs in 50–60% of cases. Electrophoresis of CSF using agar or acrylamide shows discrete bands which are not present in serum.

These bands are found in up to 95% of patients with established disease and in 50-60% after the first attack. Oligoclonal bands (OCBs) are not specific to MS but persist indefinitely, unlike other inflammatory neurological diseases (see following page).



MRI

This has contributed enormously to the diagnosis and understanding of MS. Normal white matter appears dark with low signal intensity in T2 weighted images. Myelin breakdown produces a longer relaxation time and increased signal on T2. Gliosis produces similar changes. The presence of white matter abnormalities with a periventricular distribution is suggestive but not diagnostic of MS. Paramagnetic contrast (Gadolinium) will show active inflammation. A combination of MRI and CSF (oligoclonal band) will rule out MS if both are negative. MR may predict long term outcome – following a single episode of demyelination (e.g. optic neuritis or transverse myelitis). Those with cranial MR abnormalities will relapse sooner than those without. At present MRI does not correlate well with disability, but newer techniques may be more sensitive measures of disease progression.



Periventricular lesions, most evident at frontal and occipital horns. Abnormalities are also seen in the brain stem and cerebellum. Lesions of the optic nerves and spinal cord are more difficult to detect.

Diagnosis requires two or more episodes of symptoms attributable to demyelination, at least 30 days apart at different sites in the central nervous system, and the exclusion of alternative pathologies. Research criteria have been developed for clinical studies which combine clinical features with investigation findings. The McDonald criteria (2001) allow for MRI scan evidence of the development of new lesions to lead to a diagnosis after a single clinical episode.

Primary progressive MS can be diagnosed after 1 year of progressive deficit, with brain or spine plaques along with CSF unmatched oligoclonal bands and exclusion of alternative diagnoses.

DIFFERENTIAL DIAGNOSIS

Many conditions mimic multiple sclerosis and unless strict diagnostic criteria are adhered to other treatable disorders will be missed.

Conditions with similar clinical presentations to MS

<i>Inflammatory disorders</i>	<i>Isolated cranial disorders</i>
-------------------------------	-----------------------------------

– Systemic lupus erythematosus – Polyarteritis nodosa – Behçet's disease	– AVM – Meningioma
<i>Granulomatous disorders</i> – Sarcoidosis – Wegener's granulomatosis	<i>Miscellaneous disorders</i> – Spinocerebellar degeneration – Mitochondrial disorders – Adrenoleukodystrophy – HTLV-1 myelopathy
<i>Isolated spinal cord/foramen magnum disorders</i> – Extrinsic/intrinsic tumours – Vitamin B12 disease	– Lyme disease – Acute disseminated encephalomyelitis

Conditions with similar MRI appearances to MS

– Vasculitis – Sarcoidosis – Leukodystrophies – Acute disseminated encephalomyelitis	– Small vessel vascular disease – Decompression sickness – Lyme disease – Chronic inflammatory demyelinating polyneuropathy
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Conditions with similar CSF profile to MS (presence of oligoclonal bands)

– HIV infection – Lyme disease – Syphilis	– Chronic meningitis – Neurosarcoidosis – Subacute sclerosing panencephalitis (SSPE)
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SYMPTOMATIC

Spasticity	Drugs: baclofen (GABA derivative); dantrolene (direct action on muscle); tizanidine (α_2 adrenergic agonist); botulinum toxin
	Physiotherapy and splinting
	Intrathecal baclofen
Urinary symptoms	Detrusor instability – anticholinergics (oxybutinin, tolterodine); if severe intravesical botulinum toxin injections
	Nocturia – desmopressin spray
	Incomplete bladder emptying – intermittent self catheterisation
Bowel symptoms	Dietary manipulation; laxatives; suppositories/enemas
Pain	Analgesics; anticonvulsants; antidepressants; NSAIDs; transcutaneous electrical nerve stimulation
Paroxysmal symptoms	Seizures – anticonvulsants
Fatigue	Amantadine; modafinil
Depression	Clinical psychology; antidepressants, tricyclic or SSRI
Tremor	Betablockers; primidone; if severe deep brain stimulation
Ataxia	Walking aids; physiotherapy

Symptom management will often require a coordinated multidisciplinary approach, particularly as the disease progresses.

ACUTE RELAPSE

Methylprednisone 3 g i.v. over 3 days. Check for infection beforehand, monitor blood glucose and consider an H₂ blocker for ulcer prophylaxis. Methylprednisone can also be given orally.

MODIFY NATURAL HISTORY

Relapsing remitting MS

Betainterferon and glatiramer acetate reduce the relapse rate by about 30%. The evidence that this reduction translates into reduced disability is less clear. In the UK ambulant patients with two clinically significant relapses in the last 2 years would be eligible for treatment.

Natalizumab, a monoclonal antibody, reduces the relapse rate by over 60% with reduction in disability but is associated with a risk of developing progressive multifocal leucoencephalopathy (1 in 1000). As a result it is available only for patients with aggressive disease. Mitoxantrone is a chemotherapy agent which can be used in aggressive disease with risk of cardiotoxicity and leukemia.

A range of further agents are undergoing trials.

Primary progressive MS and secondary progressive MS

There are no effective disease modifying drugs currently available.

OTHER DEMYELINATING DISEASES

NEUROMYELITIS OPTICA (Devic's disease)

A subacute disorder characterised by simultaneous or consecutive demyelination of the optic nerves and spinal cord. Whether a distinct entity or variant of MS is uncertain. Pathologically demyelination is associated with marked cavitation and necrosis (possibly due to severe oedema confined and compressed by the pia of the optic nerves and spinal cord). Systemic lupus erythematosus, Behçet's disease and sarcoidosis produce a similar picture.

Clinical features

Visual loss is rapid, bilateral and occasionally total.

Spinal cord symptoms follow – hours, days or occasionally weeks later.

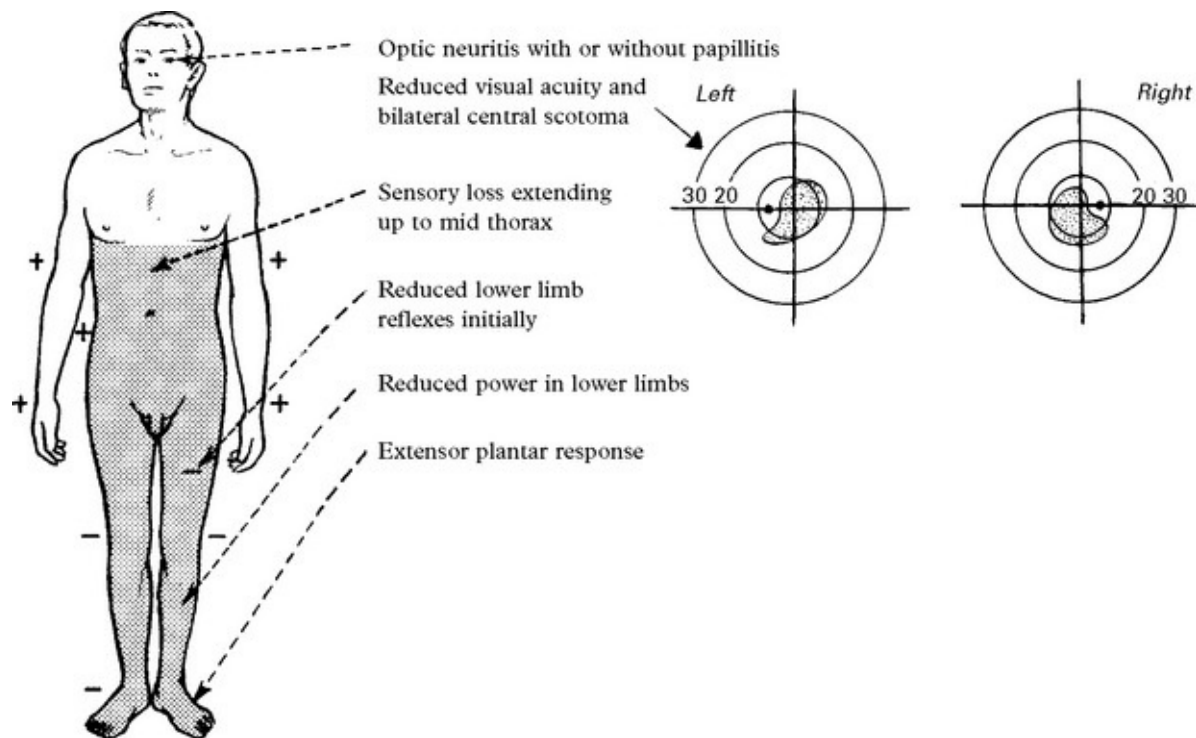
Back pain and girdle pain. Paraesthesia in lower limbs.

Paralysis may ascend to involve respiratory muscles.

Urinary retention is common.

Recovery is complete in 60–70% of patients. When recurrent attacks occur, this results in an aggressive course with a high fatality.

Examination



Investigations

Anti-aquaporin 4 antibody is positive in over 90% of patients.

Visual evoked responses are prolonged. The *CSF* shows an elevated protein with a lymphocytosis occasionally as high as 1000 cells per mm³. Gammaglobulin may be elevated and OCBs absent. MRI shows

cord swelling with enhancement over several levels.

Treatment

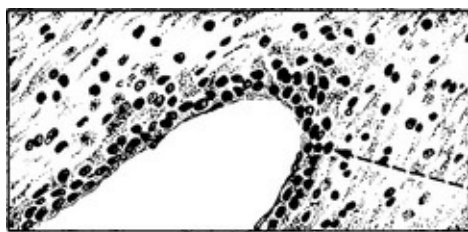
Patients may respond to i.v. methylprednisone or plasma exchange. Supportive treatment is required to minimise complications (DVT/PE, decubiti, contractures). Ventilatory support is sometimes required and may be permanent. There appears to be a case for using immunosuppressive drugs (azathioprine, cyclophosphamide) to prevent relapses though evidence is as yet limited.

TRANSVERSE MYELITIS

This occasionally occurs as the first manifestation of MS but this also occurs with viral infection (e.g. herpes virus), vasculitis and atherosclerotic vascular disease. Only 4% of patients with normal cranial MRI progress to MS. Investigations should exclude other causes of acute spinal cord syndrome – spinal cord compression – by MRI.

ACUTE DISSEMINATED ENCEPHALOMYELITIS (ADEM) (postinfectious encephalomyelitis)

ADEM is an acute immune-mediated demyelinating disorder in which small foci of demyelination with a perivenous distribution are scattered throughout the brain and spinal cord. Lesions are 0.1–1.0 mm in diameter.



Microglial, plasma cell and lymphocyte exudate around venules. Myelin becomes fragmented.

This disorder may follow upper respiratory and gastrointestinal

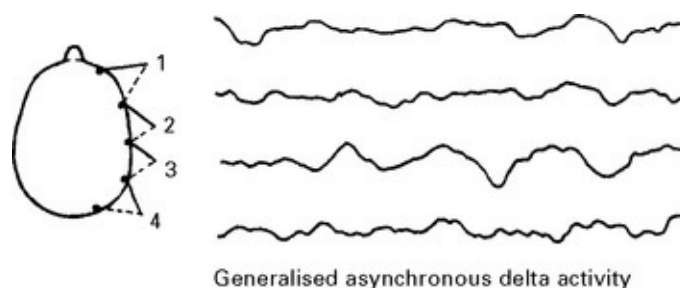
infections (viral), viral exanthems (measles, chickenpox, rubella, etc.) or immunisation with live or killed virus vaccines (influenza, rabies).

Measles is the commonest cause occurring in 1 per 1000 primary infections; next Varicella zoster (chickenpox), 1 per 2000 primary infections.

Clinical features: Within days or weeks of resolution of the viral infection, fever, headache, nausea and vomiting develop. Meningeal symptoms (neck stiffness, photophobia) are then followed by drowsiness and multifocal neurological signs and symptoms – hemisphere brain stem/cerebellar/spinal cord and optic nerve involvement. Myoclonic movements are common.

Predominantly *spinal*, *cerebral* or *cerebellar* forms occur, though usually the picture is mixed. Optic nerve involvement takes the form of optic neuritis. Rarely the peripheral nervous system is involved.

Diagnosis: No diagnosis test. CSF – 20–200 mononuclear cells. Total protein and γ globulin raised. Peripheral blood may be normal or show neutrophilia, lymphocytosis or lymphopenia.



The electroencephalogram (EEG) shows diffuse slow wave activity. CT scan is normal. MRI shows small focal white matter changes, simultaneously enhancing with contrast indicating that all are of the same degree of acuteness (unlike MS).

Diagnosis is straightforward when there is an obvious preceding viral infection or immunisation. When viral infection *immediately* precedes, distinction from acute encephalitis is often impossible.

Separation from acute MS may be difficult. Fever, meningeal signs with elevated CSF protein above 100 mg/ml with cell count greater than 50 per mm³ suggest ADEM.

Pathology: demyelination is limited to perivascular areas and lesions do not approach the same size as in MS.

Outcome: The illness is typically monophasic.

The mortality rate is 20%.

Full recovery occurs in 50%.

Poor prognosis is associated with an abrupt onset and the degree of deficit.

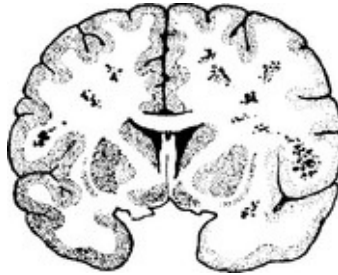
Treatment: Steroids are used, although no controlled trials have been conducted. Large dosage is recommended during the acute phase. Cyclophosphamide may be used in refractory cases.

ACUTE HAEMORRHAGIC LEUKOENCEPHALITIS

This is a rare demyelinating disease. It is regarded as a very acute form of postinfectious/acute disseminated encephalomyelitis.

Clinical picture: Antecedent viral infection, depression of conscious level and multifocal signs and symptoms. Focal features may suggest a mass lesion or even herpes simplex encephalitis.

The diagnosis is only really possible at biopsy or autopsy, but elevated CSF pressure, lymphocytosis and erythrocytes in CSF and xanthochromic appearance of fluid are all suggestive.



Pathology: Perivascular polymorph infiltration. Microscopic and macroscopic haemorrhage. Perivascular demyelination and necrotising changes in vessels.

Treatment: Steroids in high dosage should be used though evidence of value in this rare condition is scant.

PROGRESSIVE MULTIFOCAL LEUKOENCEPHALOPATHY

This is a demyelinating disease occurring in association with systemic illness in which cell-mediated and occasionally humoral immunity is depressed, *e.g.* AIDS (4% of cases), lymphoma, sarcoidosis, systemic lupus erythematosus. The disorder is due to reactivation of previous papovavirus (SV40 or JC virus) infection.

Clinical picture: Features of diffuse process – personality change, hemiparesis, cortical visual loss, seizures, *etc.* Duration of illness: 3–6 months. Non-remitting and fatal.

Pathology: Demyelination without inflammatory response, especially in subcortical white matter. Electron microscopy – papovavirus in oligodendroglia.

Diagnosis: CT scanning and MR reveal widespread multifocal white matter damage. Definitive diagnosis is made from brain biopsy. Virus can be isolated by inoculation on to glial tissue culture.

Treatment: reversal of any underlying immune deficit (for example

highly active antiretroviral therapy (HAART) in patients with AIDS) may slow progression.

LEUKODYSTROPHIES

Inborn errors of metabolism may affect the normal development of myelin. These genetic disorders usually present in infancy or childhood but occasionally produce their first manifestations in adult life.

3 specific types are recognised

- Metachromatic leukodystrophy
- Globoid cell leukodystrophy
- Adrenomyeloneuropathy or adrenoleukodystrophy (ADL).

The last condition is sex linked, characterised by adrenal insufficiency and disordered myelin in brain, spinal cord and peripheral nerve. The clinical picture is highly variable and results from a defect in beta oxidation of very long chain fatty acids (VLCFA) which build up in blood and skin fibroblasts. Dietary treatments (Lorenzo's oils) lower these and may slow progression of this fatal disorder. Heterozygote female carriers may become symptomatic with a late onset progressive myelopathy.

NEUROLOGICAL COMPLICATIONS OF DRUGS AND TOXINS

Introduction

Drugs and toxins commonly affect the nervous system. They cause a spectrum of disorders of which most are potentially reversible on withdrawal of the causal agent.

Diagnostic suspicion is especially dependent upon history:

- Availability of drugs.

- Occupational/industrial exposure to toxins.

Drug toxicity may result from:

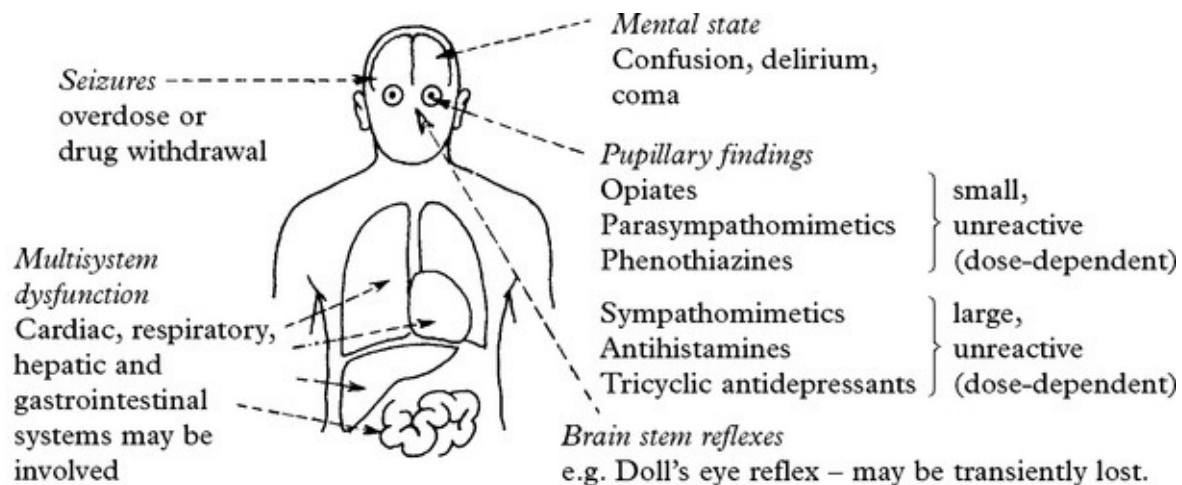
- The chronic abuse of drugs, *e.g.* barbiturates, opiates.
- The side effects of drug therapy, *e.g.* anticonvulsants, steroids.
- The wilful overdosage of drugs, *e.g.* sedatives, antidepressants.

Toxin exposure may be:

- Accidental: industrial or household poisons, *e.g.* organophosphates, carbon monoxide, turpentine.
- Wilful: solvent abuse.

History and examination

When acute intoxication is suspected, the following clinical features are supportive.



Also: Note

- Puncture marks in narcotic addicts
- The presence of a snout area rash in solvent abusers
- Rashes in barbiturate poisoning
- Respiration rate in salicylate poisoning
- Skin colour in carbon monoxide poisoning.

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- Puncture marks in narcotic addicts
- The presence of a snout area rash in solvent abusers
- Rashes in barbiturate poisoning
- Respiration rate in salicylate poisoning
- Skin colour in carbon monoxide poisoning.

Clinical features:

While the neurological picture is generally diffuse, certain pronounced symptoms occur with one drug or toxin and not with another. The following table should act as a guide to diagnosis and alert the clinician to the possible offending substance.

For treatment, the reader is advised to consult an appropriate pharmacology or general medical text.

DRUG-INDUCED NEUROLOGICAL SYNDROMES

HEADACHE

Vasodilators: antihistamines, sympathomimetics, calcium channel blockers, bronchodilators, ergotamine, cocaine

Dopamine agonists

Non-steroidal anti-inflammatories

SEIZURES

Antidepressants, antimicrobials: cycloserine, isoniazid, metronidazole, penicillin

Antineoplastics: vincristine, methotrexate.

Analgesics: pentazocine, fentanyl, opiates

Anaesthetics: ketamine, halothane, althesin

Bronchodilators. Sympathomimetics

Miscellaneous: amphetamine, baclofen, lithium, iodinated contrast media, insulin

CONFUSION/DELIRIUM

Anticholinergics. Anticonvulsants

Antimicrobials: isoniazid, rifampicin

Antineoplastics: vincristine

Dopamine agonists. Tranquillisers

Miscellaneous: cimetidine, ranitidine, lithium

PERIPHERAL NEUROPATHY

Antimicrobials: ethambutol, isoniazid, nitrofurantoin, metronidazole, dapsone

Antineoplastics: cytosine arabinoside, cisplatin, procarbazine, vincristine (and other vinca alkaloids)

Antirheumatics: colchicine, D-penicillamine, gold, indometacin

Miscellaneous: cimetidine, phenytoin

RETINOPATHY

Antimalarials: chloroquine, mepacrine

VIII NERVE DAMAGE

Aminoglycoside antibiotics: gentamicin, kanamycin, neomycin, streptomycin

Miscellaneous: cisplatin, ethyrcinic acid, quinine, salicylates

VISUAL DISTURBANCE

Phenothiazines Miscellaneous: ethambutol, indometacin, tamoxifen

OPTIC NEURITIS

Antimicrobials: chloramphenicol dapsone, isoniazid, streptomycin

Miscellaneous: chlorpropamide

MOVEMENT DISORDERS

Antiemetics: metoclopramide

Butyrophenones: haloperidol, droperidol

Dopamine agonists. Phenothiazines: chlorpromazine, thioridazine

Tricyclic antidepressants

ATAXIA

Anticonvulsants: carbamazepine, phenytoin, primidone

Antineoplastics: cytosine arabinoside, fluoracil

Phenothiazines: sedatives; barbiturates, chloral hydrate. *Tranquillisers:* diazepam

MUSCLE PAIN AND WEAKNESS

Antineoplastics: cytosine arabinoside, methotrexate, thiotepa

Statins

Miscellaneous: clofibrate, D-penicillamine, danazol, L-tryptophan.

Drug screen in suspected overdose

Too often the clinician, when managing drug or toxin overdose, requests a 'drug screen'. The techniques used in detection, *e.g.* gas chromatography, thin-layer chromatography and immunological tests, are sophisticated and time-consuming and may require samples of serum, urine or both.

The clinician must 'narrow down the field' from the history and presenting symptoms/signs and discuss with the laboratory the class of drug or toxin he suspects. A knowledge of the blood level of some drugs, *e.g.* salicylates, barbiturates, is important in deciding the approach to treatment.

SPECIFIC SYNDROMES OF DRUGS AND TOXINS

NEUROLEPTIC MALIGNANT SYNDROME

A rare life-threatening disorder induced by initiation, increase or reintroduction of neuroleptic drugs (*e.g.* chlorpromazine, haloperidol). The condition appears to result from acute dopamine receptor blockade and is characterised by *hyperpyrexia, bradykinesia, rigidity, autonomic disturbance, alteration of consciousness* and *high serum muscle enzymes (creatin kinase)*. The causal drug should be withdrawn and the patient cooled. Give dopamine agonists with dantrolene sodium to control bradykinesia and rigidity respectively. Death occurs in 15% from renal failure and/or cardiovascular collapse.

SEROTONIN SYNDROME

SSRIs can cause dystonia and occasionally low-grade fever, confusion,

autonomic disturbance, restlessness and rigidity. Early recognition and drug withdrawal is vital for good outcome.

SOLVENT ABUSE

The abuse of volatile solvents is an increasing problem especially in children. The purpose of inhalation is to achieve a state of euphoria. Habituation develops. Commonly used substances are: aerosols, cleaning fluids, nail varnish remover, lighter fluids, 'model' glue. The 'active' components of these are simple carbon-based molecules, e.g. benzene, hexane and toluene.

<i>Symptoms of acute intoxication:</i> – Euphoria – Dysarthria, ataxia, diplopia – Delusions and hallucinations occur, followed by seizures if exposure has been prolonged.	<i>Symptoms of chronic abuse:</i> – Behavioural disturbance. – Chronic ataxia. – Sensorimotor peripheral neuropathy.
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Treatment of acute intoxication is symptomatic; there are no specific antidotes.

Industrial exposure to hydrocarbons produces similar symptoms.

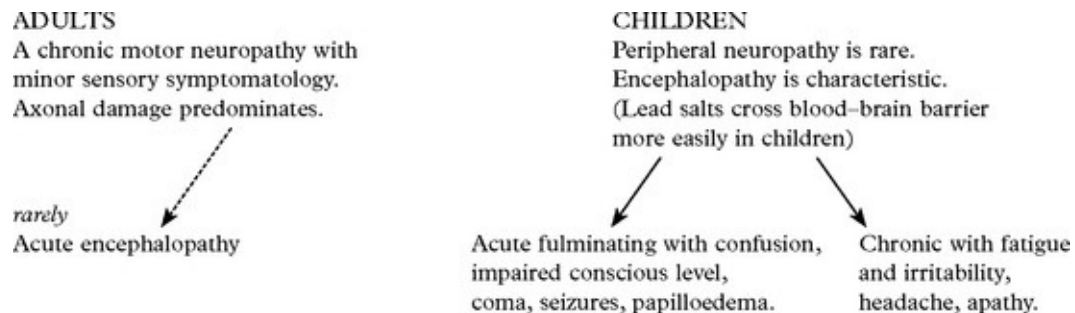
ORGANOPHOSPHATES

These are widely used as insecticides (sheep dip) and herbicides. They cause symptoms by phosphorylation of the enzyme acetyl cholinesterase. Acute intoxication produces seizures, autonomic disturbance and coma.

LEAD EXPOSURE

Lead has no biological function. It is present in normal diet as well as in the atmosphere from automobile fumes and in the water supply of old

buildings containing lead tanks and piping. Occupation exposure occurs in plumbers, burners and smelters. Lead excess interferes with *haem* synthesis. This results in the accumulation of 'blocked' metabolites such as aminolevulinic acid (ALA) in serum and urine. It also inhibits oxidative enzymes (e.g. Superoxide dismutase). Anaemia occurs with a characteristic finding in the blood film (basophilic stippling). Both the peripheral and central nervous systems are affected.



Treatment

Chelating agents (e.g. calcium disodium edetate – EDTA – or D-penicillamine) and i.v. mannitol in acute encephalopathy with papilloedema.

In acute fulminating encephalopathy the mortality has been reduced to 5%, but neurological sequelae are common.

COMPLICATIONS OF RECREATIONAL DRUG ABUSE

The problems of drug abuse are of epidemic proportions. An increasing number of neurological syndromes are recognised.

	Cocaine	Metamphetamine and Ecstasy	Heroin	Phencyclidine
<i>Origin</i>	Alkaloid from leaves of erythroxylon coca plant	Synthetic amphetamines	Alkaloid from poppy – papaver somiferin	Synthetic anaesthetic agent
<i>Clinical use</i>	Pain relief	Anorexia Narcolepsy Depression	Pain relief	Anaesthetic agent
<i>Popular name(s)</i>	'Coke', 'Snow', 'Crack' (potent pica base form)	'Speed', 'Uppers'	'Smack', 'H', 'Horse' (and many others)	'Angel dust'
<i>Method of taking</i>	Oral Intranasal Intravenous	Oral Intravenous 'high speeding'	Oral Smoked Intravenous	Oral Smoked Intranasal
<i>Mode of action</i>	Blocks reuptake of dopamine and noradrenaline and augments neurotransmission (sympathomimetic)	Increases release of dopamine and adrenaline and augments neurotransmission (sympathomimetic)	Acts as opiate receptors located on the surface of neurons	Interference with multiple neurotransmitter function
<i>Moderate dosage</i>	Alertness ↑ Euphoria Blood pressure ↑	Alert ↑ Euphoria Blood pressure ↑	Pupillary constriction Pleasurable abdominal sensation Facial flushing	Alertness ↑ Sweating Blood pressure ↑ Heart rate ↑
<i>Excessive dosage</i>	Blood pressure ↑↑ Temperature ↑ Respiration ↓ Cardiac dysrhythmia and sudden death	Blood pressure ↑↑ Temperature ↑ Respiration ↓ Cardiac dysrhythmia and sudden death	Pin-point pupils Respiration ↓ coma	Dysarthria Psychosis Nystagmus Cardiac Ataxia dysrhythmia Vigilant but and sudden unresponsive death
<i>Treatment</i>	Haloperidol (blocks dopamine reuptake) Hypotensive agents Dysrhythmic agents Anticonvulsants	As for cocaine	Naloxone (opiate antagonist) Clonidine or Methadone (for withdrawal symptoms)	Haloperidol (for psychosis)
<i>Neurological complications</i>	Headache Tremor Myoclonus Seizures	Chorea Intracranial haemorrhage (drug-induced vasculitis)	Myelitis Neuropathies and Plexopathies (immune mediated)	Dystonia Athetosis Seizures Rhabdomyolysis

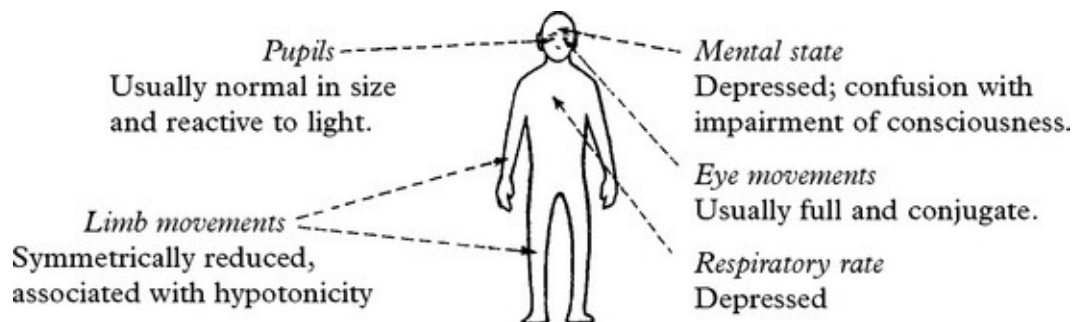
All recreational drugs are associated with increased risk of cerebral or spinal infarction or intracerebral haemorrhage. (Mechanisms are varied – drug-induced hypertension, coagulopathies, foreign body (talc) embolisation and septic emboli from infective endocarditis.)

All intravenous drug abusers are at risk of HIV infection and its

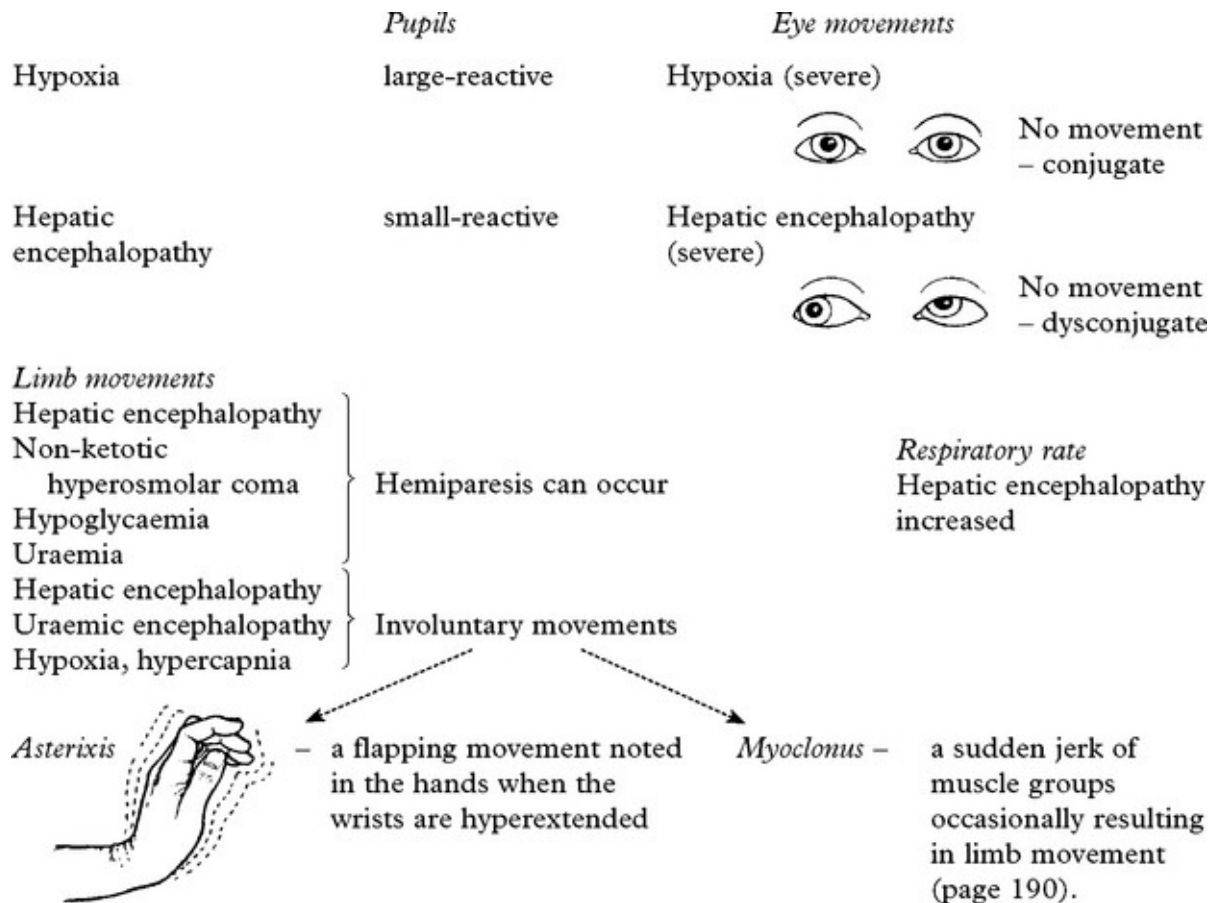
complications ([page 515](#))

METABOLIC ENCEPHALOPATHIES

In general terms, the clinical features of metabolic encephalopathy are relatively stereotyped reflecting a generalised disturbance of function of both hemispheres.



These features are characteristic but exceptions occur in specific encephalopathies –



Beware of the possibility of multiple pathology, *e.g.* an alcoholic patient with a chronic subdural haematoma may also have liver failure and thiamine deficiency.

CLASSIFICATION AND BIOCHEMICAL EVALUATION

Many metabolic disturbances cause an acquired encephalopathy in adults.

The most frequently encountered are:

– Hypoxic – Hypercapnoeic	Less commonly: – Hyponatraemia. Hypernatraemia. – Hypokalaemia. Hyperkalaemia.
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– Hypoglycaemic	– Hypocalcaemia. Hypercalcaemia.
– Hyperglycaemic	– Hypothyroidism. Lactic acidosis.
– Hepatic	– Addison's disease.
– Uraemic	

Drugs and toxins producing encephalopathy are dealt with separately (page 533).

Laboratory assessment of suspected metabolic encephalopathy

All patients should have a basic biochemical screen:

- Serum urea and electrolytes.
- Liver function (albumin, globulin, bilirubin, alkaline phosphatase and enzymes) and random blood glucose.
- Blood gases (pH, PO_2 PCO_2).
- Serum ammonia.
- Electroencephalography – slow wave activity (theta or delta) supports the diagnosis of a diffuse dysfunction: hepatic encephalopathy shows a specific triphasic slow wave configuration.
- CT scan – if the above tests are normal or coexisting structural brain disease is suspected.

Calculation of the *anion gap* may be helpful in the diagnosis of encephalopathies, especially *lactic acidosis*. The sum of the anions (Cl^- and HCO_3^-) normally equals the sum of the cations (Na^+ and K^+). An increase in the gap in the absence of ketones, salicylates and uraemia suggests lactic acidosis or ethylene glycol poisoning.

HYPOXIC ENCEPHALOPATHY

Impaired brain oxygenation results from:

- Reduced arterial oxygen pressure – lung disease.
- Reduced haemoglobin to carry oxygen – anaemia or blood loss.
- Reduced flow of blood containing oxygen (ischaemic hypoxia) – due to reduced cardiac output (with reduced cerebral blood flow).
- Biochemical block of cerebral utilisation of oxygen – rare (e.g. cyanide poisoning).

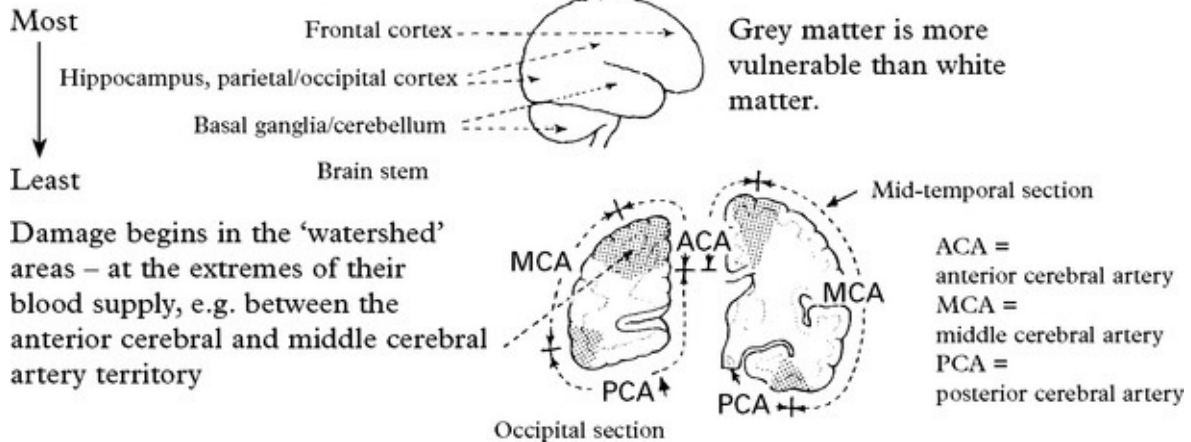
When cerebral arterial PO_2 falls below 35 mmHg (4.5 kPa), anaerobic metabolism takes over; this is not efficient and a further drop in PO_2 will result in neurological dysfunction. The extent of hypoxic damage depends not only upon the duration of hypoxia but also on other factors, e.g. body temperature – hypothermia protects against damage. The irreversibility of hypoxic damage is explained by the ‘no flow phenomenon’ – after 3–5 minutes the endothelial lining of small vessels swells – even with reversal of hypoxia, flow through these vessels is no longer possible.

SPECIFIC ENCEPHALOPATHIES

Pathology

As a consequence of high metabolic demand, some areas are more susceptible than others.

Vulnerability to hypoxia

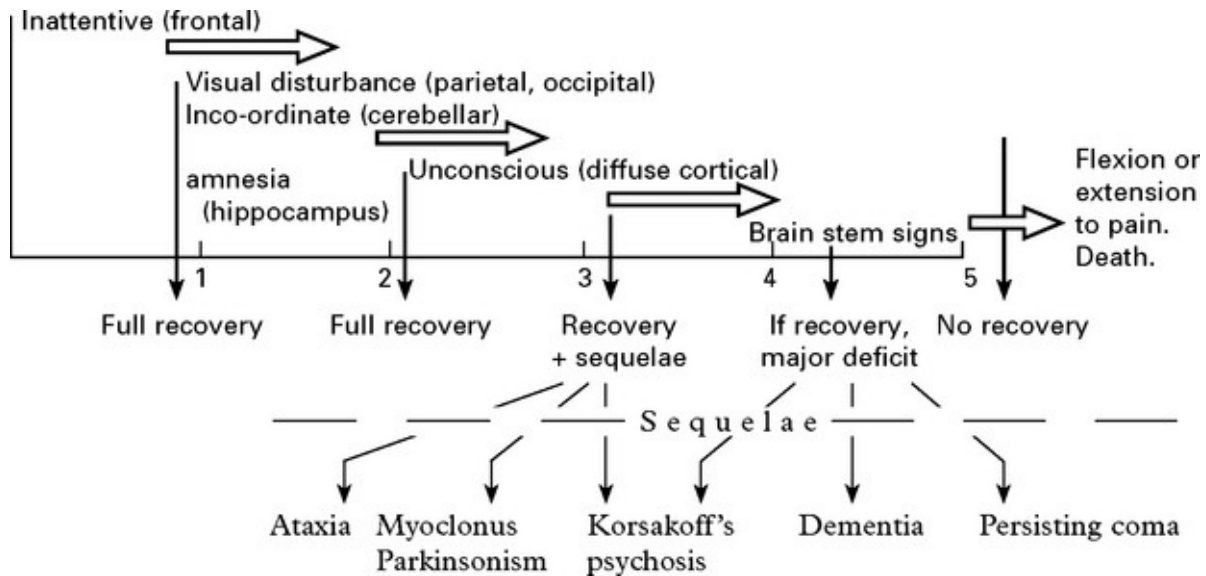


Microscopic changes depend upon the delay between the hypoxic event and death.

<i>Immediate:</i>	<i>At 48 hours:</i>	<i>At several days/weeks:</i>
Scattered petechial haemorrhages.	Cerebral oedema associated with petechial haemorrhage.	Necrosis in cortical grey matter and globus pallidus with associated astrocytic proliferation. The cerebellum and brain stem may also be affected.

Clinical features:

e.g. Severe hypoxia from circulatory arrest



Delayed hypoxic encephalopathy refers to the rare occurrence of a full clinical recovery followed after some weeks by a progressive picture → deterioration of conscious level → death. Widespread subcortical demyelination is found at autopsy.

HYPERCAPNIC ENCEPHALOPATHY: the consequence of an elevated arterial carbon dioxide level.

Clinical features:

Headache, confusion, disorientation, involuntary movements.

Papilloedema, depressed limb reflexes, extensor plantar responses.

Diagnosis: A PCO_2 greater than 50 mmHg (6 kPa) with a reduced PO_2 is found on arterial blood sampling.

The presence of headache, confusion and papilloedema may suggest intracranial tumour. If hypercapnia has not been diagnosed, such patients inevitably are referred for CT brain scan.

HYPOGLYCAEMIA ENCEPHALOPATHY: the consequence of insufficient glucose reaching the brain and may result from – overdose

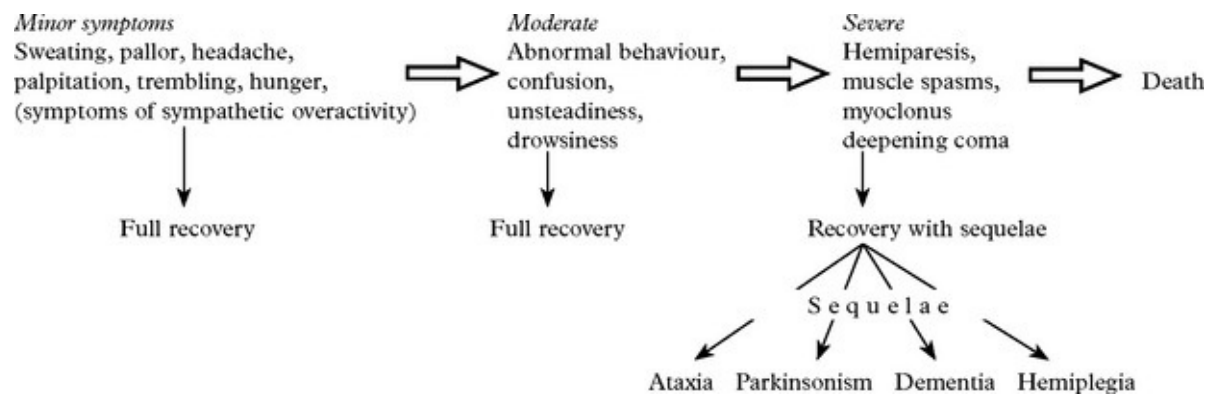
of diabetic treatment

- insulin secreting tumour – insulinoma
- hepatic disease with reduction of liver glycogen.

Serum glucose levels of 1.5 mmol/l are associated with the onset of encephalopathy. Levels at 0.5 mmol/l are associated with coma.

Pathology: Changes occur in the cerebral cortex – focal necrosis surrounded by neuronal degeneration. Subcortical grey matter (caudate nucleus) and cerebellum are vulnerable.

Clinical features: These, as with hypoxia, depend upon the duration and severity of hypoglycaemia.



Repeated mild to moderate episodes may result in a chronic cerebellar ataxia.

Repeated severe attacks may result in a mixed myelopathy/peripheral neuropathy which is distinguished from motor neuron disease by the presence of sensory signs.

HYPERGLYCAEMIC ENCEPHALOPATHY

Two types of encephalopathy develop as a consequence of hyperglycaemia:

<i>Diabetic ketoacidotic coma</i>	<i>Diabetic hyperosmolar non-ketotic coma</i>
Accumulation of acetone and ketone bodies in blood results in acidosis. Hyperventilation ensues with a reduction in $p\text{CO}_2$ and HCO_3^- . Osmotic diuresis due to hyperglycaemia results in dehydration.	This results from the hyperosmolar effect of severe hyperglycaemia. Reduction of the intracellular compartment results. Involuntary movements, seizures and hemiparesis may occur. Vascular thrombosis is not uncommon. Ketoacidosis is mild or does not occur.
The neurological presentation is that of confusion, progression to coma and, if untreated, death.	

HEPATIC ENCEPHALOPATHY

Neurological signs and symptoms secondary to hepatic dysfunction may arise in:

- acute liver failure.
- chronic liver failure complicated by infection or gastrointestinal haemorrhage.
- chronic liver failure producing characteristic *hepatocerebral degeneration*.

Clinical features:



The encephalopathy is progressive.

Pathology

Neuronal loss with gliosis is noted in the cerebral cortex as well as basal

ganglia, cerebellum and brain stem. Astrocytes with irregular and enlarged nuclei are characteristic.

Hepatocerebral degeneration (Wilson's disease) (see [page 373](#)) is an abnormality of copper metabolism and leads to a deposition of copper in the brain, particularly the basal ganglia, and the liver. This leads to a slowly progressive neurological disorder. This can vary but the predominant features include dementia, dysarthria and ataxia, primitive reflexes, choreoathetosis, myoclonus, tremor and pyramidal signs.

Consciousness is *not* impaired.

URAEMIC ENCEPHALOPATHY

Clinical features:



Pathology:

Uraemia may produce nonspecific pathological findings in the nervous system. Peripheral nervous system involvement occurs in chronic renal failure.

Dialysis encephalopathy is encountered in persons on renal dialysis exposed to high aluminium levels in the dialysate. The features are those of dementia, behavioural changes, seizures and myoclonus. The condition progresses unless aluminium levels are controlled.

Specific investigations and treatment of individual metabolic encephalopathies do not come within the scope of this book.

NUTRITIONAL DISORDERS

INTRODUCTION

Nutritional deficiency presents a massive problem in the developing world. In Western countries, alcoholism is the major cause of the neurological syndromes resulting from dietary deficiency with faddism and malabsorption disorders accounting for only a small number.

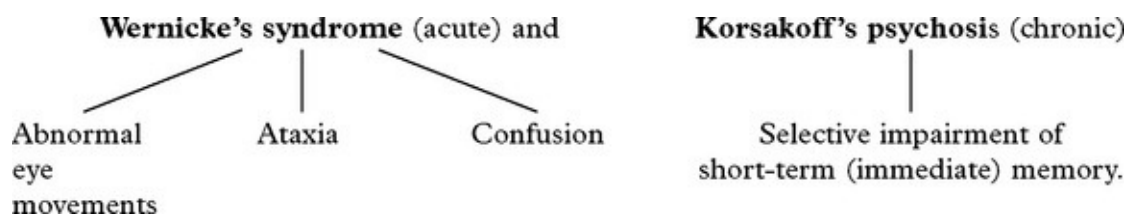
Vitamins appear important nutrients and certain disorders such as Wernicke Korsakoff syndrome (thiamine) or subacute combined degeneration (vit. B₁₂) are attributed to a single deficiency. Others such as polyneuropathy result from multiple deficiency.

Vitamin deficiency in itself does not always produce symptoms; a dietary excess of carbohydrate seems essential for the development of the neurological features of thiamine deficiency.

As a rule, nutritional disorders of the nervous system present clinically in a symmetrical manner.

WERNICKE KORSAKOFF SYNDROME

This syndrome is comprised of an acute and a chronic phase:



Patients often demonstrate additional features of nutritional deficiency – peripheral neuropathy, trophic skin changes and autonomic dysfunction (arrhythmias, postural hypotension and hypothermia). Features of acute alcohol withdrawal often coexist.

Cause

Thiamine deficiency arising from poor nutrition. Thiamine is an important coenzyme in the Krebs cycle. Deficiency results in reduced cerebral metabolism, axonal conduction, synaptic transmission and DNA synthesis.

AIDS

Hyperemesis gravidarum (continuous nausea and vomiting during pregnancy)

Thyrotoxicosis

Disseminated malignancy

Long-term dialysis

Congestive heart failure, when treated with long-term diuretic therapy

N.B. Korsakoff's psychosis may also be caused by head injury, anoxia, epilepsy, encephalitis and vascular diseases.

WERNICKE'S SYNDROME

Diagnosed in 0.1–0.4% of hospital admissions.

Pathology:

Neuronal, axonal and myelin damage occur symmetrically in the mamillary bodies, the walls of the third ventricle, thalamus and periaqueductal grey matter. Secondary vascular proliferation and haemorrhages occur within these lesions.

WERNICKE KORSAKOFF SYNDROME

Clinical features: *Acute in onset*

Ocular involvement:

Horizontal and vertical nystagmus is evident.
Unilateral or bilateral VI nerve paresis commonly occur.
Retinal haemorrhages occur.
Pupillary involvement and complete ophthalmoplegia are rare.
Polyneuropathy is present in 80% of cases.
Vestibular disturbances will occur occasionally and accentuate the ataxia.
Autonomic disturbance is common.



Confusion:

Disorientated and inattentive.
Coma may ensue
Withdrawal symptoms of alcohol:
– agitation, delusions and hallucinations – develop following admission to hospital.

Ataxia – is often the presenting symptom. May be mild or severe with inability to stand.
Lower limb (heel to shin) ataxia is modest.
Upper limbs are spared.

Investigation

- Serum B₁ levels may be low.
- Pyruvate is elevated.
- Transketolase activity is decreased (enzyme in hexose monophosphate shunt).
- MRI may show mamillary body atrophy.

Treatment

Intravenous (i.v.) Pabrinex containing thiamine (B₁), riboflavin (B₂), pyridoxine (B₆), is the only available treatment. Oral treatment is ineffective. Treatment must be given immediately to persons with a suggestive clinical picture and evidence of chronic alcohol use. Those with a history of alcohol abuse requiring IV glucose should be treated prophylactically (thiamine is critical for the metabolism of carbohydrate and levels can be exhausted by a sudden load).

With treatment – Eyes improve – in days, though nystagmus may

persist for months.

- Ataxia improves – in weeks.

Overall mortality: 15% → coma → death.

Failure to recognise and promptly treat can result in Korsakoff's syndrome or psychosis.

KORSAKOFF'S PSYCHOSIS

Sometimes encountered in traumatic or infective brain disorders, though normally overlaps with Wernicke's syndrome.

Pathology

Lesions are identical in distribution to those of Wernicke's syndrome without haemorrhages.

Clinical features

There is a disturbance of memory in which information cannot be stored. In addition the normal temporal sequence of established memories is disrupted, resulting in a semifictionalised account of the circumstances which the patient may find him/herself in (*confabulation*). This memory disturbance can only be tested for when the confusion of Wernicke's disease has cleared.

Treatment

Acute treatment is with vitamin replacement as for Wernicke's – there is no other specific treatment. The loss of short term memory causes significant disability and patients will often require closely supervised care.

B₁₂ DEFICIENCY – SUBACUTE COMBINED DEGENERATION OF THE SPINAL CORD

B₁₂ deficiency produces the specific neurological syndrome of subacute degeneration of the spinal cord (SADC). Two cobalamin-dependent enzymatic reactions occur in humans. The first reaction converts methylmalonyl-coenzyme A (CoA) to succinyl-CoA. The second involves the synthesis of methionine from homocysteine. Deficiency in B₁₂ therefore results in an accumulation of homocysteine. Despite the importance of methionine to myelin sheath phospholipid methylation, the basis of neurological damage remains uncertain.

Causes

- Inadequate diet (e.g. strict vegans)
- Increased need – pregnancy
- Defective absorption – Pernicious anaemia
 - decreased intrinsic factor (necessary for absorption)
 - malabsorption (pancreatitis, coeliac disease, gastric surgery, tapeworm infestation etc.)

Pathology

Spinal cord demyelination with eventual axon loss – affects:



posterior columns and lateral columns (corticospinal and spinocerebellar tracts). Corticospinal degeneration is most evident in the lower cord, posterior column degeneration in the upper cord. The white matter of the cerebral hemispheres can also be affected. Peripheral nerve large myelinated fibre degeneration also occurs.

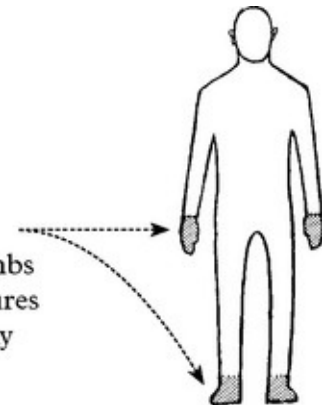
B₁₂ deficiency resulting in neurological damage is usually associated with a macrocytosis, though a normal peripheral blood film may be found.

Clinical features

Onset is subacute, though can be acutely precipitated by exposure to nitrous oxide anaesthesia.

Paraesthesia of extremities is the presenting symptom.

Numbness and distal weakness follow. Walking becomes unsteady and spasticity is evident in the lower limbs with flexor or extensor spasms. More widespread neurological features including optic atrophy, cerebral demyelination with encephalopathy and dementia develop in untreated cases.



Examination

- Gait is ataxic with positive Romberg's test (sensory ataxia).
 - Motor power is diminished distally.
 - Plantar responses are extensor.
 - Sensory loss: loss of vibration and joint position sensation in the lower limbs. Stocking/glove sensory loss is found when peripheral nerves are involved.
 - Reflex findings are variable and depend on the predominance of peripheral nerve or corticospinal tract involvement.
- Mini mental status examination (MMSE) test may suggest dementia.
- Optic pallor and centrocaecal scotoma can be demonstrated.

Diagnosis

Suspect in paraparesis with combined upper and lower motor neuron signs with 'stocking/glove' sensory loss.

Differentiate from other causes of acute myelopathy, e.g. cord compression, multiple sclerosis.

Investigation

Peripheral blood film – may show a megaloblastic anaemia

Serum B₁₂/Folate – low B₁₂. (If folate low – investigate causes: diet/drugs/malabsorption) If serum B₁₂ low (normal > 190 ng/l)

– measure intrinsic factor antibody	
– Schilling test (measure of capacity to absorb)	– if normal – dietary
	– if low – repeat with intrinsic factor (normal = pernicious anaemia, abnormal = malabsorption and investigate accordingly)

MRI – may show spinal and cerebral white matter hyperintensity on T2 images Nerve conduction studies – may show axonal neuropathy

Treatment

Consider treatment for patients who have serum B₁₂ level of less than 130 ng/l (neurological dysfunction normally occurs with levels < 100 ng/l).

Initiate treatment with vitamin B₁₂, 1000 micrograms intramuscularly given daily for 3 to 7 days, then weekly for 4 weeks.

Continue maintenance therapy for life.

Course and progression

Untreated, the disorder is progressive, the patient eventually becoming bed-bound and comatose. If diagnosed and treated early (within 2 months of onset), complete recovery can be anticipated. In established cases, only progression may be halted.

Caution: When folic acid is prescribed alone, it will improve the haematological picture of B₁₂ deficiency but cause rapid often irreversible neurological deterioration.

A clinically similar syndrome can be rarely caused by copper deficiency.

Tocopherol (vit. E).

In its active form – D α tocopherol – it acts as a membrane stabilizer and antioxidant. Deficiency occurs in chronic fat malabsorption (e.g. coeliac disease or cystic fibrosis) and results in widespread neurological disturbances – ataxia, ophthalmoplegia, seizures and corticospinal tract dysfunction. These are halted and often reversed by i.m. vit. E. It has been speculated that the antioxidant effect might make vitamin E a candidate for cytoprotection and repair within the nervous system. Studies in Parkinson's disease, multiple sclerosis and stroke are disappointing.

Abetalipoproteinaemia (Bassen-Kornzweig syndrome) predisposes to vit. E deficiency (the vitamin is transported by low-density lipoproteins). These patients have acanthocytes in the peripheral blood and a pigmentary retinopathy.

NUTRITIONAL POLYNEUROPATHY

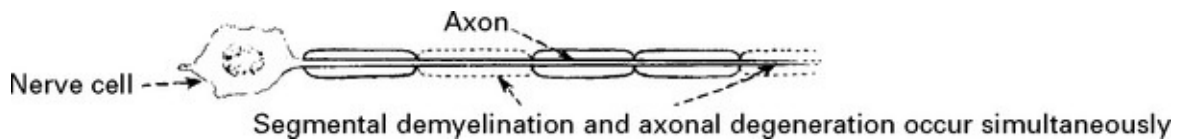
Deficiency of vitamin B complex – B₁ (Thiamine), B₂ (Riboflavin), B₃ (Nicotinic acid), B₅ (Pantothenic acid) or B₆ (Pyridoxine) – results in

peripheral nerve damage.

The combination of polyneuropathy and cardiac involvement is referred to as BERI-BERI. When oedema is also present it is termed wet beri-beri and, when absent, dry beri-beri. Beri-beri occurs in rice eating countries.

In Western countries, alcoholism is the major cause of nutritional polyneuropathy with or without cardiac involvement, otherwise world wide famine and starvation is responsible.

Pathology



The distal portions of nerves are initially affected.

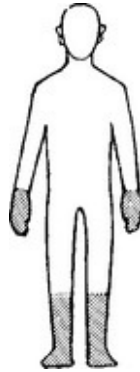
Anterior horn cells and dorsal root ganglion cells undergo chromatolysis.

Vagus nerve and sympathetic trunk involvement occurs in severe cases.

Clinical features

Symptoms: Progressive distal weakness and sensory loss with painful tingling paraesthesia involving initially lower limbs.

Autonomic complaints – impotence, dizziness (orthostatic hypotension) and disordered sweating – are common.



Signs:

- Varying degrees of *areflexia* (only ankle reflexes are lost initially).
- *Weakness* which is more marked distally than proximally and initially involves the lower limbs.
- *Sensory loss* of a 'stocking/glove' type involving all modalities of sensation.
- *Autonomic* involvement results in sweating soles of feet and postural blood pressure drop.
- *Vagus nerve* involvement results in a hoarse voice and disturbance of swallowing.

Associated signs

Shiny skin on legs with poor distal hair growth. 'Hyperpathic' painful soles of feet. Evidence of liver failure.

Diagnosis

Suggested by nutritional/alcohol history.

Supported by investigation such as peripheral blood film (elevated MCV) and disturbed liver function tests.

Nerve conduction studies reveal mildly reduced motor and sensory conduction velocities.

Differential diagnosis

Consider other causes of subacute or chronic sensorimotor neuropathy (see [page 436](#)).

Treatment

A high calorie (3000) diet should be supplemented daily with Thiamine (25 mg), Niacin (100 mg), Riboflavin (10 mg), Pantothenic acid (10 mg) and Pyridoxine (5 mg).

Burning paraesthesia may respond to gabapentin, pregabalin or carbamazepine.

Recovery may be very slow and incomplete but with the withdrawal of alcohol and adequate vitamin supplementation some improvement should occur.

TOBACCO–ALCOHOL AMBLYOPIA

A large number of toxic substances can produce impaired vision. Methyl alcohol causes sudden and permanent blindness. Chronic painless visual loss from optic neuritis develops in malnourished patients with a high tobacco consumption (Tobacco-alcohol amblyopia). This is caused by exposure to cyanide from tobacco smoking associated with low vitamin levels due to poor nutrition and absorption associated with drinking alcohol. Other potential toxins include methyl alcohol (moonshine) and ethylene glycol (antifreeze).

Pathology

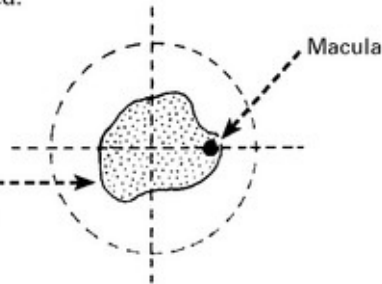
Damage involves the papillomacular bundle within the optic nerves, chiasma and optic tracts. Retinal ganglion cells in the macular region are also affected.

Clinical features

- The condition slowly develops over weeks.
- Vision in each eye becomes hazy and blurred.
- Colour vision (red/green discrimination) is involved early.

Examination

- Bilateral involvement.
- Reduced visual acuity.
- Centrocaecal scotoma (a central field defect spreading from blind spot to macula and most easily detected with a red target).
- Fundal examination is normal, though optic atrophy will occur eventually.
- Coexistent Wernicke Korsakoff syndrome or polyneuropathy are common.

**Treatment**

Vitamin B supplementation, including B₁₂, should be administered. B₁₂ possesses the capacity for cyanide detoxification. Recovery is poor if visual loss is well established.

ALCOHOL RELATED DISORDERS

ALCOHOL MYOPATHY

Muscle damage (elevated creatine phosphokinase) is not uncommon in alcoholics following acute ingestion. The cause of alcoholic myopathy is uncertain; mitochondrial disturbances, potassium depletion, rhabdomyolysis (due to seizures or local compression) have all been suggested.

There are two forms of alcoholic muscle disease

1. *Acute necrotizing myopathy* occurs after 'binge' drinking.

- Acute muscle necrosis ensues with pain/cramping and muscle tenderness/swelling.
- Myoglobin is excreted in the urine (myoglobinuria) after release from damaged muscles.
- Symptoms of alcohol withdrawal – delirium, *etc.* – coexist.
- Limb involvement may be markedly asymmetrical.
- Sometimes calf muscles are swollen and tender.
- Improvement occurs over weeks to months.
- Serum creatine phosphokinase (CPK) is elevated. Marked

myoglobinuria when present may result in renal failure.

– Elevated serum K^+ may provide cardiac arrhythmias.

2. *Chronic myopathy* Painless proximal weakness sometimes associated with cardiomyopathy. Muscle biopsy showing type 2 fibre atrophy.

Management is abstinence, vitamin supplementation and IV saline in *acute necrotizing myopathy* with myoglobinuria to prevent renal failure.

ALCOHOLIC DEMENTIA

Experimentally, chronic alcohol consumption results in neuronal loss. CT evidence of atrophy and neuropsychological impairment is common in alcoholics. However, whether or not these result from the direct toxic and dementing effect of alcohol remains uncertain.

ALCOHOLIC CEREBELLAR DEGENERATION

Probably the commonest cause of acquired ataxia, alcoholic patients may develop a chronic cerebellar syndrome either as a sequel of Wernicke's syndrome or as a distinct clinical entity. A long history of alcohol abuse is obtained. Males are predominantly affected. Onset is gradual and symptoms often stabilise. Ataxia of gait with lower limb inco-ordination predominates. The upper limbs are spared. Nystagmus is rarely present. Cerebellar dysarthria is usually mild. Coexistent signs of peripheral neuropathy are often found.

Investigations:	– Abnormal liver function tests <i>e.g.</i> elevation of enzymes – γ GT.
	– Macrocytosis in peripheral blood film.
	– CSF examination normal.
	– CT and MRI reveal cerebellar vermal atrophy.

Progression 	may evolve rapidly and reverse with improved nutrition and alcohol withdrawal.
	may evolve subacutely.
	may evolve chronically and slowly progress over many years.
Pathology:	– All the cellular elements of the cerebellar cortex are affected, but particularly Purkinje cells of the anterior and superior vermis and the anterior portion of the anterior lobes.
Pathogenesis:	– The disorder may be due to <i>nutritional deficiency</i>, especially thiamine, or else result from the direct toxic effect of alcohol or electrolyte disturbance on the cerebellum.
Differential diagnosis:	– Distinguish from hereditary and other acquired ataxias, e.g. hypothyroidism.
Treatment:	– Alcohol withdrawal, a well balanced diet and adequate vitamin supplementation.

CENTRAL PONTINE MYELINOLYSIS

Alcohol abuse, debilitating disease or rapid correction of hyponatraemia may precipitate presentation.

The lesion is one of demyelination with cavitation. Microscopically, myelin is lost, oligodendrocytes degenerate but neurons and axons are spared.

Clinically, an acute or subacute pontine lesion is suspected, evolving over a few days, with bulbar weakness and tetraparesis (locked-in syndrome).

The limbs are flaccid with extensor plantar responses.

With progression of the lesion, eye signs become evident and conscious level becomes depressed → coma → death.

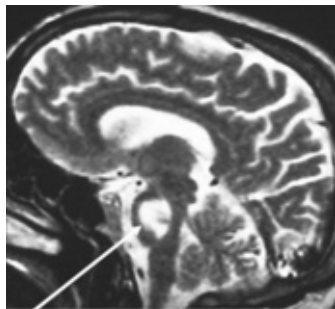
Investigations:

Electrolytic disturbances (low sodium, low phosphate) are found.

Liver function is normal. CSF examination is normal.

MRI is more sensitive than CT showing an abnormality in the pons.

Recognition of this condition before death is important in view of its reversibility, though prior to CT/MRI availability it was diagnosed at autopsy. Vigorous supportive therapy with correction of metabolic abnormalities and vitamin supplementation is advised. In patients with severe hyponatraemia (< 110 mmol/l), especially alcoholics, slow correction is essential.



MRI (T2) – increased signal filling the pons

CORPUS CALLOSUM DEMYELINATION

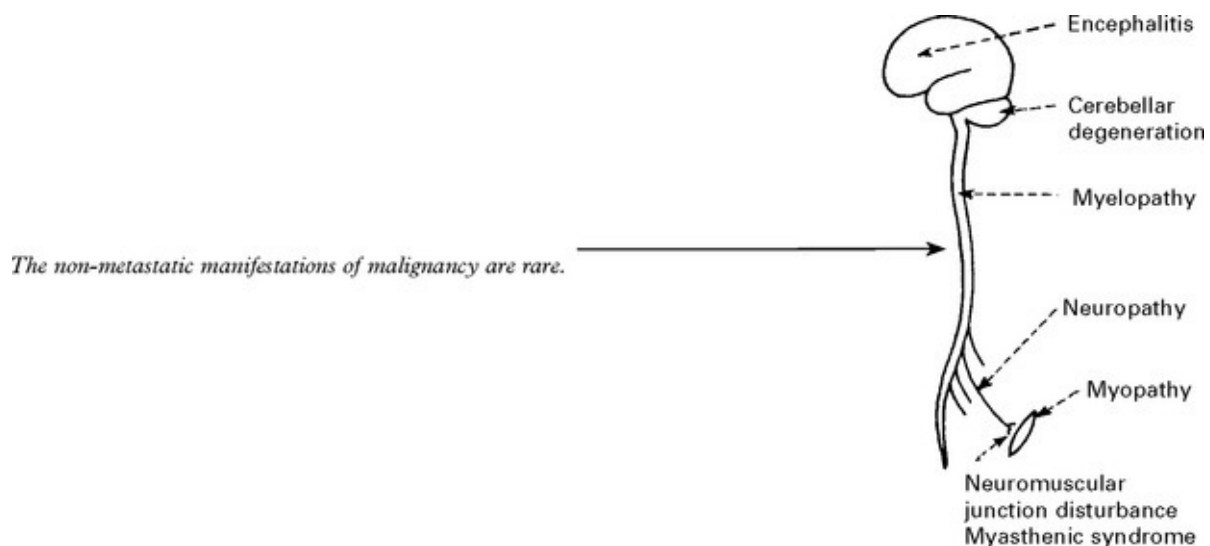
(syn: Marchiafava–Bignami disease)

This is a rare disorder occurring in malnourished alcoholics. Occasionally diagnosed premortem by MRI, progressing to death over some weeks. The clinical picture is that of personality change with signs of frontal lobe disease. The condition occurs most commonly in persons of Italian origin.

NON-METASTATIC MANIFESTATIONS OF MALIGNANT DISEASE

Disturbance of neurological function can occur in association with malignancy without evidence of metastases (0.1% of all cancer patients). Brain, spinal cord, peripheral nerve and muscle may be affected, either separately or in combination.

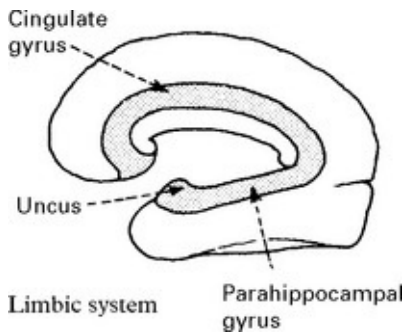
Small cell carcinoma of the lung, gynaecological malignancy and lymphoma are the commonest associated disorders. Specific antibodies (anti-neuronal), are responsible for certain syndromes. These are directed towards antigens in the nervous system and the tumour and may explain the trend toward greater life expectancy in those with, rather than those without, such non-metastatic disorders.



These are not discreet, *e.g.* neuropathy and myopathy may coexist → carcinomatous neuromyopathy; encephalitis and myelopathy → carcinomatous encephalomyelitis.

LIMBIC ENCEPHALITIS

Associated commonly with small cell lung cancer (SCLC) usually *before* this becomes clinically manifest.



Pathology

The encephalitic process selectively affects the limbic system – with neuronal loss, astrocytic proliferation and perivascular inflammatory changes.

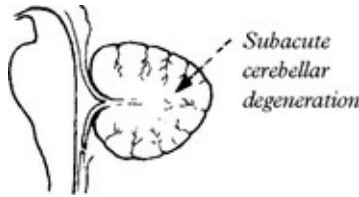
Clinical features

Disturbance in behaviour precedes the development of complex partial (temporal lobe) seizures and memory impairment. Autonomic dysfunction and sensory neuropathy often coexist. Progression is rapid.

Investigations.

Anti-voltage gated potassium channel antibodies (anti-VGKA) or anti-Hu antibodies are the most commonly found antibodies. Recently anti-NMDA receptor antibodies have been demonstrated in some patients with limbic encephalitis and prominent extrapyramidal movements. MRI may show temporal lobe abnormalities. EEG may show temporal lobe abnormalities. CSF reveals a mild lymphocytosis with protein elevation.

CEREBELLAR DEGENERATION (anti-Yo syndrome) associated with breast or ovarian carcinoma.



Pathology:

Characterised by Purkinje cell loss with some involvement of the dentate nucleus. Brain stem changes also occur.

Clinical features:

The patient presents with a rapidly developing ataxia. Brain stem involvement results in nystagmus, opsoclonus and vertigo. The course is usually rapid.

Investigations

MRI shows cortical and vermal cerebellar atrophy.

CSF is mildly abnormal and anti-Yo antibodies are present in 50% of suspected cases.

NEUROPATHY (see [page 430](#))

Sensory neuropathy: Destruction of the posterior root ganglion combined with axonal and demyelinating peripheral nerve damage causes progressive sensory symptoms. The neuropathy is subacute or chronic in evolution. Clinically dysaesthesia and numbness starts in extremities and spreads. Associated with SCLC and anti-Hu antibodies.

Sensorimotor neuropathy: A mixed neuropathy with weakness and sensory loss. The syndrome may predate the recognition of the underlying neoplasm. Rate of progression is slow and predominantly motor forms may be mistaken for ALS ([page 555](#)) associated with Hodgkin's and other lymphomas.

Rarely an acute neuropathy indistinguishable from postinfectious polyneuropathy occurs.

NECROTISING MYELOPATHY:

Flaccid paraplegia develops subacutely. Spinal MRI may show a swollen cord. Mechanism is uncertain.

MYOPATHY

Muscle weakness in malignancy takes several forms.

Proximal myopathy: A slowly progressive syndrome with weakness of proximal limb muscles.

Inflammatory myopathy (polymyositis/dermatomyositis) (see [page 474](#)):

The overall incidence of associated neoplasm in inflammatory myopathy is 15%. The typical patient is in middle age with a proximal weakness, elevated ESR and muscle enzymes with or without the skin features of dermatomyositis.

Myopathy with endocrine disturbance: Ectopic hormone production (by malignant cells) may induce a myopathy characterised by chronic progressive proximal weakness, e.g. ectopic ACTH production from small cell carcinoma of lung.

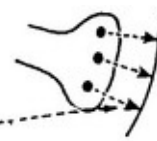
Cachetic myopathy occurs in terminally ill, wasted patients.

Investigation and treatment of non-metastatic syndromes

Successful treatment of the underlying tumour offers the only hope of improvement. The search must be exhaustive and repeated where first negative. Tumour markers (AFP, CEA, PSA etc), chest and abdominal CT, pelvic ultrasound, mammography are advised with PET (FDG) if available. Treatment with steroids, immunosuppressants (AZT, cyclosporine, etc), IVIG or plasma exchange is of uncertain benefit.

THE MYASTHENIC SYNDROME (Lambert-Eaton syndrome)

An autoimmune disorder of the neuromuscular junction. IgG voltage-gated calcium channel antibodies (VGCCAs) block the cholinergic synapse resulting in reduced acetylcholine release. The autonomic synapses are also affected.



In men the association with underlying malignancy warrants detailed investigation (see above) though a proportion of patients have no evidence of this.

Clinical features

The patient develops weakness of lower then upper limbs with a tendency to fatigue. Following brief exercise, power may paradoxically suddenly improve – second wind phenomenon. In contrast to myasthenia gravis ocular and bulbar muscles are rarely affected. Examination reveals a proximal pattern of wasting and weakness with diminished tendon reflexes. Up to 50% of patients experience symptoms of autonomic (cholinergic) dysfunction – impotence, dry mouth and visual disturbance.

Diagnosis

Confirmed electrophysiologically; the ‘second wind phenomenon’ is shown up as an incrementing response to repetitive nerve stimulation (as opposed to the decrementing response in myasthenia gravis, [page 485](#)). VGCCAs are detected in serum.

Treatment

3,4-diaminopyridine and pyridostigmine can improve symptoms. Immunosuppression with steroids, plasmapheresis or IVIG can suppress the underlying immunological abnormality.

This syndrome may respond to the removal of the underlying neoplasm if present.

DEGENERATIVE DISORDERS

Introduction

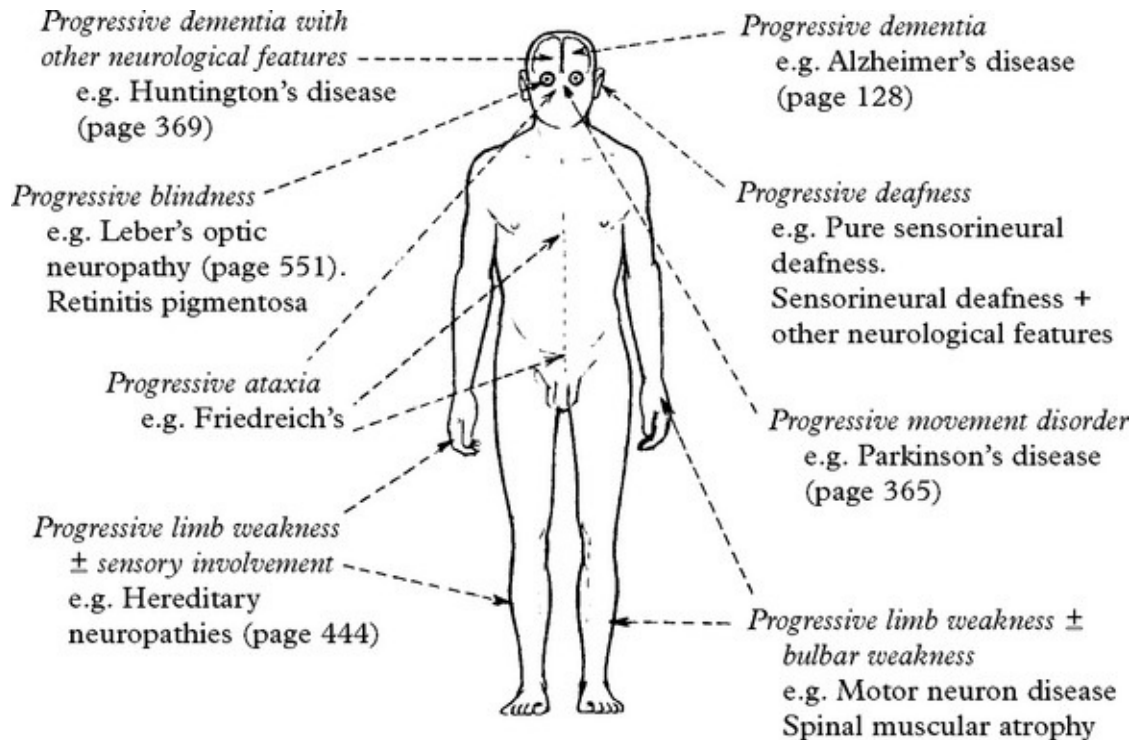
This heterogeneous group of neurological diseases characterised by selective neuronal loss, is grouped together by the lack of known aetiology. As causes of such disease are identified (e.g. metabolic, viral) they have been reclassified in their appropriate category. Of the remaining conditions many are age related or familial and in some there is an identifiable genetic basis.

Characteristically these disorders:

- are gradually progressive
- are symmetrical (bilateral symptoms and signs)
- may affect one or several specific systems of the nervous system
- may demonstrate a specific pathology or just show neuronal atrophy and eventual loss without other features.

Classification

Degenerative disorders are classified according to the specific part or parts of the central/peripheral nervous system affected and according to the ensuing clinical manifestations. These degenerative disorders may be alternatively termed the system degenerations because of their propensity to affect only part of the nervous system.



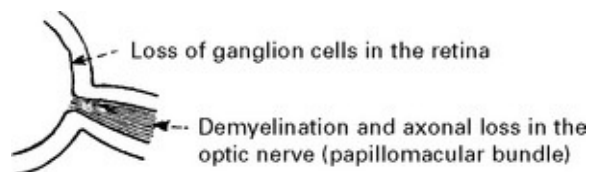
Most of these conditions are discussed in other chapters.

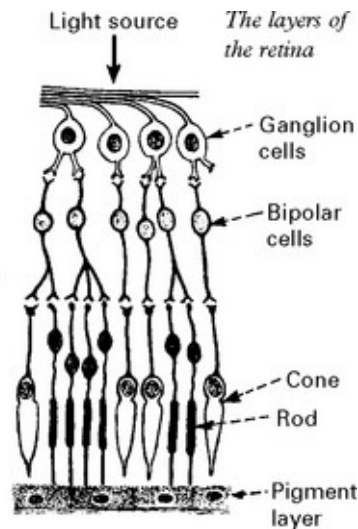
PROGRESSIVE BLINDNESS

LEBER'S HEREDITARY OPTIC NEUROPATHY (LHON)

Leber's optic neuropathy is a familial disorder of maternal inheritance with a tendency to affect males significantly more than females. It is classified as a mitochondrial disorder due to DNA mutation ([page 481](#)). Most individuals have one of three point mutations of mitochondrial DNA (mtDNA).

Pathology





Clinical features

Onset of visual loss in late teens/early twenties.

- Both eyes are simultaneously affected (rarely one eye months before the other).
- Central vision is lost with large bilateral scotomata.

Characteristically, blue/yellow colour discrimination is affected before red/green. The optic disc initially appears pink and swollen with an increase in small vessels, eventually becoming pale and atrophic.

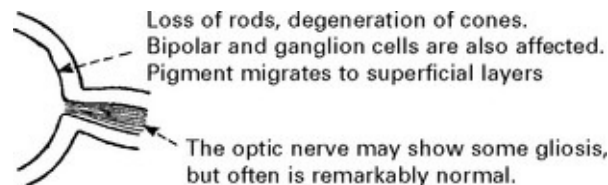
Visual impairment progresses with peripheral constriction of the fields. Complete visual loss seldom occurs.

Associated symptoms and signs of a more generalised nervous system disorder occur in a proportion of cases – dementia, ataxia, progressive spastic paraplegia – and confusion with multiple sclerosis may arise. In contrast to bilateral optic neuritis, 'leakage' occurs with fluorescein angiography. Genetic counselling for LHON is complicated by the sex and age-dependent penetrance. The mother of an affected male has the mitochondrial mutation and may or may not have symptoms. No treatment exists. Quinone analogues (ubiquinone and idebenone) may

help during periods of rapid visual worsening.

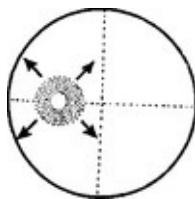
RETINITIS PIGMENTOSA

A heterogeneous hereditary disorder of the retina which may be inherited as an autosomal dominant, recessive or X-linked disorder. All layers of the retina are affected. Posterior pole cataracts and glaucoma are occasionally associated.



Clinical features

Onset of visual loss in childhood. Both eyes are simultaneously affected. Initially there is a failure of twilight vision. The patient has difficulty in making his/her way as darkness falls (nyctalopia). The retina around the macular area is first affected resulting in a characteristic ring scotoma. This gradually spreads outwards; eventually only a small 'tunnel' of central vision is left. Finally, complete blindness occurs. The majority of patients are completely blind by 50 years of age. The fundal appearance is diagnostic as a result of the superficial migration of pigment.



The *electroretinogram* – recording the electrical activity of the retina – is eventually lost.

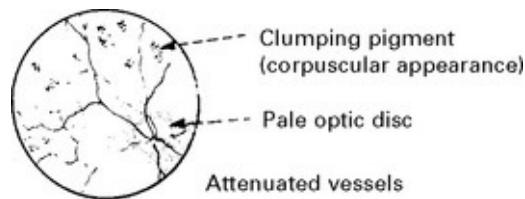
Treatment

None. Vitamins and steroids have been tried unsuccessfully.

Associated conditions in retinitis pigmentosa

Several conditions are associated with retinitis pigmentosa:

- Hypogonadism/obesity/mental deficiency
- Spinocerebellar ataxis
- Laurence Moon syndrome
- Friedreich's ataxia



The association with neuropathy and ataxia (NARP), or progressive external ophthalmoplegia and heart block (Kearns-Sayre syndrome) are due to mitochondrial disease ([page 481](#)).

PROGRESSIVE ATAXIA

The degenerative disorders manifested by progressive ataxia are termed *spinocerebellar-ataxias*.

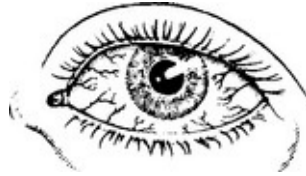
These may be classified by age of onset, presence of associated features, but increasingly by mode of inheritance.

RECESSIVELY INHERITED ATAXIAS

ATAXIA TELANGIECTASIA (Louis-Barr Syndrome)

This multisystem disorder is characterised by progressive cerebellar

ataxia, ocular and cutaneous telangiectasia and immunodeficiency.



The gene maps to chromosome 11q23 associated with mutations in the ATM gene. The ATM protein is a member of the family of proteins involved in DNA repair.

Pathologically, widespread cerebellar Purkinje and granular cell loss occurs.

A progressive ataxia develops in infancy. Telangiectasia develops later, becoming more obvious after exposure to the sun. Prevalence similar to Friedreich's ataxia.

Patients are eventually confined to a wheelchair and, because of associated low serum immunoglobulin levels are susceptible to repetitive infections.

Malignant neoplasms (lymphoreticular tumours) occur in 10%.

Patients are unusually sensitive to X-rays. Treatment of malignancy with conventional dosages of radiation can prove fatal.

Death occurs in second or third decade from infection or malignancy (often lymphoma).

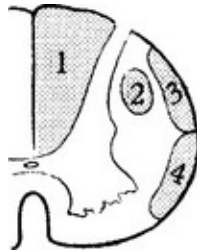
FRIEDREICH'S ATAXIA

Friedreich's ataxia is the commonest inherited ataxia with an incidence of 1/50000 in European populations and carrier frequency of 1/20.

It is caused by mutations in the FRDA gene located on chromosome 9 which encodes the protein Frataxin. It is the first autosomal recessive

disease identified in which a triplet repeat expansion (GAA) is responsible.

Pathology:



Spinal: The spinal cord is shrunken, especially in the thoracic region.

There is degeneration, demyelination and gliosis of:

1. – Posterior columns.
2. – Corticospinal tracts
3. – Dorsal spinocerebellar tracts
4. – Ventral spinocerebellar tracts.

Dorsal roots and peripheral nerves are shrunken in advanced cases.

Cerebellar: Changes in the cerebellum are less marked, there is Purkinje cell loss and atrophy of the dentate nucleus.

Peripheral nerves show loss of large myelinated axons and segmental demyelination. The corticobulbar tract and cerebrum are relatively spared.

RECESSIVELY INHERITED ATAXIA

FRIEDREICH'S ATAXIA

Clinical features

Friedreich's ataxia is characterised by progressive gait ataxia and limb incoordination, hypertrophic cardiomyopathy and increased incidence of diabetes mellitus/impaired glucose tolerance.

Sexes are equally affected. Onset of symptoms is normally around puberty, and always before 25 years of age; most patients become wheelchair bound by their late twenties. Cardio-pulmonary failure is the usual common cause of death.

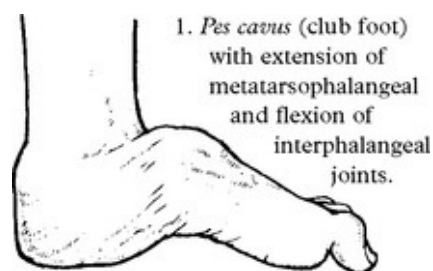
Disturbance of balance is the initial symptom, often associated with the development of scoliosis. A spastic, ataxic gait develops with incoordination of the limbs.

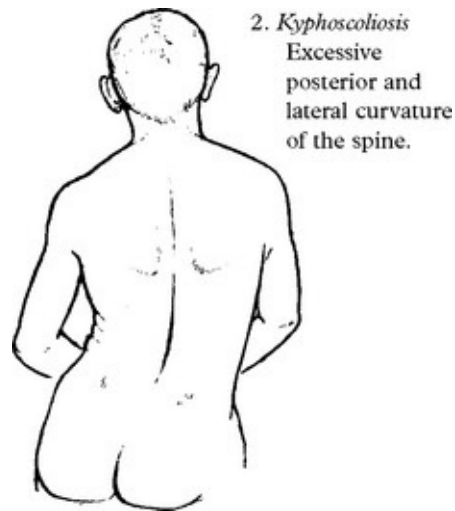
Corticospinal tract involvement results in limb weakness with absent abdominal reflexes and extensor plantar responses.

Posterior column involvement results in loss of vibration and proprioception in the extremities.

Dorsal root and peripheral nerve involvement results in absent lower limb reflexes.

Involvement of myocardial muscle (cardiomyopathy) is common and results in cardiac failure or dysrhythmias. *Musculoskeletal abnormalities* occur in 80% of cases.





Optic atrophy and deafness coexist in many cases.

There is a clinical resemblance to mitochondrial encephalopathies as well as reduced respiratory enzyme activities in some patients (Friedreich's has been suspected to involve some degree of disturbance of mitochondrial respiration).

Investigation

Identification of the gene and availability of diagnostic testing has limited the value of other ancillary investigations such as imaging and neurophysiology. Regular cardiac assessment and monitoring of blood glucose is important.

Treatment

Although there is no specific treatment for Friedreich's ataxia, many of its symptoms can be managed. Orthopaedic intervention can alleviate scoliosis, and orthopaedic appliances and physical therapy help maintain ambulation. Cardiac problems can be successfully treated pharmacologically and insulin therapy may be necessary to control diabetes mellitus.

Other causes of areflexic ataxia

Abetalipoproteinaemia (Bassen Kornzweig disease)

- Malabsorption syndrome
- Acanthocytes (thorn-shaped red blood cells)
- Low serum cholesterol, triglycerides and fatty acids
- Low vitamin E.

Hexosaminidase deficiency

- Accumulation of GM₂ gangliosides in brain and skin.

Xeroderma pigmentosum

- Sensitive to ultraviolet light
- Keratosis and skin cancer.

DOMINANTLY INHERITED AND OTHER ATAXIAS

Classification of the dominantly inherited, late-onset, cerebellar ataxias is complex and controversial. The term 'late-onset' is misleading given that these disorders may present in childhood and adolescence. Commonly other neurological features coexist: ophthalmoplegia, optic atrophy, retinal pigmentation, deafness, dysarthria, dysphagia, dementia, extra pyramidal and pyramidal signs and peripheral neuropathy. This bewildering condition is classified into 3 different clinical phenotypes.

	Autosomal dominant Cerebellar ataxia (ADCA) Type 1	Autosomal dominant Cerebellar ataxia (ADCA) Type 2	Autosomal dominant Cerebellar ataxia (ADCA) Type 3
Clinical features	Ataxia ± Ophthalmoplegia Mild dementia Optic atrophy Spasticity	Ataxia + Retinopathy (progressive visual loss) ± Dementia – extrapyramidal features	Ataxia (alone) – Age of onset > 50 years

Many different gene loci have been reported to be responsible – the spinocerebellar ataxia or SCA mutations. SCA1, SCA2, SCA3 (also known as Machado-Joseph disease), cause ADCA type 1, SCA7 causes ADCA type 2, SCA4, SCA5, SCA6 and SCA11 cause ADCA type 3, though there remains considerable phenotypic variation even within families. Causative genes have been identified as expansions of trinucleotide CAG repeat for SCA1, SCA2, SCA3, SCA6, SCA7, and SCA12, and the CTG repeat for SCA8. DNA testing is diagnostic though new loci remain to be discovered.

IDIOPATHIC LATE ONSET ATAXIA

Some may be new mutations of ADCA. For diagnosis all other causes of acquired ataxia – inflammatory, infective, nutritional, metabolic, endocrine and non-metastatic – must be excluded by appropriate investigations.

Type 1 – Age of onset 35–55 years – ataxia ± dementia, spasticity
Type 2 – Age of onset > 55 years – mid-line ataxia sparing speech/limbs
Type 3 – Age of onset 50–60 years – ataxia, titubation and tremor

THE HEREDITARY INTERMITTENT ATAXIAS

These disorders are characterized by brief paroxysmal episodes with no neurological impairment between attacks. Two types can be

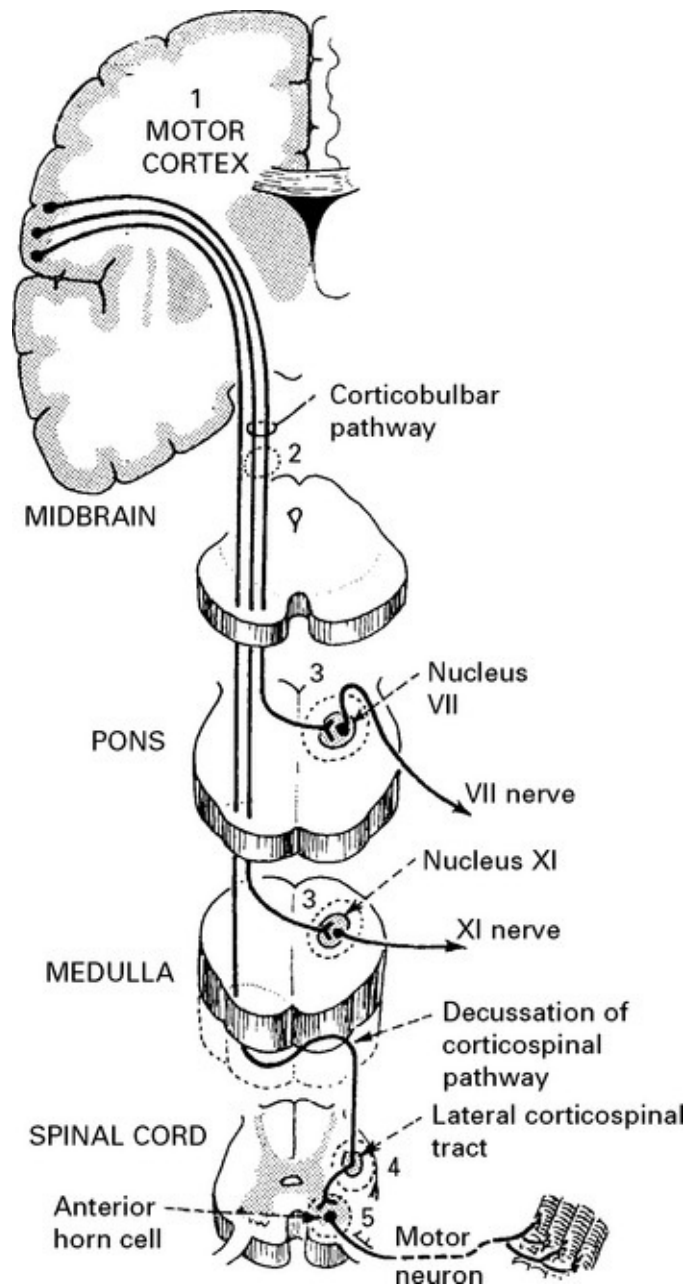
distinguished on the basis of the length of the attacks, the presence of myokymia (facial twitching), precipitating factors, response to acetazolamide and the nature of the genetic defect.

Type 1, attacks are precipitated by sudden movements, emotional stress, fatigue, exercise, or hunger. Stiffness, generalized myokymia, vertigo, nausea, diplopia and tremor also occur. The attacks last 10 minutes or less. This disorder is associated with a variety of point mutations in the voltage-gated potassium channel gene, KCNA1, located on chromosome 12p13.

Type 2, myokymia is absent, the prominent symptoms being ataxia of gait and limbs, dysarthria, and gaze-evoked nystagmus. The attacks begin abruptly and last from 15 minutes to a few hours though sometimes days. Emotional stress, physical exertion, but not movement trigger attacks. The carbonic anhydrase inhibitor acetazolamide is very effective in preventing attacks. This disorder is associated with mutations in CACNL1A4 (subunit of a voltage-gated calcium channel gene) located on chromosome 19p. SCA-6 is also associated with a small expansion of CAG repeats in this gene as is familial hemiplegic migraine, a condition sharing similar features.

MOTOR NEURON DISEASE/AMYOTROPHIC LATERAL SCLEROSIS (ALS)

Motor neuron disease (MND) is a progressive condition characterised by degeneration of upper and lower motor neurons.



Different levels of the nervous system are involved:

1. Frontal atrophy in the precentral gyrus
2. The corticobulbar pathway
3. The cranial nerve nuclei

4. The corticospinal tract

5. The anterior horn cell

The term AMYOTROPHIC LATERAL SCLEROSIS (ALS) is used synonymously with motor neuron disease.

Epidemiology

Incidence: 2 per 100 000 per year, with a prevalence of 6 per 100000. Clusters and conjugal cases have been reported.

Familial ALS accounts for 5% of cases and is usually inherited as a dominant trait.

Sex ratio: male/female – 1.5:1

Mean age of onset – 55 years.

Mean survival – 3 years (50%).

Pathology

Naked eye: Thinning of anterior roots of spinal cord. Most noticeable in cervical and lumbosacral regions.



Microscopic: Loss of neurons in motor cortex.

Loss of neurons in cranial nerve nuclei and anterior horns.

Section of brain stem: reduction of corticobulbar and corticospinal fibres.

No evidence of inflammatory response is seen in involved structures.

MOTOR NEURON DISEASE/ALS

AETIOLOGY

The cause of motor neuron disease is unknown. Several possibilities have been suggested:

- **Genetic:** Mutations in the SOD1 gene (responsible for producing the enzyme superoxide dismutase) are found in 20% of familial cases of ALS. Superoxide dismutase is important in removing toxic superoxide radicals and converting them into non-harmful substances. Defects in the enzyme lead to accumulation and anterior horn cell death. These account for about 2% of patients with ALS.
- **Viruses:** Chronic virus infection has been proposed, partly because neurotropic viruses such as polio have a devastating effect on anterior horn cells. However, no serological or virological evidence for any infection has been found.
- **Toxins:** No evidence of toxic cause has been demonstrated.
- **Minerals:** Clinical similarities between MND and neurological involvement in hyperparathyroidism and phosphate deficiency suggest a relationship with chronic calcium deficiency.

The final common pathway of anterior horn cell death, irrespective of what actually triggers the process, is a complex interaction of genetic factors, oxidative stress and glutamate excess (excitatory injury). Abnormal clumps of proteins (neurofilaments) can be found in motor neurons that may themselves be toxic or by-products of overwhelming cell injury.

CLINICAL FEATURES *At onset:*

Asymmetric weakness and wasting of extremities – 75%

Bulbar or pseudobulbar features – 25% – dysphagia or dysarthria

In both limb-onset and bulbar-onset disease the key feature is the mixture of upper and lower motor neuron involvement with normal sensation.

Frontal lobe involvement

Frontal dementia occurs in 3–5% of all patients, but is more prevalent in familial cases. *Emotional lability* – unprovoked outbursts of laughing or crying occur.

Limb-onset disease

Limb-onset ALS results from involvement of corticospinal tracts and anterior horn cells.

Signs of corticospinal tract degeneration lead to:

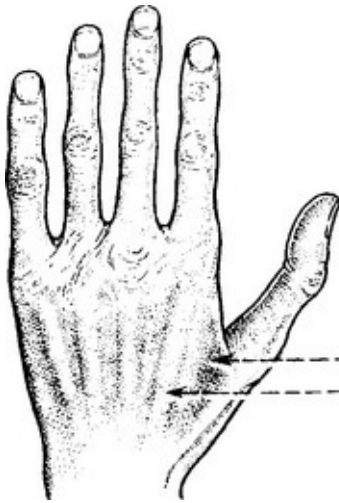
- increased tone
- brisk reflexes
- extensor plantar responses
- distinctive distribution of weakness (extensors in upper limbs; flexors in lower limbs). Spasticity is rarely severe (intact extrapyramidal inhibition). Primary lateral sclerosis is a slowly progressive form of MND restricted to the cortical spinal tract.

Anterior horn cell involvement leads to muscle atrophy, weakness and fasciculations.

The patient may be aware of fasciculation.

Muscle cramps are common. Weakness is not as severe as the degree of wasting suggests.

In the hand: wasting is evident.



1st dorsal interosseous muscle and tendons become prominent as hand muscles waste, giving 'guttered' appearance – SKELETON HAND.

MOTOR NEURON DISEASE/ALS

Bulbar-onset disease = Progressive bulbar palsy

Progressive bulbar palsy presents with a combination of corticobulbar degeneration and lower cranial nerve motor nuclei involvement.

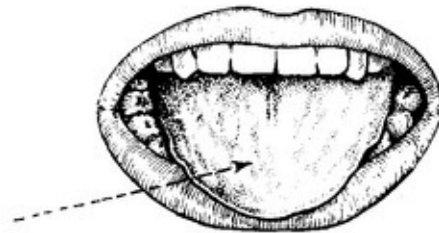
Degeneration of corticobulbar pathways to V, VII, X, XI and XII cranial nerve motor nuclei (with sparing of III, IV and VI) leads to an apparent weakness of the muscles of mastication and expression, the patient has difficulty in chewing and the face is expressionless. The jaw jerk ([page 15](#)) is exaggerated.

Food and fluid enter nasopharynx when swallowing – palatal weakness (X).



← Gag reflex is brisk when soft palate is stimulated.
Speech is drawling and monotonous.
Swallowing for solids is difficult (X).
Tongue is slow moving or immobile, pointed and cannot protrude (XII).

Degeneration of the lower cranial nerve nuclei leads to *atrophy* and *fasciculations* are present in cranial nerve innervated muscles.
Fasciculations are visible muscle twitches which occur spontaneously and represent groups of discharging motor units.
The tongue appears wasted and folded; fibrillations produce a writhing appearance.
Rarely the motor neuron disease can present with purely lower motor neuron involvement, when the progression tends to be slower.



As the disease progresses, all levels of the motor system become involved. Patients with limb-onset develop bulbar symptoms and vice versa. Respiratory muscle weakness ultimately occurs and is the usual cause of death.

Less common clinical presentations

Occasionally patients can present with:

- breathlessness from respiratory muscle failure
- repeated chest infections from occult aspiration or
- weight loss.

Uncommon clinical variants

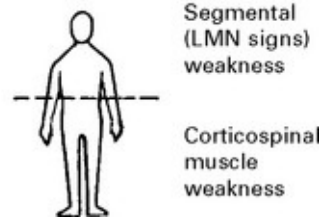
Primary lateral sclerosis is a very slowly progressive purely upper motor neuron syndrome that presents with asymmetrical spasticity.

'Flail arm' variant is when there is marked weakness and wasting of the arms with only modest weakness in the legs. This generally progresses more slowly.

Differential diagnosis includes disorders which produce combined upper and lower motor neuron signs, e.g.

Cervical spondylosis

Spinal tumours.



Hexosaminidase deficiency (autosomal recessive disorder) may mimic ALS.

An ALS like syndrome can occur with *elevated serum paraproteins*, *lymphoproliferative disease*, *lead poisoning* and *HIV infection*.

Hyperthyroidism and *hyperparathyroidism* produce muscle wasting and hyperreflexia.

Pseudobulbar palsy, a pure upper motor neuron deficit reflecting corticobulbar involvement, may result also from cerebrovascular disease or multiple sclerosis.

Progressive muscular atrophy may be confused with a spinal muscular atrophy, multifocal motor neuropathy with conduction block, limb girdle dystrophy, diabetic amyotrophy or lead neuropathy.

N.B. IN MOTOR NEURON DISEASE:

- *Sensory signs do not occur*
- *Bladder is never involved*
- *Ocular muscles are never affected.*

Investigations

EMG reveals denervation with fibrillation.

Nerve conduction studies shows normal velocities and exclude in all limbs multifocal neuropathy with conduction block.

MRI (or myelography) where appropriate excludes foramen magnum or spinal cord compression.

Thyroid and calcium studies exclude endocrine or metabolic disease.

In selected cases screen for paraproteinaemia, lymphoreticular disease and hexosaminidase deficiency.

Diagnostic criteria (El Escorial criteria for MND/ALS – World Federation of Neurology)

Presence of –

LMN signs in at least 2 limbs.

UMN signs in at least 1 region (bulbar/cervical/lumbosacral)

Progression of disease.

Absence of –

Sensory signs.

Neurogenic sphincter disturbance.

Other clinically evident CNS/PNS disease.

Exclusion of ALS-like syndromes

TREATMENT

Treatment is primarily that of managing symptoms and supporting both patient and family as these progress and their needs change.

Counselling is essential to a full understanding of the illness and its

natural history. Support from a Nurse Specialist is invaluable to meeting the challenges of each phase of illness and issues of feeding and methods of ventilatory support are best discussed well in advance so that informed decisions can be made. The comprehensive care of patients is challenging with medical, legal and ethical considerations.

Symptomatic treatment:

Anarthria and dysarthria: – Speech assessment and communication aids when indicated.

Dysphagia and aspiration: – Percutaneous endoscopic gastrostomy (PEG).

Nutrition: – Estimate calorific content and supplement diet with vitamins.

Muscle weakness: – Physiotherapy, walking aids. Splints, *etc.*

Respiratory failure: – As vital capacity drops respiratory failure becomes inevitable. Non-invasive ventilatory assistance should be considered when this falls below 75% or orthopnoea develops in patients without severe bulbar involvement. Recent trials indicate this can provide improvements to quality of life. The role for invasive mechanical ventilation is more uncertain. Rarely ALS can present with early respiratory failure before treatment issues have been discussed. This creates a major management dilemma.

Disease-modifying treatment

Riluzole is a drug with energy buffering and anti-glutamate properties. It is the only approved treatment and in a dose of 100 mg daily is safe with a marginal effect in prolonging survival by 2 months.

INHERITED MOTOR NEURON DISORDERS

SPINAL MUSCULAR ATROPHIES (SMAs)

Spinal muscular atrophy is the second most common fatal, autosomal recessive disease in Caucasians (after cystic fibrosis). The disorder is characterised by degeneration of the anterior horn cells and symmetrical muscle weakness and wasting.

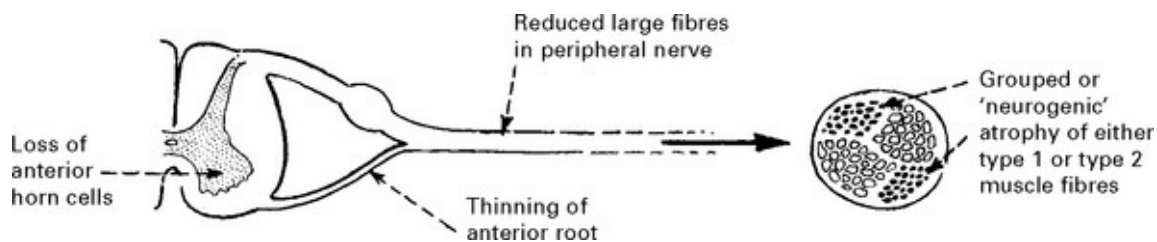
Depending on the age of onset, degree of muscular involvement and length of survival, 3 types of recessive SMA are recognised: All map to the gene locus 5q12.2-q13.3.

With an incidence of 1/10000, the offspring of patients have a disease risk of approximately 1%.

Type I – Werdnig Hoffman disease (Acute Infantile SMA)

This is an autosomal recessive disorder.

Incidence 1:25 000 births



Clinical features:

Reduced fetal movements in late pregnancy with weakness and hypotonia at birth.

Swallowing and sucking are impaired
The child lies with arms and legs abducted and externally rotated (hypotonic posture)
Contractures, wasting and fasciculation gradually become evident

} different from other causes of 'floppy' infant.

All motor milestones are delayed; 95% of all patients are dead by 18

months.

Type II – Kugelberg Welander disease (Late infantile or juvenile SMA)

Pathological features similar to Werdnig Hoffman disease.

Clinical features:

Limb girdle muscles affected.

It is slowly progressive with great variability even within the same family. Median age at death 12 years. Survival to adulthood occurs in the dominant form.

Type III (Adult onset SMA)

Onset between 2nd and 5th decade with progressive limb girdle weakness. Distinction from progressive muscular atrophy form of ALS is difficult. A benign course supports the former.

Distal and scapuloperoneal forms

Differentiation from CMT types I and II ([page 444](#)) and scapuloperoneal dystrophy ([page 470](#)) is clinically difficult and separation may only be possible on histological and neurophysiological grounds.

Spinal and bulbar muscular atrophy (Kennedy's syndrome)

X-linked adult-onset neurogenic muscular atrophy with late distal and bulbar involvement (Gene Locus: Xq11-q12). Onset of fasciculations followed by muscle weakness and wasting occur at approximately 40 years of age. Bulbar signs and facial fasciculations are characteristic. Babinski sign is negative. The disorder is compatible with long life.

Management of spinal muscular atrophies

There is no specific treatment. Care is supportive. Genetic counselling is

essential.

NEUROCUTANEOUS SYNDROMES

Previously called Phakomatoses – Phakos Greek: birthmark

These disorders are hereditary, characterised by multiorgan malformations and tumours. The literature includes many varieties of such conditions; most are extremely rare. Only the more major disorders are described below.

NEUROFIBROMATOSIS

Two distinct types occur:

Type 1 (NF1)

Characterised by café au lait spots and neurofibromas (Von Recklinghausen's disease).

Incidence: 1:4000

Inheritance: Autosomal dominant
Neurofibromin defect at 17q11

Type 2 (NF2)

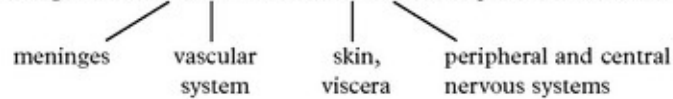
Characterised by tumours (schwannomas) of the eighth cranial nerve (vestibular division).

1:50 000

Autosomal dominant
Merlin defect at 22q.12

Pathology (type 1):

An embryological disorder in which localised overgrowth of *mesodermal* or *ectodermal* tissue produces tumours of:



Pathology (type 2): See page 399

Clinical features (type 1):

Skin manifestations: – Café au lait spots:

light brown patches on the trunk with well demarcated edges.

Subcutaneous neurofibromata lying along peripheral nerves and enlarging with age.

Mollusca fibrosa: cutaneous fibromas – large, pedunculated and pink in colour.

Plexiform neuroma: diffuse neurofibroma associated with skin and subcutaneous overgrowth and occasional underlying bony abnormality.

Skeletal manifestations: – 50% of patients exhibit scoliosis.

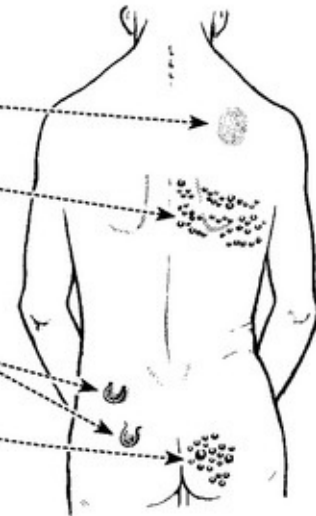
Subperiosteal neurofibromas may give rise to bone hypertrophy or rarification with pathological fractures.

Sphenoid wing dysplasia is a rare but diagnostic abnormality.

Ocular: – Lisch nodules are melanocytic hamartomas of the iris and are seen on slit-lamp examination in 90% of patients

Neoplasia: – A high incidence of leukaemia, neuroblastoma, medullary thyroid carcinoma, and multiple endocrine neoplasia occurs.

Neurological manifestations: – Mental retardation and epilepsy occur in 10–15% of patients without intracranial neoplasm.



Cerebrovascular accidents as a consequence of intimal hyperplasia are not uncommon. Three patterns of neurological neoplasia are recognised:

1. <i>Intracranial neoplasms:</i> Optic nerve glioma Multiple meningioma.	2. <i>Intraspinal neoplasms:</i> Meningioma Neurofibroma Glioma.	3. <i>Peripheral nerve neoplasms:</i> Neurofibroma – a proportion of which become sarcomatous.
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Clinical features (type 2)

Skeletal manifestations are absent. Café au lait spots rare. Posterior subcapsular cataracts occur in 50% of cases. The condition is defined by bilateral vestibular schwannomas but may present as early unilateral acoustic neuroma plus a family history of NF2. Other intracranial and intraspinal neoplasms occur.

Diagnosis

A family history is obtained in over 50% of patients. In type 1, the cutaneous manifestations are characteristic, though they may be extremely mild with only café au lait spots (more than 6 in an individual is diagnostic). As a rule, the more florid the cutaneous manifestations the less likely is there nervous system involvement. CT scanning, MRI and myelography may be necessary when nervous system involvement is suspected. Type 2 is diagnosed when imaging (MRI) confirms bilateral vestibular schwannomas. The recent cloning of the type 2 gene to chromosome 22 may lead to direct gene testing in persons at risk.

Treatment

Plexiform neuromas may be removed for cosmetic reasons. The management of intracranial and intraspinal tumours has already been discussed.

TUBEROUS SCLEROSIS

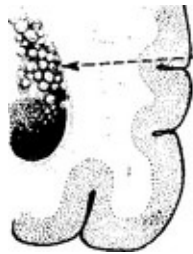
Incidence: 1:30 000.

Autosomal dominant inheritance with high sporadic mutation rate. TSC1 is caused by a mutation on chromosome 9 in the hamartin, and TSC2 on chromosome 16 in tuberin.

Characterised by cutaneous, neurologic, renal, skeletal, cardiac and

pulmonary abnormalities.

Pathology



An embryological disorder.

Hard gliotic 'tubers' arise anywhere within the hemisphere but commonly around the ventricles. Projection into the ventricles produces a typical appearance like 'dripping candle wax'.

Tubers in the brain result from astrocytic overgrowth with large vacuolated cells and loss of surrounding myelin.

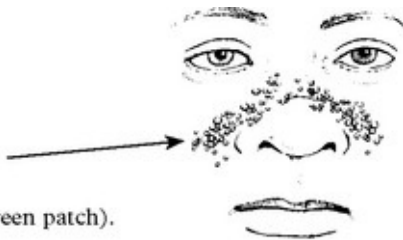
Transition may occur from gliosis to a subependymal astrocytoma.

As well as skin lesions, primitive renal tumours and cystic lung hamartomas occur.

Clinical features

Skin manifestations

The cutaneous lesions are characteristic – adenoma sebaceum, a red raised papular-like rash over the nose, cheeks and skin, appears towards the end of the 1st year, though occasionally as late as the 5th year. Depigmented areas on the trunk resembling vitiligo are common (Shagreen patch). Fibromas and café au lait spots occur occasionally. Teeth are pitted.



Neurological manifestations: – Mental retardation is present in 60% of patients, though the onset and its recognition may be delayed.

Seizures occur in almost all patients, often as early as the 1st week of life. Attacks are initially focal motor and eventually become generalised. The response to anticonvulsants is variable.

Intracranial neoplasms – astrocytomas – arise from tubers usually close to the ventricles and may result in an obstructive hydrocephalus.

Neoplasia: – Renal carcinoma occurs in 50% of patients. Retinal tumours (hamartomas) and muscle tumours (rhabdomyomas) are

common, the latter often involving the heart.

Diagnosis:

The presence of epilepsy and adenoma sebaceum is diagnostic.

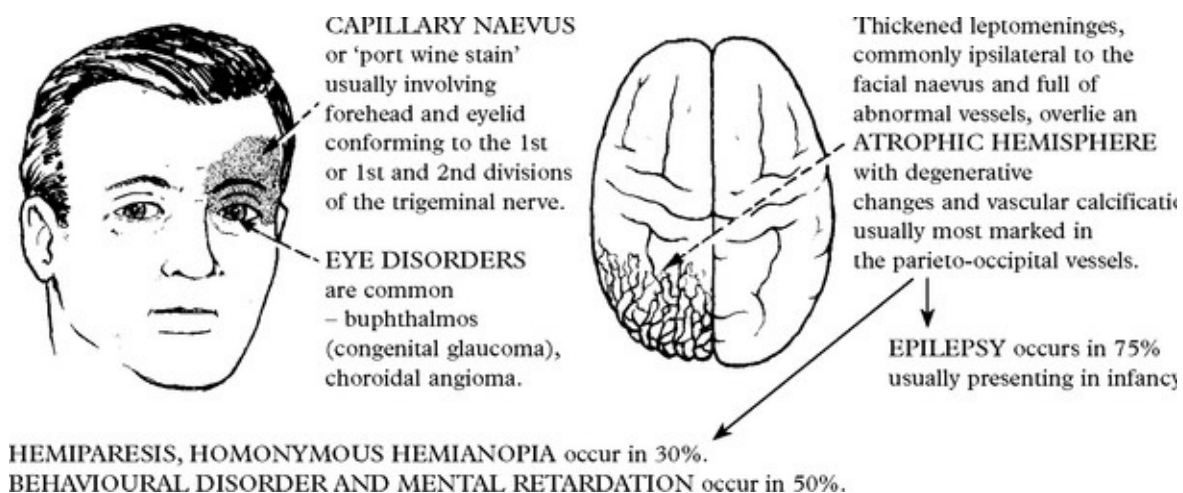
CT scan may show subependymal areas of calcium deposition. MRI shows uncalcified subependymal tubers. Other developmental abnormalities may be evident, e.g. microgyria.

Treatment:

Anticonvulsant therapy for epilepsy. Surgical removal of symptomatic lesions. High mutation rate indicates that antenatal diagnosis will not significantly reduce incidence.

STURGE-WEBER SYNDROME

This disorder is characterised by a facial angioma associated with a leptomeningeal venous angioma. There is no clear pattern of inheritance. Practically all cases are sporadic.



Skull X-rays show parallel linear calcification (tram-line sign) and CT scan, in addition, shows the associated atrophic change. Angiography demonstrates dilated deep cerebral veins with decreased cortical

drainage. Arteriovenous and dural venous sinus malformations are present in 30%.

Treatment

Intractable epilepsy may require lobectomy, or even hemispherectomy. Some recommend early excision of the surface lesion, but the rarity of the condition prevents thorough treatment evaluation.

VON HIPPEL-LINDAU (VHL) DISEASE

An autosomal dominant disorder due to mutations in VHL gene on chromosome 3 where haemangioblastomas are found in the cerebellum, spinal canal and retina, and are associated with various visceral pathologies:

- Renal angioma
- Renal cell carcinoma
- Pheochromocytoma
- Pancreatic adenoma/cyst
- Cysts and haemangiomas in liver and epididymis.

Mutation in a tumour suppressor gene is found in 60% of affected families. Any of the above may produce signs and symptoms.

Retinal haemangioblastoma is seen on fundoscopy and may produce sudden blindness. These often produce the earliest clinical manifestation of disease. Confirm with fluorescein angiography and treat with cryosurgery or photocoagulation.



Cerebellar haemangioblastoma presents with progressive ataxia.

Compression of the fourth ventricle may cause hydrocephalus with a subsequent rise in intracranial pressure.

Spinal canal haemangioblastoma – intradural or intramedullary lesion presenting with signs and symptoms of cord or root compression.

Diagnosis is established from family history and cranial imaging (MRI or CT). Renal ultrasound, abdominal CT and urinary amine estimations are required to complete the evaluation. In patients at risk, regular screening for renal, adrenal, pancreatic and intracranial tumours is recommended.

ATAXIA TELANGIECTASIA – see [page 552](#).

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